

Connectivity Bounds in Graphs and Beyond

Erdős Problem 915, from Proof to Search

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Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work is told as a journey. It begins with the variants whose answers can be proved cleanly by hand, watches those proofs grow longer and more delicate as the model changes, and at the point where a single argument no longer suffices it turns to a unified program. That program holds all twelve variants behind one common interface, so a single piece of code serves them all. It measures connectivity exactly with a maximum-flow routine, the standard way to count the independent routes between two points. It runs a prover that sweeps every graph of a small fixed size and, when none beats a target, reports that sweep as a genuine upper-bound proof. And it runs a search that, like hot metal cooling slowly into shape, settles random graphs toward the densest ones the rules allow.

I am grateful to my supervisor, Prof. Stijn Cambie, and my daily advisor for their guidance, corrections, and mathematical feedback throughout the project. One review in particular changed the shape of this thesis: Prof. Cambie put a late draft through a language model different from the one I had been working with, and it came back with a misapplied inequality I had read past many times, together with the counting idea that grew into [Proposition A.22](#) and [Theorem A.24](#). I thank my promotor, Prof. Jan Goedgebeur, whose suggestion of the `nauty` generation pipeline sharply accelerated the directed enumeration. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Above all I thank my girlfriend Sara, whose patience and encouragement carried me through the long stretches of this work. Any remaining mistakes are my own.

Contribution Statement

This thesis is my own work. I carried out the literature study, recast the twelve variants of the problem in one common language, and designed and wrote the single program that runs through all of them. I built every example, by hand or by computer search, checked each one with that program, wrote the proofs that appear in the thesis, and typeset the whole document. Where a result comes from the literature I say so and cite it. The theorems I lean on but do not claim as mine are those of Mader, Leonard, Sørensen and Thomassen, Menger, Gomory and Hu, and Baranyai, the hypergraph connectivity results of Bérczi, Chandrasekaran, Király and Kulkarni, and the standard probabilistic tools used for the random graphs.

The program has three jobs, and it keeps them separate. It *measures*: given any graph, it computes exactly, through maximum flow, how many independent routes the busiest pair of vertices has. It *proves*: for a fixed number of vertices it sweeps a finite space of graphs, and when nothing in that space beats a target, that sweep is a genuine upper-bound proof rather than a guess. And it *discovers*: a search that cools like a heated metal settling into shape wanders the space of graphs and finds dense ones, which give lower bounds and suggest conjectures but never prove an upper bound by themselves. Every general statement in the thesis rests on a proof by hand or a cited theorem. The computer searches are evidence that points the way, never a replacement for the argument. I used AI language models, including Claude Code, ChatGPT, and Gemini, as tools for literature search, L^AT_EX editing, proofreading, and trying out proof drafts. I checked every mathematical statement and proof in the thesis myself and take full responsibility for the content.

One thing about that use is worth recording, because it is a methodological point and not just a disclosure. A model that has helped write an argument is a poor reviewer of it: it reads the text the way its author does, follows the same intended reading of each sentence, and slides over the same gaps. Handing the finished draft to a *different* model, cold, behaved much more like handing it to a different person. It read each sentence as a standalone claim rather than as a step in an argument it already believed, and that is exactly how it caught a use of Whitney's inequality that runs the wrong way, in a sentence that had survived several careful readings by me and by the model that helped write it. The same pass also proposed a counting argument I had not tried. It was wrong about a good deal else, and every item it raised had to be checked against the source before anything was changed, several of them turning out to be objections to conventions the thesis states rather than to errors. But the pattern is the familiar one from working with people: the value came from the disagreement, not from the agreement, and from the fact that the second reader had no stake in the first reader's phrasing.

The genuinely new results are the following. For hyperedge networks I settle the vertex-disjoint problem completely at the two strictest route limits, $m = 2$ and $m = 3$, for every hyper-

edge size r and every number of vertices n (Theorems A.45 and A.48): the vertex-disjoint and edge-disjoint answers turn out to be equal, both $\lfloor (m-1)(n-1)/(r-1) \rfloor$. The $m = 3$ case rests on a bound for incidence graphs that I prove using Tutte's decomposition of a graph into its three-connected pieces (Lemma A.47). For one-way multigraphs at $m = 3$ I prove an unconditional attachment lemma (Lemma A.34) and an odd-step theorem (Theorem A.36, Remark A.39) that together shrink the whole problem, for every n , down to two finite statements about multigraphs on just seven vertices that a computer can check, one of which the accompanying program has already verified outright, and they do so without assuming the unproved minimum-degree conjecture the earlier route needed. For one-way simple graphs I prove an unconditional quadratic bound (Proposition A.22 and Theorem A.24): under any route limit m , such a graph has at most $\lfloor n^2/4 \rfloor + 4(m-1)(n-1)$ arcs, and deleting a comparable number of arcs always leaves a one-way bipartite graph. This fixes both the leading constant and the order of the second-order term of the standing conjecture (Conjecture 1.14) without any assumption about the shape of the extremal graph, which is what every previous route to that bound needed, and leaves only a constant factor of about eight in one coefficient. The counting idea behind it came out of the external review recorded in the foreword, and I verified it and worked out the sharpening myself. For multigraphs under the alternative counting convention I settle the vertex problem at $m = 2$ and disprove the natural guess, suggested by the data through $m = 4$, that a thickened tree stays optimal for all larger n : a bouquet of thickened cliques sharing a single vertex (Theorem A.56) beats it by a margin growing linearly in n , at a rate that is quadratic rather than linear in m (Section A.8).

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [1], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\}.$$

Mader proved that, for simple graphs and $n \geq m \geq 2$,

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, and hypergraphs), the path-separation notion (edge-disjoint or internally vertex-disjoint), and the presence or absence of directions. Sampled random graphs $G(n, p)$ run alongside throughout as a lens on the typical case rather than as a separate model.

The thesis is organised as a journey from hand proofs to computation. The early variants are settled rigorously: the multigraph value $L_m(n) = (m-1)(n-1)$, the random-graph threshold $p^* = m/n$, the equality of the edge and vertex problems for $m \leq 4$, and the first divergence at $m = 5$, where Sørensen and Thomassen prove $k_5(n) = \lfloor 8n/3 \rfloor - 3$. The directed case forces the turn: the value $\ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved by a long induction, and for $m \geq 3$ the hand proof gives way to a conjecture, after an explicit ten-vertex graph rules out the naive guess. The centrepiece is then a unified program. One multiplicity-matrix representation models almost all twelve variants, an exact maximum-flow checker measures local connectivity, a cut-counting optimisation certifies upper bounds for small sizes, and a search guided by edge sensitivity, run as both simulated annealing and tabu search, discovers extremal graphs. The program rediscovers, each in its own variant, the double star (undirected multigraphs at $m = 2$), the directed hub, the one-directional bipartite digraph, and the augmented-bipartite construction (the directed cases), certifies the directed multigraph values $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ for $n \leq 6$, and frames the remaining open problems on the directed frontier.

The interplay then pays back in the other direction. The hypergraph vertex problem is settled completely at $m = 2$ and $m = 3$: $k_m^{(r)}(n) = \lfloor (m-1)(n-1)/(r-1) \rfloor$ for every uniformity r , the $m = 3$ case through a new rank bound on incidence graphs proved by way of Tutte's triconnected decomposition. On the directed side, an unconditional attachment lemma and a conditional odd-step theorem reduce the entire directed multigraph problem at $m = 3$, value and extremal characterisation for all n , to two finite, machine-checkable statements about multigraphs on seven vertices, one of which the program has since verified outright, and directed hypergraphs

receive their first bounds, quadratic in n .

Abbreviations and Symbols

GH	Gomory-Hu
MILP	mixed-integer linear program
SA	simulated annealing
SPQR	the triconnected decomposition tree (series, parallel, rigid pieces)
$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m - 1$
r	hyperedge size (every hyperedge spans r vertices)
$\lambda_G(u, v)$	maximum number of edge-disjoint u - v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u - v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\text{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\text{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\text{dir}}(n)$	simple directed vertex-disjoint extremal function
$\ell_m^{(r)}(n), k_m^{(r)}(n)$	r -uniform hypergraph edge- and vertex-disjoint extremal functions
$M^*(n)$	the m -free directed multigraph optimum, $L_m^{\text{dir}}(n) = (m - 1) M^*(n)$
$B_{p,q}$	one-directional complete bipartite digraph, all arcs from a p -part to a q -part
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$\sigma(e)$	edge sensitivity $\lambda^{\max}(G) - \lambda^{\max}(G - e)$
$G(n, p)$	binomial random graph

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Chapter 1

The Problem and Its Base Cases

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability, and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment, and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter builds the question from nothing, settles the variants whose proofs are short enough to read in an afternoon, and then follows the same question along three axes until the proofs stretch past breaking. By the end the case for a machine has made itself.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *graph* G is a finite set of points, called *vertices*, together with a set of links, called *edges*, each joining two of the vertices. We write $V(G)$ for the set of vertices and $E(G)$ for the set of edges, so that $|E(G)|$ is the number of edges, the very quantity this thesis tries to make as large as possible. One pictures the vertices as dots and the edges as lines between them.

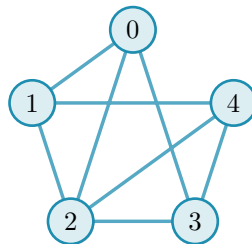


Figure 1.1. A simple graph on five vertices. It is K_5 , the *complete* graph in which every one of the $\binom{5}{2} = 10$ pairs is joined by an edge (the binomial coefficient $\binom{5}{2}$ counts the ways to choose two of the five vertices), with the two edges $\{0, 4\}$ and $\{1, 3\}$ removed, so its edge set $E(G)$ holds eight edges. The five dots are the vertex set $V(G)$ and the lines are the edges. Because at most one line joins any given pair and no edge loops back to its own endpoint, this is a simple graph. The same graph returns in [Figure 1.2](#), once we can count the independent routes between a pair.

A graph is the bare skeleton of a network: the vertices are the towns or the devices, and the edges are the roads or the links between them. We write n for the number of vertices, and the *degree* of a vertex is the number of edges meeting it. Unless we say otherwise our graphs are *simple*, meaning that at most one edge joins any given pair of vertices and its endpoints are distinct.

A *path* from a vertex u to a vertex v is a sequence of distinct vertices that starts at u , ends at v , and joins each consecutive pair by an edge. It is the formal version of a route through the network. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both, so that the loss of a single edge can break at most one of them. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u, v)$, the count of genuinely independent routes the network offers to that pair. The pair with the most independent routes sets the ceiling for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v),$$

the largest local edge-connectivity anywhere in G . A theorem of Menger [2], recalled in [Appendix A](#), gives this quantity a second face: $\lambda_G(u, v)$ also equals the least number of edges one must remove to cut every route from u to v ([Figure 1.2](#)). Many independent routes between a pair is exactly the same thing as a costly cut between them.

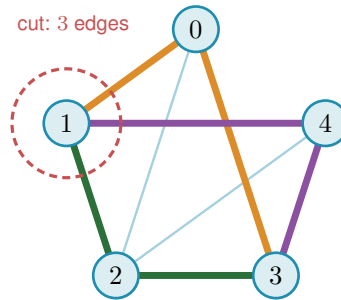


Figure 1.2. Menger’s theorem on the graph of [Figure 1.1](#). The pair 1, 3 carries no edge of its own, yet three routes still join them with no shared edge, 1-0-3 (orange), 1-2-3 (green), and 1-4-3 (purple), each through a different middle vertex, so $\lambda(1, 3) = 3$. The three edges meeting vertex 1 (inside the red dashed ring) are a smallest set whose removal cuts every 1-3 route, and no set of just two edges can cut them all, because the three routes share no edge, so each removed edge breaks at most one route and two removals leave one route intact. The cheapest cut therefore also has size three. The most independent routes a pair can muster always equals the cheapest cut between them, the fact every bound in this thesis leans on.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a ceiling on connectivity. Requiring $\lambda_G^{\max} \leq m - 1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be severed by removing fewer than m edges. The question is how many edges a graph can hold while staying under that ceiling.

Question 1.1 (Erdős Problem 915 [3]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\},$$

the problem asks for $\ell_m(n)$, the most edges possible under the ceiling. Mader settled the simple-graph case completely, proving both halves at once: no graph under the ceiling has more than $\lfloor m(n-1)/2 \rfloor$ edges (the upper bound), and some graph reaches that count (the lower bound). The brackets $\lfloor \cdot \rfloor$ round their content down to the nearest whole number, and their ceiling counterpart $\lceil \cdot \rceil$ rounds up. Both recur throughout, because an edge count is always an integer while the formulas that bound it need not be.

Theorem 1.2 (Mader [4]). *For all $n \geq m \geq 2$,*

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Mader's own proof is a direct structural induction. The proof given in full in [Appendix A](#) takes a different road, by way of the Gomory–Hu tree, found independently of Mader, together with a characterisation of the graphs that attain equality ([Theorem A.11](#)). This is the road the thesis adopts throughout, because the very same tree returns for the multigraph and hypergraph variants ([Theorem 1.3](#), [Proposition 1.15](#)), so a single argument serves all of them rather than each needing its own. Every graph has such a tree ([Theorem A.2](#)). A Gomory–Hu tree of G is a weighted tree on the same vertex set in which $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v . We single out this one identity because the same idea reappears when the program of [Chapter 2](#) stores connectivities as a tree (a graph without cycles). All $\binom{n}{2}$ local connectivities are therefore encoded by $n-1$ numbers, and $\lambda_G^{\max}$ is simply the heaviest edge of the tree.

The whole argument ([Appendix A](#)) is one act of double counting on that tree. The word *charged* below is just accounting: to charge an edge to a tree edge is to assign it there for the count, and the proof works by charging every edge of G to tree edges and then adding the charges up. Each of the $n-1$ tree edges stands for a minimum cut of G , and an original edge of G is charged once for every tree edge lying between its endpoints, so summing the tree-edge weights counts every edge of G at least once and, crucially, counts most of them at least twice. The only edges charged a single time are those joining tree-adjacent vertices, and a *simple* graph has at most $n-1$ of those, one per tree edge. Every other edge is charged twice. Since each tree weight is a local connectivity and so at most $m-1$, the two counts meet at $2|E(G)| - (n-1) \leq (m-1)(n-1)$, which rearranges to the floor. The factor of two, and with it the gap between Mader's $\lfloor m(n-1)/2 \rfloor$ and the larger value $(m-1)(n-1)$ that parallel edges will buy in [Theorem 1.3](#), is exactly the dividend of simplicity, the one place the argument uses that no pair carries a parallel edge.

One example fixes the scale. The complete graph K_n joins every pair, so every pair has $n-1$ routes that share no edge, $\lambda(u, v) = n-1$, while it carries all $\binom{n}{2}$ edges. It therefore meets the ceiling $\lambda^{\max} \leq m-1$ exactly when $m \geq n$, and from that point on the question is settled trivially: the complete graph is feasible, no simple graph on n vertices is denser, so $\ell_m(n) = \binom{n}{2}$ for every $m \geq n$ and raising m further changes nothing. For smaller m the extremal graph must be strictly sparser than complete, and pinning down exactly how much sparser is the whole problem. That

is why the theorem, and the thesis, assume $m \leq n$.

1.3 Three axes of variation

Mader's theorem is the floor we build on, and everything afterwards is a variation on it. There are three independent ways to change the rules, and they generate the landscape this thesis traverses.

The first axis is the *model*, drawn in Figure 1.3. Two independent choices generate it. The first is *arity*: an edge may join exactly two vertices, or a *hyperedge* may join any $r \geq 2$ of them, a route then passing through it between any two of its members. At $r = 2$ a hyperedge is an ordinary edge, so graphs are the $r = 2$ special case and the genuinely new setting begins at $r \geq 3$. The second is *multiplicity*: a *simple* object carries at most one copy of each edge, while a *multi* object allows parallel copies, recorded by a multiplicity $\mu(u, v)$, so a single pair can already carry several independent routes. Because the two choices are independent they form a small lattice. At the top sits the most general object, the *multi-hypergraph*, and lowering one toggle at a time restricts it downward: forbidding repeats gives the simple hypergraph, dropping the arity to two gives the multigraph, and doing both gives the simple graph at the bottom. Every theorem in this thesis is a statement about one node of this lattice, and the three rows of every comparison figure are its simple-graph, multigraph, and hypergraph levels.

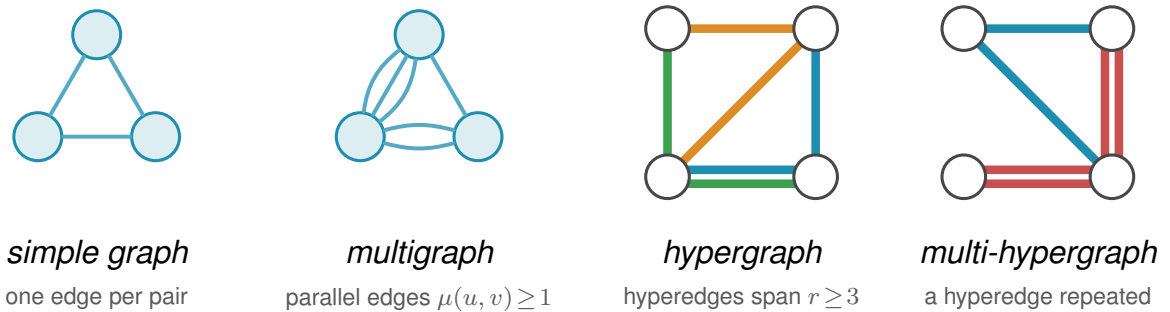


Figure 1.3. The first axis of variation, the *model*, shown for the four object classes this thesis uses. In a *simple graph* each pair of vertices is joined by at most one edge. A *multigraph* permits parallel edges, here a triple and a double, so the multiplicity $\mu(u, v)$ counts how many copies join a pair. A *hypergraph* replaces pairwise edges by hyperedges, each a set of $r \geq 2$ vertices ($r = 2$ gives back ordinary edges, and the depicted examples use $r = 3$), drawn the way a metro map draws a line, threading all of its stations, one colour per hyperedge. Where two hyperedges share a stretch between stations the two lines run side by side, one colour each, exactly as a metro map separates lines along a shared stretch of track. A *multi-hypergraph* lets a hyperedge be repeated, drawn as two parallel lines through the same stations, the hypergraph counterpart of parallel edges. When all hyperedges have the same size r the hypergraph is r -uniform. The same connectivity question, how many independent routes a pair can be forced to share, is asked of all four.

The second axis is the *direction*, a third toggle independent of the other two. Setting it at the top of the model lattice gives the single most general object in the thesis, the *directed multi-hypergraph*, of which every other class is a restriction. Figure 1.4 redraws the four models of Figure 1.3 with arcs in place of edges. An undirected object is the symmetric special case of a directed one, each edge standing for a matched pair of opposite arcs $u \rightarrow v$ and $v \rightarrow u$,

so direction strictly enlarges the family of objects. What it does *not* do is let the directed and undirected problems merge into one. The two opposite arcs offer a round trip the single edge uv does not. This is why the extremal digraphs (directed graphs) are deliberately *lopsided*, with out-degree and in-degree pulled apart in a way no symmetric object can imitate, and why the Gomory-Hu tree that settles every undirected case has no directed analogue (Appendix A). The one-way variants therefore stay distinct in both difficulty and method.

A digraph carries three degree counts at each vertex. The *out-degree* $d^+(v)$ counts the arcs leaving v , and the vertices they reach are its *out-neighbours*. The *in-degree* $d^-(v)$ counts the arcs entering v , and the vertices they come from are its *in-neighbours*. The *total degree* $d(v) = d^+(v) + d^-(v)$ adds the two. A vertex with high out-degree fans arcs outward, one with high in-degree gathers them, and keeping the two balanced is the structural idea behind several of the directed constructions to come. The directed measure itself is easiest to meet on one concrete digraph. Figure 1.5 shows a network with three arc-disjoint routes between the two chosen vertices, which equals the smallest arc-cut that separates them, and the degree bound that limits both, which is exactly how Menger reads in a one-way network.

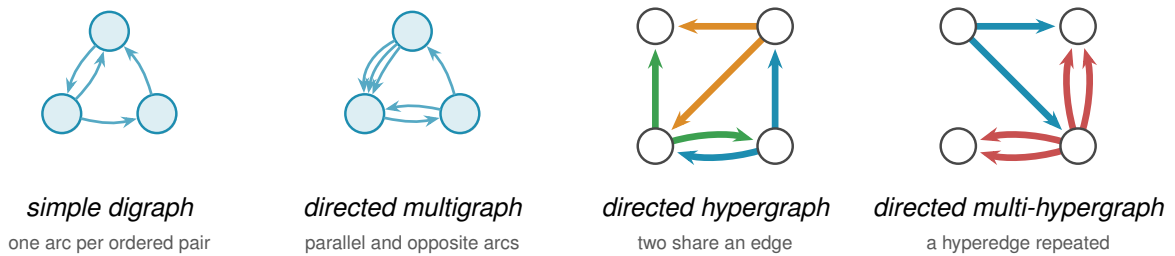


Figure 1.4. The four models of Figure 1.3 redrawn *directed*, the second axis applied to the first. Every edge becomes an arc, and a pair may now carry arcs both ways, the round trip an undirected edge cannot offer. The *simple digraph* still allows at most one arc per ordered pair, so the two-cycle on the top pair is simple. The *directed multigraph* adds parallel arcs, here a triple one way and a bidirected pair. The *directed hypergraph* draws the same three hyperedges as Figure 1.3, in the same colours, each now a metro-styled star, the tail sending one thick coloured arc to each of its heads, the star convention of Figure A.8. Taking each star’s tail to be the middle station of the undirected line keeps every arc on the same edge as before. The blue and green hyperedges still share the bottom edge, now one arc running each way, the directed counterpart of two metro lines sharing a stretch of track. The *directed multi-hypergraph* repeats the red hyperedge, so each of its arcs becomes a parallel pair. This last panel is the directed multi-hypergraph at the top of the model lattice, the most general object in the thesis.

The third axis is the *separation*: routes may be required to avoid sharing an edge, as above, or the stricter *internally vertex-disjoint*, sharing no intermediate vertex, which gives the local vertex-connectivity $\kappa_G(u, v)$ and its maximum $\kappa_G^{\max}$. Whitney’s inequality $\kappa_G(u, v) \leq \lambda_G(u, v)$ [5] records that avoiding shared vertices is at least as demanding as avoiding shared edges, and Figure 1.6 shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex.

Three models, two directions, two separations: twelve versions of one question. The thesis is the attempt to answer as many of them as honest mathematics allows, and to build a machine for the rest. Some answers are short and we prove them here. Some answers are long, and their

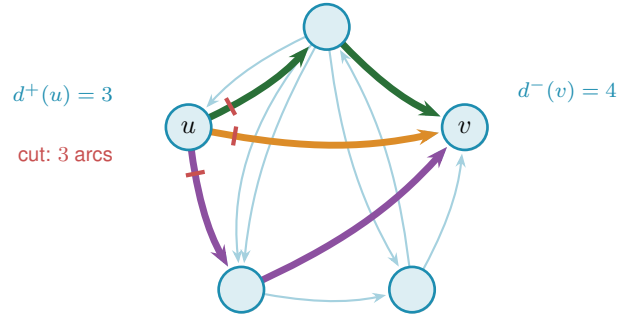


Figure 1.5. A directed multigraph on the same five vertices as Figure 1.1, drawn in the same pentagon. It is not that graph with arrows added: the pair $\{0, 4\}$ is joined here and the pair $\{0, 2\}$ carries two parallel arcs, so this is a directed multigraph rather than a simple graph. Three arc-disjoint directed routes connect u to v (orange, green, purple). The three red marks cut the arcs leaving u , the first arc of each route, so $\lambda(u, v) = 3$ by Menger's theorem. A directed cut counts only the arcs leaving u 's side, so the lone arc entering u from the top is not part of it. The out-degree $d^+(u) = 3$ and in-degree $d^-(v) = 4$ bound the value from above, and $\lambda(u, v) = \min(3, 4) = 3$ meets the smaller one, at u .

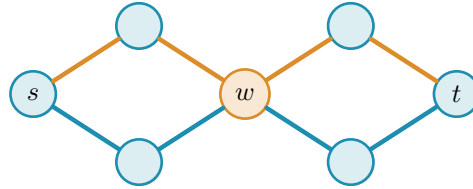


Figure 1.6. The third axis, the *separation*. The two s - t routes drawn here, one orange and one blue, share no edge, so they count as two edge-disjoint paths and $\lambda(s, t) = 2$. Both pass through the single vertex w , however, so they are *not* internally vertex-disjoint, and since every s - t route must cross w we have $\kappa(s, t) = 1$. This is Whitney's inequality $\kappa \leq \lambda$ made concrete: the gap between the two separations is exactly what the vertex-disjoint problem measures, and the rest of the chapter shows it stays shut until $m = 5$.

length is the reason Chapter 2 exists. The rest of this chapter collects the variants whose proofs are short enough to belong in the body, so that the reader meets the easy slope before the climb. Figure 1.7 draws the whole space as one tree, so that the reader has a single map to return to as the chapters visit its branches one at a time.

1.4 Cases with short proofs

Allowing parallel edges removes the parity subtlety, the dependence on whether $m(n-1)$ is even or odd, that produces the floor in Mader's theorem, and the answer becomes exactly linear. Write $L_m(n)$ for the multigraph analogue of $\ell_m(n)$.

Theorem 1.3. *For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.*

Proof. For the lower bound, take a spanning tree of K_n , that is, a connected cycle-free subgraph that touches all n vertices and so has exactly $n-1$ edges, and give every tree edge multiplicity $m-1$, meaning $m-1$ parallel copies. The result has $(m-1)(n-1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. To separate them one must cut some

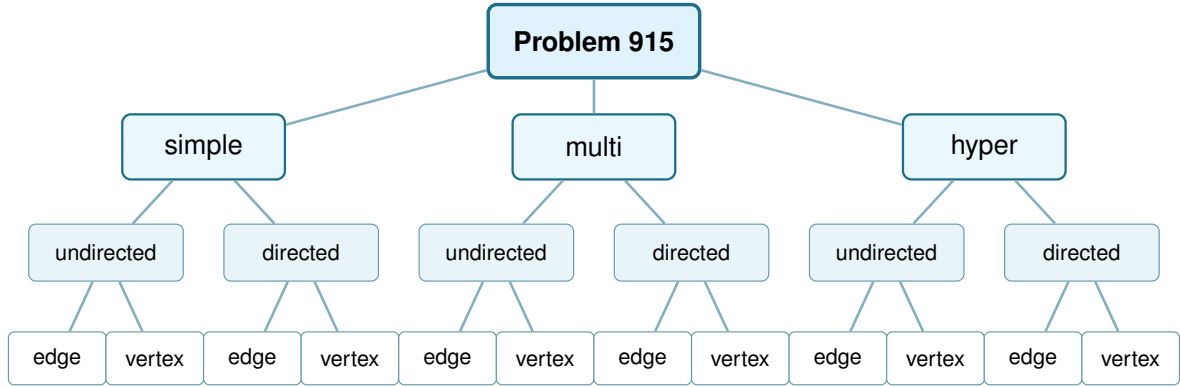


Figure 1.7. The whole space this thesis studies, drawn as one tree. Three model choices (simple graph, multigraph, r -uniform hypergraph), each splitting into undirected or directed, each in turn splitting into edge-disjoint or internally vertex-disjoint routes: twelve leaves, twelve variants. [Figure 4.7](#) redraws this exact tree once every leaf's status is known.

edge of that path, and the cheapest choice is the lightest tree edge on it, which still carries $m - 1$ copies, so $\lambda(u, v) = m - 1$ for every pair and $\lambda^{\max} = m - 1$. The graph sits right at the ceiling.

For the upper bound, suppose $\lambda_G^{\max} \leq m - 1$ and build a Gomory–Hu tree T of G ([Theorem A.2](#)), the same $(n - 1)$ -edge summary of all connectivities used for Mader's theorem. Each tree edge e stands for a minimum cut of G , of capacity $c(e) = \lambda_G(u, v) \leq m - 1$ for the pair (u, v) it represents. Now charge each edge of G to a tree edge: an edge $\{x, y\}$ crosses the cut of the lightest tree edge on the tree path from x to y , so it is counted inside that tree edge's capacity. Summing the $n - 1$ capacities therefore counts every edge of G at least once, which already gives $|E(G)| \leq \sum_{e \in T} c(e) \leq (m - 1)(n - 1)$. Unlike Mader's simple-graph count there is no second charge to halve, because parallel edges are now allowed and nothing forces an edge to be counted twice. The tree-of-multiplicity- $(m - 1)$ construction above meets this bound exactly, which is why the answer is the clean product $(m - 1)(n - 1)$ and not a floor. $\square$

The vertex-disjoint question is just as quick at the first non-trivial value. Forbidding two internally vertex-disjoint paths is a strong constraint: it forbids cycles.

Proposition 1.4. *For simple graphs, $k_2(n) = n - 1$, attained exactly by trees.*

Proof. If G contains a cycle, a closed loop of distinct vertices, then any two vertices on that cycle are joined by the two paths of the cycle, which are internally vertex-disjoint, so $\kappa_G^{\max} \geq 2$. Hence $\kappa_G^{\max} \leq 1$ forces G to be acyclic, free of cycles, and an acyclic graph on n vertices has at most $n - 1$ edges, with equality for trees. A tree indeed has $\kappa^{\max} = 1$, since removing the single intermediate vertex on any path disconnects its endpoints. $\square$

Here $k_m(n)$ denotes the vertex-disjoint extremal function, the counterpart of $\ell_m(n)$ for $\kappa^{\max}$. We have just seen $k_2(n) = n - 1 = \ell_2(n)$. The agreement of the edge and vertex problems is no accident at small m , and the moment it breaks is the first place the proofs begin to strain.

1.5 The first divergence: vertex-disjoint paths

Replacing edge-disjoint by internally vertex-disjoint routes changes nothing at first. Leonard proved that the edge and vertex problems give the same answer while m stays small.

Theorem 1.5 (Leonard [6]). *For simple graphs, all $n \geq m$ and $m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

The hypothesis $n \geq m$ is the same one Mader's theorem carries, and it is not decoration. Below it the complete graph K_n is itself feasible, since $\kappa_{K_n}^{\max} = \lambda_{K_n}^{\max} = n-1 \leq m-1$, so the answer is the trivial $\binom{n}{2}$ and the formula overshoots it. At $n=3$, $m=4$ the formula gives 4 where a graph on three vertices has only 3 edges. Every closed form in this thesis carries the same caveat, stated or inherited from its constructions, and the figures cap each curve at the trivial maximum for exactly this reason.

For $m \leq 4$ the extremal graphs for the two problems coincide: Whitney's inequality $\kappa \leq \lambda$ is tight where it matters and the harder vertex constraint costs nothing (Figure 1.9, the three overlapping blue curves). One is tempted to expect this forever. It fails at the very next value, and the failure is sharp.

Theorem 1.6 (Sørensen-Thomassen [7]). *For simple graphs and all $n \geq 13$,*

$$k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 3 > \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n).$$

The restriction $n \geq 13$ is the range over which Sørensen and Thomassen certify that $k_5(n)$ equals the formula, as the Erdős Problems database records it [3]. Below it the true $k_5(n)$ needs the case-by-case structural analysis [7] carries out rather than the closed form, which is why the stated domain begins there.

Remark 1.7 (A counting convention that shifts every quoted constant by one). The additive constant above deserves a warning, because two conventions are in circulation and they differ by exactly one. This thesis defines $\ell_m(n)$ and $k_m(n)$ as the *largest* number of edges a graph can carry while *no* pair is joined by m independent routes. The Erdős Problems database [3] states the same results the other way round, as the *smallest* number of edges that *forces* such a pair, which is one more.

The shift is visible in every entry that both sources state. The database gives $k_2(n) = n$ where a tree has $n-1$ edges, $k_3(2n) = 3n-1$ against $\lfloor 3(2n-1)/2 \rfloor = 3n-2$ here, $k_4(n) = 2n-1$ against $2n-2$, $\ell_5(2n) = 5n-2$ against $5n-3$, $\ell_6(n) = 3n-2$ against $3n-3$, and it writes the Bollobás-Erdős conjecture as $1 + \binom{m}{2}n$ where the convention here gives $\binom{m}{2}n$. Every one of those is exactly one apart, and the values in this thesis are the ones an exhaustive search returns: $k_3(4) = 4$, $k_3(5) = 6$ and $k_4(5) = 8$ were each confirmed by walking every graph of that

size.

Applied to [Theorem 1.6](#) the same shift would read $\lfloor 8n/3 \rfloor - 4$ rather than the $\lfloor 8n/3 \rfloor - 3$ printed above, which is the database's figure carried over unadjusted. Both candidates respect $k_5 \geq \ell_5$, which Whitney's inequality forces, so the thesis's own arithmetic cannot decide between them, and the original paper is the only thing that can.

One consistency point is worth recording, because it constrains how the two figures may be mixed. The threshold $n \geq 13$ was taken from the same database entry as the constant, and the two must be converted together or not at all. Left as printed, -3 with $n \geq 13$, the strict inequality of [Theorem 1.6](#) holds throughout, since $k_5(13) = 31$ against $\ell_5(13) = \lfloor 5 \cdot 12/2 \rfloor = 30$. Converting only the constant to -4 while leaving the threshold at $n \geq 13$ fails at that same first size, since $\lfloor 8 \cdot 13/3 \rfloor - 4 = 30 = \ell_5(13)$, so the divergence would begin only at $n \geq 14$ under that reading. The two admissible readings are therefore $(-3, n \geq 13)$ and $(-4, n \geq 14)$, and mixing a converted constant with an unconverted threshold is the one combination that is certainly wrong. Nothing here depends on which is right: k_5 enters only as the cited first divergence between the edge and vertex problems, and that divergence holds under either constant. A reader who needs the exact value should take it from [\[7\]](#) directly and should note which convention that paper uses before comparing it with anything above.

At $m = 5$ the vertex problem genuinely diverges from the edge problem, visible in [Figure 1.9](#) as the shaded gap that opens between the dashed blue and solid orange curves. Leonard had already found the first crack, an explicit 57-vertex, 141-edge counterexample to the general conjecture at $m = 5$ [\[8\]](#), before Sørensen and Thomassen pinned the exact value down. The reason the crack opens at all is structural. Leonard's earlier method, the one that closes $m \leq 4$, leans on the extremal graphs being assembled from cliques, vertex sets joined in all pairs, glued along shared cliques. A k -tree is built up one vertex at a time: start from a complete graph on $k + 1$ vertices, then again and again add a single new vertex and join it to all k vertices of some clique of size k that is already there. [Figure 1.8](#) grows a 2-tree, hanging each new vertex on an edge already present. These clique scaffolds are exactly the building blocks the method uses. Through $m = 4$ those scaffolds are forced, and counting their edges gives the bound.

At $m = 5$ they are no longer the unique extremal shape. Additional configurations appear, and the clean count breaks. Sørensen and Thomassen recover the exact value by a delicate structural induction, which we cite rather than reproduce. For $m \geq 6$ even that much is unknown: the vertex-disjoint problem has stood open since 1974, and the extremal structure is genuinely not understood. It is the one direction this thesis does not attempt to force, and [Chapter 4](#) explains why.

1.6 The directed case and the quadratic phenomenon

Direction breaks a symmetry that undirected graphs never had. In an undirected graph the number of edges is tied to the connectivity through every vertex's degree. In a digraph a vertex can

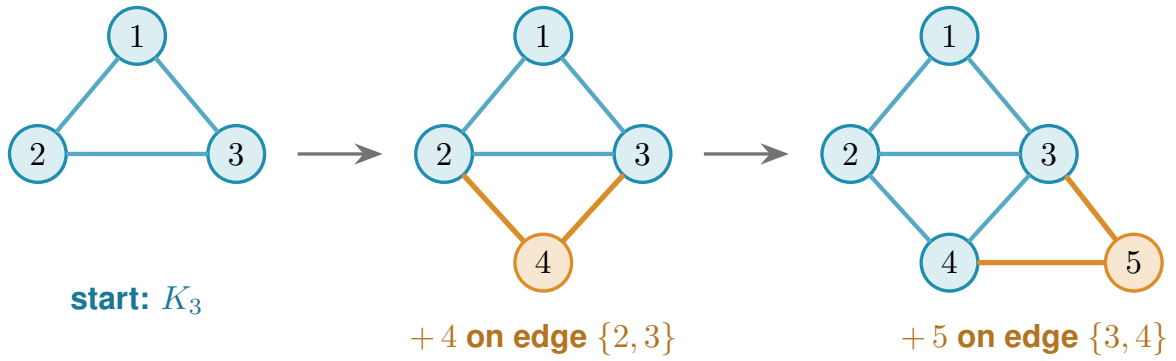


Figure 1.8. A 2-tree grown one vertex at a time, read left to right. The blue triangle on $\{1, 2, 3\}$ is the starting K_3 . Vertex 4 is then joined to the two ends of the edge $\{2, 3\}$, and vertex 5 to the two ends of the edge $\{3, 4\}$ (the new vertex and its two orange edges at each step), each new vertex hung on an edge already present. A general k -tree follows the same recipe with K_{k+1} in place of the triangle and a k -clique in place of the edge. These are the clique scaffolds Leonard’s count rests on.

have many arcs in and none out, and a whole graph can lean entirely one way. Send every arc from one half of the vertices to the other and nothing comes back: between any two vertices there is at most one route, yet the number of arcs is quadratic.

Construction 1.8 (One-directional bipartite digraph). Split V into parts A and B with $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$, and include every arc from A to B . All arcs run from A to B , and none within A , within B , or from B back to A . Since B has no outgoing arcs, a vertex in A cannot reach any other vertex of A , and two vertices of B cannot communicate at all. For any $a \in A$ and $b \in B$ the only directed path from a to b is the single arc $a \rightarrow b$, so $\lambda(a, b) = 1$ and $\lambda^{\max} = 1$. The arc count is $|A| \cdot |B| = \lfloor n^2/4 \rfloor$.

Figure 1.10 draws the construction. Every arrowhead lands in B and none leaves it, so the picture is a wall of one-way traffic: from any a on the left there is exactly one way to reach any b on the right, the single arc between them, and there is no way at all to travel between two vertices on the same side. The arc count $|A| \cdot |B|$ is quadratic, yet no pair carries more than one route. This is the lopsidedness direction buys, and it has no undirected analogue. It competes with a second construction built around a single centre.

That second construction is the directed hub. It keeps each vertex’s in-degree and out-degree balanced, the directed measure already met in Figure 1.5, and spends its arcs on a central vertex rather than a one-way wall.

Construction 1.9 (Directed hub). For $n \geq m \geq 2$, take a hub vertex h and label the remaining $N = n - 1$ vertices $0, 1, \dots, N - 1$. Add the $2N$ hub arcs $h \rightarrow i$ and $i \rightarrow h$ for every i , and on the non-hub vertices add the circulant arcs $i \rightarrow i + j \pmod{N}$ for $j = 1, \dots, m - 2$, the target label wrapping around past N back to 0 (that ring pattern is what *circulant* means, and “mod N ” is the wrap-around). Counting these gives $2N$ hub arcs (each of the N spokes joined to h in both directions) plus $(m - 2)N$ circulant arcs ($m - 2$ layers, one arc per spoke), so the total is

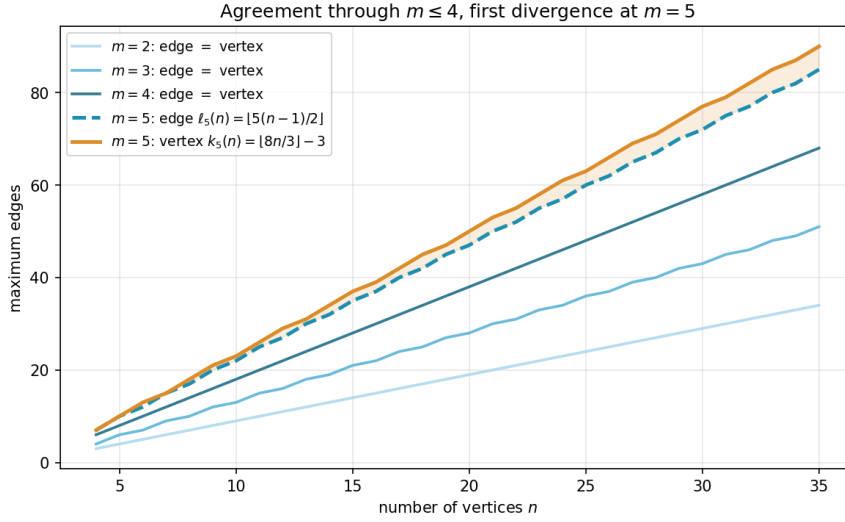


Figure 1.9. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). Through $m = 4$ the two problems give identical answers, and at $m = 5$ the vertex formula $k_5(n) = \lfloor 8n/3 \rfloor - 3$ separates above the edge formula $\ell_5(n) = \lfloor 5(n-1)/2 \rfloor$, with the gap shaded.

$2N + (m-2)N = mN = m(n-1)$ arcs. Every non-hub vertex has $d^+ = d^- = 1 + (m-2) = m-1$, and any ordered pair of vertices contains a non-hub endpoint, so the directed degree bound (Lemma A.3, which says $\lambda(u, v) \leq \min(d^+(u), d^-(v))$, a route out of u must use one of u 's out-arcs and one of v 's in-arcs) gives $\lambda^{\max} \leq m-1$.

At $m = 2$ this construction is easiest to picture: the circulant layer is empty and what remains is the *bidirected star* of Figure 1.11, a hub joined to each spoke by a matched pair of opposite arcs. Every spoke can reach every other only by going in to the hub and back out, two hops, so no pair is ever joined by two independent routes, and the arc total is exactly $2(n-1)$. For $m \geq 3$ not all $m(n-1)$ arcs can touch the hub, since a simple digraph has at most $2(n-1)$ arcs incident to one vertex: the surplus lives in the circulant layers among the spokes, tuned so that no degree exceeds the budget. The two constructions compete on different scales, linear against quadratic, and which one wins depends on n . At $m = 2$ the competition resolves exactly.

Theorem 1.10 (Directed arc value at $m = 2$). *For simple digraphs,*

$$\ell_2^{\text{dir}}(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

The hub construction leads for $n < 7$, the one-directional bipartite construction leads for $n > 7$, and the two tie at $n = 7$, where both reach 12 arcs.

The lower bound is the better of the two constructions. The upper bound is where the trouble starts.

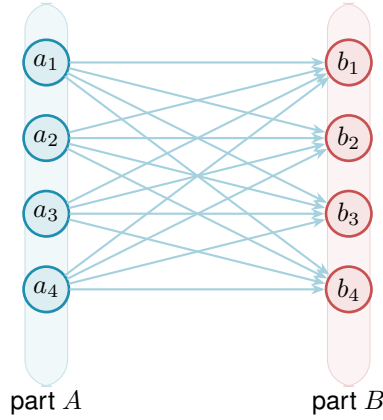


Figure 1.10. The one-directional complete bipartite digraph $B_{k,k}$ of Construction 1.8, drawn here at $n = 8$ with $|A| = |B| = 4$. Every one of the $|A| \cdot |B| = \lfloor n^2/4 \rfloor$ arcs runs left to right, from A to B . There are no arcs inside A , inside B , or back from B to A . Because B has no outgoing arcs and A no incoming ones, the only route from any a_i to any b_j is the single arc $a_i \rightarrow b_j$, so every pair has local edge-connectivity 1 while the arc count grows quadratically.

Idea of the upper bound at $m = 2$. The value $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is a contest between a linear term and a quadratic one, and which leads is pure arithmetic: $2(n-1) \geq \lfloor n^2/4 \rfloor$ precisely for $n \leq 7$, with equality at $n = 7$, where both give 12. So M follows the bidirected star up to the crossover and the bipartite digraph past it, and $n = 7$ is the single value where the two constructions tie. The proof that no digraph beats $M(n)$ is an induction on n (Appendix A). Delete a vertex of *minimum* total degree: since the degrees sum to twice the arc count, an averaging argument bounds that degree by $2a/n$, where a is the number of arcs, and the remaining digraph on $n-1$ vertices still respects the ceiling, so the inductive bound applies to it. Feeding the two facts together gives $a \leq \frac{n}{n-2} \lfloor (n-1)^2/4 \rfloor$, and a short parity check shows the right-hand side never exceeds $\lfloor n^2/4 \rfloor$ once $n \geq 8$: for even n it lands exactly on the bipartite value, and for odd n a sub-unit slack is swallowed by the fact that a is an integer. The averaging closes the step with no separate hub case, which is why the difficulty is not depth but bulk. Nothing in the argument is deep. It is a careful walk through two parity regimes, with the base cases $n \leq 7$ settled by exhaustive computer search rather than by hand. That experience of a correct but long and almost entirely mechanical case-check is exactly what motivates the chapters that follow. One genuinely wants to hand the case-checking to a machine.

The vertex-disjoint version of the same question has the same answer at $m = 2$, but it is worth being careful about why, because Whitney's inequality is easy to read backwards here. Since $\kappa \leq \lambda$ pairwise, a digraph that respects the arc ceiling automatically respects the vertex one, so the vertex-feasible family *contains* the arc-feasible one and is in general strictly larger. The free consequence therefore runs one way only: the two constructions above keep $\kappa^{\max} = 1$ as readily as $\lambda^{\max} = 1$, which makes them lower bounds for the vertex problem too. No upper bound comes with them. What settles the vertex value is that the same induction goes through unchanged, because deleting a vertex lowers $\kappa^{\max}$ for exactly the reason it lowers $\lambda^{\max}$, and because the base cases $n \leq 7$ were re-run under the stricter separation and returned the same

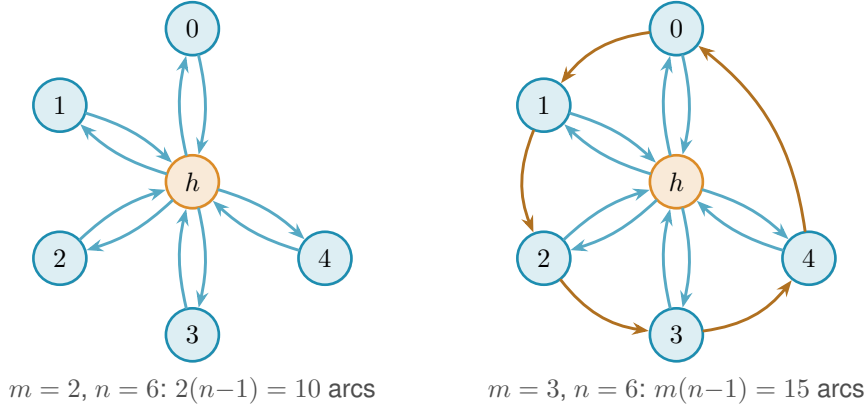


Figure 1.11. The directed hub at $n = 6$ for two values of m . *Left* ($m = 2$): the bidirected star. The hub h is joined to each spoke by one arc in each direction, giving $2(n - 1) = 10$ arcs. No circulant layer is added, so all spoke communication routes through h , so $\lambda^{\max} = 1$. *Right* ($m = 3$): a single circulant layer is added among the five spokes. The blue arcs are the same hub-spoke pairs as on the left. The orange arcs are that one circulant layer, the cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$, so each spoke has $d^+ = d^- = m - 1 = 2$. The total is $m(n - 1) = 15$ arcs and $\lambda^{\max} = 2$. For $m \geq 3$ the spokes are no longer isolated from one another, and tracing all the routes becomes the argument that fills the rest of the chapter.

six numbers. [Appendix A](#) gives both halves, and [Remark A.16](#) there spells out the direction trap in full.

Theorem 1.11 (Directed vertex value at $m = 2$). *For simple digraphs,*

$$k_2^{\text{dir}}(n) = \max\left(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor\right) = \ell_2^{\text{dir}}(n).$$

1.7 The failure of direct proof for $m \geq 3$

For $m \geq 3$ the induction no longer closes, and the natural guess is wrong. The natural initial conjecture was that the two constructions above continued to be the whole story, giving $\max(m(n - 1), \lfloor n^2/4 \rfloor)$. A single graph on ten vertices refutes it.

Remark 1.12 (A thirty-arc counterexample). Take $A = \{a_1, \dots, a_5\}$ and $B = \{b_1, \dots, b_5\}$ with all 25 arcs from A to B , and add a directed five-cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B . This digraph has 30 arcs. Every vertex of B has exactly one extra in-arc from inside B , so two arc-disjoint routes from a_i to b_j exist: the direct arc $a_i \rightarrow b_j$, and the detour $a_i \rightarrow b_{j-1} \rightarrow b_j$ using the 5-cycle predecessor of b_j (indices mod 5). For the matching upper bound, the only in-arcs of b_j reachable on a path from a_i are the direct arc $a_i \rightarrow b_j$ and the cycle arc $b_{j-1} \rightarrow b_j$. Removing both disconnects a_i from b_j , while neither alone does, so the minimum cut has size 2. Hence $\lambda(a_i, b_j) = 2$. Routes between two vertices of B never leave B and give $\lambda(b_i, b_j) = 1$. Hence $\lambda^{\max} = 2$, so the graph is feasible at $m = 3$, yet $30 > \max(3 \cdot 9, 25) = 27$.

[Figure 1.12](#) draws this thirty-arc digraph and traces the mechanism. The twenty-five faint arcs are the complete one-directional bipartite layer from A to B , the construction we already

understand. On top of them the orange five-cycle gives each vertex of B a single extra in-arc from inside B . That one extra arc is exactly what doubles the connectivity. For the marked pair a_1, b_2 the two heavy routes show how: the blue arc is the direct hop $a_1 \rightarrow b_2$, and the green path $a_1 \rightarrow b_1 \rightarrow b_2$ borrows the cycle arc $b_1 \rightarrow b_2$ for its second step. The two share no arc, so $\lambda(a_1, b_2) = 2$, and the same pair of routes exists for every a_i, b_j . Five extra arcs lift the whole digraph from one independent route per pair to two, and the thirty arcs overshoot the old prediction of twenty-seven.

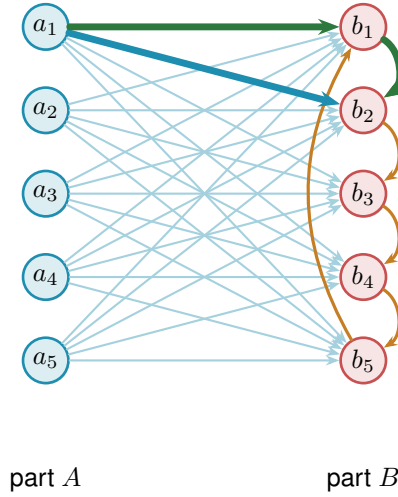


Figure 1.12. The thirty-arc counterexample of [Remark 1.12](#): the augmented bipartite digraph ([Construction 1.13](#)) at $m = 3$, $n = 10$, $|A| = |B| = 5$. The 25 faint arcs are the complete one-directional layer from A to B , and the 5 orange arcs form the directed cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B , one extra in-arc per vertex of B . The heavy blue arc and the heavy green detour show the pair a_1, b_2 carrying 2 arc-disjoint routes: the direct arc $a_1 \rightarrow b_2$ (blue) and the two-step detour $a_1 \rightarrow b_1 \rightarrow b_2$ (green) through the cycle predecessor b_1 . Hence $\lambda^{\max} = 2$, feasible at $m = 3$, while $30 > \max(3 \cdot 9, 25) = 27$.

The counterexample generalises: the five-cycle gave every vertex of B exactly one extra in-arc from inside B , and the budget for such arcs is $m - 2$ per vertex, not one. [Figure 1.12](#) is the instance $m = 3$, where that budget is one and a single cycle suffices. For larger m each b_j collects $m - 2$ cyclic predecessors and the same picture acquires more concentric chords inside B .

Construction 1.13 (Augmented bipartite digraph). For $m \geq 2$ and $n \geq m$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B| = \lfloor (n - m + 2)/2 \rfloor$, include every arc from A to B , and inside B give each vertex b_j the $m - 2$ in-arcs from its cyclic predecessors $b_{j-1}, \dots, b_{j-(m-2)}$ (indices mod $|B|$). The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs.

For the connectivity, fix $a \in A$ and $b \in B$: the direct arc and the $m - 2$ two-step detours $a \rightarrow b' \rightarrow b$ through the in-neighbours b' of b inside B are $m - 1$ arc-disjoint routes, and they are all there are, because the only in-arcs of b reachable from a are exactly those $m - 1$ arcs, which therefore form a cut. Pairs inside B are limited by the in-degree $m - 2$, pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m - 1$.

At $m = 2$ the inside of B is empty and the construction reduces to [Construction 1.8](#), with the balanced partition $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$. At $m = 3$ the formula gives $|B| = \lceil (n+1)/2 \rceil$, which coincides with $\lceil n/2 \rceil$ at odd n but gives $|B| = n/2 + 1$ at even n . Both the balanced and the shifted partition achieve the same arc count $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$ in this case, so the two are interchangeable at $m = 3$. The shift to a strictly larger B is structurally significant only at $m \geq 4$, where the balanced partition is suboptimal. At $m = 3$, $n = 10$ it is the thirty-arc counterexample of [Remark 1.12](#), drawn in [Figure 1.12](#) (using the equivalent balanced partition $|A| = |B| = 5$).

The two rival shapes invite a political picture. The hub is a single king through whom all traffic must pass, cheap to build but limited by the one throne. The bipartite family is a flat cabinet of ministers, each handling its own dossier with no central seat, and it scales quadratically. Which arrangement packs more arcs under the ceiling depends on n , and for $m \geq 3$ the cabinet overtakes the king once n is large. This is the corrected shape, and it leads to the standing conjecture for the directed arc problem.

Conjecture 1.14. *For simple digraphs and all $n \geq m \geq 2$,*

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right).$$

The hypothesis $n \geq m$ is the one both constructions already carry, and it is needed for the same reason as in [Theorem 1.5](#): below it the complete digraph is feasible and the answer is the trivial $n(n-1)$, which the formula exceeds. At $n = 3$, $m = 4$ the formula gives 8 where three vertices carry only 6 arcs. The exhaustive search confirms the value is 6 there.

For $m = 2$ this is [Theorem 1.10](#). For $m = 3$ the formula $\lfloor (n+1)^2/4 \rfloor$ equals $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$, so the conjecture reduces to the previous formula for $m = 2$ and $m = 3$. For $m = 3$, $n = 10$ the second branch gives $\lfloor 121/4 \rfloor = 30$, matching [Remark 1.12](#). For $m \geq 4$ the new formula and the old balanced-partition count $\lfloor n^2/4 \rfloor + (m-2)\lceil n/2 \rceil$ no longer agree: the quadratic branches already differ (for $m = 4$ the new one is larger by exactly one at every even n), but the hub term $m(n-1)$ dominates both at small n , so the full conjectured value first overtakes the old one at $n = 12$ for $m = 4$. The construction of [Construction 1.13](#) with the shifted partition witnesses the difference. The lower bound holds for all m , witnessed by [Construction 1.9](#) on the first branch and [Construction 1.13](#) on the second. The upper bound is open, and [Chapter 4](#) returns to it with what structural progress the machine and the hand can jointly supply.

1.8 The hypergraph variant

The third model deserves its own short treatment, not because its base case is hard but because it is the one variant whose objects are stored and manipulated differently from all the rest, and it is worth meeting that difference now rather than at the moment of computation. A hypergraph keeps no multiplicity matrix, only the list of its hyperedges, each a set of r vertices. We fix the edge size r , following the extremal literature, for a structural reason: under a connectivity ceiling

a small edge is always at least as efficient per route as a large one, so a hypergraph free to mix sizes would maximise its edge count by using pairs only, and the question would collapse back to the graph case. Fixing r is what makes the hypergraph question a new question.

The route between two vertices is a *Berge path*, which alternates vertices and hyperedges so that each consecutive pair of vertices lies in the hyperedge listed between them, and two such routes are edge-disjoint when they share no hyperedge. Concretely, in a three-uniform hypergraph a route from u to w might be the single step $u, \{u, x, w\}, w$ that crosses one hyperedge, or a longer alternation $u, \{u, x, y\}, y, \{y, z, w\}, w$ that switches hyperedge at the intermediate vertex y .

With routes redefined this way the same Gomory–Hu machinery that proved the multigraph value carries over, and the base case has the same linear shape. The picture to keep in mind is a metro map: each hyperedge is a line, and a route from one vertex to another rides a line and changes to another line at a station the two share.

A single hyperedge may serve only one route, so two Berge routes that share no hyperedge are edge-disjoint, the hypergraph counterpart of two edge-disjoint paths in a graph. Reading routes off a metro map is a matter of seeing which lines share track: two hyperedges that share several vertices run side by side along the stretch between them, as in [Figure 1.3](#), and two routes are hyperedge-disjoint exactly when the lines they ride share no track at all.

The separation axis needs its own definition here, and it is worth fixing now because the natural guess is not the one this thesis uses. Two Berge routes are called *internally disjoint* when they share neither an intermediate vertex nor a hyperedge. Both halves are required. Demanding only that the routes avoid each other’s intermediate stations would let two routes ride the same line, and a line that can run only one train at a time cannot serve two routes at once, so the resulting count would not correspond to anything the metro map can carry. Requiring both is also what makes the vertex measure the exact analogue of the graph one under the translation of [Appendix A](#), where a hypergraph becomes its incidence graph, each hyperedge becomes a node, and internally disjoint routes become internally disjoint paths in the ordinary sense. It has the pleasant side effect that Whitney’s inequality survives the move to hypergraphs by definition, since an internally disjoint family is in particular hyperedge-disjoint. Write $\kappa_{\mathcal{H}}(u, v)$ for the largest such family and $\kappa_{\mathcal{H}}^{\max}$ for its maximum over pairs.

Proposition 1.15 (Hypergraph edge bound). *An r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m - 1$ has at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. A simple hypergraph attains the bound whenever $m - 1 \leq \binom{n-2}{r-2}$, and a multihypergraph attains it whenever $(r - 1) \mid n - 1$.*

The proof, given in [Appendix A](#), runs the multigraph argument through the hypergraph Gomory–Hu theorem of Bérczi, Chandrasekaran, Király and Kulkarni [9], charging each hyperedge to the at least $r - 1$ tree cuts it crosses. The star hypertree ([Figure 1.13](#)) attains the $m = 2$ case, a hub joined to disjoint blocks of $r - 1$ vertices, with Berge connectivity one everywhere

and $\lfloor (n-1)/(r-1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. One genuine surprise is worth recording here: for $r \geq 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge (Theorem A.42). In the graph case simplicity is exactly what separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$. One edge size higher, it costs nothing.

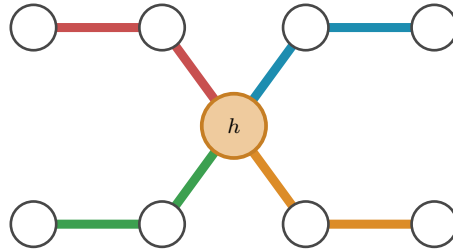


Figure 1.13. The star hypertree that attains the $m = 2$ hypergraph bound, here $r = 3$ and $n = 9$, drawn as a metro map with the hub h centred between two rows of stations. The single hub station lies on every line, and each of the $\lfloor (n-1)/(r-1) \rfloor = 4$ hyperedges is one metro line of $r = 3$ stations: it leaves the hub diagonally to an inner station and then runs horizontally to its outer station, hanging a fresh block of $r-1 = 2$ vertices off the hub. Because every line meets every other only at h , no pair has two hyperedge-disjoint routes, so the Berge connectivity is one everywhere, exactly as a spanning tree attained the multigraph bound.

The proposition counts hyperedges, but it does not say how to *measure* the Berge connectivity of a hypergraph we are handed, and that measurement is the genuinely new part. The metro picture shows why. An ordinary edge is a one-stop link between a single pair, so policing edge-disjointness means watching each pair on its own. A hyperedge is a whole line: it stops at all r of its vertices and so touches $\binom{r}{2}$ pairs at once, yet like a line that can run only one train at a time it may serve only one route. When several routes want to board the same line at different stations, the checker must wave exactly one through and turn the rest away, and it must do this for every line simultaneously. That bookkeeping is the one new piece of machinery the next chapter builds, and it is why the hypergraph rejoins the family only once the checker is in place.

1.9 Random graphs as intuition

The short proofs above hinted that the edge and vertex problems travel together for a while, agreeing through $m = 4$ before the split at $m = 5$. That separation is a statement about extremal graphs, the densest ones the ceiling allows, and it is worth asking whether the same split can be seen in ordinary graphs drawn at random, with no extremal tuning at all. The random model is the natural place to build intuition, because it shows the qualitative behaviour of the connectivity notions before any construction is pushed to its limit, and we will return to it whenever a picture of the typical graph clarifies a result. It is not a detour from the program to come, either: a sampled $G(n, p)$ is an ordinary graph, measured by the very same connectivity checker the next chapter builds for every other variant, so the random model is simply the one pipeline turned to *observe* the typical case rather than to prove or to discover the extremal one.

Each of the $\binom{n}{2}$ possible edges of $G(n, p)$ is included independently with probability p , and

on the resulting graph we read off both binding connectivities, the edge version $\lambda_G^{\max}$ and the vertex version $\kappa_G^{\max}$, from the very same sample. Whitney's inequality $\kappa_G^{\max} \leq \lambda_G^{\max}$ guarantees the vertex value never exceeds the edge one, but it is silent on how often the two actually come apart.

Figure 1.14 looks directly at the distributions, fixing $n = 16$ and histogramming both connectivities over many samples at three densities. Two things are visible at once. As the density rises the whole distribution marches rightward, from a binding connectivity around five at $p = \frac{1}{4}$ to around thirteen at $p = \frac{3}{4}$, so by $n = 16$ the typical graph already sits well past the $m = 4$ at which the proofs agreed. And comparing the edge row to the vertex row, the vertex distribution sits at or just to the left of the edge one, never to its right, which is Whitney's inequality seen across a whole distribution rather than a single graph.

The two pull apart on a fraction of samples that shrinks as the density grows, from about a fifth of the samples at $p = \frac{1}{4}$ to only a handful by $p = \frac{3}{4}$. The reason is structural. For $\kappa^{\max}$ to fall below $\lambda^{\max}$ some pair must carry more edge-disjoint routes than internally vertex-disjoint ones, which happens only when several of those routes are forced through one shared intermediate vertex. In a sparse graph routes are scarce and a single low-degree vertex can be exactly such a bottleneck, but as p grows every vertex gains degree and the graph offers many alternative intermediate vertices, so independent routes almost never need to reuse one. In the dense limit the graph approaches the complete graph, where $\kappa^{\max} = \lambda^{\max} = n - 1$ for every pair and the gap closes entirely. The divergence the thesis records at $m = 5$ is therefore not an artefact of the extremal construction. It is already there, in miniature, in the random model.

The random model also supplies a landmark of a different flavour. Below a sharp density the forbidden configuration is almost surely absent, and above it almost surely present, and the location of that threshold is the same for every one of the connectivity notions above.

Theorem 1.16 (Appearance threshold). *Let $m = m(n) \geq 2$ grow with n so that $m/\ln n \rightarrow \infty$ and $m = o(n)$, and let $c > 0$ be a constant. In the binomial random graph $G(n, p)$ with $p = c \cdot m/n$,*

$$\lim_{n \rightarrow \infty} \Pr[\lambda^{\max} \geq m] = \begin{cases} 1 & \text{if } c > 1, \\ 0 & \text{if } c < 1, \end{cases}$$

so the threshold for the appearance of a pair with local edge-connectivity at least m is $p^ = m/n$. The same threshold governs local vertex-connectivity and the directed analogues (Remark A.59). It does not govern the hypergraph models, which live on a different density scale (Remark A.60).*

The proof is a concentration argument with two halves (Appendix A). Above the threshold the edge count concentrates past Mader's bound, so Theorem 1.2 itself forces a high pair into existence. Below it the maximum degree stays under m , and the degree bound caps every local connectivity. The two hypotheses earn their keep. The first, $m/\ln n \rightarrow \infty$, says m must outgrow the natural logarithm of n , and that is what lets a union bound over the n vertices close, the union bound being nothing more than adding up the n individual failure chances and needing even their

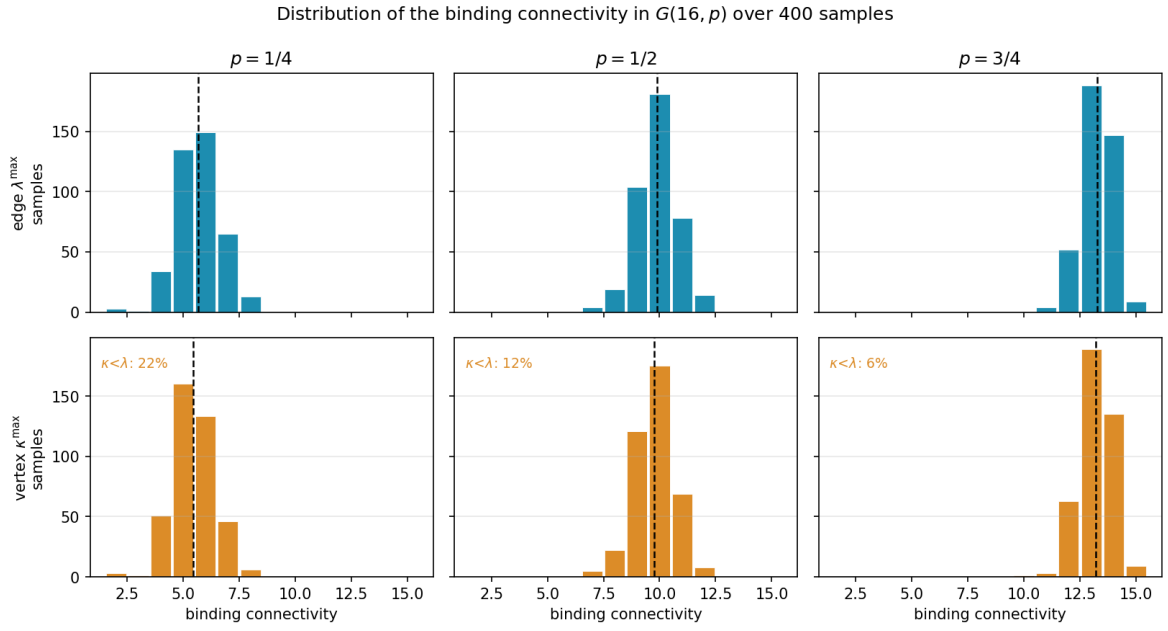


Figure 1.14. Distributions of the binding connectivity in $G(16, p)$ over 400 samples, at densities $p = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. There are $2^{\binom{16}{2}} = 2^{120}$ labelled graphs on 16 vertices, far too many to enumerate, so these are Monte Carlo samples rather than the full population, and 400 is plenty to show the shape of each distribution. The top row histograms the edge-connectivity $\lambda^{\max}$, the bottom row the vertex-connectivity $\kappa^{\max}$, measured on the same samples, with the mean marked by a dashed line and the fraction of samples on which $\kappa^{\max} < \lambda^{\max}$ printed on each vertex panel. Reading a column top to bottom, the vertex distribution sits at or to the left of the edge one, which is Whitney's inequality, and the gap is largest at the sparsest density (22% of samples) and closes as p grows (6%). All six panels share one axis range.

sum to vanish. The second, $m = o(n)$, says m stays negligible next to n , which keeps the density $p = cm/n$ below one and in the sparse regime where the threshold lives. Two caveats on scope, and the second is the one that limits how far the result may be quoted. The upper half quoted here uses Mader's bound, which is the simple undirected edge case, so it is that variant the proof settles directly. The vertex-disjoint and directed analogues sit at the same threshold $p^* = m/n$ but reach it through their own extremal bounds rather than Mader's, possibly with a different leading constant, as [Remark A.59](#) records. The hypergraph models are a different matter and are *not* covered. A threshold of this kind sits where a vertex's expected degree reaches m , and in a binomial r -uniform hypergraph a vertex lies in an expected $p \binom{n-1}{r-1}$ hyperedges, so the balance point is $m / \binom{n-1}{r-1}$, smaller than m/n by a factor of order n^{r-2} once $r \geq 3$. [Remark A.60](#) works this out and [Figure B.8](#) plots each model against its own scale. So the collapse of complexity is real but bounded: within a fixed edge size the distinctions between edges and vertices and between the two directions wash out and one density decides the matter, while changing the edge size changes the density at which everything happens.

One limitation must be stated plainly, lest the threshold be read as evidence about the problem this thesis actually studies. The growth hypothesis $m / \ln n \rightarrow \infty$ forces m to climb with n , so the theorem says nothing whatever about a fixed small m , the regime of every exact value above,

where m is 2, 3, or 4 and n runs to infinity around it. There the union bound is vacuous and the appearance of a dense pair is governed by rare low-degree configurations that need finer, Poisson-type tools. The random model therefore enters as contrast and not as evidence: it shows how cleanly the question collapses in one limit, which throws into relief how stubbornly it resists in the fixed- m limit the rest of the thesis must attack by hand and by machine.

1.10 Where this work sits

It is worth pausing, before the machine, to place the question and the method against the two traditions they descend from: the extremal graph theory that posed the problem and the computational mathematics that will help settle it.

The mathematical lineage is the older of the two. The problem belongs to the family of extremal connectivity questions opened by Bollobás and Erdős, who asked how dense a graph can be while every pair of vertices stays below a fixed connectivity [1], logged today as Erdős Problem 915 [3]. Mader’s theorem [4] closed the simple undirected edge case with the clean floor $\lfloor m(n-1)/2 \rfloor$ recalled in [Theorem 1.2](#), and it is the floor every variant in this thesis is measured against. The vertex-disjoint branch has a longer and rougher history: Leonard [6] showed the edge and vertex problems coincide through $m \leq 4$, and Sørensen and Thomassen [7] found the first divergence at $m = 5$ with $k_5(n) = \lfloor 8n/3 \rfloor - 3$, after which the exact values $k_m(n)$ remain unknown. That gap between a settled edge case and a stalled vertex case is exactly the terrain [Chapter 1](#) has been mapping, and the directed and hypergraph axes extend it further. The thesis adds to this lineage not a single new closed form but a uniform way of pushing each axis as far as exact computation allows.

The computational tradition this thesis’s pipeline belongs to is the practice of settling combinatorial questions by machine in a way a human can audit, and it splits into a few schools worth contrasting with the certify-then-prove loop built next. The oldest is *exhaustive generation*: enumerate every object of a given size up to isomorphism and check each one. McKay and Piperno’s nauty/geng toolkit [10] is the standard engine for this, and its flagship results include exact small Ramsey numbers such as $R(4, 5) = 25$ [11]. The enumeration of [Chapter 2](#) is the same idea, run on the twelve variants of Problem 915. The second school is *SAT-assisted proof*, in which a question is encoded in propositional logic and a solver emits a machine-checkable certificate: Heule, Kullmann and Marek’s resolution of the Boolean Pythagorean triples problem [12], whose verified proof runs to some two hundred terabytes, and Heule’s determination of Schur number five [13] are the canonical examples of certified solver output at scale. The third is *flag algebras*, Razborov’s framework [14] that converts asymptotic extremal bounds into semidefinite-programming certificates, the dominant tool for density problems in the large- n limit. Closest of all to the method here is the fourth, *certification by linear and integer programming* (LP/ILP), where a feasibility or optimality argument over the rationals or integers is written as a linear program, an optimisation with linear constraints, and the proof is the certificate of optimality the solved program hands back, a short list of numbers anyone can verify by direct arithmetic. This is precisely

the form the cut-counting certifier of [Chapter 2](#) takes.

Against that backdrop, what is distinctive here is consolidation rather than a new solver. A single checker measures connectivity exactly across all twelve problem variants on one data model, so search and certification share one codebase rather than living in separate tools. The certificates it emits are exact rational or integral objects, not floating-point estimates, so an upper bound it reports is a proof and not evidence. And because the same program both hunts for dense graphs and certifies the bounds they suggest, the discover step and the prove step are two modes of one loop, which is the arrangement the next chapter builds. What makes the consolidation honest is that it never blurs three kinds of knowledge: a value can be *proved* (by hand, or by exhausting every graph of a size), *certified* (by the same optimisation, reaching a vertex or two further), or only *conjectured* (witnessed by a dense graph the search exhibits, with no matching upper bound). This trichotomy, named in [Section 3.6](#) and obeyed by the summary of [Chapter 4](#), is the through-line of everything that follows.

Computer note

We are now stuck in three different ways: a fifty-year-old open problem at $m \geq 6$ for vertices, a long proof we would rather not check by hand at $m = 2$ for arcs, and a conjecture with no proof at all at $m \geq 3$. The base cases of the long induction were verified by exhaustive search, and the thirty-arc counterexample was found the same way. The next chapter builds a single program that models all of these cases at once, measures their connectivity exactly, and proves the small-case upper bounds outright.

Chapter 2

Certifying Bounds by Machine

The last chapter ended in three different kinds of stuckness, and all three share a shape. The objects are graphs of one flavour or another, the constraint is always a ceiling on connectivity, and the work is always either measuring a given graph or checking many small ones. That shared shape is what a single program can exploit. This chapter builds it, and then uses it for one purpose only: to *prove*. It gives every variant one representation, measures connectivity exactly through a max-flow checker, checks small cases by honest enumeration, and turns the finite checks the proofs of [Chapter 1](#) relied on into a single auditable certifier. The hunt for dense examples, which proves nothing on its own, waits until [Chapter 3](#).

A word on what the program is for. It is a microscope, not a source of final answers. It measures connectivity exactly, so what it reports about a given graph is a fact. It enumerates graphs of a fixed size, so for that size it can prove a theorem. It will later discover dense graphs, which are only lower bounds, but not here. Each of these has a different epistemic weight, and the program is built so that the three never get confused. [Figure 2.1](#) draws this spine: a single *model* holds every variant, one exact *measure* reads connectivity off it, and from that measurement a *prove* step bounds a fixed size from above, with the *discover* step of the next chapter and a final *observe* step for the random model added later. Every routine introduced from here on is shown as a small card carrying its name, its inputs and output, a one-line account of how it works, and a coloured badge naming its place in the spine. The function bodies are not reproduced, because the point is the interface, not the code.

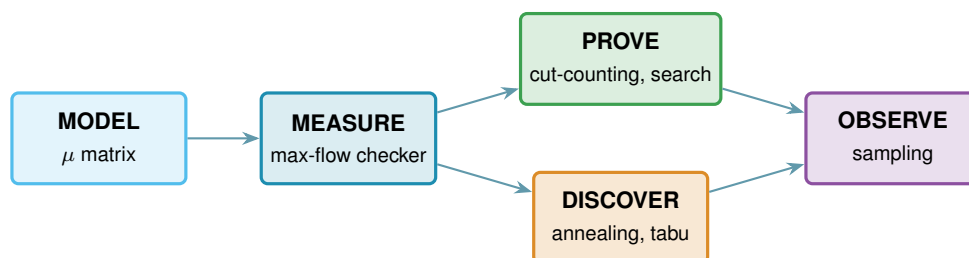


Figure 2.1. The architecture as one pipeline. A single data model, the multiplicity matrix μ , feeds an exact measurement, the max-flow checker. That measurement feeds the two bound-producing roles: a prover for upper bounds (the cut-counting optimisation and the exhaustive search), built in this chapter, and a discoverer for lower bounds (simulated annealing, or tabu search), built in [Chapter 3](#). A final sampling role studies the random model. The badge colours on the code cards below match the boxes here.

2.1 The solver

Everything in this chapter is a single function, `solve`, and it helps to hold its most basic version in mind from the start, because every later idea is a refinement of it and nothing is ever a separate program for a separate case. Given n and m , the basic strategy is as plain as it sounds. Walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ of each with a max-flow computation, and keep the densest graph that stays at or below the ceiling $m - 1$. Because it has looked at every graph, the count it returns is the exact answer, not an estimate.

That one loop already settles the smallest cases, and the rest of the chapter is built on top of it without ever leaving `solve`. We first make the measurement step trustworthy and quick, via the connectivity checker and a Gomory–Hu shortcut for undirected graphs. We then make the enumeration step skip whole families of hopeless graphs, via a pruned search. Finally, for the directed multigraph case, we replace enumeration altogether with one small optimisation that proves every m at once, which is the certifier. Each of these is the same `solve` doing less work for the same guarantee, and the choice among them is made automatically from the booleans it receives.

```
solve(n, m, directed, simple, separation, exhaustive, max_seconds)
```

DRIVER

in the variant booleans `directed` and `simple` (or hypergraph with uniformity r), the `separation` (edge or vertex), n , m , the method, and a time budget `max_seconds`

out a `SolveResult` carrying a value with an honest label: `exact`, `lower bound`, or `upper bound`

how picks the method the case allows: brute-force enumeration, the pruned search, or the cut-counting certifier when proving, or the temperature search or tabu search when discovering (Chapter 3). It always respects the budget, and it is the one function the rest of the thesis ever calls.

2.2 A single representation for all variants

Two of the three axes from Chapter 1, the model and the direction, are properties of the object, and a single data structure holds almost all of them. Of the three structural models, simple graphs and multigraphs fit the matrix directly, while the hypergraph keeps the separate storage promised in Chapter 1 and rejoins at the end of this chapter. The hypergraph could in principle be pressed into the same shape as a higher-order tensor, an r -dimensional array marking which r -sets are present, but that array is almost entirely zero and costs n^r entries, so the program keeps the compact list of hyperedges instead and pays nothing for the empty cells. Sampled $G(n, p)$ graphs are ordinary simple graphs and therefore use this same representation without any change. A graph or digraph on n vertices is stored as an integer matrix μ , where $\mu(u, v)$ is the multiplicity of the edge or arc from u to v . A simple graph is the case $\mu \in \{0, 1\}$, an undirected graph satisfies $\mu = \mu^\top$, and a multigraph allows larger entries. The third axis, edge

against vertex separation, is a property of the question rather than the object, so it lives in how we measure μ , not in μ itself. Figure 2.2 illustrates the encoding.

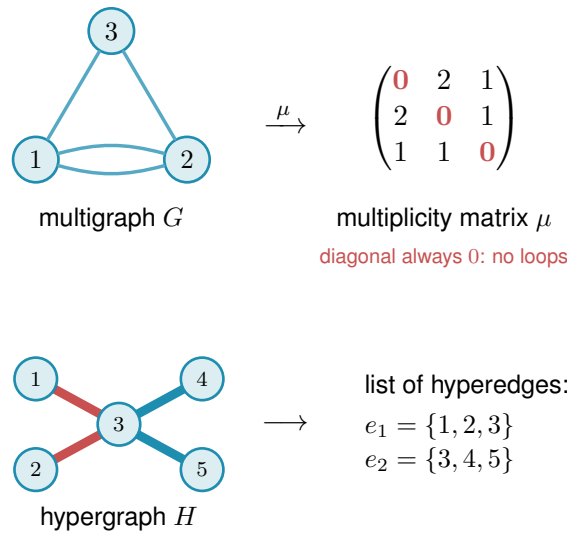


Figure 2.2. How the program stores a graph. Top: a graph or digraph on n vertices is one $n \times n$ integer matrix μ , with $\mu(u, v)$ the number of parallel edges or arcs from u to v . The double edge between 1 and 2 gives $\mu(1, 2) = 2$. The diagonal is always zero (red), since no vertex carries an edge to itself, an undirected graph keeps $\mu = \mu^\top$, a directed graph allows $\mu(u, v) \neq \mu(v, u)$, and a simple graph caps every entry at one. Bottom: the hypergraph is the one model that does not use the matrix at all, since a matrix has no room for an edge on three vertices, so it is kept as a plain list of its hyperedges, here two metro lines that share vertex 3.

```
Graph(n, Variant(directed, simple))
```

MODEL

in the vertex count n and a `Variant`, the small record holding the two booleans
out a graph whose whole state is the matrix μ , stored as the numpy array `self.mu`
how one $n \times n$ integer numpy array is the object, an undirected graph keeps $\mu = \mu^\top$, and a simple graph caps every entry at one.

Keeping a single representation is what lets one measurement routine, one certifier, and one search serve every case. Nothing downstream asks which of the twelve problems it is solving. It asks only the two booleans (directed or undirected, simple or multi) and the matrix. Everything that builds or edits a graph is a thin wrapper around μ that respects those booleans, and these operations are so routine that they need no individual exposition, and Table 2.1 lists them together.

The two booleans are all the mirroring above needs, and the whole trick fits in one short routine. Every write to μ passes through it, so this is the single place the undirected symmetry is kept, not an invariant a caller has to remember to maintain.

```
1 def _assign(self, u, v, value):
2     self.mu[u, v] = value
3     if not self.variant.directed:
```

Operation	Meaning
<code>addEdge(G, u, v, k)</code>	add k parallel edges or arcs $u \rightarrow v$ (a simple graph saturates at one)
<code>removeEdge(G, u, v, k)</code>	remove k copies, never below zero
<code>setMultiplicity(G, u, v, c)</code>	set $\mu(u, v) = c$ directly
<code>multiplicity(G, u, v)</code>	read $\mu(u, v)$
<code>hasEdge(G, u, v)</code>	whether $\mu(u, v) > 0$
<code>degree(G, v)</code>	total degree of v (in- plus out-degree if directed)
<code>edgeCount(G)</code>	number of edges or arcs, counted with multiplicity
<code>edges(G), vertices(G)</code>	iterate the present edges, and the vertex set
<code>copy(G)</code>	an independent duplicate

Table 2.1. The routine operations on the graph model. Each is a one-line wrapper that reads or writes the matrix μ while keeping an undirected graph symmetric and a simple graph binary. They are bookkeeping, and the chapter’s attention belongs to the routines that follow.

```
4 | self.mu[v, u] = value    # keep undirected graphs symmetric
```

Listing 2.1. How every write keeps an undirected graph symmetric. A directed graph simply skips the mirrored write.

2.3 The connectivity checker built on maximum flow

The whole question reduces to a flow computation. A *maximum flow* asks how much can be pushed from one point to another through a network whose links each carry a limited capacity. Menger’s theorem says the most edge-disjoint u – v paths a graph offers equals its smallest u – v cut, the fewest links whose removal severs every route between the two. The max-flow / min-cut theorem of Ford and Fulkerson [15] says that smallest cut is exactly the value of a maximum flow, once each multiplicity $\mu(u, v)$ is read as the capacity of the link $u \rightarrow v$. So the route count we are after is just a maximum-flow value: measure the flow and you have measured the connectivity. The routine that runs this flow and reports the route count is the connectivity *checker*. It is the single most important routine in the program: it is the only place graph theory meets computation, and every later claim about connectivity passes through it. Because the search of Chapter 3 calls it by the million on tiny graphs, the program ships an optional compiled C helper that runs this same flow (named later in this chapter’s reproducibility section). It is a speed-up only: it returns the identical value the pure-Python checker does, verified against it, so no number in the thesis depends on whether the helper is built.

```
local_edge_connectivity(G, s, t) → ℕ
```

MEASURE

in a graph G and an ordered pair (s, t)

out $\lambda_G(s, t)$, the maximum number of edge-disjoint s – t paths

how a single `scipy.sparse.csgraph.maximum_flow` on the network whose arc capacities are the multiplicities μ , so by Menger the flow value is the path count.

`max_edge_connectivity(G) → ℕ`

MEASURE

in a graph G

out $\lambda^{\max}(G)$, the largest local edge-connectivity over all pairs

how one max-flow per pair (ordered if directed), take the largest. This is the quantity the edge problem caps at $m - 1$.

The vertex-disjoint version follows the same idea after one construction: split each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3). Every arc $u \rightarrow v$ of the original graph becomes $u^{\text{out}} \rightarrow v^{\text{in}}$ in the split network. The construction is not special to digraphs: for an undirected graph each edge $\{u, v\}$ is first read as the two opposite arcs $u \rightarrow v$ and $v \rightarrow u$, and then the same split applies, so one routine serves the undirected and directed vertex problems alike. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint u – v paths in the original graph.

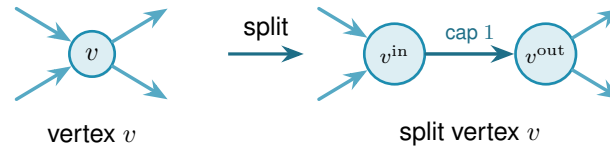


Figure 2.3. Vertex splitting for the vertex-disjoint checker. Incoming arcs attach to v^{in} and outgoing arcs leave from v^{out} , connected by a single unit-capacity arc. Any directed path that uses v as an intermediate vertex passes through this bottleneck, and two such paths would need two units of capacity and are therefore separated.

Figure 2.4 applies the split to a whole digraph. Three arc-disjoint routes are forced through one shared vertex w , so on the split network all three must cross the single unit-capacity arc inside w , which caps the number of internally vertex-disjoint routes below the edge-disjoint count.

`max_vertex_connectivity(G) → ℕ`

MEASURE

in a graph G

out $\kappa^{\max}(G)$, the largest local vertex-connectivity over all pairs

how the same sweep as the edge checker, but each flow runs on the split capacity matrix `_split_capacity_matrix(G)` of Figure 2.3, so the unit internal arcs count internally vertex-disjoint paths. Parallel edges are given capacity one here, so they never raise κ .

One convention in that codecard deserves to be said in prose, because it does not merely change how the checker computes, it changes what the multigraph vertex question asks. In vertex mode every adjacency carries capacity one, so a bundle of parallel edges counts as a single direct route. The rationale is that vertex-disjointness is about intermediate vertices, and parallel copies offer no new intermediate vertices to be disjoint over.

Remark 2.1 (What parallel edges mean in vertex mode). That choice has a consequence which

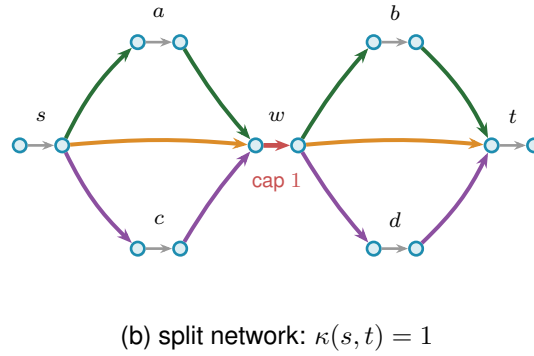
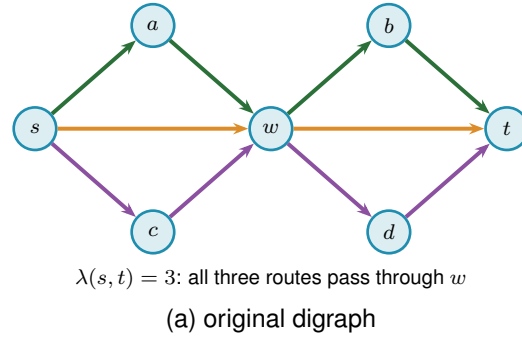


Figure 2.4. The vertex-split checker on a larger example. **(a)** A digraph in which three arc-disjoint s - t routes, $s \rightarrow w \rightarrow t$ (orange), $s \rightarrow a \rightarrow w \rightarrow b \rightarrow t$ (green), and $s \rightarrow c \rightarrow w \rightarrow d \rightarrow t$ (purple), all pass through the single vertex w , so $\lambda(s, t) = 3$. **(b)** The split network replaces every vertex V by an arc $V^{\text{in}} \rightarrow V^{\text{out}}$ of capacity one. All three routes must now traverse the single unit arc inside w (red), which carries only one unit, so at most one internally vertex-disjoint route gets through and $\kappa(s, t) = 1$. Splitting every vertex this way turns the vertex-disjoint count into an ordinary maximum flow.

has to be stated rather than left implicit. If parallel copies never raise κ , they never break feasibility either, so one could pile arbitrarily many copies onto any feasible multigraph and stay under the ceiling forever. Counted with multiplicity, the maximum number of edges would simply be infinite, and the extremal question would be empty. The two multigraph vertex variants are therefore posed here with the objective counting *adjacencies*, that is, pairs joined by at least one edge, rather than edges with multiplicity. Under that reading the question is exactly the simple question on the underlying graph, which is why those two rows of [Table 4.1](#) reduce to their simple counterparts and why every figure in this thesis reports the reduced problem for them.

The other convention is perfectly available and asks something genuinely different. A single edge is a path with no interior, so q parallel copies between the same pair are q internally vertex-disjoint paths, and feasibility caps every multiplicity at $m - 1$ by itself. Nothing runs away, and the multigraph vertex problem separates from the simple one immediately: at $n = 2$ and $m = 3$ two parallel edges are feasible where the simple graph allows one. That is a new extremal problem rather than a variant of a solved one, and [Chapter 4](#) lists it among the open problems. The appendix uses precisely this second reading whenever it argues about a general multigraph, most visibly in [Lemma A.47](#), whose hypothesis (iv) caps a multiplicity at two for exactly this

reason. The two readings never meet, because that lemma is applied only to incidence graphs, which carry no parallel edges at all, but a reader moving between the chapters and the appendix should know that the word “multigraph” is doing different work in the two places.

The two checkers are the entire interface between graph theory and computation in this thesis. As a sanity check, applied to the bidirected star of [Figure 1.11](#) on n vertices, the edge checker reports $\lambda^{\max} = 1$, and the arc count $2(n - 1)$ matches the proved values $\ell_2^{\text{dir}}(n) = 2, 4, 6, 8$ for $n = 2, 3, 4, 5$.

The same checker answers two different questions. Throughout this chapter it runs in its exact-value mode, returning $\lambda^{\max}$ or $\kappa^{\max}$ on the nose, and every proved or reported number rests on that exact reading. The search of [Chapter 3](#) needs only the cheaper yes-or-no question “is the binding connectivity above the bound?”, so there the same flow runs in a capped mode that stops as soon as it has seen one route too many. The capped mode never changes a reported value: it is a faster way to ask for a single bit, and [Chapter 3](#) shows that the search built on it reproduces the exact search step for step.

The Gomory–Hu tree from [Chapter 1](#) reappears here as an efficiency, and it is the honest kind. For undirected graphs all $\binom{n}{2}$ edge-connectivities are read off a single weighted tree, so $\lambda^{\max}$ is the heaviest tree edge rather than the maximum over $\binom{n}{2}$ separate flows. The speed-up is real, but it is bought with a theorem, not a heuristic.

`max_edge_connectivity_via_tree(G) → ℕ`

MEASURE

in an undirected graph G

out $\lambda^{\max}(G)$, identical to the pairwise sweep

how build one Gomory–Hu tree with `networkx.gomory_hu_tree`, then return its heaviest edge, which costs $n - 1$ flows in place of $\binom{n}{2}$ and cross-checks the checker.

The hypergraph is the one model that never used the multiplicity matrix, since it is stored as a plain list of hyperedges, each one a set of r vertices. The checker still has to read it, and the trick is to run the same flow computation as before on a slightly larger network that we build on the spot. The difficulty is that one hyperedge ties several vertices together at once. By the edge-disjointness convention of [Chapter 1](#), only one route may be counted through a hyperedge at a time, so we cannot just treat it as a bundle of ordinary links. Instead, for every hyperedge we add one extra helper point and connect each of that hyperedge’s member vertices to it ([Figure 2.5](#)). A route that wants to use the hyperedge now enters through one member vertex, passes through the helper point, and leaves through another.

The key is to make the helper point carry a single unit of capacity, built by the very same device as the vertex bottleneck of [Figure 2.3](#): the helper point is split into an in-copy and an out-copy joined by one unit-capacity arc, and every route boarding the hyperedge must squeeze through that one arc. A second route would need a second unit of capacity there and so is shut out. That one unit is what captures the rule that one hyperedge serves one route.

Counting routes between two vertices is then the very same maximum flow as before, run on this enlarged network, and when a hyperedge happens to hold just two vertices the helper point collapses back to an ordinary link, so the answer agrees with the graph case. That the flow on the enlarged network counts exactly the hyperedge-disjoint Berge routes is the hypergraph version of Menger’s theorem, proved in [Appendix A \(Theorem A.40\)](#), so the checker rests on a theorem rather than an analogy.

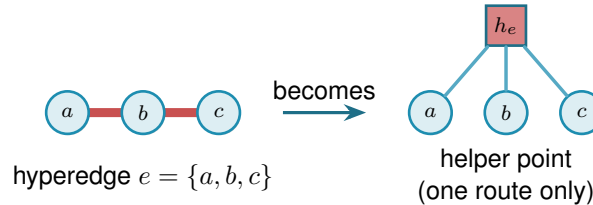


Figure 2.5. Measuring Berge connectivity by maximum flow. A hyperedge e , drawn as a metro line through its members (left), is replaced by a helper point h_e joined to each member (right). The helper point holds a single unit of capacity inside it, so at most one route may pass through h_e and the line runs one train at a time, which is what makes the hyperedge serve only one route. Counting independent routes between two vertices is then an ordinary maximum flow on this enlarged network, and a two-vertex hyperedge collapses to a single ordinary edge.

[Figure 2.5](#) shows one hyperedge in isolation. [Figure 2.6](#) carries the idea to a whole hypergraph in two panels. Panel (a) is the hypergraph drawn as a metro map: each of the eight hyperedges is one coloured line threading its three member stations, vertex to vertex, and a station on several hyperedges shows several coloured lines running through it. Panel (b) reads the same map as the flow network. For one pair of stations, u and v , it adds the helper point of every hyperedge a route boards, drawn as a small gate h_e in that line’s colour, and traces the edge-disjoint Berge routes between u and v as thin black lines passing through those gates, with every hyperedge rail faded to the background. The checker reads $\lambda(u, v) = 4$: two routes run straight through the two hyperedges that contain both u and v , and two longer ones change lines at an intermediate station, four in all, while a fifth would have to reuse a gate.

One way to picture the helper point is the metro map of [Chapter 1](#): each hyperedge is a metro line, its member vertices are the stations on it, and the helper is the line itself. Boarding at any station counts as taking that line. Because only one route may pass through the helper, only one train runs on the line at a time, so a second route must reach its destination without ever boarding it. The same flow computation that counts edge-disjoint paths through ordinary links now counts routes that share no line, which is exactly what Berge edge-disjointness means.

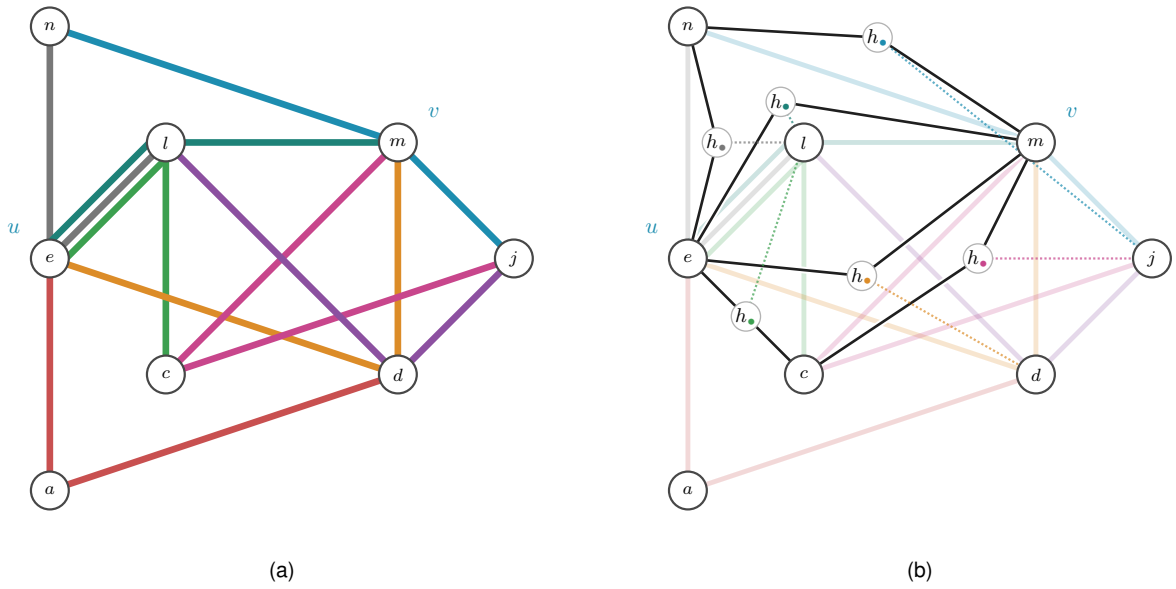


Figure 2.6. A 3-uniform hypergraph on eight stations, with the pair u, v singled out. Panel (a) draws it as a metro map: every hyperedge is one coloured line threading its three members, vertex to vertex, the metro reading of [Chapter 1](#). Panel (b) reads the same picture as the helper-point flow network. Each hyperedge a route boards becomes its one-route gate $h_\bullet$ of [Figure 2.5](#), shown in that line's colour and placed on the route, and the thin black lines are the edge-disjoint Berge routes from u to v threading those gates. A dotted leg ties each gate to the hyperedge's third member, the station that route does not use, and every hyperedge rail is faded to the background so the black routes read clearly. Two routes pass straight through the two hyperedges that hold both u and v , and two longer ones transfer at an intermediate station, so the checker reads $\lambda(u, v) = 4$. A fifth route would have to reuse a gate.

`max_hyperedge_connectivity(H) → ℕ`

MEASURE

in an r -uniform hypergraph H

out $\lambda^{\max}(H)$, the largest Berge edge-connectivity over all pairs

how build the enlarged capacity matrix `_hyper_capacity_matrix(H)` in which every hyper-edge becomes a helper point that only one route may pass through, then take the same maximum flow as the graph checker. On two-vertex hyperedges it matches ordinary edge-connectivity.

On the complete graph K_n the hypergraph checker returns $n - 1$, and on the complete three-uniform hypergraph $K_4^{(3)}$ it returns 3, larger than the two hyperedges through each pair because longer Berge routes also carry flow. The self-test confirms both of these values, so the construction is checked rather than merely asserted. The same helper-point network also answers the vertex-disjoint hypergraph question without alteration: splitting each ordinary vertex as in [Figure 2.3](#) counts internally vertex-disjoint Berge routes. The one remaining direction, the *directed* hypergraph, needs the helper point to learn which way is which, and that is the subject of the next subsection.

2.3.1 The directed hypergraph checker

A *directed* hyperedge carries an orientation: it is a single *tail* vertex together with $r - 1$ *head* vertices, drawn as the star of [Figure A.8](#), and a directed Berge route may only enter it at the tail and leave it at one of its heads. The undirected helper point treated a hyperedge as a line any route could board at any station. The directed one must instead be a one-way turnstile: a route may step onto the hyperedge only at its tail and step off only at a head. The gadget that enforces this is the helper point made directional. For a hyperedge $e = (a; b, c, \dots)$ we add a gate node g_e , send one capacity-one arc from the tail into it, $a \rightarrow g_e$, and send an arc out of it to each head, $g_e \rightarrow b$, $g_e \rightarrow c$, and so on ([Figure 2.7](#)). A route reaches g_e only through the tail a , and the single unit of capacity on $a \rightarrow g_e$ lets just one route use the hyperedge, exactly as the undirected turnstile did, but now it can leave only towards a head. The gate is built once per hyperedge, directed or not, by the same loop:

```
1 gate_in, gate_out = base + 2 * index, base + 2 * index + 1
2 cap[gate_in, gate_out] = 1 # one route through this hyperedge
3 if hypergraph.directed:
4     tails, heads = _dir_tails_heads(edge)
5     for tail in tails:
6         cap[leave(tail), gate_in] = _UNBOUNDED
7     for head in heads:
8         cap[gate_out, enter(head)] = _UNBOUNDED
```

Listing 2.2. The gate loop behind [Figure 2.5](#) and [Figure 2.7](#): one capacity-one gate per hyperedge, wired one-way for a directed hyperedge and both ways for an undirected one.

[Figure 2.8](#) reads the same hypergraph directed, again in two panels. Panel (a) gives the whole

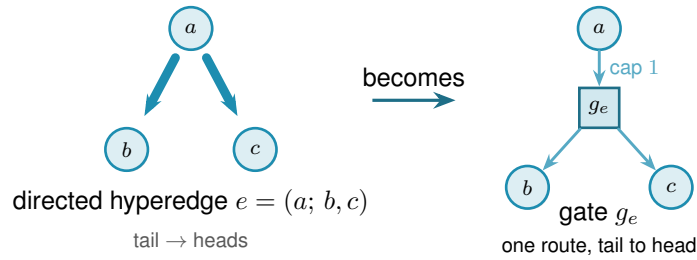


Figure 2.7. The directed helper point. A directed hyperedge $e = (a; b, c)$, the tail a fanning to its heads b, c (left), becomes a gate node g_e fed by one capacity-one arc from the tail and feeding an arc to each head (right). A route may reach g_e only through the tail and may leave only towards a head, so the gate is a one-way version of the turnstile of Figure 2.5, and the single unit of capacity still lets only one route use the hyperedge.

hypergraph oriented, every hyperedge a one-way line whose arrows point away from its tail, so the middle-tailed lines fan out from their centre and the end-tailed ones run one way along their length. Panel (b) traces the routes from v to u , the gate rule of Figure 2.7, fading every hyperedge rail to the background and threading a black arrow from the route's entry station through the one-route gate $h_\bullet$, set in a clear gap off the rails, to its exit, with a dotted leg to the hyperedge's third member. Orientation costs connectivity. Of the four undirected routes between u and v only two survive the arrows, $\lambda(v, u) = 2$, because v can set out only through the two hyperedges of which it is the tail, and the two survivors cross at the interchange l so they share no hyperedge. The same vertex-splitting trick of Figure 2.3 laid on top then counts internally vertex-disjoint directed routes, so this one construction measures all four hypergraph variants and `solve` reaches every one of the twelve problems through a single checker.

The gate is indifferent to how a hyperedge's vertices are split between tails and heads. It draws one capacity-one arc in from every tail and one out to every head, so the single forward tail it was built for is only the simplest case: give it a hyperedge with several tails and one head, or several of each, and it still counts the routes that enter at a tail and leave at a head. The same checker therefore measures every *orientation model* of a directed Berge edge, not just the one-tail one, and Section 4.3.1 takes up which of those models are worth distinguishing and what the program finds for them.

With the checker in place the proposed value of Proposition 1.15 can be tested on any small hypergraph the way every other variant is, which is the subject of the next section.

2.4 Checking every graph of a fixed size

The checker measures one graph. To bound every graph of a given size, the most direct method is the one the base cases of Chapter 1 already needed: enumerate them all. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal cell, a multigraph a choice in $\{0, \dots, m-1\}$, and a digraph varies every ordered pair rather than only the upper triangle. Walking that product, measuring each candidate with the checker, and keeping the densest feasible one settles the value exactly, and because it has looked at every graph it is a genuine upper

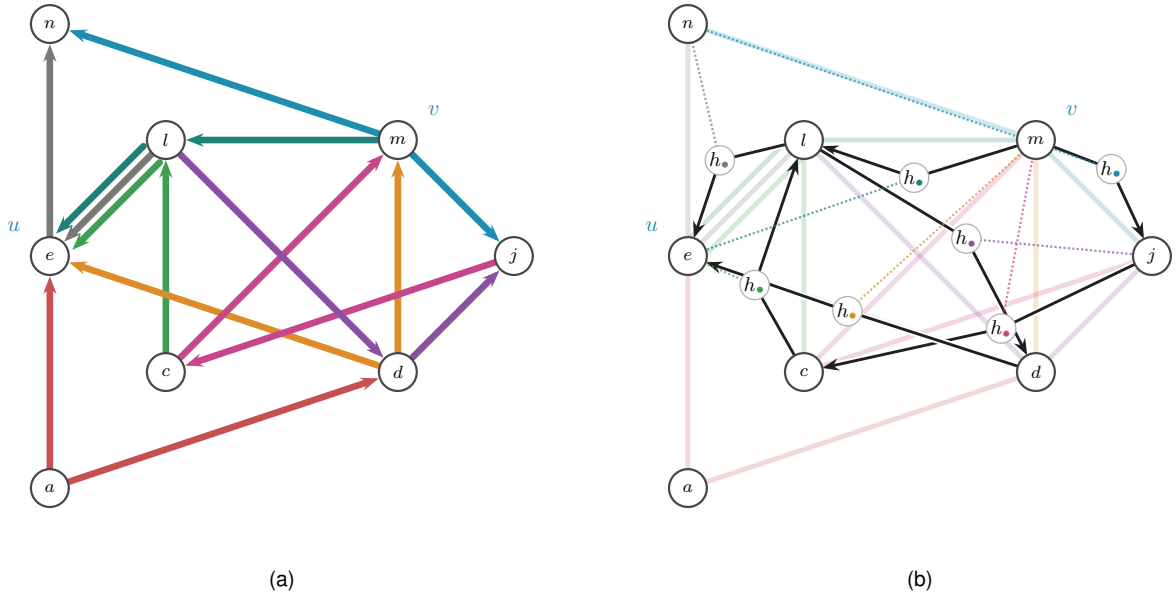


Figure 2.8. The hypergraph of Figure 2.6 read directed. Panel (a) gives the whole directed hypergraph, every hyperedge a one-way line whose arrows all run away from its tail station, so a route may join a line at its tail and ride it to a head. Because the map keeps each hyperedge on the track it already occupied, a line whose tail is an end station is oriented along its own length rather than fanned out as the star of Figure A.8, so both renderings appear here. One rule reads either of them off the picture: the tail of a coloured line is the single station of that line into which none of its arrows points, and the other $r - 1$ stations are its heads. Applied here it gives the eight tail-and-heads splits $a \rightarrow ed$, $m \rightarrow nj$, $c \rightarrow le$, $d \rightarrow em$, $l \rightarrow en$, $m \rightarrow le$, $j \rightarrow cm$ and $l \rightarrow dj$. Panel (b) traces the routes from v to u . Every hyperedge rail is faded to the background, and a black arrow runs from the route's entry station through the hyperedge's one-route gate $h_\bullet$ to its exit, the directed turnstile of Figure 2.7, with the gate set in a clear gap off the rails. A dotted leg in the line's colour ties each gate to the hyperedge's third member, the station the route does not visit. Two edge-disjoint directed Berge routes from v to u survive. The long route $v \rightarrow j \rightarrow c \rightarrow l \rightarrow u$ rides nmj , mcj , elc and nel in turn, and the short route $v \rightarrow l \rightarrow d \rightarrow u$ rides elm , ldj and edm . They cross only at the interchange l , so they share no hyperedge, and the checker reads $\lambda(v, u) = 2$, since v can set out only through the two hyperedges of which it is the tail. Against the undirected four of Figure 2.6, the arrows have halved the count.

bound, not evidence.

This is the first thing `solve` does when it is asked to prove a small case. With its exhaustive flag set it walks that product, measures each candidate with the checker, and keeps the densest feasible one. The value it returns is exact and exhausted, a true upper bound for that n , but only while the count of graphs stays small. No new routine is involved, only `solve` choosing to enumerate.

Run on the small cases this confirms the proved values directly, with no induction and no cleverness. [Table 2.2](#) collects a spread of variants where the enumeration finishes: in every row the machine returns exactly the value [Chapter 1](#) proved by hand. This is the first payoff of the uniform representation. One driver, `solve`, pointed at six different problems, settles all six.

Variant	m	n	<code>solve</code>	Theory
simple undirected, edge	2	5	4	$\lfloor m(n-1)/2 \rfloor = 4$
simple undirected, edge	3	5	6	$\lfloor m(n-1)/2 \rfloor = 6$
multigraph undirected, edge	3	5	8	$(m-1)(n-1) = 8$
simple undirected, vertex	2	6	5	$k_2(n) = n-1 = 5$
simple directed, arc	2	5	8	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 8$
simple directed, arc	2	6	10	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 10$

Table 2.2. Variants that converge under plain enumeration. For each small case `solve`, in its exhaustive mode, walks every graph of the variant, measures it with the checker, and returns the densest feasible one. Every value matches the hand proof of [Chapter 1](#) exactly.

The honesty of the method is also its limit. The number of graphs it must visit grows as $2^{\binom{n}{2}}$ for simple undirected graphs and $2^{n(n-1)}$ for digraphs, and capping multiplicities at $m-1$, so that each cell ranges over the m values $\{0, \dots, m-1\}$, raises the base from two to m , giving $m^{\binom{n}{2}}$ undirected multigraphs and $m^{n(n-1)}$ directed ones. The enumeration is exact wherever it finishes, and it finishes only for the smallest sizes. [Chapter 3](#) opens with exactly how fast this wall arrives. For now the point is that the wall arrives precisely where [Chapter 1](#) needed help, at the directed base cases $n = 6, 7$, so the rest of this chapter is about reaching a little further than blind enumeration can.

2.5 Reaching further: a pruned search for the directed base cases

The base cases $n \leq 7$ of the directed $m = 2$ theorem ([Theorem 1.10](#)) are the finite checks the induction of [Appendix A](#) depends on, and they sit just past where blind enumeration of digraphs is tractable. They can be reached by enumerating more cleverly. Feasibility is monotone: adding an arc can never lower $\lambda^{\max}$, so once a partial digraph is infeasible every completion of it is infeasible too and the whole branch is abandoned at once. A counting bound prunes further. At any partial digraph, count the arcs already placed and add the most the still-undecided cells could ever contribute. That sum is a ceiling on every completion of the branch, so if even the ceiling cannot beat the best feasible digraph found so far, the branch is dropped unexplored ([Figure 2.9](#)). What remains is a branch and bound over simple digraphs with the connectivity checker as the feasibility test, exact and complete at these sizes where the blind walk is not.

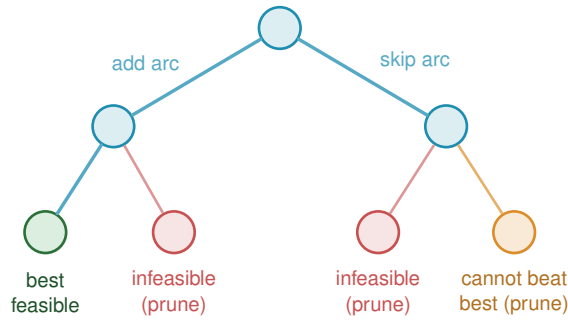


Figure 2.9. Branch and bound over digraphs. Each node fixes one more cell, arc present or absent, so the leaves are whole digraphs. Two rules cut entire subtrees before they are walked. A branch whose partial digraph already breaks the connectivity ceiling is infeasible, and so is every completion of it (red). A branch whose best possible completion still cannot beat the densest feasible digraph found so far is dropped on the counting bound (orange). Only the surviving branches are explored, which is what lets the search reach the base cases $n \leq 7$ the blind walk cannot.

Here `solve` changes its implementation without changing its interface. For a simple digraph it runs this branch and bound in place of the blind walk, still with the connectivity checker as the feasibility test, and so reaches the base cases $n \leq 7$ that blind enumeration cannot. The booleans alone decide which of the two it uses, and the caller never asks.

Run on the directed $m = 2$ problem `solve` returns the complete sequence 2, 4, 6, 8, 10, 12 for $n = 2, \dots, 7$, each proved complete at its size. The densest digraph it finds at each n is exactly the extremal shape [Chapter 1](#) named, the bidirected star (the hub) at every $n < 7$ and, at the crossover $n = 7$ itself, a graph of the tying value 12 that both the hub and the one-directional bipartite digraph attain. The sweep stops there, so the bipartite side of the crossover is witnessed by the construction and the checker rather than by this search ([Chapter 3](#) records that the cooled walk in fact stalls on the hub at $n = 8$). Within its range the search therefore confirms not just the values but the winning construction. These are exactly the base cases the induction requires, and with them the long proof of [Theorem 1.10](#) rests on a single auditable search rather than on anything done by hand. The exhaustive-search transcript is in [Appendix A](#).

2.6 Generating the directed cases faster

The $m = 3$ directed multigraph result reduces, through [Remark A.39](#), to a handful of finite statements about multigraphs on six and seven vertices, and proving them means listing every such multigraph up to a renaming of its vertices. The program does this in two independent ways, on purpose: when two different methods return the same list, that agreement is itself a check.

The first way is a search written from scratch. It is a depth-first walk over multiplicity matrices that abandons a branch the moment it turns infeasible or can no longer beat the best graph found, and it keeps just one representative of each isomorphism class by reducing every finished graph to a canonical relabelling, the smallest matrix in dictionary order over all relabellings. The canonical form keeps the memory small, but the time is spent walking labelled partial graphs, and time is what runs out: the full classification at $n = 5$ costs this in-house search about 456

seconds, and it climbs steeply from there.

The second way (Table 4.2 names it) does not search at all, it generates, and it was suggested by Prof. Jan Goedgebeur. The idea is to hand the hard part, listing one graph per isomorphism class, to a tool that already does it supremely well: the `geng` program from the `nauty` toolkit of McKay and Piperno [10], invoked through a subprocess and decoded from its `graph6` output (a compact one-line text encoding of a graph). `geng` lists the non-isomorphic *undirected* graphs on n vertices, and the program then decorates each one with the directed multiplicities the problem needs, so the isomorphism reduction is inherited for free from the support graph. The decoration itself is plain to state. For one support graph, walk its edges in a fixed order and assign each edge $\{u, v\}$ an ordered pair of multiplicities $\mu(u, v), \mu(v, u)$, each between 0 and $m-1$ but not both zero, since an edge of the support must carry at least one arc. Along the way the checker measures the partly decorated multigraph, and the moment some pair already exceeds the connectivity ceiling the whole branch of assignments is abandoned, since adding further arcs can never lower a connectivity, the same monotonicity prune as Figure 2.9. A decoration that survives to the end with the target arc count is kept, reduced to its canonical form so that each isomorphism class is reported once. Different supports share nothing, so they are processed independently across the processor cores. The companion orientation tools `directg` and `watercluster2` are not used, because each gives only one orientation per edge and so cannot express the bidirected arcs and repeated multiplicities the extremal graphs are built from. Layering the multiplicities in-house on top of `geng` is what fits the problem. To be precise about credit: Prof. Goedgebeur is not an author of `nauty`, which is the work of McKay and Piperno cited above. He suggested the generation route, and it is his suggestion that the speed-up rests on.

The two ways agree to the last graph. The generated pipeline reproduces the in-house classification exactly at every settled size, two non-isomorphic extremal families at $n = 4$ and three at $n = 5$, and it runs substantially faster, since the isomorphism work it never repeats is precisely what `geng` has already done. One difference decides which to trust further out: the in-house search leans on a conjectured counting bound once $n \geq 7$, while the generated enumeration prunes only with proved bounds, so it is a sound and complete search at every n . That soundness has now paid off. Pointed at the degree-bounded classification at $n = 7$, fact (b) of Remark A.39, the generation enumerator completed the sweep in about seven hours across ten processor cores and returned exactly the three predicted extremal families, settling fact (b) outright and leaving the value statement, fact (a), as the $m = 3$ argument's one remaining computation.

2.7 Proving every m at once

The pruned search proves one (n, m) at a time. For the directed multigraph arc problem `solve` has a sharper implementation, one that proves every m at a fixed n in a single pass, and it is worth the extra machinery because it closes the directed multigraph case outright for the sizes it reaches.

The key is that m can be scaled out of the problem completely, so that it never appears in the

computation at all. Start with any directed multigraph that is feasible at level m , so $\lambda^{\max} \leq m - 1$, and divide every multiplicity by $m - 1$. Two things happen at once. The multiplicities become weights $w(u, v)$ between 0 and 1, and since dividing every capacity by $m - 1$ divides every flow by the same factor, the maximum flow between each pair drops to at most one. The arc count, meanwhile, turns into the total weight $\sum_{u \neq v} w(u, v) = |A|/(m - 1)$. So a dense feasible multigraph becomes a heavy feasible weight matrix, and the other way round. Write

$$M^*(n) = \max \left\{ \sum_{u \neq v} w(u, v) : w(u, v) \in [0, 1], \text{ maxflow}(s, t; w) \leq 1 \text{ for all } s, t \right\}$$

for the largest total weight any such matrix can carry. It is one number for each n , with no m inside it, and scaling a feasible multigraph down gives

$$L_m^{\text{dir}}(n) \leq (m - 1) M^*(n) \quad \text{for every } m \geq 2.$$

That inequality holds always, and it is the direction that does the proving. The reverse does not come for free, and it is worth being exact about why, because the two are easy to run together. Scaling *up* needs an optimal weight matrix whose entries are whole multiples of $1/(m - 1)$, and a real optimum has no reason to oblige. What closes the gap is not an argument but an observation about the answer the solve returns: the heaviest matrix it finds is the *double star*, the bidirected star carrying weight one on both arcs $h \rightarrow v$ and $v \rightarrow h$ between the hub and every other vertex. Its weights are all 0 or 1, so multiplying them by $m - 1$ gives an honest integer multigraph, not a fractional one, with $2(n - 1)(m - 1)$ arcs and $\lambda^{\max} = m - 1$, for every m at once. Whenever the optimum is attained by such a $\{0, 1\}$ matrix, the two bounds meet and

$$L_m^{\text{dir}}(n) = (m - 1) M^*(n) \quad \text{for every } m \geq 2,$$

so that one solve settles infinitely many values. This is exactly what happens at every $n \leq 6$, which is what [Theorem 2.2](#) rests on. It is a verified property of those optima rather than a theorem about all n , and where it is not verified the safe reading is the inequality: an integral question such as fact (a) of [Remark A.39](#) is then asked of the integral program `prove_integral_arc_bound` directly, not deduced from a fractional value.

What remains is to compute $M^*(n)$, and the difficulty is that the constraint “maximum flow at most one” is itself the answer to a flow problem rather than a plain inequality on w . The max-flow / min-cut theorem removes it. The maximum s – t flow equals the smallest capacity of any *cut* separating s from t . A cut just splits the vertices into a source side holding s and a sink side holding t , and its capacity is the total weight of the arcs that cross from the source side to the sink side. So $\text{maxflow}(s, t) \leq 1$ holds exactly when some such cut has capacity at most one. That lets us drop the flow entirely: instead, for every ordered pair (s, t) , the optimisation picks one cut and we require its capacity to stay at or below one.

A cut is described by a yes/no label x_u^{st} on each vertex, equal to one when u is on the source side, with s fixed on the source side and t on the sink side. An arc $u \rightarrow v$ crosses the cut exactly

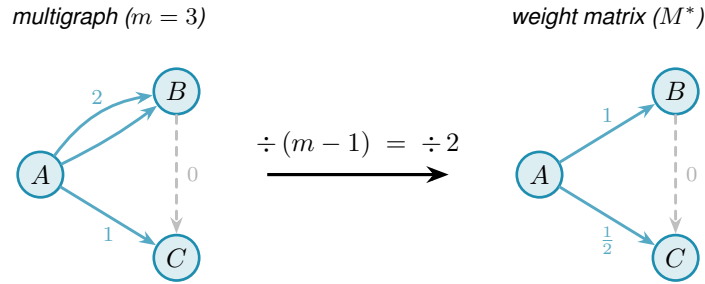


Figure 2.10. The scaling step for $m = 3$. Dividing each multiplicity by $m - 1 = 2$ maps the integer multigraph (left) to a weight matrix with all values in $[0, 1]$ and max-flow at most one between every pair (right). The double arc $A \rightarrow B$ (multiplicity 2) becomes weight 1, the single arc $A \rightarrow C$ (multiplicity 1) becomes weight $\frac{1}{2}$, and the absent arc $B \rightarrow C$ (multiplicity 0, shown dashed) stays absent. Crucially, m drops out on the right, so a single optimisation over all weight matrices bounds $L_m^{\text{dir}}(n) \leq (m - 1) M^*(n)$ for every m simultaneously. The reverse inequality is not automatic and needs the optimum to be attained by a $\{0, 1\}$ matrix, which is a verified property of the optima at $n \leq 6$ rather than a theorem about all n , as the text above sets out.

when u is on the source side and v is not, and a helper variable p_{uv}^{st} then picks up its weight $w(u, v)$. Forcing the crossing weights to sum to at most one for every pair forces every pair to admit a cut of capacity one, hence maximum flow at most one. This is the exact feasible region, not a relaxation of it, a relaxation being a loosened version that would also admit weight matrices no multigraph realises, and that exactness is what turns the result into a proof. The optimisation reports both the best total weight it has found and a value it has shown no matrix can exceed, and when those two meet the gap is zero and $M^*(n)$ is certified.

For the directed multigraph arc problem, then, `solve` runs this cut-counting optimisation in place of any enumeration: it chooses the minimum cut for every ordered pair, and a zero gap makes the reported $M^*(n)$ a proved upper bound, from which $L_m^{\text{dir}}(n) \leq (m - 1)M^*(n)$ delivers a ceiling for every m at once, met from below by the double star whenever the reported optimum is integral.

`prove_directed_multigraph(n) → ProofResult`

PROVE

in the vertex count n (and tightening flags such as `use_deletion_cuts`)

out $M^*(n)$ with a proof status, giving $L_m^{\text{dir}}(n) \leq (m - 1)M^*(n)$ for every m , with equality when the optimum found is integral

how builds the cut-counting optimisation, the mixed-integer linear program written out below, once with `_cut_counting_model` as a pulp model, then `_pick_solver` runs it on CBC, the free solver bundled with pulp, or on the commercial Gurobi solver by a one-line switch. A zero optimality gap is a proof. The integer-weight companion `prove_integral_arc_bound(n, m, target)` answers the single feasibility question behind fact (a) of [Remark A.39](#).

`solve` assembles this optimisation from n alone, and a reader content to take it on the strength of the cut argument just given can skip to [Theorem 2.2](#) without loss. Written out it

reads more heavily than it behaves, so we first name every symbol in plain words, then show the one inequality that looks cryptic is a simple yes-or-no switch.

The optimisation is a *Mixed-Integer Linear Program*, or MILP for short. “Linear” means every constraint and the objective are linear functions of the variables: no products, no squares, no exponentials. “Mixed-integer” means the variables split into two kinds: most are ordinary real numbers that may take any value in a continuous range, but a few are forced to be whole numbers, either 0 or 1. The reason for the split is conceptual. Arc weights w are naturally continuous: they are scaled multiplicities and can sit anywhere in $[0, 1]$. But the cut assignment for a vertex is a hard binary choice: a vertex is either on the source side of a given cut or on the sink side, with no meaningful notion of being “half on each side”. Allowing the cut labels to be fractional would describe a different, weaker condition and could report an optimistic value that no integer-weight multigraph can actually achieve. Keeping those labels in $\{0, 1\}$ is what keeps the optimal gap honest, and it is what makes the program’s result a proof rather than an estimate.

Three quantities appear, and only three. The first is already familiar.

- ★ $w(u, v) \in [0, 1]$ is the *scaled multiplicity* of the arc $u \rightarrow v$, the amount of arc weight the matrix places on $u \rightarrow v$ after the division by $m - 1$ of the previous section. It is what we are maximising the total of. This is a continuous variable: it may be any real number in $[0, 1]$.
- ★ $x_u^{st} \in \{0, 1\}$ is the *side label* for vertex u with respect to the ordered pair (s, t) . The optimisation chooses one cut for each pair, and x_u^{st} records whether u is on the source side (1) or the sink side (0). The two endpoints are pinned: $x_s^{st} = 1$ and $x_t^{st} = 0$. This is the integer variable: a vertex cannot straddle the cut.
- ★ $p_{uv}^{st} \geq 0$ is the *crossing weight*, the part of $w(u, v)$ that the pair (s, t) ’s chosen cut is forced to count. It equals $w(u, v)$ when the arc $u \rightarrow v$ crosses from source to sink, and is otherwise left at zero. This is a continuous variable.

With those three named, the program is short:

$$\begin{aligned}
 &\text{maximise} && \sum_{u \neq v} w(u, v) \\
 &\text{subject to} && p_{uv}^{st} \geq w(u, v) + x_u^{st} - x_v^{st} - 1 && \text{for all } (s, t), (u, v), \\
 &&& \sum_{u \neq v} p_{uv}^{st} \leq 1 && \text{for all } (s, t), \\
 &&& w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1 && \text{for all } (s, t), \\
 &&& d(v_0) \leq d(v_1) \leq \dots \leq d(v_{n-1}), && \text{with } d(v) = \sum_{u \neq v} (w(u, v) + w(v, u)), \\
 &&& w \in [0, 1], \quad x \in \{0, 1\}.
 \end{aligned}$$

The first line is the only one that looks cryptic, and it is just an indicator written as an inequality. Its right-hand side $w(u, v) + x_u^{st} - x_v^{st} - 1$ takes one of four shapes according to which side the

cut puts the two endpoints u and v on, and Table 2.3 works all four out. Because $w(u, v) \leq 1$, the right-hand side is positive in one case only, the crossing case, where it equals $w(u, v)$ and so forces $p_{uv}^{st} \geq w(u, v)$. In the other three cases it is zero or negative, so the constraint reads $p_{uv}^{st} \geq (\text{something} \leq 0)$ and leaves p_{uv}^{st} free to stay at zero, which the maximisation is happy to let it do.

x_u^{st}	x_v^{st}	meaning	$w + x_u^{st} - x_v^{st} - 1$	forces
1	0	u source side, v sink side: arc <i>crosses</i>	$w(u, v)$	$p_{uv}^{st} \geq w(u, v)$
1	1	both on the source side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	0	both on the sink side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	1	u sink side, v source side	$w(u, v) - 2 \leq 0$	$p_{uv}^{st} \geq 0$

Table 2.3. The first constraint, case by case. This is where the bound $w \leq 1$ earns its keep: it makes the three non-crossing rows land at or below zero, so the helper p_{uv}^{st} is pushed up to the arc weight in exactly one case, the crossing one. The constraint is therefore an exact indicator, “count $w(u, v)$ when the arc crosses this cut, and nothing otherwise”, with no slack and no fudge.

Figure 2.11 shows one chosen cut and the weights it counts. The second line of the program now reads plainly. It says the total crossing weight of this pair’s cut, $\sum_{u \neq v} p_{uv}^{st}$, is at most one. By max-flow / min-cut a cut of capacity at most one exists for (s, t) exactly when $\text{maxflow}(s, t) \leq 1$, so the two lines together say “every ordered pair admits a cut of capacity at most one”, which is precisely the feasibility we want. This is the true feasible region, not a relaxation of it, so an optimal solution reported with zero gap is a proof and not merely evidence.

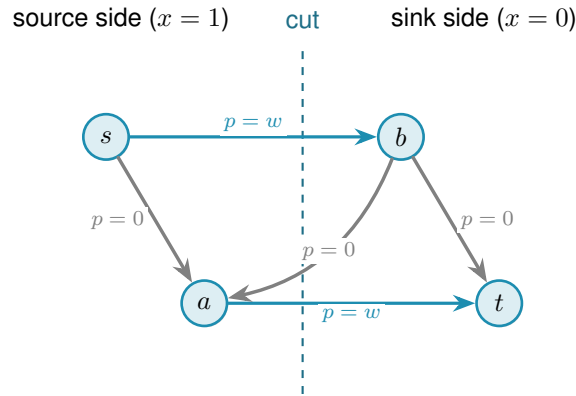


Figure 2.11. One ordered pair’s chosen cut. The dashed line splits the vertices into the source side, where $x_u^{st} = 1$, and the sink side, where $x_u^{st} = 0$. Only arcs running from source side to sink side cross the cut, and the first constraint forces their helper p_{uv}^{st} up to the arc weight w (the two thick arcs). Arcs that stay on one side, and arcs that run the other way across the line, contribute nothing. The second constraint then caps the total counted weight, the thick arcs, at one.

The last two lines change no answer. They only help the solver finish in time. The third is a redundant shortcut. The direct arc $s \rightarrow t$, together with one two-step detour $s \rightarrow x \rightarrow t$ through each intermediate vertex x , already forms that many routes between s and t . So $w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1$ holds of any feasible matrix anyway, and stating it outright just sharpens the continuous picture the solver reasons from.

The $\min$ in that line is not a linear function, so it cannot appear in the MILP as written. To keep

the program linear, each $\min(w(s, x), w(x, t))$ is replaced by a fresh helper variable z_x^{st} together with a binary selector $b_x^{st} \in \{0, 1\}$. The selector records which of the two arguments is smaller: $b_x^{st} = 1$ means $w(s, x)$ is no larger than $w(x, t)$. Four linear inequalities then pin z_x^{st} exactly to the minimum. Two upper bounds, $z_x^{st} \leq w(s, x)$ and $z_x^{st} \leq w(x, t)$, prevent z from exceeding either argument. Two lower bounds, one of the form $z_x^{st} \geq w(s, x) - (1 - b_x^{st})$ and the other $z_x^{st} \geq w(x, t) - b_x^{st}$, each become active only when the selector chooses that argument: at any integer value of b exactly one lower bound bites and forces z up to the chosen weight. Together with the upper bounds this pins $z_x^{st} = \min(w(s, x), w(x, t))$ exactly. This technique is called the McCormick linearisation of a minimum. It adds one continuous variable z and one binary variable b per intermediate vertex per ordered pair, all of them appearing only in the tightening constraint and nowhere else, so the proof structure of the program is unchanged.

The fourth line orders the vertices by weighted degree. Renaming vertices cannot change any count, so insisting on this order rules out no graph, and it spares the solver the many relabelled copies of a graph that differ only in vertex names.

Theorem 2.2 (Directed multigraph value, small n). *For $n \leq 6$, $M^*(n) = 2(n - 1)$, and the optimum is attained by a $\{0, 1\}$ weight matrix. Hence for all $m \geq 2$, $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$, attained by the double star.*

It is worth being clear about what “for all m ” rests on, because the theorem proves infinitely many values from finitely many solves. It is not that each m was tested. In this mode `solve` never sees m at all. It proves the single m -free number $M^*(n)$, and because that optimum turns out to be attained by a $\{0, 1\}$ matrix, $L_m^{\text{dir}}(n) = (m - 1)M^*(n)$ then carries that one fact to every m at once. The only quantity that is checked case by case, and the only genuine limit on how far the result reaches, is n .

`solve` proves [Theorem 2.2](#) by returning $M^*(3) = 4$, $M^*(4) = 6$, $M^*(5) = 8$, and $M^*(6) = 10$, each optimal with zero gap, so that $M^*(n) = 2(n - 1)$ over this range. The first three finish quickly, $M^*(3)$ and $M^*(4)$ in well under a second and $M^*(5)$ in a few seconds, while $n = 6$ takes about twenty minutes. At $n = 7$ the same encoding does not close, and we do not pretend otherwise: it is a finite obstacle of scale, not a new idea, and it marks the current edge of certified knowledge for this variant. The solver transcript is in [Appendix A](#).

2.8 What the solver may claim

Now that the pieces inside `solve` are built, it is worth stating plainly, because the rest of the thesis depends on it, what each is allowed to claim. The checker *measures*: its connectivity values are exact. Its three proving implementations, the exhaustive enumeration, the pruned search, and the cut-counting, *prove*: an exhausted enumeration or an optimal, zero-gap solve is an upper bound for that fixed n . The search of [Chapter 3](#) will *discover*: it produces concrete graphs, hence lower bounds, and never an upper bound. These three claims are exactly what `solve` labels its answer with, so that an exact value, a lower bound, and an upper bound can never be quietly

confused with one another.

With the microscope built and aimed at proof, the variants with small answers are settled and the directed base cases the hand proof needed are in hand. What proof by enumeration cannot do is reach the sizes where the answers are only conjectured, because the number of graphs explodes long before the interesting cases arrive. The next chapter measures that explosion exactly, and then turns the same program loose to *discover* the dense graphs that blind enumeration can never reach.

2.9 Reproducibility

Every claim in this thesis that rests on computation can be re-run from a single Python driver. All of the program's logic lives in `program/erdos915_unified.py`, one self-contained file. Its core needs only `numpy` and `scipy`, because every connectivity value it reports is a single `scipy` integer maximum flow on a capacity matrix. The model, the checker, the search, the enumeration, and the self-test all run on those two libraries alone.

A few more libraries sit on top, each for one job. The certifier uses `pulp`, a Python library for writing an optimisation once and handing it to any solver, and runs it on CBC, the free solver `pulp` bundles, or on the commercial Gurobi solver by a one-line switch. Two others are optional and only for presentation: `matplotlib` draws the figures, and `networkx` supplies the Gomory–Hu tree behind one of them. An optional compiled helper, `_erdos_fast.so` (built from `_erdos_fast.c` by `build_fast.sh`), speeds up the hot-path `_tiny_maxflow` and `_canonical_form` calls, but the pure-Python path returns the same answers.

A handful of named entry points reach the whole pipeline. `solve` drives every variant. The measures are `max_edge_connectivity` and `max_vertex_connectivity`, with the hypergraph and Gomory–Hu views beside them. The provers are `prove_directed_multigraph` and `enumerate_extremal_directed_multigraphs`, and `search_for_dense_graph` is the discoverer. Running `python erdos915_unified.py` executes the built-in self-test, the suite of invariant checks that confirms the connectivity values quoted throughout this chapter, including the sanity checks on the bidirected star and on K_n and $K_4^{(3)}$.

The figures and proofs are just as reproducible. Every figure is regenerated by the companion driver `program/make_figures.py`, which imports the same file and fixes every random seed, so the pictures are reproduced exactly rather than merely resembled, and the table in `program/README.md` records which routine produces which figure. The solver transcripts behind this chapter's certified values, the pruned-search log for [Theorem 1.10](#) and the cut-counting certificate for [Theorem 2.2](#), are reproduced verbatim in [Appendix A](#), so a reader may check the proofs without running anything at all.

2.9.1 What a machine-checked claim in this thesis means

Reproducibility is not the same as certification, and it is worth stating plainly what standard the computed results here meet, since a solver reporting OPTIMAL is not by itself a mathematical certificate. Four practices carry the weight.

First, the exhaustive searches are self-certifying in a way the optimisations are not. A finished include-or-exclude sweep has visited a superset of the objects it needed to visit, and the two prunes are proved sound in [Chapter 3](#), so its verdict depends on no solver and no floating-point arithmetic, only on the checker. Where a value in this thesis is called *proved*, that is usually the reason.

Second, every reported connectivity is exact and integral. The checker is a single integer maximum flow, so no extremal value in the thesis is a rounded floating-point quantity. Real numbers enter in exactly two places, and neither carries a reported extremal value on its own. The first is the fractional optimisation of [Theorem 2.2](#), where the reported optimum is a $\{0, 1\}$ matrix and is checked exactly. The second is the fractional hill-climb behind [Remark A.33](#), which runs a floating-point maximum flow and is reported as numerical evidence for a conjecture, never as a value.

Third, the load-bearing computations are run twice, by implementations that share no code. The base cases of [Theorem 1.10](#) and [Theorem 1.11](#) were each confirmed by the program and by separately written checkers, agreeing on every size. The values of [Section A.8](#) were computed by the program's routine and by an independent script, agreeing on every cell. Fact (b) of [Remark A.39](#) was produced by the generation enumerator and re-verified graph by graph with the exact checker. Two implementations agreeing is not a proof, but it is the practical guard against the failure mode a certificate would catch, namely a correct method with a wrong program.

Fourth, the self-test at the foot of the file is a standing regression check on every invariant the text quotes, and it is run after every change to the program.

What this does not amount to is a formally checkable certificate of the kind a SAT solver can emit, such as a resolution proof a small independent verifier could replay. For the one claim here that *would* rest on a solver's infeasibility verdict rather than on an exhaustion, fact (a) of [Remark A.39](#), that would be the right standard to aim at. Fact (a) is not settled either way at present, so nothing in this thesis currently rests on such a verdict, and it is recorded among the open problems rather than claimed.

Chapter 3

Discovering Bounds by Search

The previous chapter proved what could be proved by looking at every graph of a fixed size. That method is honest and exact, and it stops almost at once. This chapter begins by measuring exactly how fast it stops, because the speed of that collapse is the whole reason a second kind of tool is needed. Then it builds that tool. Where the certifier walks all graphs, the search walks a few of them well, cooling like a physical system toward the densest feasible shape. What it returns is never a proof. It is a concrete graph, hence a lower bound and a conjecture, a stepping stone toward an upper bound that a later proof or certificate must supply. The chapter ends by asking the question any such tool must face: how good is it, and how far from the truth might its answers be.

3.1 The wall: how enumeration explodes

The brute force of [Chapter 2](#) visits every graph of a given size. The trouble is how many that is. A simple undirected graph on n vertices is a yes-or-no choice on each of the $\binom{n}{2}$ possible edges, so there are $2^{\binom{n}{2}}$ of them. A digraph chooses on each of the $n(n-1)$ ordered pairs, giving $2^{n(n-1)}$, and allowing multiplicities up to a cap c raises the base from two to $c+1$. These are not numbers that merely grow. They detonate. [Figure 3.1](#) draws the growth on a logarithmic scale, where each step up the axis multiplies the count tenfold, and the message is the same in every panel: the count passes any reachable budget while n is still in the single digits.

Three readings of the figure matter for what follows. Adding vertices is the sharpest cost: each new vertex multiplies the count by an exponential in n , so a single extra vertex can move a problem from seconds to centuries. Raising the connectivity budget is the next cost, lifting the base of the exponent rather than its height, which is why the multigraph curves sit above the simple one in every panel. Direction is the third: a digraph has twice as many cells to fill as the undirected graph on the same vertices (and a directed 3-uniform hyperedge, a tail with two heads, has three times as many), so the directed panel climbs fastest of all, and those are precisely the variants whose answers [Chapter 1](#) could not prove. The separation, by contrast, costs nothing here, because edge and vertex disjointness search the same graphs and differ only in what the checker measures, which is why a single pair of panels suffices for all of them. The certifier of [Chapter 2](#) reaches a few vertices further than blind enumeration, because its cut-counting optimisation is polynomial in the size of the encoding rather than in the number of graphs, but solving that encoding is itself hard and the gain is measured in single vertices, not orders of magnitude. Past that, no exhaustive method reaches the interesting sizes. Something

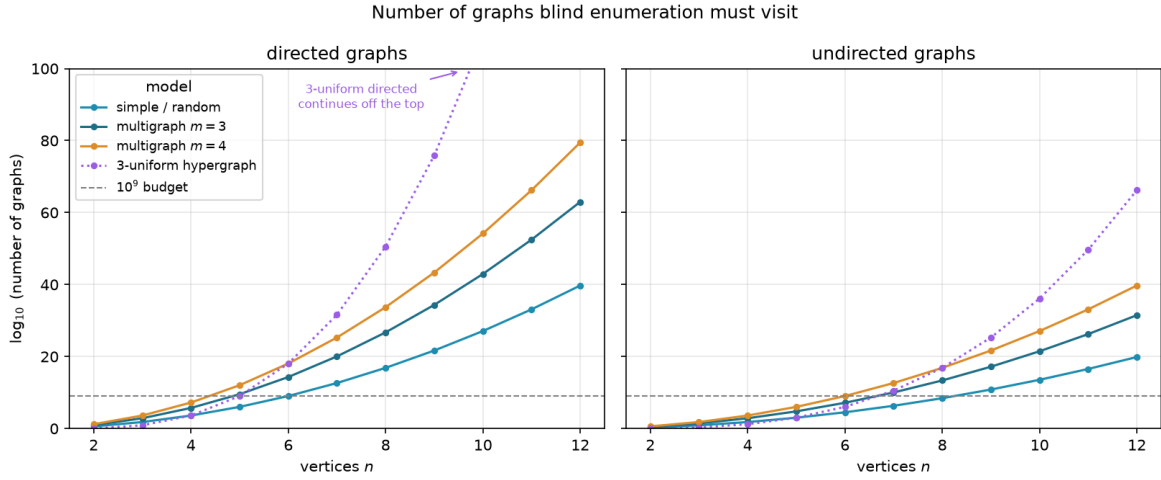


Figure 3.1. The number of graphs blind enumeration must visit, on a logarithmic vertical axis, for directed graphs (left) and undirected graphs (right). Each panel shows the simple model, two multigraph connectivity caps, and the three-uniform hypergraph (dotted), with a dashed line at a generous budget of 10^9 candidates, and every curve crosses that budget while n is still in the single digits. Writing E for the number of off-diagonal cells a model may fill, a simple graph offers 2^E candidates, a multigraph with multiplicity cap $m - 1$ (the connectivity ceiling) offers m^E (one choice per cell from $\{0, \dots, m - 1\}$), and the 3-uniform hypergraph offers $2^{\binom{n}{3}}$ undirected or $2^{n \binom{n-1}{2}}$ directed, since a directed hyperedge is a tail together with two heads. The count depends only on the model and the direction, never on whether one later asks for edge- or vertex-disjoint routes, so a single pair of panels covers all of those variants at once. Direction multiplies the number of cells to fill, by two for graphs (ordered versus unordered pairs) but by three for the 3-uniform hypergraph, so the directed panel sits far above the undirected one, and the directed cases, exactly the ones [Chapter 1](#) left open, are the first to go.

that does not look at every graph is required.

3.2 The temperature search

We want, for fixed n and m , a graph with as many edges as possible subject to $\lambda^{\max} \leq m - 1$ (or $\kappa^{\max} \leq m - 1$ for the vertex variants). The space of graphs on n vertices is enormous and its feasible region has no helpful shape: adding an edge can be harmless until, together with edges added long before, it suddenly creates a forbidden pair. A greedy walk that only ever accepts an improving move stalls at the first such collision, in a graph that is locally maximal but globally far from extremal.

The classical method for this kind of landscape is *simulated annealing* (SA), named after the metallurgical process of cooling a heated material slowly so that its atoms settle into a low-energy crystal rather than a disordered glass. Applied to graphs, annealing escapes local optima by sometimes accepting a worsening move, and by accepting fewer of them as the run proceeds. Each graph G is a state with energy

$$E(G) = -|E(G)| + \Lambda \cdot \max(0, \lambda^{\max}(G) - (m - 1)),$$

Algorithm 1: Temperature search for a dense feasible graph

Input: vertices n , value m , start T_0 , cooling α , penalty Λ , steps S
Output: the densest feasible graph encountered

```
1  $G \leftarrow$  empty graph on  $n$  vertices;  
2  $T \leftarrow T_0$ ;  
3 for  $S$  steps do  
4   propose  $G'$  by toggling one adjacency, biasing removals toward low-sensitivity edges;  
5    $\Delta \leftarrow E(G') - E(G)$ ;  
6   if  $\Delta \leq 0$  or  $\text{rand}() < e^{-\Delta/T}$  then  
7      $G \leftarrow G'$ ;  
8   if  $G$  is feasible and denser than the best so far then  
9     record  $G$  as the best;  
10   $T \leftarrow \alpha T$ ;  
11 return the best recorded graph;
```

which rewards edges and penalises any excess connectivity. The penalty is a soft constraint rather than a hard one, so the walk may pass briefly through infeasible graphs rather than being walled inside the feasible region, and we return below to why that freedom is worth having. A move toggles one adjacency. The Metropolis rule accepts an improving move always and a worsening move only with probability $e^{-\Delta E/T}$, a chance that shrinks both as the move gets worse and as the temperature drops, and the temperature T falls geometrically, multiplied by a fixed factor just below one at every step. Algorithm 1 is the whole loop, the one place in this chapter where a body is worth showing, because the loop *is* the method.

This loop is what `solve` runs in its discovery mode. With the exhaustive flag off it anneals instead of enumerating, following the Metropolis walk of Algorithm 1 under geometric cooling, removals biased toward low-sensitivity edges, and returns the densest feasible graph it encountered together with the full temperature trace. What it returns is a witness, hence a lower bound, never a proof.

`search_for_dense_graph(n, m, variant, separation) → SearchResult`

DISCOVER

in the size n , the ceiling m , the `Variant`, the separation, and a step budget

out a `SearchResult`: the densest feasible graph found, with the full step-by-step trace

how one Metropolis cooling run of Algorithm 1. The driver `best_of_searches` dispatches several independent runs across processor cores and keeps the densest, so a single local optimum never decides the answer.

It is fair to ask why the walk is allowed above the ceiling at all. Reaching every feasible graph does not require it. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the ceiling, and reading that path backward builds the target up the same way. The

feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is.

We let it go infeasible anyway, because reachability and efficiency are different things. A feasible graph can be locally maximal, meaning every edge we might add creates a forbidden pair, and yet still sit far below the densest feasible graph. To improve such a graph the walk has to add an edge and then remove a different one, a swap whose first half is infeasible. A walk confined to feasible graphs could only find that better graph by retreating most of the way back to the empty graph and climbing a different slope, a detour so long that the cooling schedule usually ends first. Allowing a brief excursion above the ceiling lets the walk tunnel straight across instead, trading one load-bearing edge for a better one in two moves rather than dozens. The soft penalty also turns the search into a single smooth landscape with no hard walls for the Metropolis rule to bounce off, and the same machinery then carries over unchanged to variants whose feasible region is not so conveniently connected. Going above the bound is never the goal, then. It is the shortcut the annealer takes on its way back down to it.

3.3 Temperature and edge sensitivity

The phrase “biasing removals toward low-sensitivity edges” in [Algorithm 1](#) is where the temperature acquires its meaning, and it is the heart of the method. For an edge e define its *sensitivity*

$$\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e),$$

the amount by which deleting e relaxes the binding connectivity. A load-bearing edge sits on a tightest cut and has $\sigma(e) > 0$. Removing it would drop $\lambda^{\max}$ and so loosen the very constraint that limits how many edges the graph may hold. A free edge has $\sigma(e) = 0$: it can be removed without easing the binding pair, which usually means it can be put to better use elsewhere.

`edge_sensitivity(G, u, v) → ℕ`

MEASURE

in a graph G and one of its edges $e = (u, v)$

out $\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e)$

how delete e and let the checker remeasure, where a positive value marks a load-bearing edge. The whole-graph version `sensitivity_map(G)` reuses one shared baseline measurement. This single number is the dial the cooling turns.

When the search proposes a removal it draws the edge with probability proportional to $e^{-\sigma(e)/T}$. The dial is now explicit. At high temperature these weights flatten and the walk toggles edges regardless of how load-bearing they are, exploring widely and willing to dismantle essential structure. As the temperature falls the weights concentrate on $\sigma = 0$, so the walk almost never disturbs a load-bearing edge and instead rearranges only the free ones, until the graph crystallises onto an extremal shape. This is exactly the behaviour one wants, and it is the behaviour the formulation asks for: the temperature tracks how strongly removing an edge

changes the bound. Figure 3.2 shows one such run settling onto a certified optimum. Section B.2 collects five further traces, covering both matrix models in both directions and one run under the vertex separation, and the behaviour is the same in each.

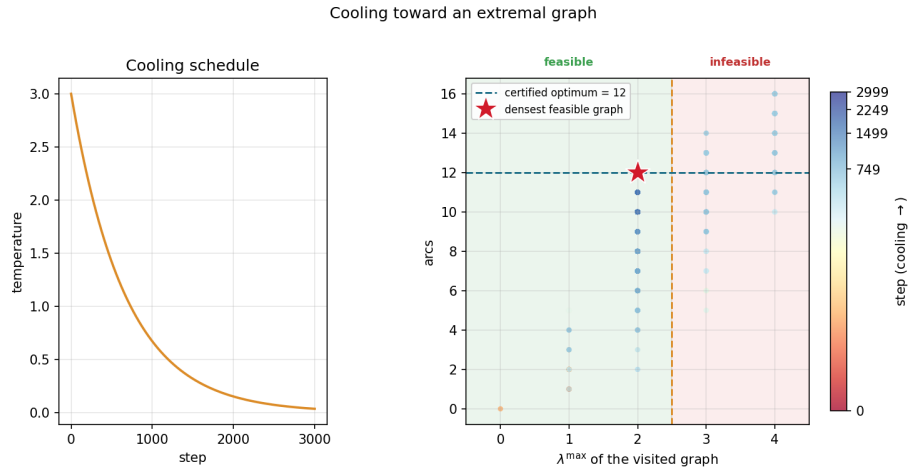


Figure 3.2. A single cooling run on the directed multigraph problem at $n = 4$, $m = 3$. *Left:* the temperature schedule, cooling geometrically from its starting value. *Right:* every visited graph plotted by its binding connectivity $\lambda^{\max}$ (horizontal) against its arc count (vertical), coloured by the step at which it was visited. Annealing uses a soft penalty rather than a hard wall, so while hot (dark points) the walk strays into the infeasible region $\lambda^{\max} > m - 1 = 2$, picking up extra arcs it is not allowed to keep. As the temperature falls (bright points) the penalty drives the walk back across the feasibility boundary and onto the densest graph that stays feasible, the certified optimum of 12 arcs at $\lambda^{\max} = 2$ (star).

The same sensitivities also explain a finished graph. Colouring a graph's arcs by σ shows at a glance which arcs are structural and which are slack. Figure 3.3 illustrates all four sensitivity levels on one directed multigraph: the darkest arcs carry the most flow and lower $\lambda^{\max}$ the most when removed, the cool arc contributes nothing. The picture is not decoration: it is the certificate of tightness made visible, and it tells the search exactly which arcs to leave alone as the temperature falls.

A single cooling schedule can still settle into a local optimum, so in discovery mode `solve` runs several independent schedules from different starting seeds and keeps the densest result across all of them. Because the restarts share no state and none depends on the outcome of another, they run in parallel. The program dispatches them across the available processor cores, so the wall-clock cost of k restarts is a small multiple of a single run rather than k times it, bounded below by the slowest restart and by the cost of starting the worker processes. A handful of parallel restarts is enough to reach the known optima at the sizes we examine. This is a reliability measure, not a change of method, and the parallelism is invisible to every number the thesis reports: the same best graph would be found by running the schedules sequentially, just more slowly.

Both statements hold verbatim for the vertex measure $\kappa^{\max}$.

Proof. Fix an ordered pair (s, t) . By the max-flow / min-cut theorem $\lambda_G(s, t)$ is the smallest capacity of any cut separating s from t , where the capacity of a cut is the total multiplicity of the arcs that cross it from the source side to the sink side.

For part (1), reducing one adjacency by a unit lowers the capacity of every cut that the changed arc crosses and leaves all other cut capacities untouched, so no cut capacity rises. The smallest cut capacity therefore cannot rise either, which gives $\lambda_{G'}(s, t) \leq \lambda_G(s, t)$. Taking the maximum over all pairs yields $\lambda^{\max}(G') \leq \lambda^{\max}(G)$. Adding a unit is the same statement read backwards.

For part (2), write G'' for G with one unit added to a single adjacency. That extra unit of capacity crosses any fixed cut at most once, so every cut has capacity at most one greater in G'' than in G . The smallest cut capacity thus grows by at most one, giving $\lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$, and together with part (1) we have $\lambda_G(s, t) \leq \lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$. The maximum over pairs gives $\lambda^{\max}(G'') \leq \lambda^{\max}(G) + 1$.

For $\kappa^{\max}$ the argument is identical on the vertex-split network of [Figure 2.3](#), where changing an adjacency changes the capacity of its single arc $u^{\text{out}} \rightarrow v^{\text{in}}$ in the same way. Parallel copies leave that capacity at one and so change $\kappa^{\max}$ not at all, which only strengthens both parts. $\square$

These two facts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0, \lambda^{\max} - (m - 1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \leq m - 2$, stays feasible by part (2), again with excess zero and no measurement. Only an addition made right at the boundary $\lambda^{\max} = m - 1$ can tip the graph over, and only there does the search run a single flow, capped so that it stops the instant it finds one route too many. The exact connectivity value is recomputed only on the rare step whose proposal is genuinely infeasible, where the penalty term needs it. The three cases of [Proposition 3.1](#) read directly off the code:

```

1 if (is_add and lam_ub <= m - 2) or (not is_add and feasible_current):
2     excess = 0                                # part (2) or part (1): provably feasible
3 elif not exceeds_bound(proposal, m - 1, separation=separation):
4     excess = 0                                # at the boundary, but the flow clears it
5 else:
6     excess = measure_full(proposal) - (m - 1) # infeasible: exact penalty

```

Listing 3.1. The three cases behind [Proposition 3.1](#), in the order the proof gives them: removal from a feasible graph (part 1), addition below the bound (part 2), and only otherwise a single capped flow.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at the cadence where it already pays for flows to refresh

Construction	n	m	Search	Theory
spanning tree (simple undirected, $m=2$)	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected, $m=3$)	6	3	10	10
bidirected star (simple directed, $m=2$)	4	2	6	6
bidirected star (simple directed, $m=2$)	6	2	10	10
hub-bipartite tie (simple directed, $m=2$)	7	2	12	12
double star (multi directed, $m=3$)	4	3	12	12
double star (multi directed, $m=3$)	5	3	16	16

Table 3.1. Validation by rediscovery. For each variant the temperature search, run from scratch with a handful of restarts, recovers exactly the extremal value proved or certified in the earlier chapters. The “Search” column is an honest lower bound found by annealing, and the “Theory” column is the proved or certified value.

the sensitivity map. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

The key point is that the energy returned on every step is, by [Proposition 3.1](#), exactly the value the naive implementation would have computed. Every accept-or-reject decision, every random draw, and the graph the walk finally returns are therefore unchanged, and the speed-up is invisible to every number the thesis reports. This is verified rather than asserted. A regression test runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. Nothing about the optimisation hides inside the bound: the witness the search reports is always certified by an exact measurement, exactly as in the slower search.

It is worth noting where this method earns its keep. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu tree of [Chapter 2](#), which encodes every pairwise edge-connectivity in $n - 1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let one annealer serve every variant at speed.

3.5 Rediscovering the known extremisers

A search method is only worth trusting if it recovers what is already known before it is pointed at what is not. Run from the empty graph on the cases whose answers [Chapter 1](#) and [Chapter 2](#) settled, and told only to pack in edges without breaching the connectivity ceiling, the cooled search arrives unprompted at the named constructions. [Table 3.1](#) reports the densest graph it found against the value the theory predicts, and in every case the two agree.

What the table hides is that the graphs themselves are the named constructions, not merely graphs of the right size. The simple undirected run at $m = 2$ settles on a spanning tree, the only shape with $n - 1$ edges and no cycle. The undirected multigraph run settles on a star with every

edge at multiplicity $m - 1$, an instance of the multi-tree extremiser behind [Theorem 1.3](#). The directed multigraph runs settle on the double star of [Chapter 2](#), both directions of every spoke at full multiplicity. The simple directed runs find the bidirected star at $n = 4$ and, at $n = 7$, land on twelve arcs, the value at which the hub and the one-directional wall tie, exactly the crossover [Theorem 1.10](#) predicts. The cooled search does not know these names. It rediscovers the structures because, under the connectivity ceiling, there is nowhere denser to go.

Some extremisers are past the size at which a quick search lands reliably, but the exact checker of [Chapter 2](#) settles them immediately by measuring their connectivity. The one-directional bipartite digraph on eight vertices has 16 arcs at $\lambda^{\max} = 1$, and 16 already exceeds the linear hub value $2(n - 1) = 14$: the smallest size at which the quadratic phenomenon strictly wins, made concrete. It is also a measured illustration of the search's limit: repeated cooling runs at $n = 8$, $m = 2$ stall at the hub value of 14, because dismantling a finished hub and rebuilding a one-directional wall is far more than the single-arc moves of the walk can tunnel through. The known answer there comes from the construction and the checker, not from the search, which is exactly the division of labour this chapter argues for. The augmented bipartite digraph on ten vertices at $m = 3$ has 30 arcs at $\lambda^{\max} = 2$, the thirty-arc counterexample of [Remark 1.12](#), confirmed pair by pair against the naive bound of 27 it refutes. The directed double star on seven vertices at $m = 4$ has 36 arcs at $\lambda^{\max} = 3$, matching $2(n - 1)(m - 1)$. None of these is a proof of optimality. Each is an exactly measured feasible graph, a lower bound the checker stamps as genuine.

3.5.1 Simulated annealing versus tabu search

The annealer is not the only way to walk this landscape. A natural alternative, suggested by Jan Goedgebeur, is *tabu search* ([Table 4.2](#) names the implementation), which keeps the energy and the neighbourhood of [Algorithm 1](#) but replaces the cooling schedule with a deterministic best-improving step. At each step it evaluates every single-edge move from the current graph and takes the one that lowers the energy most, then forbids the pair it just changed for a fixed number of steps, the *tabu tenure*, so the walk cannot immediately undo its last move and stall in a two-step cycle. When no improving move is left and the search stalls, it perturbs the incumbent with a few random moves and carries on. There is no temperature and no accept-worse coin: apart from the perturbation kicks the walk is deterministic. Where annealing explores by occasionally stepping uphill and trusts the cooling to settle it, tabu climbs greedily and trusts the tabu list to keep it from retracing its path.

The two engines rediscover the same values, but at very different speeds, and on the harder directed cases the difference is decisive. [Table 3.2](#) runs them head to head at an equal budget of six seconds each, both restarted within that budget. On the simple undirected problem both reach the optimum at once. On the two directed problems past the certified range tabu reaches the optimum within the budget while the annealer plateaus below it, assembling the eighteen-arc simple-directed extremiser and the full twenty-four-arc double star at $n = 7$ where the annealer stalls at sixteen and twenty. The cause is the one already met at the $n = 8$, $m = 2$ hub stall above. The annealer cools onto whatever dense shape it first reaches and then cannot dismantle

it within the budget, whereas tabu’s best-improving step follows the steepest density gradient and the tabu list stops it sliding back, so it assembles the structured extremiser the cooled walk struggles to find.

Variant	n	m	optimum	best in 6 s		time to optimum	
				SA	tabu	SA	tabu
simple undirected	7	3	9	9	9	1.3 s	0.1 s
simple directed	7	3	18	16	18	—	2.6 s
directed multigraph	7	3	24	20	24	—	2.5 s

Table 3.2. Simulated annealing (SA) against tabu search at an equal six-second budget per method, all at $m = 3$, the annealer restarted from fresh seeds and tabu restarted on stalls. Only the simple undirected value is a proved optimum (Mader). The two directed values are the best known rather than proved, found by search and construction past the certified range that stops below $n = 7$ for those variants (Section 3.6), so here the “optimum” column is the target each engine is trying to reach. Both reach the simple undirected optimum. On the two directed problems tabu reaches the best known value while the annealer plateaus below it. The last two columns give the wall-clock time each method first reached that value on the reference machine, with a dash where it did not reach it in the budget.

Figure 3.4 shows the divergence directly, plotting the densest feasible graph each engine has found against wall-clock time on the two directed multigraph cases. Tabu climbs in a few decisive steps and holds at the optimum, while a single annealing schedule rises quickly and then flattens a few arcs below it. The annealer’s independent restarts recover the optimum on the smaller case, as Table 3.1 records, so a single plateau is not the whole story there. But at $n = 7$ the plateau persists within the budget, and one tabu schedule already reaches what the restarted annealer does not.

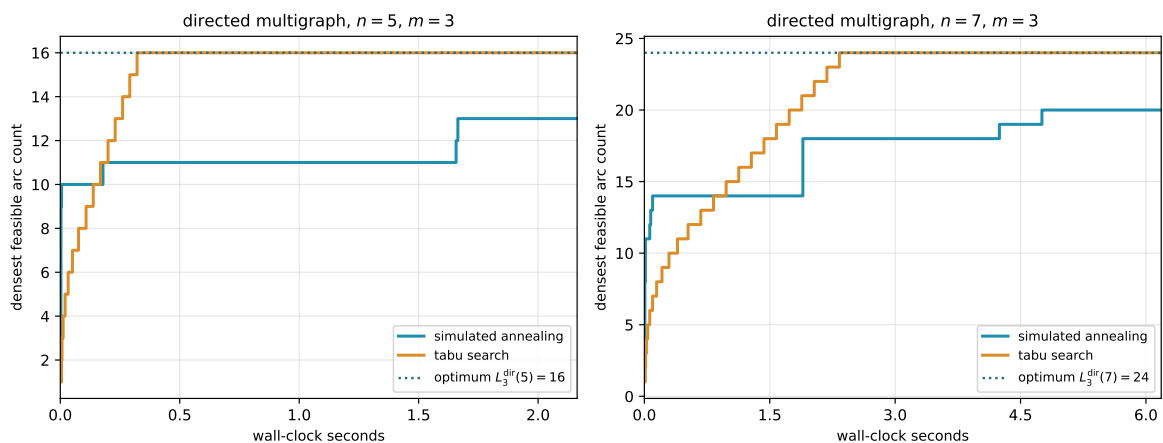


Figure 3.4. Densest feasible graph found against wall-clock time, for one simulated-annealing schedule and one tabu schedule on the directed multigraph at $n = 5$ and $n = 7$, $m = 3$. Tabu climbs in a few best-improving steps to the optimum (dotted) and holds, while a single annealing schedule rises quickly and then flattens below it. The time axis is wall-clock on the reference machine, so unlike the seed-reproducible figures elsewhere this one is a representative timed run.

This advantage comes at a cost the equal-time comparison already absorbs but is worth naming: each tabu step scans the whole neighbourhood, so it is far more expensive than the an-

nealer's single random proposal, and tabu takes many fewer but better-aimed steps in the same wall-clock. Tabu search is therefore the program's default for solving the problems, reaching every value this thesis reports, the rediscovery of [Table 3.1](#) and the conjectural lower bounds alike, and reaching the harder directed optima past the certified range where the annealer plateaus. Simulated annealing is kept for the self-check and verification, where the small values fall at once, the lighter per-step cost is all that is needed, and its independent restarts parallelise trivially across cores.

3.6 The trichotomy: proved, certified, conjectured

It is worth stopping to lay out, for every variant, the three kinds of knowledge the computation can hold about it, because the whole value of the pipeline is that they are kept apart and never quietly merged. About any given value the program can do one of three things. It can *prove* the value, closing a fixed size from above by exhausting every graph of that size, an upper bound with the standing of a hand proof. It can *certify* it, the same kind of upper bound but obtained by the cut-counting optimisation rather than blind exhaustion, reaching a vertex or two further. Or, where proof and certificate both run out, it can only *conjecture* the value, exhibiting a concrete dense graph that pins down a lower bound and leaving the matching upper bound to a later argument.

Proof and certificate bound the answer from above. The search bounds it only from below. Where a closed *formula* exists it can tie the two together, threading through the certified small cases and the conjectured large ones as a single curve. This is the discipline the whole thesis runs on, and it is worth stating as one: the search proposes a lower bound, a certificate or a hand proof disposes the upper bound, and only when both meet is a value called settled.

We begin with the single variant where all three kinds of knowledge are non-trivial and visibly distinct, the directed simple arc problem at a fixed $m \geq 3$, and then sweep the whole family. For that variant the knowledge splits cleanly by the size of the graph.

For small n the answer is not conjectured at all. The pruned exhaustion inside `solve` walks every simple digraph on n vertices and returns the exact maximum, complete and certain, an upper bound with the standing of a hand proof. At $m = 3$ it certifies 6, 9, 12 for $n = 3, 4, 5$, and there the exhaustion stops within a reasonable budget, the pruned walk no longer closing $n = 6$ in tractable time. That is the certified region, finite and closed.

Beyond it the ceiling is gone and only lower bounds remain, each witnessed by a concrete feasible graph. The search finds the extremal 15 arcs at $n = 6$ on its own. Past that size it falls behind: across six restarts it reaches 17, 17, 20, 23 for $n = 7, \dots, 10$, while the named constructions of [Chapter 1](#), measured exactly by the checker, supply 18, 21, 25, 30, the last being the thirty-arc graph of [Remark 1.12](#). The reported lower bound at each size is the better of the two, exactly as the rediscovery section combined search and checker by hand. These are conjectures with an honest floor and no matching ceiling. That is the searched region, unbounded and forever one-sided.

The two regions are tied together by a single closed form. The conjectured value

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right) \quad (n \geq m)$$

of [Conjecture 1.14](#) is not a fresh guess bolted on past the certified range. It is one formula that already passes through every certified value at small n and then continues, without a seam, through every searched value at large n . The directed-arc panels of [Figure 3.5](#) show this directly, the certified squares at the small sizes and the open lower-bound circles beyond both sitting on the one red conjecture curve, proof below the formula and witnessed graphs above it. The formula is the through-line that makes the small certified cases and the large conjectured ones a single statement rather than two.

Two features of those directed-arc panels are worth naming. The crossover from the linear hub term to the quadratic bipartite term happens at $n = 9$, where the second argument of the maximum first wins, and that is past the last certified size. The shape of the answer therefore changes in exactly the region where only the search can see it, which is precisely why a discovery tool was needed and a certifier alone would not do. And at $n = 10$ the formula reads 30, the count of the very counterexample that overturned the natural initial conjecture, so the closed form is not a smooth fit to easy data but one that survives the case that broke the previous guess.

Asking the same three questions of every variant, the answers sort into three regimes, and [Figure 3.5](#) shows all twelve at once, one panel per variant, in exactly this proved, conjectured, and guessed language.

In the first regime the closed form is already a theorem for every n , proved by hand in [Chapter 1](#) or cited, and the computation only confirms it. The undirected edge families live here for every m , together with the directed problems at $m = 2$ and, up to a threshold in m that differs by model, the undirected vertex families as well: through $m \leq 4$ for simple graphs and multi-graphs ([Theorem 1.5](#)) and through $m \leq 3$ for hypergraphs ([Theorems A.45](#) and [A.48](#)). Past those thresholds the same rows drop into the third regime below, which is why the leaf colours of [Figure 4.7](#) are stated at general m and [Table 4.1](#) carries the exact ranges. For these the certified small cases and the rediscovered constructions of [Table 3.1](#) are validation, not new knowledge: the formula was never in doubt, and the program's role is to check that the pipeline reproduces it. That the two multigraph vertex rows measure exactly as their underlying simple graphs is a matter of the counting convention of [Remark 2.1](#) rather than a theorem, so those rows need no computation of their own at all. The directed vertex rows do need their own, since Whitney's inequality gives a lower bound there but no upper one, and they get it: every exact point on those panels comes from the search and the exhaustion run under the vertex test.

In the second regime the three columns genuinely differ and the computation is load-bearing. This is the regime of the worked example above. The directed simple arc and vertex problems at $m \geq 3$ are certified only at the small n the exhaustion reaches and conjectured by the search beyond, with the formula of [Conjecture 1.14](#) the through-line.

The directed multigraph arc problem sits here too, with one extra economy: in its cut-counting mode `solve` proves the m -free number $M^*(n) = 2(n-1)$ outright for $n \leq 6$, and the scaling identity $L_m^{\text{dir}}(n) = (m-1)M^*(n)$ of [Chapter 2](#) carries that one certified number to every m at once, so the closed form $2(n-1)(m-1)$ is proved across the whole m axis at those sizes. Past the crossover at $n = 7$ the conjectured value switches to the bipartite branch $(m-1) \lfloor n^2/4 \rfloor$, where the even sizes are proved conditional on the odd ones and only a single mixed-regime minimum-degree statement keeps the induction from closing in general ([Appendix A](#)). For $m = 3$ specifically this route is bypassed entirely, reducing the whole problem to one finite computation, as [Chapter 4](#) details.

In every row of this regime the formula agrees with proof where proof reaches and with the search where it does not, exactly the linkage the worked example displayed.

In the third regime there is no formula at all, and the honesty is in drawing the guess as a guess. The undirected vertex problem at $m \geq 6$ has been open since 1974 with its extremal structure genuinely unknown. The undirected hypergraph vertex problem joins it, but only from $m \geq 4$: at $m \leq 3$ its value is settled for every uniformity and every n ([Theorems A.45](#) and [A.48](#)), which is why that panel of [Figure 3.5](#) carries a proved blue curve while its counterpart in [Figure 3.6](#) carries a yellow guess. The two *directed* hypergraph variants are in the third regime at every m , and they need a one-tail, many-head reading of a Berge edge that this thesis defines itself only so that `solve` can measure it at all. Here the program still computes exact values at the few sizes that finish, and its search still exhibits feasible graphs at a handful more, but no closed form is in sight. The panels of [Figure 3.5](#) carry, instead of a red conjecture curve, a yellow line that simply interpolates the machine points in straight segments, trusted no further than the eye. We draw it *through* the points rather than fitting a smooth curve above them: the points are honest lower bounds, the truth lies on or above them, and a line riding above the points would claim more than the search ever found.

Two cautions govern how to read the dotted curves. At small n the complete graph or hypergraph is itself feasible, its binding connectivity being only $n-1$, so the earliest points merely trace the trivial maximum and reveal nothing of the open behaviour: the undirected-vertex points run up $\binom{n}{2}$ until n passes m , and only then bend toward the linear growth the proved edge value predicts.

Past that bend the dotted line is as strong as the best lower bound behind it, which at each size is the better of the quick search and a named construction the checker verifies. The search alone stalls, exactly as it stalled on the quadratic wall in the rediscovery above: where it cannot build a denser construction its own reports climb only linearly. The clearest case is the directed hypergraph, whose bipartite family is quadratic, reaching $\sim (m-1)n^2/(4(r-1))$ hyperarcs by [Proposition A.50](#): there the walk slips below that construction at larger n , so it is the construction that carries the circles, with the search free to beat it wherever it can. We therefore read the dotted trends as honest lower bounds and not as fitted laws, and the one closed-form hypothesis we record, the quadratic for the directed hypergraph, comes from the construction and not from

the shape of the dots.

So to the question of whether one formula links what is proved to what is conjectured, the answer is three-valued and depends on the variant, and the grid of [Figure 3.5](#) shows all three at a glance. Where the problem is already solved the curve is blue, a theorem the computation merely agrees with. Where the problem is open but tractable the curve is red, a conjecture the certified squares confirm below and the searched circles confirm above, and that is the case the program was built for. Where the problem is genuinely hard the curve is yellow, a guess interpolated through the search points and an open question of what, if anything, the dots truly trace.

One feature stands out in the bottom row: the directed hypergraph panels, both arc and vertex, climb steeply at $m = 3$, the yellow guess curving upward rather than running straight. This is genuine and not search instability: the directed Berge edge packs a tail and $r - 1$ heads into a single object, and the bipartite construction of [Proposition A.50](#) already reaches roughly $(m - 1)n^2/(4(r - 1))$ hyperarcs, so the value grows with n^2 . The same quadratic shape sits one column over, in the directed arc and vertex panels, whose conjectured red curves carry the $\left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor$ term and rise just as fast. Super-linear growth is the signature of *direction* across the grid, not of the hypergraph alone: every directed column bends upward while the two undirected columns stay linear. [Figure 3.6](#) redraws the same grid at $m = 6$: the proved curves climb linearly with m exactly as their formulas predict, while the open variants' search points and their yellow interpolation climb higher without a proved curve to meet, so raising the threshold does not change which regime a variant lives in, only how far the red and yellow curves have travelled.

The same grid answers the harder question the trichotomy raises, which is what the search is worth where no upper bound exists to check it against. It says two things at once. First, wherever an upper bound exists, proved or certified, the search lands exactly on it and never above it at every plotted size, which is strong evidence that the search is not leaving edges on the table, with the one instructive exception recorded above: the directed $m = 2$ run at $n = 8$, where it stalls on the hub and the bipartite optimum must be supplied by the construction instead.

Second, on the directed arc problem at $m \geq 3$, where no upper bound is known past the certified range, the plotted lower bounds land exactly on the conjectured value $\max(m(n - 1), \lfloor (n + m - 2)^2/4 \rfloor)$ at every reachable n , and the certified points agree with it as far as they go, but past the first open size it is the constructions, not the cooled walk, that carry the bounds there. Run from scratch the search confirms the conjecture at $n = 6$ and then plateaus a few arcs short (17, 17, 20, 23 against 18, 21, 25, 30 at $n = 7, \dots, 10$, across six restarts), for the same reason as the $m = 2$ stall recorded earlier: the extremal shapes past the crossover are bipartite walls, and a walk that has crystallised onto a hub cannot rebuild itself into a wall by single-arc moves within the cooling budget.

So to the question “do we believe the reported lower bounds are best possible,” the grid answers: wherever a proof or a certificate can check them, yes, and on the open sizes they sit on the only value anyone has conjectured, with the search confirming the first open size and

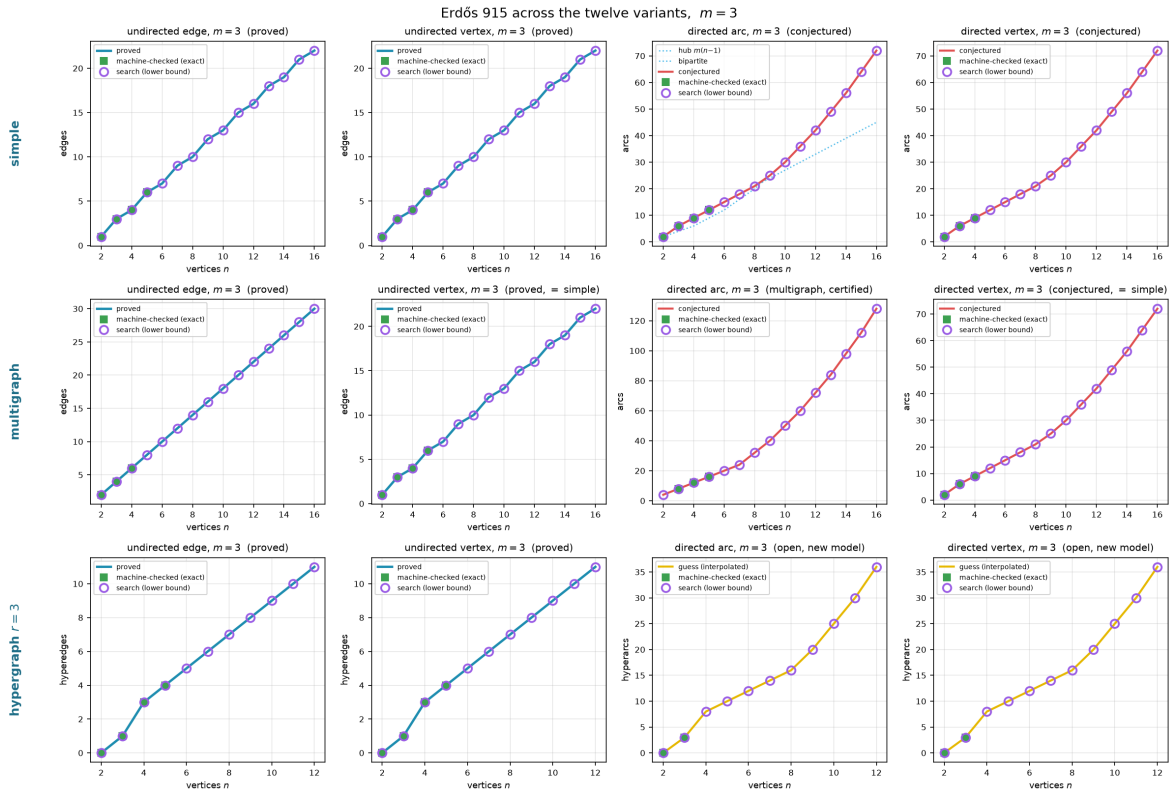


Figure 3.5. All twelve variants at $m = 3$, edges against n , laid out by model (rows: simple, multigraph, 3-uniform hypergraph) and by separation and direction (columns: undirected edge, undirected vertex, directed arc, directed vertex). A *blue* curve is a value proved for all n , a *red* curve a conjectured formula, and a *yellow* curve a bare interpolation of the machine points where no formula is known, the three told apart by colour rather than by dash pattern. Filled squares are the sizes `solve` proved or certified exactly. Open circles are lower bounds at larger n , the better at each size of the search's densest find and a named construction measured by the checker, reported as a running maximum so a feasible graph at one size is never lost at the next. On the open variants the yellow curve is the piecewise-linear interpolation *through* those circles, never above them, and the axes are scaled to the data rather than to a loose certain interval. The directed hypergraph panels in the bottom row rise steeply because their value is quadratic in n , not linear.

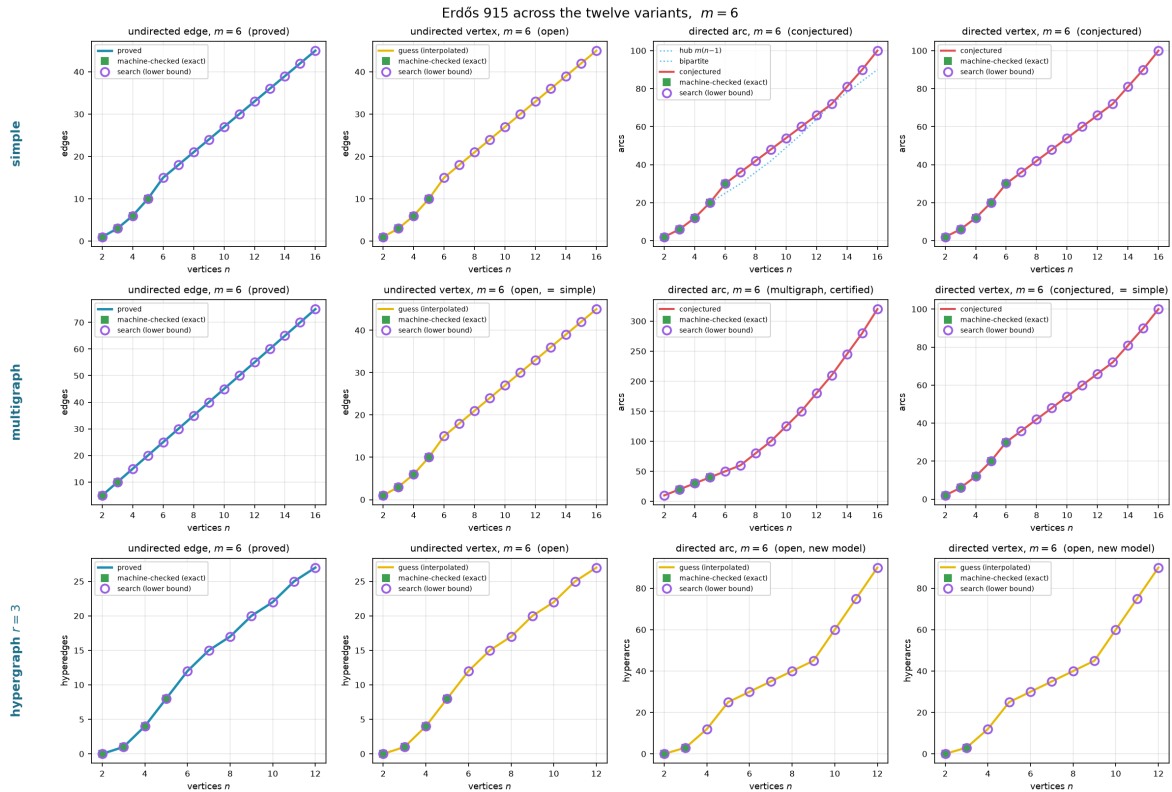


Figure 3.6. The same twelve variants at $m = 6$. The undirected edge and multigraph edge cases remain proved (Mader’s theorem holds for all m), while the undirected vertex separation is the famous open problem at $m \geq 5$ and the directed cases carry the same conjectured formula scaled to $m = 6$. On the open vertex panels the circles never sit below the proved edge value at the same size, because every graph feasible for the edge problem is feasible for the vertex problem (Whitney), so the edge extremiser is an honest lower bound there, and the search is free to climb past that floor where the vertex problem is genuinely richer. Comparing this figure with the $m = 3$ version above shows that the proved cases grow linearly with m as the formulas predict, while on the open cases the circles and their yellow interpolation climb steadily without a proved curve to meet.

the constructions carrying the rest. The gap between those lower bounds and what is proved is exactly the gap between the conjecture and the missing upper bound, which [Chapter 4](#) names precisely. The search has not closed that gap, and it cannot. What the pipeline supplies is reproducible evidence for which side of the gap the truth lies on.

Honesty

A number found by the search is reported as a lower bound, and is marked as certified or proved only when a separate cut-counting optimisation or a hand proof supplies the matching upper bound. No extremal value is ever asserted on the strength of the search alone. The figures in this chapter use fixed random seeds, so every search point is reproducible.

Chapter 4

Synthesis, Results, and Open Problems

The journey set out from one clean theorem and followed it across twelve variants, proving what proved cleanly, building a machine where the proofs ran out, certifying the small cases the machine could close, and using the search to discover the dense graphs the proofs had singled out and to point at the ones still open. This closing chapter draws the threads together. It maps the directed frontier, where proof and certificate both still fall short of the full answer, names the two genuine surprises the landscape contains, gathers the twelve answers into one picture, and says plainly what the program changed and where it stops.

4.1 The directed frontier and the single open lemma

Two of the open directions are directed, and they share a structure worth stating exactly rather than hiding. The directed arc problem is a race between two constructions on different scales. The directed hub ([Construction 1.9](#)) reaches $m(n-1)$ arcs and grows linearly. The augmented bipartite construction ([Construction 1.13](#)) packs $\lfloor (n+m-2)^2/4 \rfloor$ arcs and grows quadratically. Which leads depends on n , and the two cross at an order linear in m , near $n = 9$ at the $m = 3$ of [Figure 4.1](#). [Conjecture 1.14](#) asserts that together they are the whole story.

[Conjecture 1.14](#) is proved as a lower bound for all m , and it is a theorem at $m = 2$ by [Theorem 1.10](#). Past $m = 2$ it sits squarely in the second regime of the trichotomy of [Section 3.6](#): certified by exhaustion at the small sizes `solve` can close for $m = 3$, conjectured by the temperature search beyond, and never beaten anywhere the search reaches. What is missing is a general upper bound for $m \geq 3$, and the obstacle is structural rather than computational.

The route to the upper bound is to show that an extremal digraph, once it is past the hub regime, must look like the augmented bipartite construction: its vertices split into a part A and a part B with every arc running $A \rightarrow B$, and B thickened only mildly. Three structural facts, each proved in [Appendix A](#), push in this direction. At $m = 2$ out-neighbourhoods are mutually unreachable and second out-neighbourhoods are disjoint, and for every m a complete $A \rightarrow B$ partition forces each vertex of B to absorb at most $m-2$ arcs from inside B . With the partition free to vary, the total arc count $|B|(|A| + m - 2) = |B|(n - |B| + m - 2)$ is maximised at $|B| = \lceil (n + m - 2)/2 \rceil$, which gives $\lfloor (n + m - 2)^2/4 \rfloor$, precisely the conjectured upper bound.

The reduction is clean except at a single point, and it is worth naming the point exactly rather

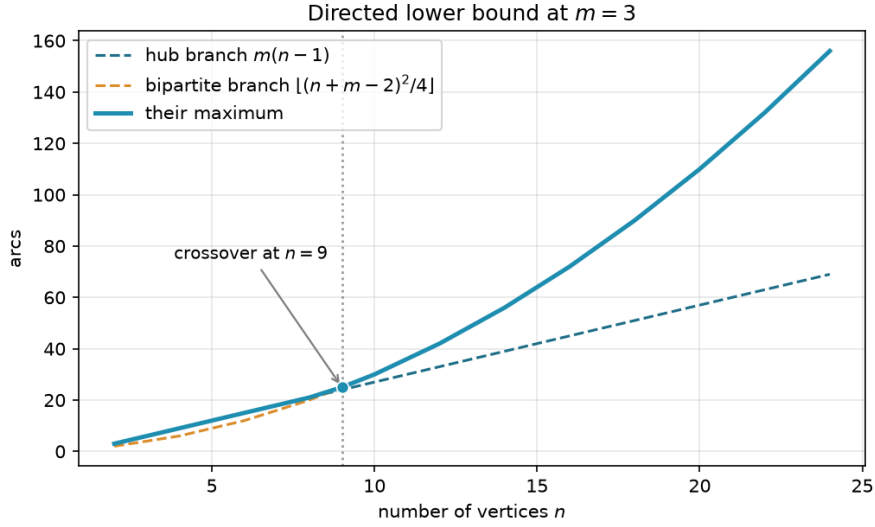


Figure 4.1. The two directed lower bounds at $m = 3$. The linear hub branch leads for small n . The quadratic augmented-bipartite branch overtakes it near $n = 9$ and leads thereafter. [Conjecture 1.14](#) asserts their maximum is the exact value.

than hiding it. The count goes through provided the extremal digraph decomposes the way [Figure 4.2\(a\)](#) draws it, and that shape is four conditions rather than one: a partition $V = A \cup B$ exists, every arc from A to B is present, no arc lies inside A , and no arc runs from B back to A . The last condition is the one with a name and a picture. Under it a route between two vertices of B can never escape B , so the connectivity inside B is governed entirely by B 's own arcs and the two-hop budget applies, and [Figure 4.2\(b\)](#) shows how a single backward arc destroys exactly that. It is also the condition where the natural delete-and-compensate exchange fails to be monotone, which is why it is the one usually named as the obstacle.

Naming it alone would nonetheless overstate the position, so the appendix states the whole hypothesis as [Conjecture A.21](#): an extremal non-hub digraph admits such a partition, complete forwards, empty inside A , and with nothing coming back. Nothing proved here forces an arbitrary extremal digraph into that shape, and the small-case evidence for it is exactly the kind of thing the search of [Chapter 3](#) can probe. So the honest summary is that [Conjecture 1.14](#) for $m \geq 3$ awaits one structural decomposition theorem, of which the backward-arc clause is the hardest quarter.

What does not wait is the leading constant, and neither, it turns out, does the order of the term after it. [Proposition A.22](#) proves, with no hypothesis at all beyond feasibility, that a digraph with $\lambda^{\max} \leq m - 1$ has at most $\lfloor n^2/4 \rfloor + \sqrt{m} n^{3/2}$ arcs, and that deleting at most $\sqrt{m} n^{3/2}$ of them leaves a one-directional bipartite digraph. Its engine is a single observation the rest of the chapter has been circling: for any ordered pair, the direct arc and the two-step detours through distinct midpoints are automatically arc-disjoint, so feasibility caps their number, and summing that cap over every pair caps $\sum_x d^+(x) d^-(x)$. A vertex with both a large in-degree and a large out-degree is therefore expensive, every vertex is nearly a pure source or a pure sink, and throwing away each vertex's smaller side leaves the bipartite wall standing.

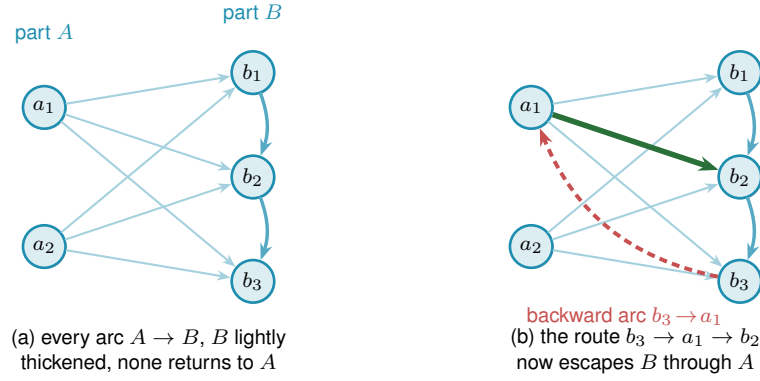


Figure 4.2. The directed frontier reduction and the single obstruction. **(a)** The structure the upper bound wants. Every arc runs from A to B (faint), B carries only a light internal thickening (the $m - 2$ budget, dark blue), and no arc returns to A , so a route between two vertices of B never leaves B and the count $|A||B| + (m - 2)|B|$ is complete. **(b)** A single backward arc $b_3 \rightarrow a_1$ (red) breaks this. The route $b_3 \rightarrow a_1 \rightarrow b_2$ (red leg then green leg) now escapes B through A , an extra route the budget never counted. Ruling out backward arcs in an extremal digraph is the one missing lemma behind the $m \geq 3$ upper bound, and the checker refuted a proposed counterexample by finding exactly such a hidden route.

That error term is then sharpened all the way to the right order. The $\sqrt{n}$ is wasted precisely when every vertex carries a middling degree, which a digraph with $\Theta(n^2)$ arcs cannot do, so a dichotomy on the minimum degree removes it: [Theorem A.24](#) proves, still with no structural hypothesis, that

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m - 1)(n - 1), \quad \text{hence} \quad \ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n).$$

So both the leading constant $\frac{1}{4}$ of [Conjecture 1.14](#) and the *order* of its second-order term are unconditional theorems. What the conjecture still claims beyond them is one constant: it says the second-order coefficient is exactly $(m - 2)/2$, and the theorem pins it between that and $4(m - 1)$, a factor of roughly eight. That constant, and not the shape of the answer, is what [Conjecture A.21](#) is now needed for, and [Remark A.25](#) shows that the obvious way to close it, reading the same two inequalities more carefully, provably cannot work.

The machine has been useful here in the negative direction too: a proposed counterexample to the conjecture, a backward arc bolted onto the thirty-arc graph, was refuted by the checker, which found the third route the construction’s author had missed.

The directed vertex-disjoint problem shares the constructions but not the proofs, and the distinction matters more than it looks. Whitney’s inequality $\kappa \leq \lambda$ says a digraph obeying the arc ceiling obeys the vertex one, so the arc constructions are vertex-feasible and every lower bound carries across for free. It says nothing in the other direction, and the vertex-feasible family is the larger of the two, so no arc upper bound restricts it. At $m = 2$ the vertex value is nonetheless the same, and [Theorem 1.11](#) proves it by running the induction a second time under the stricter separation, with its own exhaustive base cases through $n = 7$. For $m \geq 3$ the vertex value is conjectured equal to the arc value, on the strength of the same constructions and of exhaustive

small-case checks run under the vertex test rather than inherited from the arc one. What it will not do is follow from the arc case, even once [Conjecture A.21](#) is supplied, because an upper bound proved over the smaller family says nothing over the larger one. The vertex problem needs its own upper bound at $m \geq 3$, and [Table 4.1](#) lists it as a separate conjecture rather than a corollary. The directed multigraph problem has its own two-branch story, parallel to the simple one. The conjectured value is

$$L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor),$$

where the double star realises the linear branch and the one-directional bipartite multigraph, every arc at multiplicity $m-1$, realises the quadratic one.

The quadratic branch here uses the balanced partition $|A| = \lfloor n/2 \rfloor$, $|B| = \lceil n/2 \rceil$, giving $(m-1) \lfloor n^2/4 \rfloor$ arcs. The multigraph and the simple digraph disagree about the best partition, and the reason is worth spelling out. In a multigraph the $m-1$ routes per pair are supplied directly by $m-1$ parallel copies of every $A \rightarrow B$ arc, so nothing needs to be added inside B , and the arc count $(m-1)|A||B|$ is largest at the balanced split. In a simple digraph each ordered pair carries at most one arc, so the extra routes must instead come from arcs inside B , and feeding B those in-arcs is what pushes its optimal size up to $|B| = \lceil (n+m-2)/2 \rceil$ once $m \geq 4$ (at $m \leq 3$ the balanced and shifted splits happen to give the same count, as [Construction 1.13](#) records). One consequence is that the multigraph's two branches cross at $n = 7$ for every m , because the factor $m-1$ multiplies both branches and cancels, whereas the simple crossing point grows with m and already sits near $n = 9$ at $m = 3$.

On the linear side the certifier closes the problem, proving $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ for $n \leq 6$ and every m at once ([Theorem 2.2](#)), and at $m = 2$ the whole problem collapses onto the simple digraph theorem for every n , because two parallel arcs are already two routes ([Theorem A.31](#)). On the quadratic side, a scaling-and-deletion induction proves the bipartite bound for even $n \geq 8$, conditional on the bound one vertex below. The odd case once reduced to a single minimum-degree statement ([Conjecture A.29](#)), which the program confirms in the purely saturated and purely unsaturated regimes but leaves open in the mixed one, where multiplicities strictly between 0 and $m-1$ defeat the integrality the two pure regimes lean on ([Appendix A](#)).

A sharper route now bypasses that conjecture at $m = 3$, and the way it does so is the cleanest instance in the thesis of the proof handing its remaining work to the machine. An *attachment lemma* ([Lemma A.34](#)) bounds the degree of a vertex glued onto the bipartite extremiser, and a *conditional odd step* built on it ([Theorem A.36](#), for $m \in \{3, 4\}$) carries the value and extremal characterisation from each even level to the next odd one. At $m = 3$ that characterisation propagates through both parities ([Remark A.39](#)), so the entire problem for all $n \geq 8$ collapses onto two finite, integral statements at $n = 7$:

- (a) $L_3^{\text{dir}}(7) = 24$, and
- (b) the only 24-arc feasible multigraphs on 7 vertices with maximum degree at most 8 are $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 .

[Conjecture A.29](#) is thereby off the critical path for $m = 3$, replaced by two machine-checkable targets: a MILP infeasibility check for fact (a), and an exhaustive enumeration of degree-bounded multigraphs for fact (b). The enumeration has since closed fact (b): the sound generation enumerator of [Chapter 2](#) completed the sweep and returns exactly the three predicted extremisers ([Remark A.39](#)). The value statement (a), $L_3^{\text{dir}}(7) = 24$, is now the one computation still standing between the $m = 3$ problem and a complete theorem, and there the MILP remains runtime-limited on the bundled solver.

The linear branch turns out to be richer than the double star first suggested. Every doubled bidirected spanning tree is extremal, confirmed exhaustively at $n = 4$ and 5 , and the saturated attachment lemma ([Lemma A.37](#)) explains why: the doubled trees are exactly the closure of a single doubled edge under maximal attachment. Several distinct extremisers tie at 24 arcs for $n = 7$, and sifting and classifying them is precisely what the enumeration must do.

For $m \geq 4$ the conditional step still yields the value but uniqueness no longer propagates through odd levels, and for $m \geq 5$ even the value step stalls. These residual gaps are stated in full among the open problems below. They are finite, identified targets rather than mysteries, and natural goals for the next round of certified computation.

4.2 Two principal phenomena

Across all the variation, two phenomena stand out as the ones a first guess would not have predicted.

The first is the divergence at $m = 5$. The edge and vertex problems agree for $m \leq 4$, and it would be natural to expect them to agree forever. Instead they part company at the very next value, with $k_5(n) = \lfloor 8n/3 \rfloor - 3$ pulling strictly above $\ell_5(n)$ ([Remark 1.7](#)), and beyond it the vertex problem becomes genuinely hard and has stayed open since 1974.

The second is the quadratic bipartite phenomenon. In an undirected graph every edge contributes to the degree of both its endpoints, and Mader's theorem turns that symmetry into a hard ceiling: $\lfloor m(n-1)/2 \rfloor$ edges, linear in n . A digraph breaks the symmetry. Place all n vertices into two equal parts A and B and draw every arc from A to B , one direction only. The result has $\lfloor n^2/4 \rfloor$ arcs, quadratic in n , yet $\lambda^{\max} = 1$: every pair has at most one arc-disjoint route, because every route must cross the A -to- B wall exactly once and no arc crosses the other way to offer a second path. A graph whose arc count grows like n^2 can sit at connectivity one, something no undirected graph can do past a handful of vertices.

The initial conjecture for the directed case was linear, matching the undirected intuition. The one-directional bipartite construction refutes it outright, already at $m = 2$: for $n = 8$ the wall holds 16 arcs while the linear hub holds only 14. At $m = 3$ the refutation is the thirty-arc augmented bipartite graph of [Remark 1.12](#), which packs 30 arcs at $\lambda^{\max} = 2$ against the naive prediction of 27. The construction generalises to every m via [Construction 1.13](#), thickening the larger part B slightly so that each vertex of B absorbs $m - 2$ extra in-arcs from inside B : those extra arcs add

exactly $m - 2$ short detours from A to each $b \in B$, bringing $\lambda^{\max}$ from 1 to $m - 1$ while the arc count climbs to $\lfloor (n + m - 2)^2/4 \rfloor$.

The quadratic term is what makes the directed cases the deepest in the thesis. The upper bound argument must explain why no configuration beats the bipartite wall, which requires producing the partition in the first place, ruling out backward arcs, and controlling the internal structure of the large part B . That structural task is exactly what [Conjecture A.21](#) has not yet resolved for $m \geq 3$. What [Proposition A.22](#) and [Theorem A.24](#) show is that the wall is at least the right shape: every feasible digraph is within $O_m(n)$ arcs of being one, so the difficulty is a constant inside the second-order term rather than the phenomenon. The undirected problem has no analogous obstacle, because Mader's degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore not just a curiosity: it is the geometric reason the directed and undirected problems live on different levels of difficulty.

That same scarcity of extremal graphs is the thread through the remaining figures. [Figure 4.3](#) shows where the frontier runs: it plots every labelled graph at small n as a single point, its binding connectivity $\lambda^{\max}$ on the horizontal axis and its edge count on the vertical, so the whole population of graphs is visible at once. The points crowd into the low-connectivity, low-edge corner and thin out upward, where the blue extremal envelope, an integer staircase carrying one dot per connectivity level, traces the densest graph available at each level. The empty region above that staircase is precisely the part the main theorems forbid a graph to occupy. The extremal problem is the question of where that staircase runs, and the figure shows it is a thin frontier with almost nothing pressed against it from below.

[Figure 4.4](#) steps inside each graph to the level of individual pairs. For every graph in the full enumeration it records the connectivity of every vertex pair and pools all the observations, so the histogram counts not graphs but (graph, pair) observations. Each bar is split and stacked by the connectivity of the graph the pair came from: blue for pairs drawn from the lower-connectivity graphs, red for pairs from the higher-connectivity ones. The boundary between the two is set per variant at the midpoint of the connectivity range that variant's enumeration can reach, since the twelve variants span very different ranges and a single fixed boundary would leave some panels entirely one colour. A pair whose own connectivity exceeds the boundary can only come from a graph above it, so the bars to the right of the dashed line are wholly red, while to the left the two colours mix, since a high-connectivity graph still carries many low-connectivity pairs. The blue mass sits almost entirely at the lowest levels: high-connectivity pairs are rare even inside the graphs that have one.

[Figure 4.5](#) then shows how the same graphs distribute by sheer edge count, with the same per-variant stacked colouring. At each edge count the blue part counts the lower-connectivity graphs and the red part the higher-connectivity ones, so the bar height is the total number of graphs there and the split shows how the two populations separate. The red mass sits to the right, since denser graphs tend to carry higher connectivity, and the dotted line marks the densest graph still on the low-connectivity side, landing far out in the sparse tail. The densest low-connectivity

Separation	Model	Value	Status
<i>Undirected</i>			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (cited, Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 3$ ($m = 5, n \geq 13$, see Remark 1.7)	cited, open $m \geq 6$
vertex	multi	adjacency count, so the simple value $\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$)	convention (Remark 2.1)
vertex	hyper	$\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$ ($m \leq 3$), $\geq$ that for all m	proved $m \leq 3$ (Theorems A.45 and A.48), open $m \geq 4$
<i>Directed</i>			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$)	proved, conj. $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$	proved $n \leq 6$ and $m=2$, cond. $m=3$
arc	hyper	quadratic in n , exact value open	lower bd. (Proposition A.50)
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$)	proved separately (Theorem 1.11), conj. $m \geq 3$
vertex	multi	adjacency count, so the simple digraph value	convention (Remark 2.1)
vertex	hyper	quadratic in n , exact value open	lower bd. (Proposition A.50)

Table 4.1. The twelve structural variants of Problem 915 and their status. Every simple-graph closed form here is stated for $n \geq m$, the hypothesis [Theorem 1.2](#), [Theorem 1.5](#) and [Conjecture 1.14](#) carry: below it the complete graph or digraph is itself feasible and the answer is the trivial maximum, which the formulas exceed. “Certified” means proved by the cut-counting optimisation for the stated sizes. “conj.” means a conjecture with a proved matching lower bound. “cond.” means proved conditional on a finite, identified statement (for $m = 3$ directed multigraphs, the one remaining $n = 7$ computation, fact (a) of [Remark A.39](#)). “lower bd.” means only a proved lower bound is known, with the exact value open. “Convention” marks the two rows whose value is fixed by the counting convention of [Remark 2.1](#) rather than by a theorem. The appearance threshold $p^* = m/n$ of [Theorem 1.16](#) is deliberately not a column here: it is proved for $G(n, p)$, extends to the vertex and directed analogues by [Remark A.59](#), and does *not* transfer to the hypergraph models, whose natural density scale is different ([Remark A.60](#)).

graphs, the ones the extremal problem is about, are exceptional in every variant.

Finally, [Figure 4.6](#) lifts the whole study onto the (n, m) grid at once, drawing the best feasible edge count as a bar surface across every variant. The three proved variants rise as smooth linear or quadratic sheets that the formulas describe exactly, while the open variants show a visible step between the machine-proved blue bars and the search’s purple lower bounds, so the surface makes the proved-versus-open divide of the trichotomy a single readable shape. Reading along the m axis recovers the linear scaling in the threshold, and reading along n recovers each variant’s growth rate, which is the whole of [Table 4.1](#) compressed into one object.

4.3 Summary of the twelve variants

[Table 4.1](#) records the status of all twelve structural variants. Each entry is marked by how it is known: proved by hand here or in [Appendix A](#), cited to the literature, certified by the cut-counting optimisation for the stated sizes, or conjectured with a proved lower bound. The distinction is the honesty contract of [Chapter 2](#) carried through to the summary. [Figure 4.7](#) shows the same distinction as a picture before the table spells out every case.

[Table 4.2](#) answers a different question about the same twelve variants: not what is known, but which routine of the program knows it. Most variants share machinery rather than each needing its own, so the table is grouped by family, and every name in it has its own codecard box where it is built in [Chapters 2](#) and [3](#).

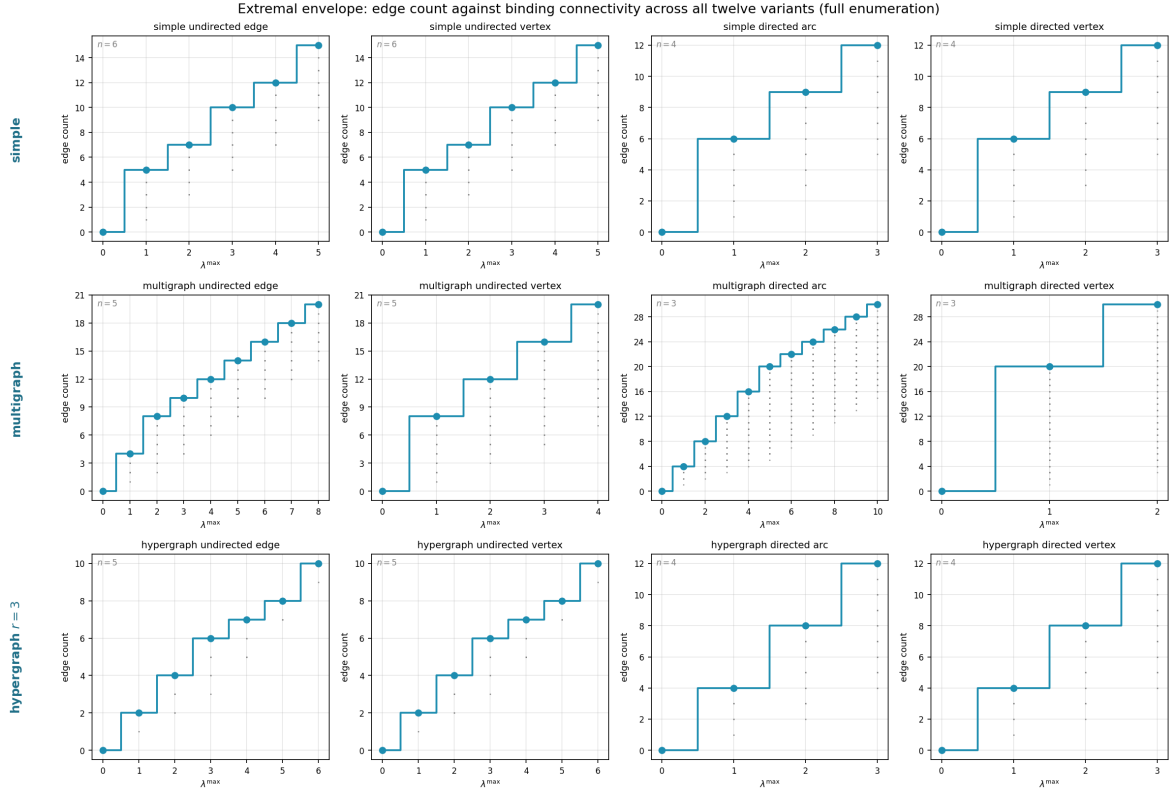


Figure 4.3. The complete landscape of all labelled graphs at small n (see panel annotations for each variant's n): every point is one graph, plotted at its binding connectivity $\lambda^{\max}$ horizontally and its edge count vertically. The blue staircase marking the densest graph at each connectivity level is the extremal envelope, the frontier the extremal problem asks about, with one dot per level so its exact value is legible. Graphs crowd the low-connectivity, low-edge region. The extremal graphs live at the top of the staircase. The emptiness above the envelope is exactly the region the main theorems forbid. The horizontal reach differs by panel because the connectivity that fits in a variant's enumeration n differs: arc-disjoint multigraph routes grow with edge multiplicity (the multigraph directed arc panel reaches $\lambda^{\max} = 10$ at $n = 3$), whereas vertex-disjoint routes are capped by the available internal vertices, so the multigraph directed *vertex* panel reaches only $\kappa^{\max} = 2$ at the same $n = 3$ (with a single internal vertex, a pair has at most the direct route plus one detour). This is a property of the separation, not of the search.

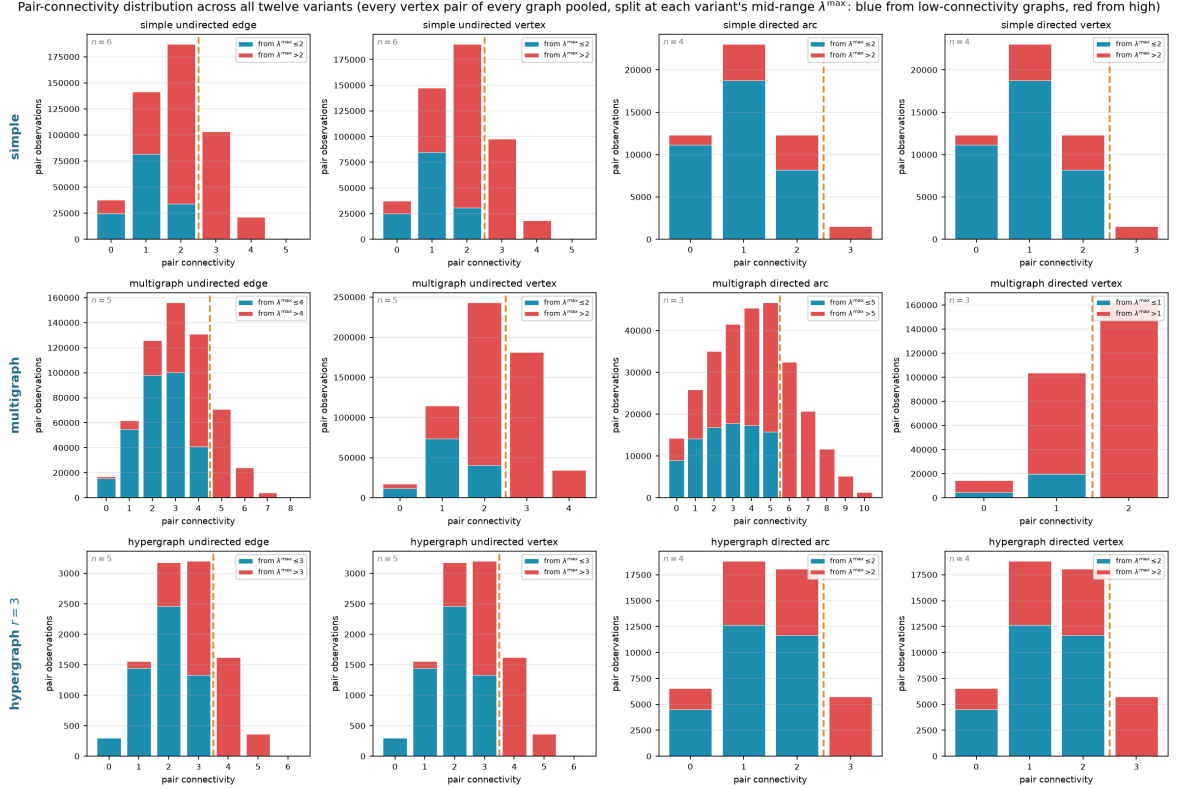


Figure 4.4. Pair-connectivity distribution across all twelve variants, pooling every vertex pair of every graph in the full enumeration. The horizontal axis is the connectivity of a single pair, and a bar's height is the number of (graph, pair) observations at that level. Each bar is split and stacked by the connectivity of the graph the pair was drawn from, blue from graphs at or below a per-panel boundary T and red from graphs above it, with the dashed line at T . The boundary is set per variant to the midpoint of the connectivity range that variant's enumeration reaches (each panel's legend gives its T), so the split stays informative in every panel instead of collapsing to one colour. Bars to the right of the dashed line are wholly red, since a pair of connectivity above T forces its graph above T . To the left the colours mix.

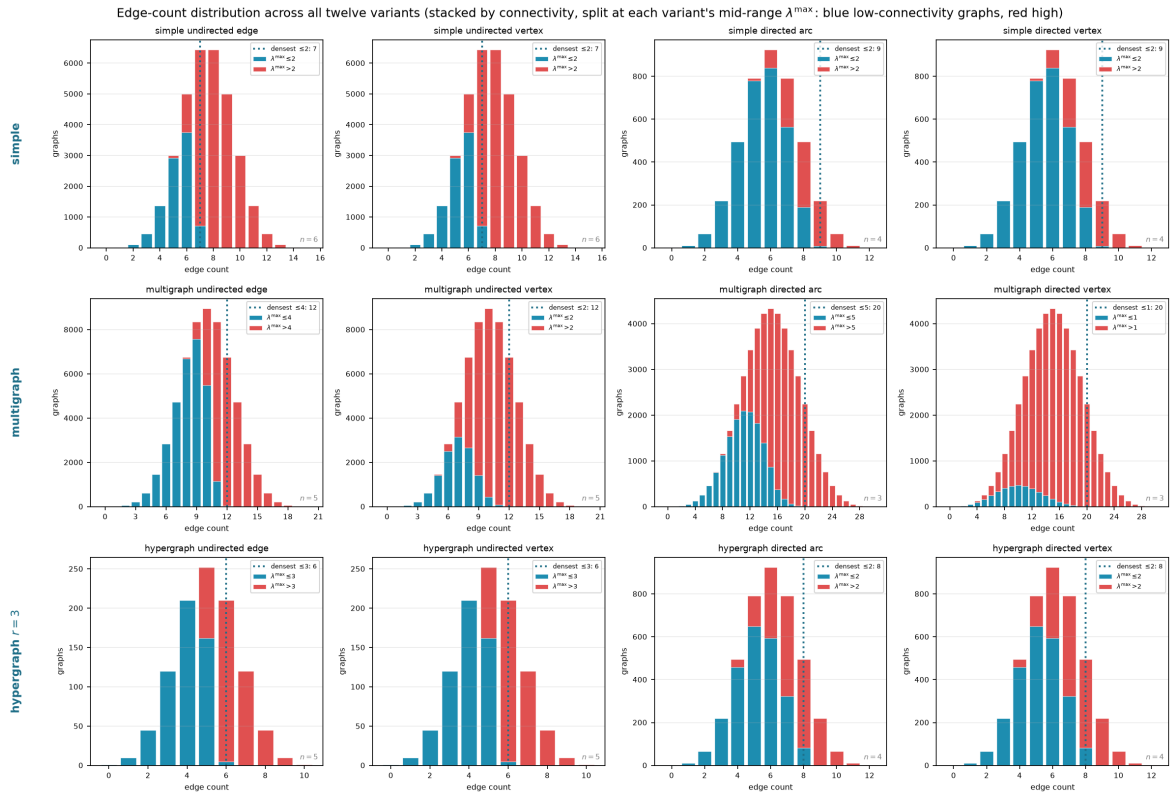


Figure 4.5. Edge-count distribution across all twelve variants, the same enumeration viewed by edge count rather than connectivity. Each bar is split and stacked by graph connectivity, blue for graphs at or below the per-panel boundary T and red above it (each panel's legend gives its T), so the full height is the total number of graphs at that edge count. The dotted vertical line marks the densest graph on the low-connectivity side. The reader can see directly how the two populations separate by edge count and how far low-connectivity graphs sit from the bulk.

Erdős 915: optimal bound over (n, m) , blue where proved, purple where only a search lower bound is known

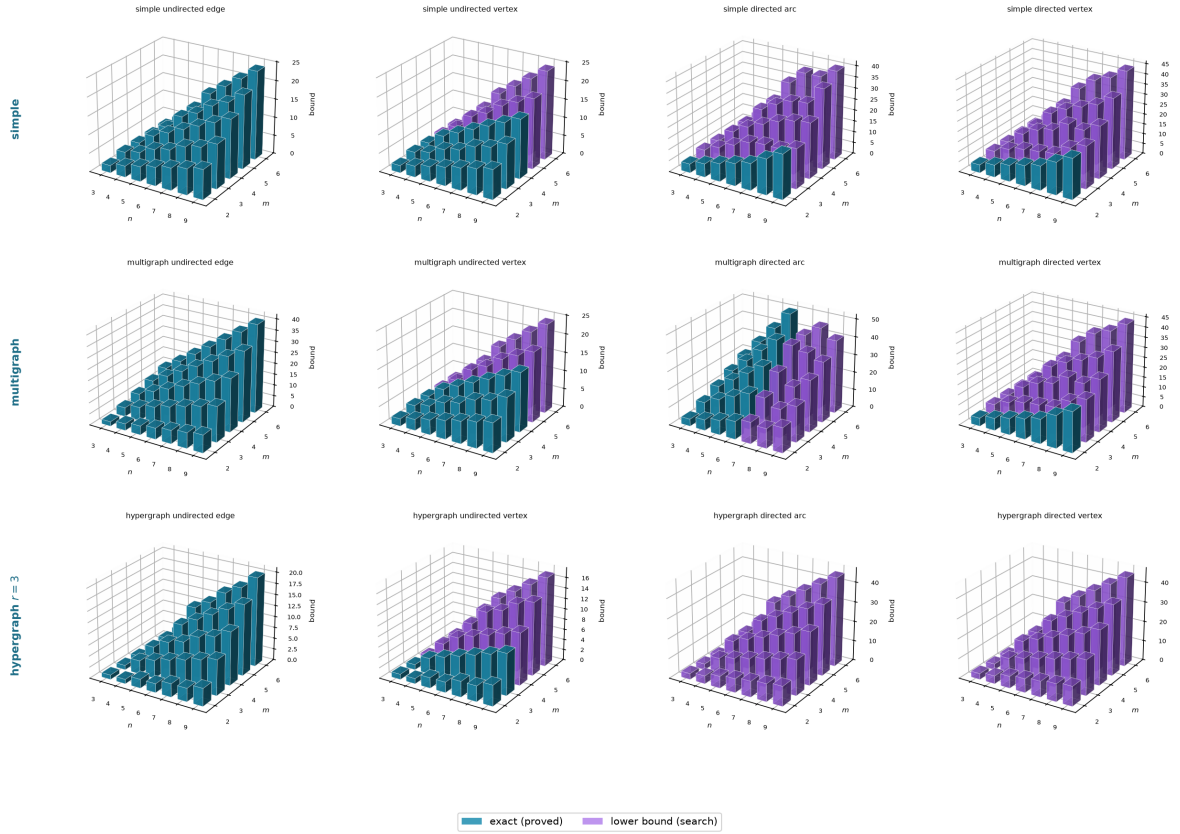


Figure 4.6. The optimal bound as a surface over the (n, m) grid for all twelve variants, every panel drawn from the same camera so the shapes compare directly. Each bar's height is the largest feasible edge count found by `solve` at that vertex count and forbidden threshold m . The single thing to read off is colour: a bar is *blue* exactly where the value is machine-proved and *purple* exactly where only a search lower bound is known, so each panel is a little map of what is settled. The three fully proved variants (Mader's undirected edge cases and the multigraph analogue) are solid blue sheets, rising linearly or quadratically as their formulas predict, while the open variants turn purple precisely where proof runs out. Reading along the m axis shows how the bound scales with the forbidden threshold, and reading along the n axis shows the growth rate for a fixed variant.

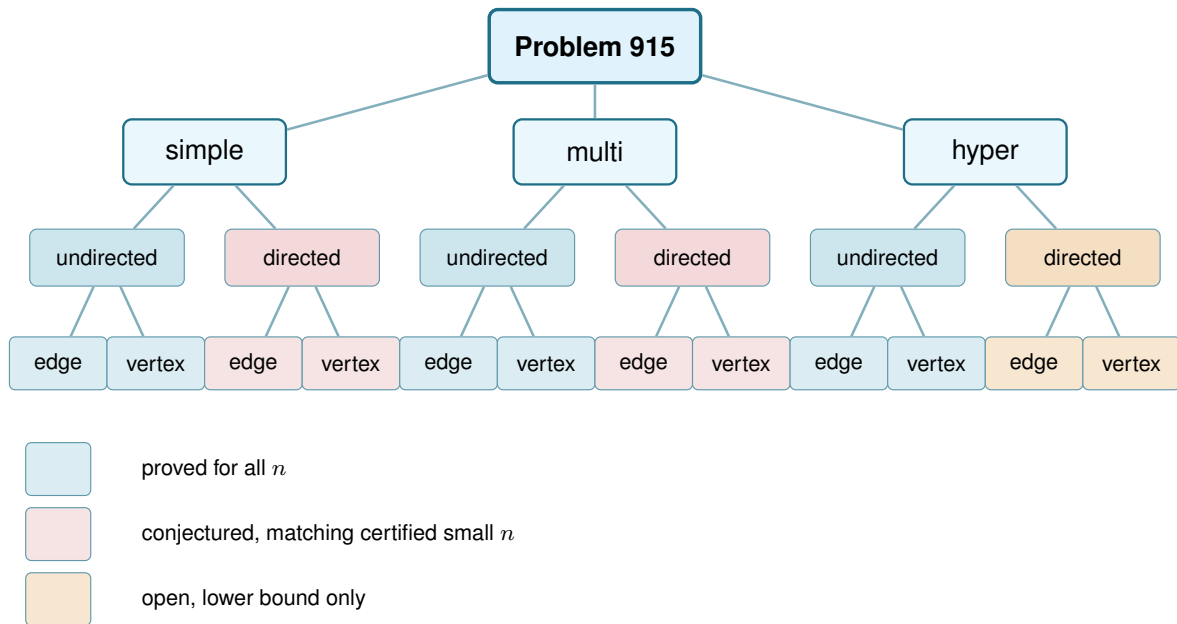


Figure 4.7. The same branching as Figure 1.7, now with every leaf resolved by the trichotomy of Section 3.6: blue is proved for all n , red is a conjectured formula backed by certified small cases and a matching search, and amber is open with only a lower bound known. Colour marks the status at general m , the harder regime this thesis mostly studies, and hides m -dependent exceptions and one convention that Table 4.1 states precisely. Every red leaf is a proved theorem at $m = 2$ and becomes conjectured only at $m \geq 3$, the directed edge leaves by Theorem 1.10 and the directed vertex leaves by the separate Theorem 1.11, which does not follow from it. Three blue leaves are proved only up to a finite m : the undirected vertex problem for simple graphs and for multigraphs ($m \leq 4$) and for hypergraphs ($m \leq 3$), open beyond. The two multigraph vertex leaves inherit whatever their simple counterparts have, because the counting convention of Remark 2.1 makes those two questions the same question.

4.3.1 Orientation models for the directed hypergraph

The directed rows of Table 4.1 read a directed Berge edge in one way, the *forward* one of a single tail fanning to $r - 1$ heads. It is not the only reading. In general a directed r -uniform hyperedge is a split of its vertices into a non-empty tail set T and head set H , a route entering at a tail and leaving at a head, and the gate checker of Chapter 2 measures any such split without change. Three models are worth naming, *forward* ($|T| = 1$), *backward* ($|H| = 1$), and *general* (any split, with genuinely mixed ones appearing once $r \geq 4$). The natural question, and one a reader of the directed figures tends to ask, is whether they multiply the table.

They do not, and two results say why. First, backward is not a new problem. Reversing every hyperarc turns a backward hypergraph into a forward one and reverses every route along the way, so the two models share every extremal number, edge and vertex, for all r , m and n (Lemma A.51). The backward column is the forward column read upside down. Second, the general model, though it admits more hyperedges on a single vertex set, obeys the same first-order bound. A hyperedge (T, H) occupies $|T| |H| \geq r - 1$ ordered tail-head pairs, each usable by at most $m - 1$ disjoint one-step routes, so $(r - 1)|E| \leq (m - 1)n(n - 1)$, exactly the forward cap of Proposition A.50 (Proposition A.52). The forward construction is itself a set of general

Variant family	Measure	Prove	Discover
Undirected, edge (all three models)	max_edge_connectivity max_hyperedge_connectivity	exhaustive enumeration (solve), Gomory–Hu cross-check	search_for_dense_graph
Undirected, vertex (all three models)	max_vertex_connectivity max_hyper_vertex_connectivity	exhaustive enumeration (solve)	search_for_dense_graph
Directed arc, simple	max_edge_connectivity	_exhaustive_directed: a pruned branch-and-bound	tabu_search_for_dense_graph
Directed arc, multigraph	max_edge_connectivity	the cut-counting optimisation (prove_directed_multigraph, prove_integral_arc_bound); the $n = 7$ classification (enumerate_extremal_directed_multigraphs_via_generation)	tabu_search_for_dense_graph
Directed vertex (simple and multi)	max_vertex_connectivity	reduces to the arc problem above	tabu_search_for_dense_graph
Directed hyper (arc and vertex)	hyper_connectivity	max_feasible_hyperedges: a pruned branch-and-bound	search_for_dense_graph

Table 4.2. The machinery of [Chapters 2 and 3](#) indexed by variant family rather than by function name. This is the cross-reference from a variant to the routine that handles it; the routine’s own signature, inputs, and outputs are in its codecard box.

hyperedges, so general is squeezed between the same lower and upper bounds as forward, and its value too is quadratic in n . Those two bounds differ by a factor of four, so what this fixes is the order and not the leading constant, and it does not by itself force the two models to agree asymptotically.

Where the models part company is only at small sizes, and the program shows exactly where. [Table 4.3](#) lists the extremal hyperedge counts the branch-and-bound checker ([Table 4.2](#)) proves for the forward and general models. The method is the include-or-exclude search of [Chapter 2](#) ([Figure 2.9](#)) walked over the candidate hyperedges instead of the arc cells: a hyperedge is included only while the checker confirms the growing hypergraph stays feasible, a branch is dropped once even including everything that remains cannot beat the best count already found, and a sweep that finishes has considered every candidate set, so the value it reports is proved exact rather than searched. The general one is strictly denser precisely when vertices are scarce, $n \approx r$, where the richer set of orientations lets a single vertex-set carry more hyperedges. At $n = r = 4$ it reaches eight hyperarcs against forward’s four, and the surplus grows with the connectivity budget. Once n exceeds r the computed gap closes, proved equal already at $r = 3$, $n = 4$, and larger searches find no separation thereafter. The reading that suggests itself is that once the hypergraph spreads past a single edge-set, the one-tail model already saturates the Menger budget and extra orientations buy nothing. That reading is computational evidence and a conjecture, not a theorem: what is proved is that both models obey the same upper bound and admit the same quadratic construction ([Proposition A.52](#)), and those two bounds differ by a factor of four, which is too coarse to force the values to agree. On the strength of the evidence the orientation axis is treated as collapsing, which is why Problem 915 stays a twelve-variant question, and the general model enters as a refinement at the margin rather than a thirteenth

column. Proving that it collapses, or exhibiting the first $n > r$ where it does not, is left open.

r	m	n	forward	general
3	3	3	3	4
3	3	4	8	8
4	3	4	4	6
4	4	4	4	8

Table 4.3. Extremal hyperedge counts the branch-and-bound checker proves exact for the forward and general directed r -uniform models, at the small sizes where they can differ. The general model is strictly denser only when $n \approx r$, doubling the forward count at $n = r = 4$, $m = 4$, and the gap closes once n exceeds r (proved equal at $r = 3$, $n = 4$). Edge- and vertex-disjoint counts agree throughout, and backward equals forward by [Lemma A.51](#).

4.4 Contributions and limitations of the program

The program turned a per-case craft into a uniform pipeline. Before it, each variant came with its own one-off computation and its own hand argument. After it, one representation holds every variant, one exact checker measures connectivity, one certifier proves small-case upper bounds, and one cooled search discovers extremisers and the lower bounds they witness. The rediscovery of [Chapter 3](#) is the evidence that the pipeline is faithful: pointed at solved cases, it returns the solved answers.

Its limits are as clear as its reach, and stating them is part of the honesty the thesis insists on. The search proves no upper bounds. It only ever exhibits graphs. The certifier proves upper bounds, but only for a fixed size, and the directed multigraph encoding closed unconditionally only up to $n = 6$. Of the two $m = 3$ computations at $n = 7$, the generation enumerator has completed the classification, fact (b), while the value statement, fact (a), remains runtime-limited on the bundled solver and is the one certificate still outstanding. And no amount of either replaces the genuinely structural gaps the thesis leaves open: the decomposition of [Conjecture A.21](#) that would turn [Conjecture 1.14](#) into a theorem, of which the backward-arc clause is the hardest part but not the whole, the directed vertex problem at $m \geq 3$, which Whitney’s inequality pointedly does not hand over with the arc case, and the undirected vertex-disjoint problem at $m \geq 6$, which has resisted for half a century and which the program can probe but not crack.

4.5 Open problems

- ★ The decomposition of [Conjecture A.21](#): prove that an extremal non-hub digraph admits a partition $A \cup B$ with every arc from A to B present, no arc inside A , and no arc from B back to A . That last clause is the backward-arc condition, and it is the one where the natural delete-and-compensate exchange fails to be monotone, but the other three are hypotheses too, and no argument here forces an arbitrary extremiser into that shape. The decomposition completes the upper bound in [Conjecture 1.14](#) for $m \geq 3$. Unconditionally, [Theorem A.24](#) already gives $\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$, so neither the leading constant nor the order of

the second-order term is at stake any more. What the decomposition is needed for is the *coefficient* of that linear term, which the conjecture puts at $(m-2)/2$ and the theorem bounds above by $4(m-1)$. Closing that factor of roughly eight is the smaller target, and [Remark A.25](#) narrows how it can be approached: the bound is already the exact maximum obtainable from the two inequalities the proof uses, so a sharper reading of them cannot help and a genuinely independent third inequality is required, one constraining the interference among the few expensive near-balanced vertices themselves.

- ★ The directed *vertex* problem at $m \geq 3$, which is a separate question and not a corollary of the arc one. Whitney's inequality makes the vertex-feasible family the larger of the two, so an arc upper bound does not restrict it, and [Conjecture A.21](#) would not settle it either. At $m = 2$ the two values agree ([Theorem 1.11](#)), proved by re-running the induction and its base cases under the stricter separation rather than by transfer. Whether they agree for $m \geq 3$ is open. The exhaustive search run under both tests at $m = 3$ returns the same value at every n it reaches, tabulated in [Remark A.17](#), so there is no small counterexample, but that is evidence and not an argument. The useful next step is structural rather than numerical: find the first digraph in which some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* that slack buys extra arcs, or show that it never can.
- ★ The multigraph vertex problem under the other counting convention, worked out in [Section A.8](#). Reading parallel copies as distinct routes makes this a genuinely new question, and the section settles $m = 2$ and exhibits several competing constructions. The natural guess from the $m \leq 4$ data, that a thickened tree becomes optimal once $n \geq m + 2$, is false for every $m \geq 5$: [Theorem A.56](#) shows that a bouquet of thickened cliques sharing one vertex beats the tree by an amount that grows linearly in n , and the true growth rate in m is $\Theta(m^2)$ rather than the $\Theta(m)$ the tree would give. That construction is a lower bound of the right order and not the answer: at $m = 5$, $n = 7$ it gives 27 where an unfinished exhaustive search has already found 28, so the exact value of $K_m^{\text{multi}}(n)$ is open.
- ★ The last finite $n = 7$ computation that closes the directed multigraph problem at $m = 3$. By the attachment lemma and the conditional odd step ([Lemma A.34](#), [Corollary A.35](#), [Theorem A.36](#)), the entire $m = 3$ problem (value *and* extremal uniqueness for all $n \geq 8$) reduces along a single induction to two integral statements about multigraphs with multiplicities in $\{0, 1, 2\}$ ([Remark A.39](#)): (a) $L_3^{\text{dir}}(7) = 24$, and (b) every 24-arc feasible multigraph on 7 vertices with maximum degree at most 8 is $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, or the doubled bidirected path P_7 . Fact (b) is settled: the sound generation enumeration completed and returns exactly those three multigraphs, each re-verified by the exact checker ([Remark A.39](#)). What remains is fact (a) alone, where the MILP infeasibility check still reaches its time limit on the bundled CBC solver, so the gap is a single finite computation rather than a structural one. It is also a computation a reader can run: `prove_integral_arc_bound(7, 3, 25, use_gurobi=True)` hands the identical model to Gurobi, which is freely licensed for academic use and belongs to a different class of solver, and an INFEASIBLE verdict there is fact (a) outright. [Conjecture A.29](#) is thereby

bypassed for $m = 3$, though it remains open and of independent interest as the natural minimum-degree statement for the directed multigraph problem.

- ★ Extremal uniqueness at odd levels for directed multigraphs with $m \geq 4$. The conditional odd step ([Theorem A.36](#)) and the propagation of [Remark A.39](#) are stated for $m \in \{3, 4\}$. For $m = 4$ the *value* step goes through, but at an odd extremal level the counting no longer forces a vertex of degree $(m - 1)k$ (all degrees $\geq (m - 1)k + 1$ are arithmetically consistent), so the uniqueness statement does not yet propagate. For $m \geq 5$ the same escape climbs one level and even the *value* step stalls: a surplus extremiser could be degree-regular (every vertex at one and the same degree) strictly above the deletion bound, so no vertex deletion is forced to land on an extremal multigraph. Closing either hole asks for an arithmetic or structural input the present averaging argument does not supply.
- ★ The undirected vertex-disjoint problem for $m \geq 6$, open since 1974, where even the extremal structure is unknown.
- ★ The hypergraph vertex problem at $m \geq 4$. The problem is now settled at $m = 2$ and $m = 3$ for every uniformity and all n , with $k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor = \ell_m^{(r)}(n)$ ([Theorems A.45](#) and [A.48](#)). The incidence reduction stands for every m , but the rank bound that drives it ([Lemma A.47](#)) leans on the triconnected (Tutte/SPQR) decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$ it would need a 4-connectivity analogue, where no comparably clean decomposition is available. Exhaustive computation finds no counterexample at $m = 4$ at any of eleven sizes, confirming both that $\lfloor 3(n-1)/(r-1) \rfloor$ is attained and that one more hyperedge is infeasible, so the agreement $k_m^{(r)} = \ell_m^{(r)}$ survives well past the theorem. That it must break eventually is not a guess: at $r = 2$ this problem is exactly the multigraph vertex problem of [Section A.8](#), where [Theorem A.56](#) breaks the formula for every $m \geq 5$. So the question is not whether the pattern fails but whether $r \geq 3$ postpones the failure past $m = 4$, which leaves the $m = 4$ row sitting on the boundary, and where it first breaks for $r \geq 3$ is unknown.
- ★ A machine-checkable certificate for fact (a) of [Remark A.39](#). It is the one computed claim in this thesis that would rest on a solver's infeasibility verdict rather than on a finished exhaustion ([Section 2.9.1](#)), so it is the one place where a replayable proof object, of the kind a pseudo-Boolean or SAT solver emits, would raise the standard rather than merely repeat the work. Encoding the question as a pseudo-Boolean constraint problem over the 42 multiplicities in $\{0, 1, 2\}$, with the cut conditions written out per ordered pair, is the natural route.
- ★ The directed hypergraph variants. The tail-and-heads model has first bounds and a quadratic lower-bound family ([Proposition A.50](#)): counting how many hyperedges may serve the same ordered pair of vertices caps the total at $(m - 1)n(n - 1)/(r - 1)$, and a bipartite construction reaches $\approx (m - 1)n^2/(4(r - 1))$, so the value is quadratic in n . The two bounds differ by a factor of four, so the order is settled and the leading constant $\lim_{n \rightarrow \infty} \text{ex}(n)/n^2$ is not, for any of the orientation models. That constant is the right first target, ahead of exact small values, and [Conjecture A.53](#) says which way it should go: the factor of four is exactly the gap between

the n^2 ordered pairs the upper bound allows and the $n^2/4$ the bipartite construction uses, the same gap that appears in the simple directed problem, where [Theorem A.24](#) now proves the bipartite pattern right to first order. So the upper bound is the side that should improve. The appendix records where the proof of that theorem fails to transfer, which is that a hyperedge carries $r - 1$ heads at once, so two-step routes through different midpoints need not be edge-disjoint. The orientation axis does not visibly widen the gap: the backward model coincides with forward by reversal ([Lemma A.51](#)) and the general model obeys the same cap ([Proposition A.52](#)), beating forward only in the scarce-vertex regime $n \approx r$ ([Section 4.3.1](#)). That the three models share a leading constant is a conjecture supported by small-case computation, not a consequence of sharing two bounds four apart.

The map is filled in where honest mathematics could fill it, and where it could not, the blanks are marked precisely, with a machine standing ready at each of them. That, finally, is the use of building the microscope: not that it answers every question, but that it shows exactly which questions are left, and hands the next reader a clean place to start.

Appendix A

Full Proofs and Computational Transcripts

This appendix holds the arguments deferred from the body: the ones that are either standard machinery or long case-checking, and whose length, not whose difficulty, kept them out of the narrative. The body carries the ideas. Here they are carried out in full. Because graphs are visual objects, the longer arguments here are accompanied by figures: each is drawn rather than merely described, and the prose points the reader to the picture at the moment it is needed.

How to read this appendix. The sections divide into standard machinery and the thesis's own proofs, and a reader chasing one thread of the body need not read all of them. The first two sections, on [Theorem A.1](#), [Theorem A.2](#), and Mader's theorem, are classical results included only so the body can cite them in the exact form it uses, and a reader who already trusts Menger, Gomory–Hu, and Mader may skip them. Three sections carry the load-bearing original arguments. The directed arc value at $m = 2$ supplies the finite base cases the body's induction rests on. The directed multigraph problem at large n is the longest argument here and the one behind the main directed contribution. The hypergraph vertex problem proves the incidence-rank lemma that settles the $m = 3$ hypergraph case. The remaining sections, on the hypergraph edge bounds and the random threshold, each support a single claim of the body and can be read on their own when that claim is reached. Finally, everything the program asserts is reproduced verbatim in the computational transcripts and the self-test section, so the machine proofs can be audited without running anything.

A.1 Menger, Gomory–Hu, and the degree bound

The two classical theorems below are the foundation of everything the thesis measures, and we state them at the level of generality at which the program actually applies them. A *capacitated graph*, or *weighted graph*, is a graph, directed or not, carrying a nonnegative capacity $c(e)$ on each edge or arc. A plain graph is read as the *unit-capacity* case $c \equiv 1$, and a multigraph as the integer-capacity case, one unit per parallel copy. A *flow* from u to v sends units along edges without exceeding any capacity, and the *capacity of a cut* is the total capacity of the edges crossing it. Both theorems below are theorems about capacitated graphs, and the thesis uses them mostly at unit and integer capacities, where the maximum flow is exactly the connectivity λ .

Theorem A.1 (Menger, as max-flow min-cut [2]). *In any capacitated digraph, and for distinct*

vertices u, v , the maximum flow from u to v equals the minimum capacity of a cut separating u from v . At unit capacities this is the combinatorial statement we cite throughout: the maximum number of pairwise edge-disjoint u – v paths equals the minimum number of edges whose removal separates them,

$$\lambda_G(u, v) = \min\{|C| : C \subseteq E(G), G - C \text{ has no } u\text{--}v \text{ path}\}.$$

A directed capacitated graph is the most general object here, and every graph model the thesis uses is a special case of it. An undirected graph is the symmetric case, each edge a pair of opposite unit-capacity arcs. A multigraph is the integer-capacity case, with $\mu(u, v)$ units on the pair uv . The hypergraph version ([Theorem A.40](#)) is this very theorem applied to the helper-point network of [Figure 2.5](#), a capacitated digraph. The internally vertex-disjoint version is obtained not by changing the theorem but by changing the graph it runs on, a device called vertex splitting: replace each internal vertex w by two copies joined by a single unit-capacity arc $w^- \rightarrow w^+$, send every arc that entered w into w^- and every arc that left w out of w^+ . A flow may now pass through w at most once, so edge-disjoint paths in the split graph are exactly internally vertex-disjoint paths in the original, and a minimum edge cut there is a minimum vertex cut here. This split graph is literally what the checker of [Chapter 2](#) builds, which is why a single maximum-flow routine measures all four separations.

Theorem A.2 (Gomory–Hu [[16](#)]). Every undirected capacitated graph G has a weighted tree T on the same vertex set, its Gomory–Hu tree, such that for all u, v the value $\lambda_G(u, v)$ equals the minimum edge weight on the u – v path in T , and each tree edge e corresponds to a minimum cut of G whose capacity is the weight of e . All $\binom{n}{2}$ pairwise values are thereby stored in the $n - 1$ weights of a single tree.

Two features of the statement fix how far it reaches, and both are used below. Being a theorem about capacities, it applies verbatim to a simple graph (unit weights), to a multigraph (integer weights, the multiplicity μ as capacity), and, in the form of [Theorem A.41](#), to the hyperedge-connectivity of a hypergraph. But it needs the cut function to be *symmetric*, which forces G to be undirected: a digraph has $\lambda(u, v) \neq \lambda(v, u)$ in general and admits no such tree, and the vertex-connectivity function fails symmetry for the same reason. So the Gomory–Hu tree is precisely the tool for the undirected edge problems, where it drives Mader’s theorem and its multigraph and hypergraph cousins ([Theorems 1.2](#) and [1.3](#) and [proposition 1.15](#)) through one argument, and precisely the tool that vanishes once arcs turn one-way.

Primer: why a Gomory–Hu tree exists

The theorem is far from obvious, so here is why such a tree can always be built, in enough detail that nothing is taken on faith. Everything rests on one property of cuts. For a set of vertices S write $d(S)$ for the total capacity of the edges with exactly one endpoint in S , which is the capacity of the cut $(S, V \setminus S)$. The cut function is *submodular*: for any two vertex sets

A and B ,

$$d(A) + d(B) \geq d(A \cap B) + d(A \cup B).$$

One checks this one edge at a time. An edge contributes to each side of the inequality according to where its two endpoints fall relative to A and B , and a short case check over the possible placements shows its contribution on the right never exceeds its contribution on the left. The only edges that could contribute more on the right have one endpoint in $A \setminus B$ and the other in $B \setminus A$, and those contribute 0 on the right and 2 on the left, so the inequality only gains.

Submodularity buys the single lemma the construction needs, the *uncrossing* lemma: if $(S, V \setminus S)$ is any cut and u, v lie on the same side of it, then some minimum $u-v$ cut keeps all of S (or all of $V \setminus S$) on one side, that is, it does not cross S . The picture is that a cheapest $u-v$ cut which sliced through S could be pushed clear of S without ever gaining capacity, and submodularity is exactly the inequality guaranteeing the push is free.

With that lemma the tree is built in $n - 1$ rounds, one minimum-cut computation each. Begin with the trivial tree, a single node holding all n vertices. In each round pick a node of the current tree that still holds two or more vertices, choose two of its vertices u and v , and compute a minimum $u-v$ cut in the graph in which every neighbouring tree node, itself already a bag of several vertices, is shrunk to one vertex placed on whichever side the cut assigns it. Split the chosen node along the two sides of the cut and join the two halves by a new tree edge weighted by the cut's capacity. The uncrossing lemma is what makes the shrinking legitimate, because it says the minimum $u-v$ cut can be chosen to respect every cut found in earlier rounds, so no earlier tree edge is spoiled. After $n - 1$ rounds every node holds a single vertex and T is a weighted tree on V with $n - 1$ edges.

Two facts finish the argument, and they are exactly what the body uses. Each tree edge is by construction the capacity of an actual cut of G , so its weight is a genuine minimum $u-v$ cut value $\lambda_G(u, v)$ for the pair it separates. And for any u, v the value $\lambda_G(u, v)$ equals the lightest edge on the $u-v$ path of T . One inequality holds because every tree edge on that path is itself a cut separating u from v , so none can be lighter than the global minimum $u-v$ cut. The other holds because the lightest tree edge on the path is a genuine $u-v$ cut of exactly that weight. The reader who wants the uncrossing induction spelled out in full will find it in Gomory and Hu [16]. The point here is only that the tree is an honest theorem, computed by $n - 1$ maximum-flow calls, and not a convenient fiction.

The most basic constraint on local connectivity is local: a pair cannot have more independent routes than either endpoint has edges. The constructions below use it constantly.

Lemma A.3 (Degree bound). *For every pair of vertices of a graph G , $\lambda_G(u, v) \leq \min(\deg u, \deg v)$. In a digraph D , $\lambda_D(u, v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint $u-v$ path uses a distinct edge at u , so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct

out-arc and enters v by a distinct in-arc. □

A.2 Mader's theorem in full

The upper bound of [Theorem 1.2](#) rests on three short lemmas about a Gomory–Hu tree T of G . For an edge $e = \{u, v\}$ of G , write $\text{dist}_T(e)$ for the number of tree edges on the unique $u-v$ path in T . Since T is a tree on the very same vertex set, $\text{dist}_T(e) \geq 1$ for every edge. [Figure A.1](#) shows a small graph beside one of its Gomory–Hu trees and marks which edges have $\text{dist}_T(e) = 1$ and which have $\text{dist}_T(e) \geq 2$. It is worth keeping in view, since the three lemmas below are exactly statements about that picture. The first lemma is the one place the proof uses that G is *simple*, and it is exactly where the simple-graph bound improves on the multigraph bound.

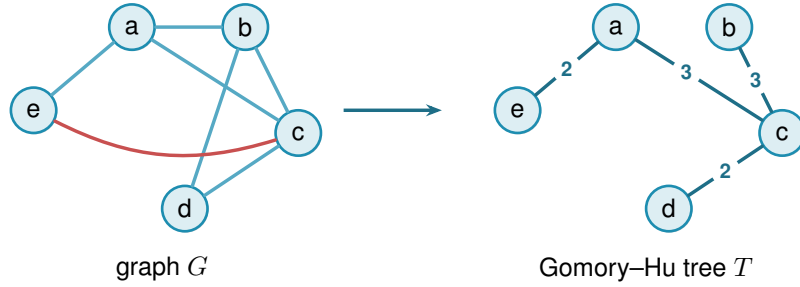


Figure A.1. A simple graph G (left) and a Gomory–Hu tree T of it (right). Both live on the same five vertices. The tree T has edges ea (2), ac (3), cb (3), cd (2), where each weight equals the local edge-connectivity of its endpoints in G . These four edges are also edges of G , so each has $\text{dist}_T(e) = 1$: these are the E_1 edges of [Lemma A.4](#), and a *simple* graph admits at most $n - 1 = 4$ of them, one per tree edge. The remaining edges ab , bd , and the red chord ec all join tree-distance-2 pairs: the path in T from e to c is $e-a-c$, so $\text{dist}_T(ec) = 2$. Reading $\text{dist}_T(e)$ as the number of tree cuts an edge crosses makes $\sum_e \text{dist}_T(e) = \sum_t c(t)$ ([Lemma A.6](#)). Every edge crosses at least one cut, but only the $n - 1$ tree edges cross exactly one, which is the surplus that drives [Lemma A.5](#) and ultimately the factor of two in Mader's bound.

Lemma A.4 (Distance-one edges). *At most $n - 1$ edges of a simple graph G satisfy $\text{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\text{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T . The tree T has exactly $n - 1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G . □

Lemma A.5 (Sum of tree distances). $\sum_{e \in E(G)} \text{dist}_T(e) \geq 2|E(G)| - (n - 1)$.

Proof. Partition $E(G)$ into $E_1 = \{e : \text{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \text{dist}_T(e) \geq 2\}$. Then

$$\sum_{e \in E(G)} \text{dist}_T(e) \geq |E_1| + 2|E_{\geq 2}| = 2|E(G)| - |E_1| \geq 2|E(G)| - (n - 1),$$

using [Lemma A.4](#) in the last step. □

Lemma A.6 (Double-counting identity). *Let G carry unit capacities, so that each weight $c(t)$ of a Gomory–Hu tree T is the plain number of edges of G crossing the cut that t induces. Then*

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \text{dist}_T(e).$$

This is the form used for Mader’s theorem, where G is a simple graph and so unit-capacitated by default.

Proof. By [Theorem A.2](#) each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e \in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \sum_{t \in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e \in E(G)} \text{dist}_T(e),$$

where the last equality holds because an edge $\{u, v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u – v path in T . $\square$

Proof of [Theorem 1.2](#), upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m - 1$ and let T be a Gomory–Hu tree of G . Each tree edge $t = \{x, y\}$ has weight $c(t) = \lambda_G(x, y) \leq m - 1$, so summing over the $n - 1$ tree edges, $\sum_t c(t) \leq (m - 1)(n - 1)$. Combining with [Lemmas A.5](#) and [A.6](#),

$$2|E(G)| - (n - 1) \leq \sum_{e \in E(G)} \text{dist}_T(e) = \sum_{t \in E(T)} c(t) \leq (m - 1)(n - 1).$$

Rearranging gives $2|E(G)| \leq m(n - 1)$, and since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. $\square$

Note where the factor of two came from: every edge crosses at least one tree cut, but a simple graph can afford at most $n - 1$ edges that cross only one, so on average each edge is charged closer to twice. For multigraphs that refinement is unavailable, since parallel edges between tree-adjacent pairs each have $\text{dist}_T = 1$, and the same chain of inequalities stops at $|E| \leq \sum_e \text{dist}_T(e) \leq (m - 1)(n - 1)$, which is exactly the multigraph bound of [Theorem 1.3](#).

The lower bound is a pair of constructions, drawn side by side in [Figure A.2](#). The first is the shape the bound was guessed from, and works when $m - 1$ divides $n - 1$. The second generalises it to every $n \geq m \geq 2$.

Construction A.7 (Shared-clique graph). For $m \geq 3$ with $(m - 1) \mid (n - 1)$, take $k = (n - 1)/(m - 1)$ copies of the complete graph K_m and identify one vertex of each copy into a single shared hub vertex h . The result has $k(m - 1) + 1 = n$ vertices and $k \binom{m}{2} = \frac{m(n-1)}{2}$ edges. Every non-hub vertex has degree $m - 1$, and any pair of vertices contains a non-hub vertex, so $\lambda^{\max} \leq m - 1$ by [Lemma A.3](#).

Construction A.8 (Hub-and-spokes graph). For $n \geq m \geq 2$, let $H_{m,n}$ have vertex set $\{h, v_1, \dots, v_{n-1}\}$ with (i) all $n - 1$ *hub edges* $\{h, v_i\}$, and (ii) a *spoke graph* on $\{v_1, \dots, v_{n-1}\}$ with $\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor$ edges in which every v_i has spoke degree at most $m - 2$. Such a spoke graph exists by Lemma A.9.

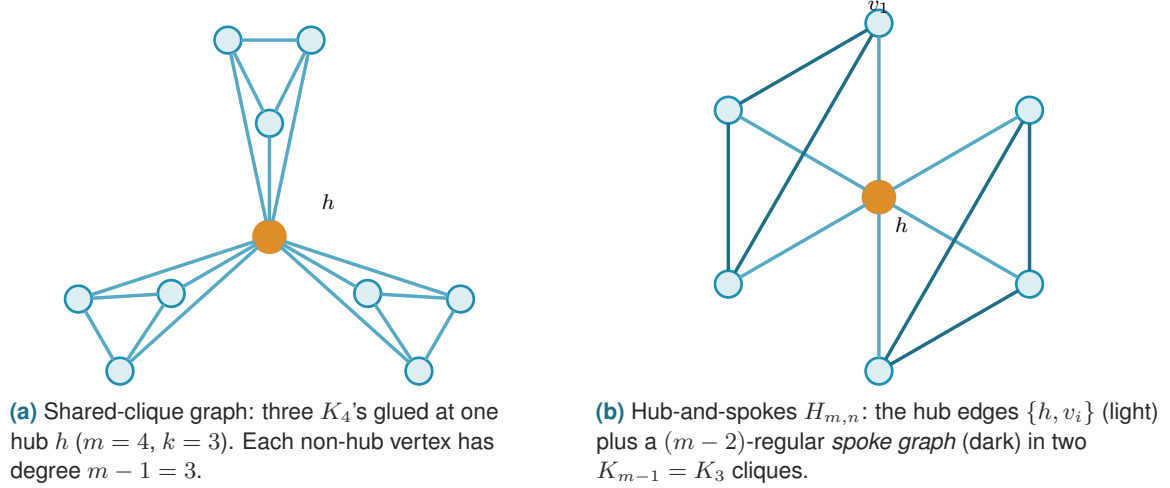


Figure A.2. The two extremal families. (a) The shared-clique graph of Construction A.7 works when $(m - 1) \mid (n - 1)$: every pair contains a non-hub vertex of degree exactly $m - 1$, so $\lambda^{\max} \leq m - 1$ by the degree bound, while the edge count hits $\frac{m(n-1)}{2}$. (b) The hub-and-spokes graph $H_{m,n}$ of Construction A.8 reaches the same bound for every $n \geq m$: each spoke vertex v_i has total degree $1 + (m - 2) = m - 1$, and choosing the spoke graph as disjoint cliques K_{m-1} recovers (a). The hub is drawn in orange in both.

Lemma A.9 (Near-regular graphs exist). For $N \geq 1$ and $0 \leq r \leq N - 1$ there is a graph on N vertices with $\left\lfloor \frac{rN}{2} \right\rfloor$ edges and maximum degree at most r .

Proof. If rN is even we build an r -regular circulant graph on vertex set $\{0, \dots, N - 1\}$: for even r connect each i to $i \pm j \pmod{N}$ for $j = 1, \dots, r/2$, and for odd r (then N is even) use $j = 1, \dots, (r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both are r -regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r - 1)$ -regular circulant with distances $1, \dots, (r - 1)/2$ and add the near-perfect matching that pairs vertex i with $i + (N + 1)/2 \pmod{N}$ for $i = 0, \dots, (N - 3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2} + \frac{N-1}{2} = \frac{rN-1}{2} = \left\lfloor \frac{rN}{2} \right\rfloor$ edges, with one vertex of degree $r - 1$ and all others of degree r . $\square$

Theorem A.10 (Extremal graphs exist). For all $n \geq m \geq 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \leq m - 1$.

Proof. The edge count is $(n - 1) + \left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, since $(m - 2)(n - 1)$ and $m(n - 1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i , whose total degree is $1 + (m - 2) = m - 1$. Lemma A.3 then gives $\lambda(u, v) \leq m - 1$. When $(m - 1) \mid (n - 1)$, choosing the spoke graph as k disjoint cliques K_{m-1} recovers Construction A.7, so the hub-and-spokes family genuinely generalises the shared-clique one. $\square$

Together the upper bound and [Theorem A.10](#) prove [Theorem 1.2](#). The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.11 (Characterisation of equality). *Let G be simple with $\lambda_G^{\max} \leq m - 1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G . If $m(n - 1)$ is even, then every tree edge has weight exactly $m - 1$, every tree edge is also an edge of G , and every edge of G joins vertices at tree distance at most 2. If $m(n - 1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m - 2$, or one tree edge is not an edge of G , or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three nonnegative slacks. The first, $S_1 = (m - 1)(n - 1) - \sum_t c(t)$, counts tree weights below $m - 1$. The second, $S_2 = \sum_{e: \text{dist}_T(e) \geq 2} (\text{dist}_T(e) - 2)$, counts edges beyond tree distance two. The third, $S_3 = (n - 1) - |E_1|$, counts tree edges that support no edge of G . Retracing the proof, $\sum_e \text{dist}_T(e) = 2|E| - |E_1| + S_2$ and $\sum_e \text{dist}_T(e) = \sum_t c(t) = (m - 1)(n - 1) - S_1$, so altogether $2|E| = m(n - 1) - (S_1 + S_2 + S_3)$. If $m(n - 1)$ is even, equality $|E| = m(n - 1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n - 1)$ is odd the floor absorbs exactly one unit, so $S_1 + S_2 + S_3 = 1$ and exactly one of the three slacks equals one. $\square$

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G , so G contains its own Gomory–Hu tree and has only $n - 1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod{3}$ it is consistent with the trees of K_4 blocks glued at shared cut vertices, the case of [Construction A.7](#) that the search of [Chapter 3](#) rediscovers.

A.3 The directed arc value at $m = 2$

Three short lemmas feed the induction. The first shows a hub forces a linear bound for every m , and is the proof behind the hub branch of [Conjecture 1.14](#). The three lemmas below and the proof of [Theorem 1.10](#) that they support are the author’s own, and its finite base cases $2 \leq n \leq 7$ were verified by the author’s program, through the exhaustive search of [Chapter 2](#) whose transcript is reproduced at the end of this appendix.

Lemma A.12 (Two-hop lemma). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$ containing a vertex r with $d^+(r) = n - 1$. Then every $v \neq r$ has at most $m - 2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \dots, u_{m-1}$ were distinct in-neighbours of v other than r , that is, distinct vertices each sending an arc to v , the paths $r \rightarrow v$ and $r \rightarrow u_i \rightarrow v$ for $i = 1, \dots, m - 1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n - 1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \leq m - 1$. $\square$

Theorem A.13 (Hub upper bound). *If a simple digraph D with $\lambda_D^{\max} \leq m - 1$ has a vertex r with $d^+(r) = n - 1$, then $|A(D)| \leq m(n - 1)$.*

Proof. Split the arcs into those leaving r (exactly $n - 1$), those entering r (at most $n - 1$), and the internal arcs avoiding r . By [Lemma A.12](#) each of the $n - 1$ other vertices receives at most $m - 2$ internal arcs, so there are at most $(m - 2)(n - 1)$ internal arcs, and the total is at most $m(n - 1)$. $\square$

Lemma A.14 (No transitive triangle). *A simple digraph with $\lambda^{\max} \leq 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \rightarrow z$ and $x \rightarrow y \rightarrow z$ would be two arc-disjoint routes.*

Lemma A.15 (Deletion monotonicity). *Write $D - v_0$ for the digraph obtained from D by deleting the vertex v_0 together with all arcs incident to it. For any digraph D and vertex v_0 ,*

$$\lambda^{\max}(D - v_0) \leq \lambda^{\max}(D) \quad \text{and} \quad \kappa^{\max}(D - v_0) \leq \kappa^{\max}(D),$$

since arc-disjoint paths in $D - v_0$ are still arc-disjoint paths in D , and internally vertex-disjoint paths in $D - v_0$ are still internally vertex-disjoint paths in D . The two statements are proved by the same one-line observation because deleting a vertex only removes routes, never creates one, and it cannot make two surviving routes share anything they did not share before.

Proof of Theorem 1.10. Write $M(n) = \max(2(n - 1), \lfloor n^2/4 \rfloor)$. A direct computation shows $2(n - 1) \geq \lfloor n^2/4 \rfloor$ exactly for $n \leq 7$, with equality at $n = 7$, so $M(n) = 2(n - 1)$ for $n \leq 7$ and $M(n) = \lfloor n^2/4 \rfloor$ for $n \geq 7$. The lower bound $\ell_2^{\text{dir}}(n) \geq M(n)$ is witnessed by the directed hub construction ([Construction 1.9](#) at $m = 2$, the bidirected star) and the one-directional bipartite construction ([Construction 1.8](#)).

For the upper bound we show by strong induction on n that $\lambda_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$.

Base cases $2 \leq n \leq 7$. All six are settled uniformly by the pruned exhaustive search of [Chapter 2](#), which walks every simple digraph with $\lambda^{\max} \leq 1$ on up to seven vertices and reports the maxima 2, 4, 6, 8, 10, 12, matching $M(n)$ at each size. The transcript is reproduced below, and the cut-counting certifier independently confirms the values within its reach through the $m = 2$ reduction of [Theorem A.31](#). The two smallest are also immediate by hand, as a check on the machine rather than a separate argument: at $n = 2$ there are at most 2 arcs, and at $n = 3$ a fifth arc would leave at most one arc of the complete bidirected triangle missing, so three of the rest form a transitive triangle, which [Lemma A.14](#) forbids. The point of the theorem is precisely that $n = 4, 5, 6, 7$ are *not* immediate by hand, which is what makes handing the case-check to the program the right move.

Inductive step $n \geq 8$. Let D be a simple digraph on n vertices with $\lambda_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 minimising the total degree $d(v) = d^+(v) + d^-(v)$. Since $\sum_v d(v) = 2a$, averaging gives $d(v_0) \leq 2a/n$. By [Lemma A.15](#) the digraph $D - v_0$ on $n - 1 \geq 7$ vertices satisfies the induction hypothesis, so $a - d(v_0) \leq M(n - 1) = \lfloor \frac{(n-1)^2}{4} \rfloor$. Combining,

$$a \leq \frac{2a}{n} + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \quad \implies \quad a \leq \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

It remains to check that this right-hand side never exceeds $\lfloor n^2/4 \rfloor$, and the two parities behave differently enough to treat them in turn.

For even $n = 2k$ with $k \geq 4$, first simplify $\lfloor (n-1)^2/4 \rfloor$. Expanding $(2k-1)^2 = 4k^2 - 4k + 1$ and dividing by 4 gives $k^2 - k + \frac{1}{4}$, whose floor is $k(k-1)$. The bound then reads

$$a \leq \frac{2k}{2k-2} k(k-1) = \frac{k}{k-1} k(k-1) = k^2,$$

and since $\lfloor n^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$ as well, the bound lands exactly on the target with no slack left over.

For odd $n = 2k+1$ with $k \geq 4$, the same simplification gives $\lfloor (n-1)^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$, so the bound is $a \leq \frac{2k+1}{2k-1} k^2$. To compare this fraction with the target, split off its integer part by dividing $(2k+1)k^2$ by $2k-1$. The remainder is exactly k , because $(2k-1)(k^2+k) = 2k^3 + k^2 - k$ and $(2k+1)k^2 = 2k^3 + k^2$ differ by k , so

$$\frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \frac{(2k+1)k^2}{2k-1} = k(k+1) + \frac{k}{2k-1}.$$

The leftover fraction $\frac{k}{2k-1}$ lies strictly between 0 and 1 for every $k \geq 1$, so the bound puts a at most an integer $k(k+1)$ plus a positive amount below one. Since a is itself an integer it cannot reach the next integer up, so $a \leq k(k+1)$. The target $\lfloor n^2/4 \rfloor = \lfloor (2k+1)^2/4 \rfloor = k^2 + k = k(k+1)$ agrees, so for odd n it is integrality, not the raw arithmetic, that closes the last fraction of a unit. In both parities $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction. $\square$

Two remarks on the shape of this proof. First, the minimum-degree deletion closes the step on its own, and no separate hub case is needed, because the averaging bound holds whether or not D contains a hub, although [Theorem A.13](#) shows directly that a hub digraph never exceeds the linear branch. Second, for even n the bound is exactly tight against the one-directional bipartite construction, which is $\frac{n}{2}$ -regular in total degree, so every step of the chain is achieved by the conjectured extremiser, and for odd n the slack $\frac{k}{2k-1} < 1$ is absorbed by integrality. The proof is long not because it is deep but because the two regimes of $M(n)$ must be carried through the arithmetic separately, and that bookkeeping is precisely what [Chapter 2](#) hands to the machine at the base cases.

Remark A.16 (Which way Whitney runs). It is worth pausing on the direction of Whitney's inequality before using it, because the tempting reading is the wrong one. From $\kappa_D(u, v) \leq \lambda_D(u, v)$ for every pair it follows that $\kappa_D^{\max} \leq \lambda_D^{\max}$, hence that

$$\{D : \lambda_D^{\max} \leq m-1\} \subseteq \{D : \kappa_D^{\max} \leq m-1\}.$$

Every digraph feasible for the *arc* problem is feasible for the *vertex* problem, and not the other way round. The vertex-feasible family is the larger of the two, so the inequality between the extremal

numbers it hands over for free is

$$k_m^{\text{dir}}(n) \geq \ell_m^{\text{dir}}(n),$$

and an upper bound for the arc problem says nothing at all about the vertex problem. The gap is not vacuous: two routes that share an interior vertex but no arc are counted by λ and not by κ , so there really are digraphs in the difference of the two families. What Whitney gives is therefore exactly one thing, that an arc construction is an honest vertex *lower* bound, and the vertex upper bound has to be earned separately. The proof below earns it, and [Chapter 4](#) keeps the same discipline for $m \geq 3$, where the vertex problem stays open even though the arc conjecture is stated.

Proof of Theorem 1.11. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ as before.

Lower bound. The directed hub and the one-directional bipartite construction both have $\lambda^{\max} = 1$, so by [Remark A.16](#) both have $\kappa^{\max} \leq 1$ and are feasible for the vertex problem. Hence $k_2^{\text{dir}}(n) \geq M(n)$.

Upper bound. We show by strong induction on n that $\kappa_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$. The induction is the one that proved [Theorem 1.10](#), and every step of it survives the change of separation, because the only two properties of the ceiling that argument uses are that it is inherited by vertex deletion and that it holds at the base sizes.

The base cases $2 \leq n \leq 7$ are settled by the same pruned exhaustive search of [Chapter 2](#), run with the vertex feasibility test in place of the arc one. It walks every simple digraph with $\kappa^{\max} \leq 1$ on up to seven vertices and reports the maxima 2, 4, 6, 8, 10, 12, matching $M(n)$ at each size. The transcript is reproduced in [Section A.10](#) beside the arc one.

For the inductive step $n \geq 8$, let D be a simple digraph on n vertices with $\kappa_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 of minimum total degree, so $d(v_0) \leq 2a/n$ by averaging. By [Lemma A.15](#), which holds for κ exactly as it does for λ , the digraph $D - v_0$ on $n-1 \geq 7$ vertices again satisfies $\kappa^{\max} \leq 1$, so the induction hypothesis gives $a - d(v_0) \leq M(n-1) = \lfloor (n-1)^2/4 \rfloor$. From here the two parity computations are verbatim those of [Theorem 1.10](#): they use only the two displayed inequalities and the integrality of a , no property of the separation. They yield $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction.

Hence $k_2^{\text{dir}}(n) = M(n) = \ell_2^{\text{dir}}(n)$. The two problems have the same answer at $m = 2$, but this is a computed coincidence of the two searches meeting at every base size, not a formal consequence of Whitney's inequality. $\square$

Remark A.17 (The directed vertex problem at $m \geq 3$). The proof above transfers the $m = 2$ induction but nothing more, and for $m \geq 3$ the vertex problem is genuinely separate: the arc conjecture, and the decomposition of [Conjecture A.21](#) that would prove it, both concern the smaller family and leave $k_m^{\text{dir}}(n)$ untouched. What can be said is computational. Running the

exhaustive search of [Chapter 2](#) under both feasibility tests at $m = 3$ gives

n	2	3	4	5	6
$\ell_3^{\text{dir}}(n)$	2	6	9	12	15
$k_3^{\text{dir}}(n)$	2	6	9	12	15

with every entry proved complete, so the two separations agree at every size the exhaustion reaches, and from $n = 3$ onwards both match the conjectured $\max(m(n-1), \lfloor (n+m-2)^2/4 \rfloor)$. At $n = 2$ the conjecture's formula gives 3 where only 2 arcs exist at all, the usual degeneracy of the formula below $n = m$. Agreement over five sizes is evidence and not a proof, and the natural next question is structural rather than numerical: to separate the two problems one needs a digraph where some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* where that slack pays for extra arcs. The two demands pull against each other, which is perhaps why no small example does both.

A.4 Structural propositions on the directed frontier

These are the proved ingredients of the reduction discussed in [Chapter 4](#). The section ends with what the reduction still assumes, stated as [Conjecture A.21](#), and with the one quadratic bound that assumes nothing, [Proposition A.22](#).

Proposition A.18 (Mutually unreachable out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$. For any vertex u and distinct out-neighbours v, w of u , there is no directed path from v to w in $D - u$.*

Proof. If P were a directed v – w path avoiding u , then $u \rightarrow w$ and $u \rightarrow v \xrightarrow{P} w$ would be two arc-disjoint u -to- w routes: the first uses only the arc (u, w) , the second starts with (u, v) and continues along P , none of whose arcs touch u . This contradicts $\lambda_D(u, w) \leq 1$. The restriction to $D - u$ is not cosmetic, and [Figure A.3](#) draws both the argument and the reason for the restriction: in the digraph with arcs (u, v) , (v, u) , (u, w) every local connectivity is 1, yet $v \rightarrow u \rightarrow w$ reaches w from v through u . $\square$

Proposition A.19 (Disjoint second out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$, let $u \in V$, and let $v_1 \neq v_2$ be out-neighbours of u . Then $(N^+(v_1) \setminus \{u\}) \cap (N^+(v_2) \setminus \{u\}) = \emptyset$.*

Proof. A common out-neighbour $w \neq u$ would give the two arc-disjoint routes $u \rightarrow v_1 \rightarrow w$ and $u \rightarrow v_2 \rightarrow w$, contradicting $\lambda_D(u, w) \leq 1$. $\square$

Proposition A.20 (Two-hop budget on bipartite partitions). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$. If V admits a partition $A \cup B$ with every arc from A to B present, then every $b \in B$ has in-degree at most $m - 2$ from inside B , and consequently the number of arcs inside B is at most $(m - 2)|B|$.*



(a) If a directed $v \rightsquigarrow w$ path (red) existed in $D - u$, then $u \rightarrow w$ and $u \rightarrow v \rightsquigarrow w$ would be two arc-disjoint u - w routes, against $\lambda(u, w) \leq 1$.

(b) The counterexample $\{(u, v), (v, u), (u, w)\}$: $\lambda^{\max} = 1$, yet w is reachable from v , but only *through* u .

Figure A.3. Why [Proposition A.18](#) must exclude paths through u . (a) For out-neighbours v, w of u in a digraph with $\lambda^{\max} \leq 1$, no $v \rightsquigarrow w$ path can avoid u : such a path would combine with the lone arc $u \rightarrow w$ into two arc-disjoint routes. (b) The restriction to $D - u$ is essential. In the three-vertex digraph $\{(u, v), (v, u), (u, w)\}$ we have $\lambda^{\max} = 1$, yet v reaches w via $v \rightarrow u \rightarrow w$, a route that reuses the vertex u , which the proposition allows.

Proof. Fix $a \in A$ and $b \in B$, and let $b'_1, \dots, b'_d$ be the in-neighbours of b inside B , where $d = \deg_B^-(b)$. The direct arc $a \rightarrow b$ together with the two-step detours $a \rightarrow b'_i \rightarrow b$ are $1 + d$ arc-disjoint a -to- b routes, all present because $A \rightarrow B$ is complete. Feasibility forces $1 + d \leq m - 1$, so $d \leq m - 2$, and summing over B bounds the internal arcs by $(m - 2)|B|$. $\square$

We now combine [Proposition A.20](#) with the best choice of partition. Suppose every arc of D either runs from A to B or stays inside B : the partition $A \rightarrow B$ is complete, no arc lies inside A , and crucially *no arc runs from B back to A* . Write $b = |B|$, so that $|A| = n - b$. The arcs then come in exactly two kinds. The arcs crossing the partition number $|A||B| = (n - b)b$, one for each ordered A - B pair. The arcs inside B number at most $(m - 2)b$: since no route between two vertices of B ever needs to leave B (there is no arc back to A to leave by), [Proposition A.20](#) applies to B on its own and caps the in-degree of each $b \in B$ from inside B at $m - 2$. There are no other arcs, so, adding the two kinds,

$$|A(D)| \leq (n - b)b + (m - 2)b = b(n + m - 2 - b).$$

The right-hand side is a downward parabola in b . Writing $s = n + m - 2$, it is $b(s - b) = -b^2 + sb$, maximised at $b = s/2$, so over integers the best partition takes $b = \lceil s/2 \rceil$ (equivalently $\lfloor s/2 \rfloor$), and the maximum value is

$$\max_b b(s - b) = \left\lfloor \frac{s^2}{4} \right\rfloor = \left\lfloor \frac{(n + m - 2)^2}{4} \right\rfloor.$$

This is the quadratic branch of [Conjecture 1.14](#), reached uniformly in m . It is worth being exact about how much was assumed, because the count is often summarised as resting on the absence of backward arcs alone, and that summary is too generous. Four things were granted at once: that a partition $V = A \cup B$ exists at all, that every arc from A to B is present, that no arc lies inside A , and that no arc runs from B back to A . Only the last of the four is the backward-arc condition. The first three describe the shape of the extremiser, and nothing proved so far forces

an arbitrary extremal digraph to have that shape. The honest statement of the gap is therefore a structural one:

Conjecture A.21 (The missing decomposition). *Let n be large enough that the quadratic branch of [Conjecture 1.14](#) strictly exceeds the linear one, that is, $\lfloor (n + m - 2)^2/4 \rfloor > m(n - 1)$. Then every extremal digraph D on n vertices with $\lambda_D^{\max} \leq m - 1$ admits a partition $V(D) = A \cup B$ such that every arc from A to B is present, no arc lies inside A , and no arc runs from B to A .*

The hypothesis on n has to be there, and it is worth seeing why, because the obvious phrasing is too narrow. On the linear branch the extremisers are *not* just the hub. At $m = 2$, every bidirected spanning tree is feasible and has exactly $2(n - 1)$ arcs, since between two vertices the only route follows the unique tree path, so the whole extremal set on that branch is the set of bidirected trees. An exhaustive classification confirms it: at $n = 5$ and $n = 6$ there are exactly 3 and 6 extremal digraphs up to isomorphism, which are precisely the numbers of trees on 5 and 6 vertices, and none of them admits the decomposition. Excluding “the hub” would leave all the other trees behind, so the clean hypothesis is on the size rather than on the shape.

Given [Conjecture A.21](#) the count above closes [Conjecture 1.14](#) for $m \geq 3$. The backward-arc clause is the part where the natural delete-and-compensate exchange fails to be monotone, which is why it is the clause usually named, but the partition itself is not free either, and [Chapter 4](#) states the position that way. Testing the conjecture by exhaustion is harder than it looks: the crossover sits at $n = 8$ for $m = 2$ and at $n = 9$ for $m = 3$, so every size an exhaustive sweep can reach at $m = 3$ still lies on the linear branch, where the statement says nothing. That is the practical reason this conjecture has resisted a computational probe rather than a proof.

What can be proved without any of the four is the leading constant, and the argument is short enough to give in full. It also says something the conjecture does not: that every feasible digraph is within a lower-order number of arcs of being one-directional bipartite.

Proposition A.22 (Unconditional quadratic bound and stability). *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m - 1$. Then*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + \sqrt{m} n^{3/2}.$$

Moreover D can be made one-directional bipartite, meaning every remaining arc runs from one part to the other and none returns, by deleting at most $\sqrt{m} n^{3/2}$ arcs. Consequently, for each fixed m ,

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + O_m(n^{3/2}),$$

so the leading constant $\frac{1}{4}$ of [Conjecture 1.14](#) holds unconditionally, with no hypothesis on the structure of the extremiser.

Proof. Step 1, a budget on two-step routes. Fix an ordered pair (u, v) with $u \neq v$, and let $p_2(u, v)$ count the vertices x with both $u \rightarrow x$ and $x \rightarrow v$ present. Such an x is automatically distinct from

u and from v , since D has no loops, so each one gives a genuine two-step route $u \rightarrow x \rightarrow v$. Two of these routes with different midpoints $x \neq x'$ share no arc, since the first uses (u, x) and (x, v) and the second uses (u, x') and (x', v) , and neither uses the arc (u, v) , which has no midpoint at all. So the direct arc, when present, together with all the two-step routes, is a family of pairwise arc-disjoint u -to- v routes, and feasibility caps its size:

$$\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v) \leq \lambda_D(u, v) \leq m - 1.$$

Summing over all $n(n - 1)$ ordered pairs gives $|A(D)| + \sum_{u \neq v} p_2(u, v) \leq (m - 1) n(n - 1)$.

Step 2, from pairs to degrees. Counting directed walks of length two by their midpoint, $\sum_{u,v} p_2(u, v) = \sum_x d^-(x) d^+(x)$, where this sum runs over *all* ordered pairs, the diagonal $u = v$ included. The diagonal terms count the closed walks $u \rightarrow x \rightarrow u$, that is, twice the number of digons of D . Distinct digons use disjoint pairs of arcs, so twice the number of digons is at most $|A(D)|$, and that is exactly the slack Step 1 left over. Combining the two,

$$\sum_{x \in V} d^+(x) d^-(x) \leq [(m - 1) n(n - 1) - |A(D)|] + |A(D)| = (m - 1) n(n - 1).$$

This is the one inequality the whole section runs on, and it is worth reading in words: a vertex with a large in-degree *and* a large out-degree is expensive, because it manufactures two-step routes for many pairs at once, and the budget for those routes is only quadratic in n .

Step 3, the smaller half-degree is small on average. Since $\min(a, b) \leq \sqrt{ab}$ for non-negative a, b , and by the Cauchy-Schwarz inequality in the form $\sum_x \sqrt{a_x} \leq \sqrt{n \sum_x a_x}$,

$$\begin{aligned} \sum_{x \in V} \min(d^+(x), d^-(x)) &\leq \sum_{x \in V} \sqrt{d^+(x) d^-(x)} \leq \sqrt{n \sum_{x \in V} d^+(x) d^-(x)} \\ &\leq \sqrt{n \cdot (m - 1) n(n - 1)} = n \sqrt{(m - 1)(n - 1)} \leq \sqrt{m} n^{3/2}. \end{aligned}$$

Step 4, deleting the smaller half of every vertex. Call x *source-like* if $d^+(x) \geq d^-(x)$ and *sink-like* otherwise, and write S and T for the two classes, so $V = S \cup T$. Delete every in-arc of a source-like vertex and every out-arc of a sink-like vertex. A source-like x loses $d^-(x) = \min(d^+(x), d^-(x))$ arcs and a sink-like x loses $d^+(x) = \min(d^+(x), d^-(x))$ arcs, so the number of deleted arcs is at most $\sum_x \min(d^+(x), d^-(x))$, which Step 3 bounds by $\sqrt{m} n^{3/2}$. An arc (a, b) survives only if a is not sink-like and b is not source-like, that is, only if $a \in S$ and $b \in T$. So what remains runs entirely from S to T , which is the one-directional bipartite shape, and it has at most $|S| |T| \leq \lfloor n^2/4 \rfloor$ arcs. Adding the two counts gives the bound.

The asymptotic statement follows because the augmented bipartite construction ([Construction 1.13](#)) already gives $\ell_m^{\text{dir}}(n) \geq \lfloor (n + m - 2)^2/4 \rfloor = n^2/4 + O_m(n)$, and $n^{3/2}$ dominates n . $\square$

The error term $n^{3/2}$ is larger than the $(m - 2)n/2$ that [Conjecture 1.14](#) predicts, so the proposition as it stands does not separate the conjectured formula from its near neighbours. It can be

sharpened to the right order, and the extra idea is small enough to be worth giving, because it explains where the $\sqrt{n}$ was being wasted.

Step 3 is lossy exactly when every vertex carries a middling degree. The Cauchy-Schwarz step is tight only if $d^+(x)d^-(x)$ is about $(m-1)n$ at every vertex, which forces every degree down to about $\sqrt{mn}$. But a digraph with $\Theta(n^2)$ arcs has average degree $\Theta(n)$, and at a vertex of large degree the constraint $d^+(x)d^-(x) \leq (m-1)n$ bites much harder: with $d^+ + d^-$ of order n , the smaller of the two must be of order m rather than $\sqrt{mn}$. The dense case and the tight case of Step 3 are therefore incompatible, and the way to exploit that is a dichotomy on the minimum degree, with vertex deletion to handle the sparse side.

Lemma A.23 (Small side of a high-degree vertex). *For any vertex x of a digraph with total degree $d(x) = d^+(x) + d^-(x) > 0$,*

$$\min(d^+(x), d^-(x)) \leq \frac{2 d^+(x) d^-(x)}{d(x)}.$$

Proof. Write $\mu = \min(d^+(x), d^-(x))$ and $M = \max(d^+(x), d^-(x))$, so $\mu + M = d(x)$ and $\mu M = d^+(x)d^-(x)$. Since $\mu \leq M$ we have $M \geq d(x)/2$, hence $d^+(x)d^-(x) = \mu M \geq \mu d(x)/2$, which rearranges to the claim. $\square$

Theorem A.24 (The quadratic bound with a linear error). *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ satisfies*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1),$$

so for each fixed m ,

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n),$$

and both the leading constant $\frac{1}{4}$ and the order of the second-order term of [Conjecture 1.14](#) hold unconditionally.

Proof. Induction on n . For $n = 2$ a simple digraph has at most 2 arcs and the right-hand side is $1 + 4(m-1) \geq 5$, so the base holds. Let $n \geq 3$ and split on the minimum total degree.

Case 1: some vertex v has $d(v) < n/2$. Since $d(v)$ is an integer, $d(v) \leq \lfloor n/2 \rfloor$. By [Lemma A.15](#) the digraph $D - v$ on $n-1$ vertices is still feasible, so the induction hypothesis applies to it and

$$|A(D)| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4(m-1)(n-2) + \left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-2),$$

using [Lemma A.27](#) for the last step, and this is below the claimed bound.

Case 2: every vertex has $d(x) \geq n/2$. By [Lemma A.23](#) and Step 2 of [Proposition A.22](#),

$$\sum_{x \in V} \min(d^+(x), d^-(x)) \leq \sum_{x \in V} \frac{2d^+(x)d^-(x)}{d(x)} \leq \frac{4}{n} \sum_{x \in V} d^+(x)d^-(x) \leq \frac{4}{n} (m-1)n(n-1),$$

which is $4(m-1)(n-1)$. Step 4 of [Proposition A.22](#) deletes exactly that many arcs and leaves at most $\lfloor n^2/4 \rfloor$, so the bound holds in this case too.

For the two-sided asymptotic statement, [Construction 1.13](#) gives

$$\ell_m^{\text{dir}}(n) \geq \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor = \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1),$$

so the second-order term is linear in n from both sides. □

What is left open is therefore a constant factor in one coefficient, not a shape. [Conjecture 1.14](#) says the second-order coefficient is exactly $(m-2)/2$, and the theorem pins it between that and $4(m-1)$, a factor of roughly eight. Closing that gap is where the structure of [Conjecture A.21](#) lives, and [Remark A.25](#) rules out the most obvious way to try.

Remark A.25 (Case 2 cannot be sharpened without a new inequality). One might hope to cut Case 2's constant using the observation that in the conjectured extremiser, one whole side has $\min(d^+, d^-) = 0$, so summing [Lemma A.23](#)'s worst case at every vertex looks wasteful. It is not: Case 2's bound is already the exact maximum obtainable from the two facts it uses, the budget $\sum_x d^+(x)d^-(x) \leq (m-1)n(n-1)$ of Step 2 in [Proposition A.22](#) and the floor $d(x) \geq n/2$, and no cleverer use of those two facts alone can do better.

At a vertex with $d(x) = s \geq n/2$, achieving $\min(d^+(x), d^-(x)) = t$ costs $d^+(x)d^-(x) = t(s-t)$ at least, cheapest at $s = n/2$, giving the cost function $c(t) = t(n/2 - t)$ on $t \in [0, n/4]$. Since c is concave with $c(0) = 0$, for any two vertices with values $0 < a \leq b$ strictly below the cap, shifting mass from a to b preserves $a+b$ while lowering $c(a) + c(b)$, because c' is decreasing. Repeating this exchange shows that the true maximum of $\sum_x t_x$ under a shared budget Q is attained by concentrating the budget on as few vertices as the cap allows, each at $t_x = n/4$, and the rest at $t_x = 0$, exactly the qualitative picture the conjectured extremiser suggests, and it already gives $4Q/n = 4(m-1)(n-1)$ once the number of vertices at the cap is below n . The crude per-vertex bound loses nothing because it is tight at exactly the same point. [Lemma A.23](#) is an equality only when $d^+(x) = d^-(x)$, and the floor substitution $d(x) \rightarrow n/2$ is an equality only when $d(x) = n/2$. Both conditions hold at once exactly when $d^+(x) = d^-(x) = n/4$, which is the vertex profile the true optimum concentrates on, so nothing is lost there. Cutting the constant therefore needs a second, independent inequality, one that constrains interference among the handful of expensive near-balanced vertices themselves rather than a sharper reading of the one already in hand.

A.5 The directed multigraph problem at large n

The body proves $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ for $n \leq 6$ (Theorem 2.2). This section records what is known beyond, where the one-directional bipartite construction takes over. Throughout, the scaling reduction of Chapter 2 is used in the following form.

Lemma A.26 (Scaling reduction). *Let D be a directed multigraph with $\mu(u, v) \leq m-1$ on every ordered pair and $\lambda^{\max}(D) \leq m-1$. Then $w := \mu/(m-1)$ takes values in $[0, 1]$, satisfies $\max\text{flow}(s, t; w) \leq 1$ for all $s \neq t$, and $|A(D)| = (m-1) \sum_{u \neq v} w(u, v)$.*

Proof. Set $w = \mu/(m-1)$. Every multiplicity obeys $0 \leq \mu(u, v) \leq m-1$, so each $w(u, v)$ lies in $[0, 1]$. Dividing every capacity by the constant $m-1$ divides every flow value by $m-1$, and by Theorem A.1 the maximum flow under the capacities μ is $\lambda_D(s, t) \leq m-1$, so under w it is at most 1. For the arc count, recall that the number of arcs is the total multiplicity, $|A(D)| = \sum_{u \neq v} \mu(u, v)$, since each parallel copy is one arc. Substituting $\mu = (m-1)w$,

$$|A(D)| = \sum_{u \neq v} \mu(u, v) = \sum_{u \neq v} (m-1) w(u, v) = (m-1) \sum_{u \neq v} w(u, v),$$

which is the claimed identity. $\square$

Lemma A.27 (Arithmetic identity). *For all $n \geq 2$, $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \lfloor n/2 \rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$.*

Proof. For $n = 2k$ the left side is $k(k-1) + k = k^2$, and for $n = 2k+1$ it is $k^2 + k = k(k+1)$. Both match the right side. $\square$

Proposition A.28 (Bipartite upper bound for even $n \geq 8$, conditional). *Let $n = 2k \geq 8$ be even. Suppose every fractional weighting w' on $n-1$ vertices with values in $[0, 1]$ and all pairwise maximum flows at most one has total weight at most $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor$. Then every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq m-1$ satisfies $|A(D)| \leq (m-1) \left\lfloor \frac{n^2}{4} \right\rfloor$.*

Proof. By Lemma A.26 the arc count is $|A(D)| = (m-1)M$ with $M := \sum_{u \neq v} w(u, v)$, so it suffices to bound $M \leq \lfloor n^2/4 \rfloor$ for the scaled weighting w , all of whose pairwise maximum flows are at most 1.

Averaging. Give each vertex its weighted total degree $d(v) = \sum_u (w(u, v) + w(v, u))$. Summing over all vertices counts the weight of each ordered pair exactly twice, once at its tail and once at its head, so $\sum_v d(v) = 2M$, and some vertex v_0 has $d(v_0) \leq 2M/n$.

Deletion. The number $d(v_0)$ is exactly the total weight carried by pairs touching v_0 , so deleting v_0 leaves a weighting on $n-1$ vertices of total $M - d(v_0)$. Removing a vertex cannot raise any maximum flow, so that restricted weighting still has all pairwise flows at most 1, and the hypothesis applies to it:

$$M - d(v_0) \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \left\lfloor \frac{(2k-1)^2}{4} \right\rfloor = k(k-1),$$

the last step because $(2k - 1)^2/4 = k^2 - k + \frac{1}{4}$ floors to $k(k - 1)$.

Combining. Since $n = 2k$ we have $d(v_0) \leq 2M/n = M/k$, hence

$$M - \frac{M}{k} \leq M - d(v_0) \leq k(k - 1), \quad \text{that is} \quad M \cdot \frac{k - 1}{k} \leq k(k - 1).$$

Dividing by $(k - 1)/k > 0$ gives $M \leq k^2 = \lfloor n^2/4 \rfloor$, and multiplying back by $m - 1$ gives $|A(D)| \leq (m - 1) \lfloor n^2/4 \rfloor$. $\square$

The small cases $n \in \{4, 6\}$ genuinely do not satisfy the conclusion, and must not: there $2(n - 1) > \lfloor n^2/4 \rfloor$ and the double star beats the bipartite weighting, which is why the certified values of [Theorem 2.2](#) sit on the linear branch. For odd n the same deletion gives $M \leq k(k + 1) + \frac{k}{2k - 1}$, an overshoot of less than one that integrality no longer absorbs, because fractional weights need not have integer totals. The induction would close if some vertex always had weighted degree at most $(m - 1)k$ before scaling, which is the missing minimum-degree statement.

Conjecture A.29 (Minimum-degree lemma for odd n). *In any directed multigraph D on $n = 2k + 1$ vertices ($k \geq 3$) with $\lambda^{\max}(D) \leq m - 1$ and $m \geq 3$, some vertex has total degree $d(v) \leq (m - 1)k$, counting arcs with multiplicity.*

At $m = 2$ the statement is a proposition rather than a conjecture, through a reduction that also settles the whole multigraph problem at $m = 2$.

Proposition A.30 (Parallel-arc elimination at $m = 2$). *A directed multigraph with $\lambda^{\max} \leq 1$ has no parallel arcs, since two parallel arcs are already two arc-disjoint routes. Hence it is a simple digraph.*

Theorem A.31 (Directed multigraphs at $m = 2$). *For all $n \geq 2$, $L_2^{\text{dir}}(n) = \ell_2^{\text{dir}}(n) = \max(2(n - 1), \lfloor \frac{n^2}{4} \rfloor)$.*

Proof. Immediate from [Proposition A.30](#) and [Theorem 1.10](#). $\square$

Proposition A.32 (Minimum degree at $m = 2$, odd n). *In any directed multigraph on $n = 2k + 1 \geq 7$ vertices with $\lambda^{\max} \leq 1$, some vertex has $d(v) \leq k$.*

Proof. By [Proposition A.30](#) and [theorem 1.10](#), $|A(D)| \leq \max(2(n - 1), \lfloor n^2/4 \rfloor) = \lfloor n^2/4 \rfloor = k(k + 1)$, the maximum being the quadratic branch because $n \geq 7$. The average total degree is $\frac{2k(k+1)}{2k+1} < k + 1$, and some integer degree is at most k . The hypothesis $n \geq 7$ is needed: on the linear branch the statement can fail, as the bidirected star on three vertices has minimum total degree $2 > 1$. $\square$

For $m \geq 3$ the two extreme regimes of [Conjecture A.29](#) are settled. In the *purely saturated* regime, where every multiplicity is 0 or $m - 1$, the scaled weighting is the indicator of a simple digraph with $\lambda^{\max} \leq 1$, its degrees are integers, and the argument of [Proposition A.32](#) applies

verbatim after multiplying back by $m - 1$. In the *all-unsaturated* simple regime at $n = 7$, $m = 3$, the program certifies by a cut-counting infeasibility that no simple digraph satisfies all two-hop inequalities with every total degree at least 7. The open case is the *mixed* regime, where some multiplicity lies strictly between 0 and $m - 1$: there the scaled degrees are genuinely fractional, the averaging bound $\frac{2k(k+1)}{2k+1}$ lands strictly between k and $k + 1$, and no integrality rescues it. This is the single identified gap that separates the certified values of [Theorem 2.2](#) from the full conjecture

$$L_m^{\text{dir}}(n) = (m - 1) \max(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor),$$

whose lower bound is unconditional: the double star realises $2(n - 1)(m - 1)$ and the one-directional bipartite multigraph, every arc at multiplicity $m - 1$, realises $(m - 1) \lfloor n^2/4 \rfloor$ with $\lambda^{\max} = m - 1$.

Remark A.33 (Why the mixed regime resists, and its kinship with the back-arc gap). Two facts explain why the mixed regime is genuinely hard rather than merely unfinished, and both are worth recording so the conjecture is not mistaken for a routine averaging exercise.

First, averaging alone cannot close it, even granted the strongest input available. Work in the scaled weighting $w = \mu/(m - 1)$ of [Lemma A.26](#), with total weight $M = \sum_{u \neq v} w(u, v)$ and weighted total degree $d(v)$, so the conjecture asks for a vertex with $d(v) \leq k$. Granting the fractional even bound that drives [Proposition A.28](#), the odd deletion already gives $M \leq k(k + 1) + \frac{k}{2k-1} = \frac{(2k+1)k^2}{2k-1}$. Since the minimum degree never exceeds the average $\frac{2M}{2k+1}$,

$$\min_v d(v) \leq \frac{2M}{2k+1} \leq \frac{2k^2}{2k-1} = k + \frac{k}{2k-1} < k + 1.$$

In the saturated and $m = 2$ regimes the degrees are integers, so this rounds down to k and the conjecture follows, exactly as in [Proposition A.32](#). In the mixed regime $d(v)$ is genuinely fractional and the leftover $\frac{k}{2k-1}$ cannot be rounded away. The obstruction is not the particular bound chosen. To force $\min_v d(v) \leq k$ by averaging one would need $M \leq k^2 + \frac{k}{2}$, yet the one-directional bipartite weighting with $|A| = k$ and $|B| = k + 1$ already carries $M = k(k + 1) = k^2 + k$, so no averaging argument can finish for any m . Closing the gap needs a structural step that exhibits a specific low-degree vertex, which is precisely what [Lemma A.34](#) supplies at $m = 3$ in order to *bypass* the conjecture rather than prove it.

Second, the conjecture is tight, and the natural way to break it fails for a reason that ties this gap to the back-arc obstruction of [Conjecture 1.14](#). At the bipartite extremiser the minimum degree is exactly k , attained on the B side. The obvious attempt to raise it bolts a little backward weight onto the saturated bipartite point, sending $w(b, a_0) = \epsilon$ from every $b \in B$ to one fixed source $a_0 \in A$, which would lift each B -degree to $k + \epsilon$. It is infeasible for every $\epsilon > 0$. With the forward layer at full weight the backward arcs open the recirculating routes $a \rightarrow b \rightarrow a_0 \rightarrow b'$, so the flow from a source $a \neq a_0$ to a sink b' rises to $1 + k\epsilon > 1$. A single backward arc laid on a saturated forward layer manufactures an extra route, the same mechanism that keeps the upper bound of [Conjecture 1.14](#) open and the reason the bipartite extremisers are locally rigid under

the searches of [Chapter 3](#). The minimum-degree gap and the back-arc gap are two faces of one obstruction. A systematic fractional hill-climb (`fractional_search`), a search that repeatedly nudges one weight at a time and keeps a change only when the objective improves while every pairwise flow stays at most one, run with three initialisation types and eight seeds each at $n = 9, 11, \text{ and } 13$, finds no feasible weighting with minimum weighted degree above k , confirming the bipartite point's rigidity for $k = 4, 5, \text{ and } 6$. Feasibility of a fractional weighting is checked by `fractional_flows_feasible`, which calls `_float_maxflow_value`, a textbook maximum-flow routine (Edmonds–Karp [17]) run directly on the fractional capacities, since `scipy's csgraph` max-flow accepts only integer capacities.

Lemma A.34 (Attachment lemma). *Let $m \geq 2$ and let D_0 be the one-directional complete bipartite multigraph with parts A and B , $|A| = p \geq 2$, $|B| = q \geq 2$, and $\mu(a, b) = m - 1$ for every $a \in A, b \in B$, with no other arcs. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v , a multiset because parallel copies are allowed (multiplicities at most $m - 1$). If $\lambda_D^{\max} \leq m - 1$, then $d(v) \leq (m - 1) \max(p, q)$.*

The proof is a bookkeeping argument on the arcs at v , and [Figure A.4](#) is its scaffold. Sort those arcs into the four classes i_A, o_A, i_B, o_B of the figure, by direction and by which part they touch. Three combinations are impossible, because each completes an m -th arc-disjoint route between two old vertices on top of the $m - 1$ parallel arcs they already share. An i_A arc with an o_B arc makes $a \rightarrow v \rightarrow b$. An i_A arc with an o_A arc to a *different* source makes $a \rightarrow v \rightarrow a' \rightarrow b$. An i_B arc with an o_B arc to a *different* sink makes $a \rightarrow b \rightarrow v \rightarrow b'$. Forbidding these forces v into one of a few rigid shapes, a pure sink touching a single source, a pure source touching a single sink, or a small mixed remnant, and in each shape a count of two-step routes through v caps its degree. The three cases below are exactly those shapes, and the bound $(m - 1) \max(p, q)$ is whichever of them is largest.

Proof. The strategy is to sort the arcs at the new vertex v into four groups by their direction and which side they touch, rule out the combinations that would manufacture an m -th route between two old vertices, and then bound whatever survives. Write $i_A = \sum_a \mu(a, v)$, $i_B = \sum_b \mu(b, v)$, $o_A = \sum_a \mu(v, a)$, $o_B = \sum_b \mu(v, b)$ for the total multiplicities into v from A , into v from B , out of v to A , and out of v to B , so that $d(v) = i_A + i_B + o_A + o_B$. Three route patterns are forbidden outright, because each adds an m -th arc-disjoint route on top of the $m - 1$ parallel arcs of a bipartite pair:

- (1) $\mu(a, v) \geq 1$ and $\mu(v, b) \geq 1$: the path $a \rightarrow v \rightarrow b$ together with the $m - 1$ arcs $a \rightarrow b$ gives $\lambda(a, b) \geq m$.
- (2) $\mu(a, v) \geq 1$ and $\mu(v, a') \geq 1$ with $a' \neq a$: the path $a \rightarrow v \rightarrow a' \rightarrow b$ (any $b \in B$) gives $\lambda(a, b) \geq m$.
- (3) $\mu(b, v) \geq 1$ and $\mu(v, b') \geq 1$ with $b' \neq b$: the path $a \rightarrow b \rightarrow v \rightarrow b'$ (any $a \in A$) gives $\lambda(a, b') \geq m$.

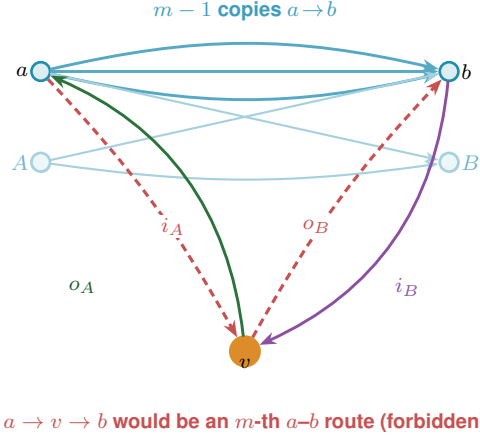


Figure A.4. The attachment lemma (Lemma A.34): a new vertex v is glued onto $(m - 1) \cdot B_{p,q}$, the one-directional complete bipartite multigraph in which every $a \in A$ sends $m - 1$ parallel arcs to every $b \in B$ (the bundle along the top, faint for the pairs other than a, b). The arcs at v fall into four classes by direction and side: i_A into v from A , o_A out of v to A , i_B into v from B , and o_B out of v to B . The two red legs i_A and o_B are drawn together because they are the worst case: the route $a \rightarrow v \rightarrow b$ laid on top of the $m - 1$ existing $a \rightarrow b$ arcs is an m -th arc-disjoint a - b route, so $\lambda(a, b) \geq m$, which feasibility forbids. Ruling this out, with two siblings like it, caps $d(v) \leq (m - 1) \max(p, q)$.

Each bound below rests on one principle. The local connectivity $\lambda(s, t)$ is at least the number of pairwise arc-disjoint s -to- t routes we can exhibit, because any such family of routes is an integral flow from s to t . Feasibility says $\lambda \leq m - 1$, so the moment we can name m arc-disjoint routes between two vertices we have a contradiction.

Case I: $i_A \geq 1$. Some arc enters v from a vertex $a \in A$. Pattern (1) then forces $o_B = 0$, because any arc $v \rightarrow b$ would complete the route $a \rightarrow v \rightarrow b$, an m -th route on the pair (a, b) laid on top of its $m - 1$ parallel arcs.

Suppose first that $o_A \geq 1$, so v also sends an arc to some vertex of A . Pattern (2) makes this rigid: an arc into v from a and an arc out of v to a different $a' \neq a$ would give the route $a \rightarrow v \rightarrow a' \rightarrow b$ for any b , again an m -th route. So every arc between v and A , in either direction, must touch one single vertex a^* . Counting arc-disjoint routes from a^* to v , there are the $\mu(a^*, v)$ direct arcs, and for each $b \in B$ the two-step route $a^* \rightarrow b \rightarrow v$, which can be run $\min(\mu(a^*, b), \mu(b, v))$ times on disjoint arcs. Different b use different arcs, and none reuse a direct arc, so

$$\lambda(a^*, v) \geq \mu(a^*, v) + \sum_b \min(\mu(a^*, b), \mu(b, v)) = \mu(a^*, v) + i_B,$$

the last equality because $\mu(a^*, b) = m - 1$ is at least $\mu(b, v)$, so each minimum is $\mu(b, v)$ and the sum is i_B . Feasibility caps the left side at $m - 1$, so $\mu(a^*, v) + i_B \leq m - 1$. Every arc at v now touches either a^* or B , so $d(v) = \mu(a^*, v) + i_B + \mu(v, a^*)$, and with the first two summing to at most $m - 1$ and the last at most $m - 1$ we get $d(v) \leq 2(m - 1)$.

Suppose instead $o_A = 0$. With $o_B = 0$ already, v is now a pure sink. The same two-step count applies to every $a \in A$ at once, giving $\lambda(a, v) \geq \mu(a, v) + i_B$, hence $\mu(a, v) \leq (m - 1) - i_B$ for each of the p vertices of A . Summing over A bounds the in-flow from A by $i_A \leq p(m - 1) - p i_B$,

so

$$d(v) = i_A + i_B \leq p(m-1) - (p-1)i_B \leq p(m-1),$$

the final step because $(p-1)i_B \geq 0$.

Case II: $i_A = 0$ and $o_B \geq 1$. This is Case I read backwards. Reversing every arc turns $(m-1) \cdot B_{p,q}$ into $(m-1) \cdot B_{q,p}$, swaps in-arcs with out-arcs, and sends pattern (2) to pattern (3), so the same argument applies with the roles of A and B exchanged. It yields either $d(v) \leq 2(m-1)$, or v is a pure source for which $\lambda(v, b) \geq \mu(v, b) + o_A \leq m-1$ holds for every b , and summing over the q vertices of B gives $d(v) \leq q(m-1) - (q-1)o_A \leq q(m-1)$.

Case III: $i_A = o_B = 0$. Now v receives nothing from A and sends nothing to B , so its only arcs are the i_B in-arcs from B and the o_A out-arcs to A . For any $a \in A$ the two-step routes $a \rightarrow b \rightarrow v$ over all b give $\lambda(a, v) \geq i_B$, so $i_B \leq m-1$, and for any $b \in B$ the routes $v \rightarrow a \rightarrow b$ over all a give $\lambda(v, b) \geq o_A$, so $o_A \leq m-1$. Hence $d(v) = i_B + o_A \leq 2(m-1)$.

Every case bounds $d(v)$ by either $2(m-1)$ or $\max(p, q)(m-1)$. Since $p, q \geq 2$ we have $2(m-1) \leq (m-1)\max(p, q)$, so in all cases $d(v) \leq (m-1)\max(p, q)$. $\square$

Corollary A.35 (Attachment equality). *In the setting of Lemma A.34 with $\max(p, q) \geq 3$, suppose $d(v) = (m-1)\max(p, q)$. If $q > p$, then v is a pure source with $\mu(v, b) = m-1$ for every $b \in B$, so $D = (m-1) \cdot B_{p+1, q}$ with v joining the source side. If $p > q$, then v is symmetrically a pure sink at full multiplicity, and if $p = q$, it is one of the two.*

Proof. Rerun the cases of Lemma A.34. The mixed cases cap $d(v)$ at $2(m-1) < (m-1)\max(p, q)$. The pure-sink case gives $d(v) \leq p(m-1) - (p-1)i_B$, which reaches $(m-1)\max(p, q)$ only when $p \geq q$, $i_B = 0$ and every $\mu(a, v) = m-1$. The pure-source case symmetrically forces $o_A = 0$ and every $\mu(v, b) = m-1$ when $q \geq p$. Feasibility of the resulting digraph is the bipartite construction itself. $\square$

Theorem A.36 (Conditional odd step, $m \in \{3, 4\}$). *Fix $k \geq 4$ and $m \in \{3, 4\}$, and assume (i) $L_m^{\text{dir}}(2k) = (m-1)k^2$, and (ii) every directed multigraph attaining it is the one-directional complete bipartite multigraph $(m-1) \cdot B_{k, k}$. Then $L_m^{\text{dir}}(2k+1) = (m-1)k(k+1)$.*

Proof. The lower bound is $(m-1) \cdot B_{k, k+1}$. The upper bound is a counting squeeze. If the arc count were even one over the target, then deleting any single vertex would still leave a feasible multigraph on $2k$ vertices, which by (i) cannot exceed $(m-1)k^2$, so every vertex would have to carry a large degree. Yet the average degree is only barely above that floor, so some vertex must sit exactly on it, and deleting that one lands precisely on the even extremiser $(m-1) \cdot B_{k, k}$ of (ii). The attachment lemma then caps how much that vertex could have carried, below what the surplus requires, and the contradiction forces the surplus to vanish. We now make this precise.

For the upper bound let D on $n = 2k+1$ vertices have $\lambda^{\max} \leq m-1$ and $|A(D)| = (m-1)k(k+1) + j$ with $j \geq 1$. Deleting any vertex v leaves a feasible multigraph on $2k$ vertices,

so by (i) $d(v) \geq |A(D)| - (m-1)k^2 = (m-1)k + j$ for every v . The average degree is

$$\frac{2|A(D)|}{2k+1} = (m-1)k + j + \frac{(m-1)k - (2k-1)j}{2k+1},$$

by direct computation. If $(m-1)k - (2k-1)j < 0$ the average lies below the minimum, a contradiction. Otherwise the correction term is nonnegative and, because $m \leq 4$ and $j \geq 1$, strictly below one: $(m-1)k - (2k-1)j \leq (m-3)k + 1 \leq k + 1 < 2k + 1$. The average therefore lies in $[(m-1)k + j, (m-1)k + j + 1)$ while every degree is at least $(m-1)k + j$, so some vertex v has $d(v) = (m-1)k + j$ exactly, and $D - v$ has exactly $(m-1)k^2$ arcs. By (ii), $D - v = (m-1) \cdot B_{k,k}$, and [Lemma A.34](#) caps $d(v) \leq (m-1)k < (m-1)k + j$, the final contradiction. Hence $j \leq 0$. $\square$

The attachment lemma bounds the degree of a vertex glued onto the *bipartite* extremiser. The linear branch has an exact analogue, worth stating once for the whole family it governs. Call a directed multigraph *everywhere-saturated* at level m when every ordered pair of distinct vertices has local connectivity exactly $m-1$. The doubled bidirected spanning trees at multiplicity $m-1$, the extremisers of the linear branch, are exactly such graphs: the $m-1$ routes between two vertices run along the doubled tree path, and cutting one doubled edge of that path is a cut of $m-1$ arcs.

Lemma A.37 (Saturated attachment). *Let $m \geq 2$ and let D_0 be a directed multigraph on at least two vertices with $\lambda_{D_0}(x, y) = m-1$ for every ordered pair of distinct vertices. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v . If $\lambda_D^{\max} \leq m-1$, then $d(v) \leq 2(m-1)$. If $d(v) = 2(m-1)$, then every arc at v joins v to one single vertex u , with $\mu(v, u) = \mu(u, v) = m-1$, and D is again everywhere-saturated.*

Proof. No mixed pair. Suppose v has an in-arc from u and an out-arc to w with $u \neq w$. By Menger and saturation D_0 carries $m-1$ pairwise arc-disjoint u -to- w routes, and the route $u \rightarrow v \rightarrow w$ uses only arcs incident to v , so it is arc-disjoint from all of them and $\lambda_D(u, w) \geq m$, a contradiction. Hence v is a pure source, a pure sink, or every arc at v touches one single vertex u .

A pure source has $d(v) \leq m-1$. Let $t = \sum_x \mu(v, x)$ and fix u with $\mu(v, u) \geq 1$. Any cut $(S, \bar{S})$ with $v \in S$ and $u \in \bar{S}$ has capacity $\sum_{x \in \bar{S}} \mu(v, x) + e_{D_0}(S \setminus \{v\}, \bar{S})$. If S contains a vertex x of D_0 , the second term is at least $\lambda_{D_0}(x, u) = m-1$ and the first is at least $\mu(v, u)$. If $S = \{v\}$, the capacity is t . Every cut therefore has capacity at least $\mu(v, u) + \min(t - \mu(v, u), m-1)$, and by max-flow min-cut so does $\lambda_D(v, u)$, which feasibility caps at $m-1$. Were $t - \mu(v, u) \geq m-1$, this would force $\mu(v, u) = 0$, a contradiction, so $t - \mu(v, u) < m-1$ and the bound reads $t \leq m-1$.

A pure sink has $d(v) \leq m-1$. Reverse every arc: the reverse of an everywhere-saturated multigraph is everywhere-saturated, a pure sink becomes a pure source, and the previous step applies.

A single partner. If every arc at v touches one vertex u , then parallel arcs are already that many arc-disjoint one-step routes, so $\mu(v, u) \leq m-1$ and $\mu(u, v) \leq m-1$, giving $d(v) \leq$

$2(m - 1)$.

Equality. Since $2(m - 1) > m - 1$, a degree of $2(m - 1)$ rules out the pure cases, so v has a single partner at full multiplicity both ways. The result is feasible and again everywhere-saturated: a route through v must enter and leave through u , so it adds nothing to any flow between old vertices, and $\lambda_D(x, v) = \min(\lambda_{D_0}(x, u), \mu(u, v)) = m - 1$ for every old x , symmetrically for $\lambda_D(v, x)$. $\square$

Remark A.38 (The linear branch grows one leaf at a time). The equality case of [Lemma A.37](#) attaches a new leaf to an arbitrary vertex at full multiplicity, and that is the only attachment reaching $2(m - 1)$. Iterating from a single doubled edge, itself everywhere-saturated on two vertices, generates exactly the doubled bidirected spanning trees, so the extremal family of the linear branch is closed under maximal attachment and grows one leaf at a time. The program confirms the lemma exhaustively at $m = 3$: over every doubled tree on five and six vertices and every one of the 3^{2n} possible attachment patterns, the densest feasible attachment has degree exactly $4 = 2(m - 1)$, and the degree-4 patterns are precisely the single-partner attachments at full multiplicity, one per choice of partner.

Remark A.39 (The $m = 3$ chain, sharpened). [Theorem A.36](#) removes [Conjecture A.29](#) from the critical path: the odd case no longer needs a minimum-degree statement, only the extremal characterisation one level down, and for $m = 3$ that characterisation propagates along the same induction in both parities.

Odd levels ($n = 2k + 1 \geq 9$): an extremiser has $|A| = 2k(k + 1)$ arcs and every degree at least $2k$ (deletion bound). All degrees $\geq 2k + 1$ is impossible since $(2k + 1)^2 > 4k(k + 1) = \sum_v d(v)$, so some $d(v) = 2k$ exactly, $D - v$ is the even extremiser $2 \cdot B_{k,k}$, and [Corollary A.35](#) (the $p = q$ case) forces v to rejoin as a full-multiplicity pure source or sink: $D = 2 \cdot B_{k+1,k}$ or $2 \cdot B_{k,k+1}$.

Even levels ($n = 2k \geq 10$): equality in the averaging step forces $2k$ -regularity, every vertex at degree exactly $2k$, hence $D - v$ extremal at $2k - 1$ for every v . Odd uniqueness one level down leaves $2 \cdot B_{k,k-1}$ or $2 \cdot B_{k-1,k}$, and [Corollary A.35](#) with $d(v) = 2k = 2 \max(k - 1, k)$ (the $q > p$ case for $2 \cdot B_{k-1,k}$, its $p > q$ mirror for $2 \cdot B_{k,k-1}$) rebuilds exactly $2 \cdot B_{k,k}$ either way.

The seam at $n = 8$: the same regularity argument runs from $n = 7$, where the branches tie at 24 arcs and several extremisers coexist. On the linear branch the extremisers are richer than the double star alone. Every *bidirected spanning tree at full multiplicity $m - 1$* is feasible with $2(n - 1)(m - 1)$ arcs: deleting a tree edge splits the vertices in two, the only arcs from one side to the other are that edge's own $m - 1$ copies, and the $m - 1$ routes along the tree path meet that cut, so every local connectivity is exactly $m - 1$ and the doubled trees are everywhere-saturated in the sense of [Lemma A.37](#). Exhaustive enumeration confirms that at $m = 3$ these doubled trees are the *only* extremisers for $n = 4, 5, 6$ alike, with 2, 3, and 6 classes matching the tree shapes exactly: the in-house enumerator (`enumerate_extremal_directed_multigraphs`) settles $n \leq 5$, and the sound generation enumerator has now completed the uncapped $n = 6$ classification as well. A deleted vertex of the 8-regular D leaves a 24-arc extremiser with maximum degree at most 8: among doubled trees only the doubled path qualifies (tree degree ≤ 2), the double star

(hub degree 24) being excluded. The doubled path P_7 is everywhere-saturated, so [Lemma A.37](#) caps any feasible attachment to it at $d(v) \leq 2(m-1) = 4$, while 8-regularity demands $d(v) = 8$. [Figure A.5](#) draws what a degree-8 attachment would have to be, weight 4 on each path end, closing the path into the doubled bidirected cycle C_8 with $\lambda = 4 > 2$: the collapse the lemma rules out wholesale. So every deletion is bipartite-type, the attachment equality applies, and $D = 2 \cdot B_{4,4}$ again.

The whole $m = 3$ problem therefore rests on two finite integral statements about multigraphs with multiplicities in $\{0, 1, 2\}$:

- (a) $L_3^{\text{dir}}(7) = 24$, which $M^*(7) = 12$ would imply but which is not implied by it in reverse, so it is posed to the integral program directly, and
- (b) every 24-arc feasible multigraph on 7 vertices with maximum degree at most 8 is $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, or the doubled bidirected path P_7 .

Both targets sit at sizes the certifier and the enumerator can realistically reach. Fact (a) is the single feasibility query `prove_integral_arc_bound(7, 3, 25)`: the cut-counting model of [Chapter 2](#) with the weights read as integer multiplicities and the maximisation replaced by the fixed arc total 25, so that an INFEASIBLE verdict is a proof that no 25-arc feasible multigraph on seven vertices exists. The model is run with two extra families of rows (`use_deletion_cuts`), and each is a valid inequality, satisfied by every feasible weighting, so adding them speeds the solver without ever excluding a witness. The deletion bounds say that restricting a feasible weighting to any five or six of the seven vertices leaves a feasible weighting, whose total is therefore capped by the certified $M^*(5)$ or $M^*(6)$. The degree-pair inequality $d^+(s) + d^-(t) - w(s, t) \leq n - 1$ holds for every ordered pair because the direct arc and the two-step detours are arc-disjoint routes, so $\text{maxflow}(s, t) \geq w(s, t) + \sum_x \min(w(s, x), w(x, t)) \geq d^+(s) + d^-(t) - w(s, t) - (n - 2)$, the last step using $\min(a, b) \geq a + b - 1$ for weights in $[0, 1]$, and feasibility caps the left side at one. Fact (b) is the run `enumerate_extremal_directed_multigraphs_via_generation(7, 3, 24, max_degree=8)` on the generation enumerator of [Chapter 2](#), which prunes with proved bounds only and is therefore a sound complete search at $n = 7$, where the in-house depth-first twin is not. That run has now completed, in about seven hours across ten processor cores, and it returns exactly three isomorphism classes: $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 , each re-verified by the exact checker. *Fact (b) is therefore settled by machine*, and the whole $m = 3$ directed multigraph problem now rests on fact (a) alone. The $n = 6$ companion run (target 20, maximum degree 8) completed the same way and returns exactly one class, the doubled bidirected path P_6 , the pattern the seam argument uses one level up. The uncapped $n = 6$ classification also pays directly into fact (a): a hypothetical 25-arc feasible multigraph on 7 vertices has every degree at least 5 by the deletion bound against $L_3^{\text{dir}}(6) = 20$, and a vertex of degree exactly 5 would delete to a 20-arc extremiser on six vertices, a doubled tree, to which [Lemma A.37](#) forbids any attachment of more than 4 arcs. Any witness against fact (a) therefore has minimum total degree at least 6.

Fact (a) is stated so that a reader can settle it independently, and the practical route is the mixed-integer program rather than the enumeration. The call is `prove_integral_arc_bound(7, 3, 25)`, which asks whether any directed multigraph on seven vertices with $\lambda^{\max} \leq 2$ carries 25 arcs, so an INFEASIBLE verdict is exactly fact (a). The free CBC solver bundled with `pulp` does not close it, and neither does the uncapped generation enumerator within any budget tried here, but the same model handed to Gurobi is a different proposition: the identical call with `use_gurobi=True` runs the model on a solver strong enough for this class of problem, and Gurobi is freely licensed for academic use. That switch is the whole change required, since `_pick_solver` routes the same `pulp` model to whichever backend is available. The verdict is a proof either way, because the cut formulation is exact rather than a relaxation.

For $m \geq 4$ one hole remains: at an odd extremal level the counting no longer forces a vertex of degree $(m-1)k$ (all degrees $\geq (m-1)k+1$ is arithmetically consistent once $k(m-3) \geq 1$), so the uniqueness statement does not yet propagate through odd levels, although the *value* at odd levels follows from even uniqueness whenever that is available. For $m \geq 5$ the same escape climbs one level: a hypothetical extremiser carrying $j \geq 1$ surplus arcs could have *every* degree strictly above the deletion bound (all degrees $\geq (m-1)k+j+1$ is consistent exactly when $(2k-1)j \leq (m-1)k-2k-1$, which admits $j=1$ once $m \geq 5$), so no vertex deletion is forced to land on an extremal multigraph and even the *value* step stalls. This is why [Theorem A.36](#) is stated for $m \in \{3, 4\}$.

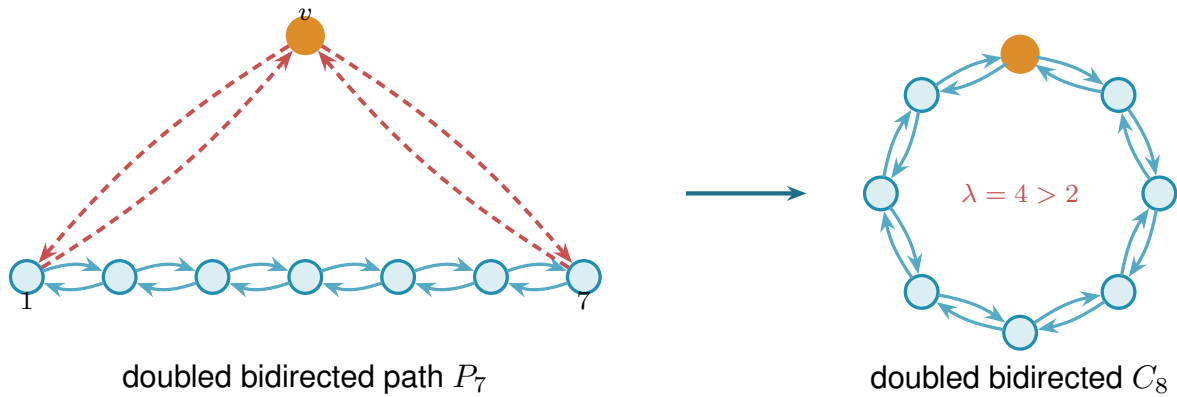


Figure A.5. The seam argument of [Remark A.39](#). A degree count forces a deleted vertex of the 8-regular extremiser D to leave the doubled bidirected path P_7 (left, each path edge a pair of opposing arcs). To reach degree 8 the new vertex v must attach with full multiplicity to the two ends 1 and 7 (red), bidirected. That closes the path into the doubled bidirected cycle C_8 (right), which is *infeasible*: the two directions around the cycle give two arc-disjoint routes each way, so $\lambda = 4 > 2$ between any pair. The contradiction rules out the path branch, forcing every deletion to be bipartite-type and hence $D = 2 \cdot B_{4,4}$.

A.6 The hypergraph bounds

The body's hypergraph claims rest on two theorems: a Menger theorem for Berge routes, which also certifies the helper-point checker of [Chapter 2](#), and a hypergraph Gomory–Hu theorem from the recent literature.

Theorem A.40 (Menger for hypergraphs). *For a hypergraph $\mathcal{H}$ and vertices u, v , the maximum number of hyperedge-disjoint Berge u – v paths equals the minimum size of a set C of hyperedges whose removal separates u from v .*

Proof. Build the helper-point network of Figure 2.5: for each hyperedge e a helper arc $e^- \rightarrow e^+$ of capacity one, and arcs $x \rightarrow e^-$ and $e^+ \rightarrow x$ of unbounded capacity for each member $x \in e$. We match the two sides by translating in each direction.

Berge paths become flow. A Berge path $u, e_1, v_1, \dots, e_k, v$ becomes the directed walk $u \rightarrow e_1^- \rightarrow e_1^+ \rightarrow v_1 \rightarrow \dots \rightarrow v$, crossing one helper arc per hyperedge it uses. The e_i along a single Berge path are distinct, and two hyperedge-disjoint Berge paths share no hyperedge at all, so they use disjoint sets of helper arcs and translate to *arc-disjoint* flow paths, paths that share no arc, each carrying one unit of flow.

Flow becomes Berge paths. Conversely, because every capacity is an integer, a maximum flow may be taken integral, and an integral flow of value k splits into k arc-disjoint unit paths. Reading a helper crossing $\dots x \rightarrow e^- \rightarrow e^+ \rightarrow y \dots$ back as the single hyperedge step x, e, y , that is, *suppressing* the two helper nodes, turns each unit path into a Berge path, and no two of them can share a hyperedge, since that would send two units across some capacity-one helper arc.

Cuts. Finally, the member arcs are uncapped, so any cut of finite capacity uses helper arcs only, and a set of helper arcs separates u from v in the network exactly when the corresponding hyperedges separate them in $\mathcal{H}$. The max-flow min-cut theorem equates the largest number of arc-disjoint flow paths with the smallest helper-arc cut, which is precisely the claim. $\square$

Theorem A.41 (Hypergraph Gomory–Hu, Bérczi et al. [9]). *Every r -uniform hypergraph $\mathcal{H}$ on vertex set V has a weighted tree T^* on V , its hypergraph Gomory–Hu tree, such that $\lambda_{\mathcal{H}}(u, v)$ equals the minimum weight on the u – v tree path, and each tree edge t induces a vertex partition $(S_t, V \setminus S_t)$ with $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, where $\delta_{\mathcal{H}}(S)$ is the set of hyperedges incident to both sides, that is, having at least one vertex in S and at least one outside it.*

This is the exact hypergraph echo of Theorem A.2, the same tree-of-cuts summary of all pairwise connectivities, with the single change that the capacity of a cut now counts the hyperedges straddling it, $|\delta_{\mathcal{H}}(S)|$, in place of the edges. Bérczi, Chandrasekaran, Király and Kulkarni [9] obtain the tree by the same uncrossing strategy sketched for the graph case above, applied now to the submodular hyperedge-cut function $|\delta_{\mathcal{H}}(\cdot)|$. We use only the two properties in the statement, the path-minimum reading of $\lambda_{\mathcal{H}}$ and the identity $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$.

Proof of Proposition 1.15. Upper bound. Let $\mathcal{H}$ be r -uniform with $\lambda_{\mathcal{H}}^{\max} \leq m - 1$, and let T^* be its hypergraph Gomory–Hu tree (Theorem A.41), a weighted tree on the same n vertices with weights c^* . The plan is to charge each hyperedge to tree edges and count the charges two ways, exactly as in the Mader proof, with the one change that a hyperedge spans r vertices rather than two.

Step 1: each hyperedge crosses at least $r - 1$ tree cuts. Fix a hyperedge e , a set of r vertices, and let $\text{ST}(e)$ be its *smallest connected subtree*, the minimal subtree of T^* that contains all r of them (its Steiner tree). A tree spanning at least r vertices has at least $r - 1$ edges, so $\text{ST}(e)$ has at least $r - 1$ of them. Each such edge t is a genuine cut for e : deleting t from T^* splits the vertices into the two sides $(S_t, V \setminus S_t)$, and because t lies on $\text{ST}(e)$ the hyperedge e has at least one vertex on each side. So e touches both sides, that is, $e \in \delta_{\mathcal{H}}(S_t)$. Therefore

$$|\{t \in E(T^*) : e \in \delta_{\mathcal{H}}(S_t)\}| \geq r - 1 \quad \text{for every hyperedge } e.$$

Step 2: count the incidences two ways. Sum that inequality over all hyperedges, then exchange the order of summation. This is legitimate because both of the middle sums below simply count the same pairs (e, t) with $e \in \delta_{\mathcal{H}}(S_t)$, once grouped by the hyperedge and once by the tree edge:

$$(r - 1) |E(\mathcal{H})| \leq \sum_e |\{t : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\{e : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\delta_{\mathcal{H}}(S_t)|.$$

Step 3: read off the bound. By the Gomory–Hu property each tree edge has $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, and each weight $c^*(t)$ is a local connectivity $\lambda_{\mathcal{H}}$ of its endpoints, hence at most $m - 1$ by feasibility. Summing over the $n - 1$ tree edges, $\sum_t |\delta_{\mathcal{H}}(S_t)| = \sum_t c^*(t) \leq (m - 1)(n - 1)$. Chaining Steps 2 and 3,

$$(r - 1) |E(\mathcal{H})| \leq (m - 1)(n - 1),$$

and dividing by $r - 1$, then rounding down because $|E(\mathcal{H})|$ is an integer, gives $|E(\mathcal{H})| \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. *Lower bound.* When $(r - 1) \mid (n - 1)$, take a hub h and split the other $n - 1$ vertices into groups $G_1, G_2, \dots$ of size $r - 1$ each. Form the star hypertree whose hyperedges are $\{h\} \cup G_i$, one per group, and give each hyperedge multiplicity $m - 1$. Every non-hub vertex lies in exactly $m - 1$ hyperedges, which form a cut isolating it, so by [Theorem A.40](#) every pair has Berge connectivity at most $m - 1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m - 1 \leq \binom{n-2}{r-2}$, [Theorem A.42](#) below attains the floor exactly, without even needing repeated hyperedges. $\square$

The multiplicities in the lower bound can in fact be avoided: for $r \geq 3$, *simple* r -uniform hypergraphs already attain the same bound, in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader's $\lfloor m(n - 1)/2 \rfloor$ from the multigraph value $(m - 1)(n - 1)$.

Theorem A.42 (Simplicity is free for $r \geq 3$). *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m - 1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a simple r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m - 1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is [Proposition 1.15](#), since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By [Lemma A.44](#) applied with rank $r - 1$ and

degree ceiling $d = m - 1$ on the $n - 1$ vertices of V' , there is an $(r - 1)$ -uniform simple hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m - 1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r -sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \leq m - 1$ hyperedges, which form a cut isolating w . By [Theorem A.40](#), $\lambda_{\mathcal{H}}^{\max} \leq m - 1$. $\square$

The sparse hypergraph the construction needs is a corollary of a classical partition theorem, which we state before using it.

Theorem A.43 (Baranyai [18]). *If r divides n , the $\binom{n}{r}$ distinct r -subsets of an n -element set can be partitioned into $\binom{n-1}{r-1}$ classes, each a perfect matching, meaning a family of n/r disjoint r -sets that together cover every vertex exactly once. More generally, for any prescribed class sizes $a_1 + \dots + a_t = \binom{n}{r}$, and with no divisibility assumed, the r -subsets can be partitioned into classes of exactly those sizes so that in the class of size a_i every vertex lies in either $\lfloor ra_i/n \rfloor$ or $\lceil ra_i/n \rceil$ of the sets, as evenly balanced as the size permits.*

The first statement is the memorable one. When r divides n the complete r -uniform hypergraph, the one holding every r -set, comes apart into $\binom{n-1}{r-1}$ perfect matchings, the exact hypergraph analogue of splitting a regular bipartite graph into perfect matchings by König's theorem. The proof is an induction that keeps the partial partition as balanced as possible at every step, an integer-flow argument in the spirit of Hall's marriage theorem rather than an explicit construction, and its full form is in Baranyai [18]. We need only the general equitable form, and only its conclusion about degrees.

Lemma A.44 (Sparse uniform hypergraphs exist). *Let $r \geq 2$, $n \geq r$, and $0 \leq d \leq \binom{n-1}{r-1}$. There exists an r -uniform simple hypergraph on n vertices with maximum degree at most d and exactly $\left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges.*

Proof. Put $e := \left\lfloor \frac{dn}{r} \right\rfloor \leq \frac{n}{r} \binom{n-1}{r-1} = \binom{n}{r}$. Apply the equitable form of [Theorem A.43](#) with two classes, of sizes e and $\binom{n}{r} - e$. Within each class every vertex is covered either $\lfloor re/n \rfloor$ or $\lceil re/n \rceil$ times, so the class of size e is a simple r -uniform hypergraph with $e = \left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges and maximum degree at most $\lceil re/n \rceil \leq \lceil d \rceil = d$, the last step because $re/n \leq r(dn/r)/n = d$ and d is an integer. $\square$

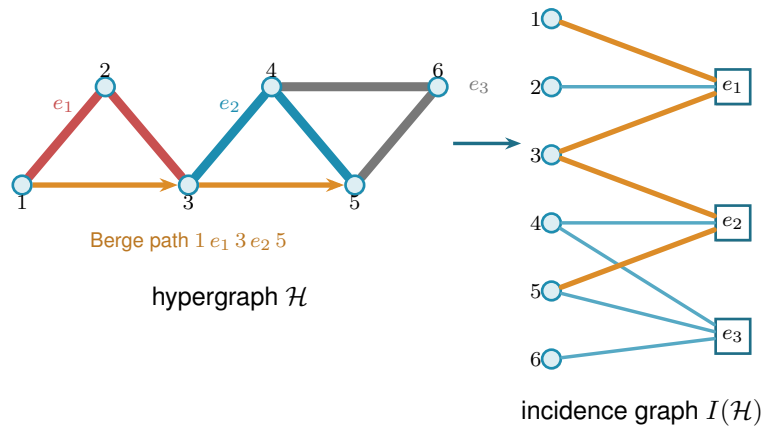
A.7 The hypergraph vertex problem

The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which [Figure A.6](#) draws in full. Write $I(\mathcal{H})$ for the bipartite incidence graph of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes. The interior of such a path consists of both kinds of node, the hyperedges it rides and the vertices where it changes hyperedge, so two paths of $I(\mathcal{H})$ are

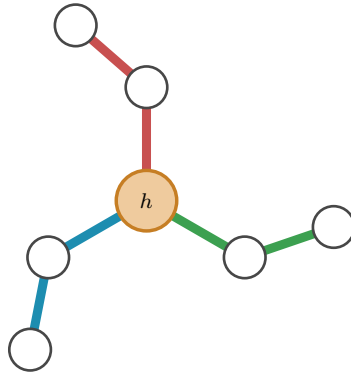
internally disjoint in the ordinary graph sense exactly when the two Berge routes share neither a hyperedge nor an intermediate vertex. That is precisely the separation fixed in [Chapter 1](#), and it is the reason both halves were required there. Hence

$$\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v),$$

the maximum number of internally disjoint $u-v$ paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree- r nodes on one side of a bipartite graph whose other side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths. [Figure A.6a](#) traces a single Berge path through both pictures, and [Figure A.6b](#) shows the star hypertree of [Theorem A.45](#), whose incidence graph is a tree.



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1-e_1-3-e_2-5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2 , changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1-e_1-3-e_2-5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r-1=2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.6. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of [Theorem A.45](#), whose incidence graph is a tree.

Theorem A.45 (Hypergraph vertex value at $m = 2$). *For $r \geq 2$ and $n \geq 1$, the maximum number*

of hyperedges of an r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is

$$k_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree. In particular $k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Proof. We claim $\kappa_{\mathcal{H}}^{\max} \leq 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \geq 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \dots, e_2, v_2$ are two Berge v_1 - v_2 paths with disjoint hyperedge sets and disjoint interior vertex sets, so $\kappa(v_1, v_2) \geq 2$. (At $\ell = 2$ this is the repeated pair: two hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$. This proves the claim. A forest on $n + q$ nodes has at most $n + q - 1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \leq n + q - 1$, i.e. $q \leq (n-1)/(r-1)$, and q is an integer. The star hypertree has $\lfloor (n-1)/(r-1) \rfloor$ hyperedges and a tree as incidence graph (each block vertex lies in one hyperedge, every cycle would need two hyperedges sharing two vertices), so the floor is attained. $\square$

Proposition A.46 (General lower bound). *For $r \geq 3$, $m \geq 2$ and $m-1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \geq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of [Theorem A.42](#) has Berge edge-connectivity at most $m-1$, and Whitney's inequality $\kappa \leq \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. $\square$

The $m = 3$ case also closes completely. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\text{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Primer: global connectivity and the pieces of a graph

The lemma below speaks of a graph being 2-connected or 3-connected and of taking it apart at small separators. These are the global cousins of the local $\kappa(u, v)$ from [Chapter 1](#), so we fix the words once. A graph on more than k vertices is k -connected if it stays connected

after the removal of any $k - 1$ vertices. By Menger this is the same as asking that every pair of vertices be joined by at least k internally vertex-disjoint paths, so k -connected means $\kappa(u, v) \geq k$ for all pairs at once, the uniform version of the local measure. Thus 2-connected means no single vertex disconnects the graph, and 3-connected means no two vertices do. Three named obstructions go with this. A *cut vertex* is a vertex whose removal disconnects the graph, so a connected graph on at least three vertices is 2-connected exactly when it has no cut vertex. A *separating pair* is a set of two vertices whose removal disconnects the graph, the obstruction a 2-connected graph must lack in order to be 3-connected. Finally a *block* is a maximal piece with no cut vertex of its own, hence either a single bridge edge or a maximal 2-connected subgraph. Two blocks overlap in at most one vertex, always a cut vertex, and the blocks and cut vertices hang together in a tree, the *block-cut tree*, with one node per block and one per cut vertex. Step 1 of the proof is precisely an induction over that tree.

Lemma A.47 (Incidence rank lemma). *Let G be a connected multigraph with vertex set $X \cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then $\text{rank}(G) \leq |X| - 1$.*

Proof. Induction on $|V| + |E|$. The proof peels G down to a core that cannot exist. Step 1 handles a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z -vertex or X -vertex without changing the bound. What survives is 2-connected with minimum degree at least three, and Step 4 shows that such a graph already violates the connectivity ceiling (iv), so it never arises. *Base* $|V| \leq 2$. A single vertex lies in X by (iii), and $0 \leq |X| - 1$. On two vertices, (i)–(iii) force both into X , parallel edges are internally disjoint paths, so (iv) caps the multiplicity at 2 and $\text{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its *blocks* $B_1, \dots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, either a single bridge edge or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G . No cycle can pass through two different blocks, so the cycle rank adds up over blocks, $\text{rank}(G) = \sum_i \text{rank}(B_i)$, and it is enough to bound each $\text{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Each block inherits the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X , giving $0 \leq |X \cap B| - 1$. Every other block is 2-connected, hence has minimum degree at least 2, so (i) to (iv) hold inside it and the induction hypothesis gives $\text{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v , so $b(v) = 1$ except at cut vertices. Counting each X -vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block–cut tree identity $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$, which holds because the blocks and cut vertices form a tree in which each cut vertex v is joined to the $b(v)$ blocks that contain it. Subtracting this count of blocks from the previous display and separating the cut vertices into their X - and Z -parts leaves

$$\text{rank}(G) = \sum_i \text{rank}(B_i) \leq \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \leq |X| - 1,$$

where the last inequality holds because each $b(z) - 1 \geq 0$, so the Z -cut sum only pushes the bound further below $|X| - 1$.

Step 2: a Z -vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0 \in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z -vertex, which (ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\text{rank} = |E| - |V| + 1$ is unchanged. The new edge runs X to X , so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z -vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1, x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G .

Step 3: an X -vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z -vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X| \geq 2$ to begin with, for if $|X| \leq 1$ then by (i) and (ii) every z would send all its edges to one X -vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\text{rank}(G) - 1 \leq (|X| - 1) - 1$, that is $\text{rank}(G) \leq |X| - 1$.

Step 4: the remaining case is impossible. Now G is 2-connected with $|V| \geq 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x, x') \geq 2$ then a third internally disjoint route exists: walk from x to any third vertex w inside $G - x'$, then from w to x' inside $G - x$. The concatenation visits x only first and x' only last, so it contains an x – x' path with at least one interior vertex, giving $\kappa(x, x') \geq 3$, against (iv). If G is 3-connected, Menger puts every pair at $\kappa \geq 3$, so $|X| \leq 1$ and every z has degree at most 1 by (i), absurd.

Otherwise G is 2-connected but not 3-connected, so it can be taken apart at its 2-vertex separators.

Primer: the triconnected (SPQR) decomposition

A 2-connected graph that is not 3-connected must have a separating pair, and cutting along every such pair sorts any 2-connected graph into a few standard shapes. This is a classical

theorem of Tutte [19], made into a linear-time algorithm by Hopcroft and Tarjan [20].

Theorem (triconnected decomposition). Every 2-connected graph G on at least three vertices decomposes into a *tree of pieces*, unique up to relabelling, in which each piece is one of three kinds: a *cycle* of length at least three (an S -piece, for *series*), a *bond*, meaning two vertices joined by three or more parallel edges (a P -piece, for *parallel*), or a 3-connected simple graph on at least four vertices (an R -piece, for *rigid*). Two adjacent pieces share exactly one *virtual edge* $\{a, b\}$, a placeholder that records a separating pair of G at which the two pieces were split apart. The three initials S, P, R , together with Q for a trivial single-edge piece, give the structure its usual name, the *SPQR tree*.

The tree is built by the recursion that also defines it. If G is already a cycle, a bond, or 3-connected, it is a single piece and the tree is one node. Otherwise choose a separating pair $\{a, b\}$, split G into the parts that this pair separates, add the virtual edge $\{a, b\}$ to each part so the split is remembered, and recurse on the parts. Every split makes the parts strictly smaller, so the recursion halts, and Tutte's theorem is that the resulting collection of pieces does not depend on the order in which the separating pairs are taken. The one fact the proof draws from all this is that a virtual edge $\{a, b\}$ always stands for a genuine a – b path through the rest of G , so an argument that must cross from a to b outside a given piece can always do so on the far side of that piece's virtual edge.

The decomposition just stated is sketched in Figure A.7, which also marks the leaf at which the argument bites. Two properties are all the proof needs from it, and both are visible in the figure. First, a vertex of a piece that touches none of that piece's virtual edges belongs to that piece alone, so it keeps in G exactly the degree it has in the piece. Second, for any virtual edge $\{a, b\}$ the part of G on the far side of the cut contains a genuine a – b path whose interior avoids the piece, so the virtual edge can always be replaced by a real route.

Now G itself is none of the three piece types, since it is not a single cycle (its degrees are at least three), not a bond (it is simple), and not 3-connected (the case just handled), so the tree has at least two leaves. Let L be one, with unique virtual edge $\{a, b\}$. If L is a cycle, a vertex of L off $\{a, b\}$ has degree 2 in G , absurd. If L is a bond, it holds at least two real parallel edges, absurd in a simple graph. If L is 3-connected, then any two of its vertices are joined by three internally disjoint paths inside L , at most one through the virtual edge, which the far-side path replaces: so $\kappa_G(u, v) \geq 3$ for all $u, v \in V(L)$, forcing $|X \cap V(L)| \leq 1$ by (iv). But $|V(L)| \geq 4$ leaves a Z -vertex $z^* \notin \{a, b\}$. Its neighbours lie in $X \cap V(L)$ by (i), at most one vertex, reached by at most one edge since G is simple, giving degree at most 1, which is absurd. $\square$

Theorem A.48 (Hypergraph vertex value at $m = 3$). *For $r \geq 2$, every r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $r \geq 3$ with $2 \leq \binom{n-2}{r-2}$ the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) = \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor = \ell_3^{(r)}(n).$$

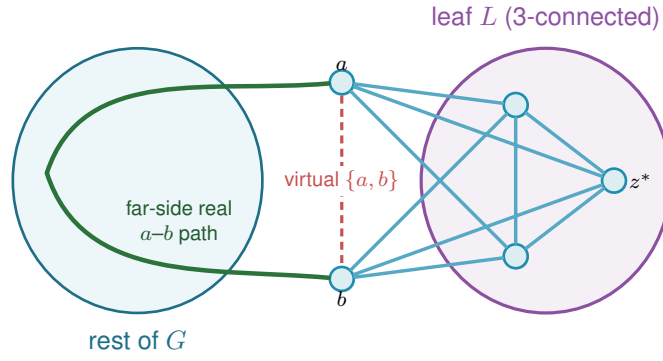


Figure A.7. The triconnected (Tutte/SPQR) decomposition in Step 4 of [Lemma A.47](#), schematic: G splits into a tree of pieces (blobs) glued along virtual edges, and the analysis works at a *leaf* L with its single virtual edge $\{a, b\}$ (dashed red). The far side of the tree supplies a *real* a - b path (green) that internally avoids L . If L is 3-connected, any two of its vertices are joined by three internally disjoint paths inside L , at most one using the virtual edge, and the far-side path replaces it, so $\kappa_G \geq 3$ for every pair in L . With $|V(L)| \geq 4$ that forces a hyperedge-node $z^* \notin \{a, b\}$ of degree at most one, which is the contradiction that closes the lemma.

Proof. Apply [Lemma A.47](#) to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X -node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum rank $\leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r incidences) and $n + q$ nodes, so $\sum_c \text{rank} = rq - (n + q) + C$, while $\sum_c (|X_c| - 1) = n - C$. The inequality reads $rq - (n + q) + C \leq n - C$, hence $q(r - 1) \leq 2(n - 1) + 2(1 - C) \leq 2(n - 1)$, the last step using $C \geq 1$. The lower bound is [Proposition A.46](#) at $m = 3$. $\square$

Remark A.49 (Edges as gates at $r = 2$, and the boundary of the method). At $r = 2$ the theorem speaks of multigraphs under the hyperedge-as-gate convention, in which parallel edges are distinct one-step routes, giving the multigraph value $2(n - 1)$ of [Theorem 1.3](#), complementing the body's convention for graphs, where parallel adjacencies count once. The proof of [Lemma A.47](#) leans on the triconnected decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$, the analogous question asks for a rank bound relative to $\kappa \leq 3$, where no comparably clean decomposition exists. That the pattern $k_m^{(r)} = \ell_m^{(r)}$ must fail eventually is not merely an analogy with the graph case ($m = 5$, Sørensen-Thomassen), which is stated for simple graphs under the body's convention. It already fails inside this convention, at $r = 2$. The identification in the first sentence of this remark runs both ways: the $r = 2$ case of the present problem *is* the multigraph vertex problem of [Section A.8](#), since a route there is a family of edge-disjoint and internally disjoint paths, which is exactly [Lemma A.54](#). So [Theorem A.56](#) applies to it, and the value exceeds $\lfloor (m - 1)(n - 1)/(r - 1) \rfloor = (m - 1)(n - 1)$ for every $m \geq 5$ and every $n \geq 3$, by a margin that grows with n . The formula is therefore not a candidate for all m and r together, and the live question is only whether $r \geq 3$ postpones the failure past $m = 4$.

For $m = 4$ it does not fail anywhere within reach. An exhaustive search over r -uniform multihypergraphs confirms both halves of the value, that $\lfloor 3(n - 1)/(r - 1) \rfloor$ is attained and that

one more hyperedge is infeasible, at $(n, r) = (4, 3), (5, 3), (5, 4), (6, 4), (6, 5), (7, 5), (7, 6), (8, 6), (8, 7), (9, 7)$ and $(9, 8)$. That is a wider net than the single cell $k_4^{(3)}(5) = 6$, and it is evidence rather than proof: the missing ingredient remains the 4-connectivity rank bound, and the $r = 2$ failure at $m = 5$ shows the surrounding claim has a genuine boundary rather than extending indefinitely.

Proposition A.50 (Directed hypergraphs: first bounds). *Model a directed r -uniform hyperedge as a tail vertex together with $r - 1$ head vertices, a Berge route entering at the tail and leaving at a head (the program's convention, drawn in Figure A.8). If every pair of vertices has at most $m - 1$ arc-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \leq \frac{(m-1)n(n-1)}{r-1},$$

and for $m-1 \leq \binom{n-\lceil n/2 \rceil - 1}{r-2}$ there are feasible constructions with $\max_{1 \leq \alpha \leq n-1} \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor \approx (m-1)n^2/(4(r-1))$ hyperedges, so the value is quadratic in n .

Proof. Upper bound: a hyperedge with tail u serves the $r - 1$ ordered pairs (u, h) , h a head. If some ordered pair (u, v) were served by m distinct hyperedges, those would be m arc-disjoint one-step routes, so each ordered pair is served at most $m - 1$ times and $(r-1)|E| \leq (m-1)n(n-1)$. Lower bound: split V into tails A , $|A| = \alpha$, and heads B . By Lemma A.44 there is a simple $(r - 1)$ -uniform hypergraph $\mathcal{G}^*$ on B with maximum degree $m - 1$ and $\lfloor (m-1)|B|/(r-1) \rfloor$ edges. Give every tail $a \in A$ the hyperedges (a, H) , $H \in \mathcal{G}^*$. No vertex of B is a tail, so every route is a single step and $\lambda(a, b) = \deg_{\mathcal{G}^*}(b) \leq m - 1$, while pairs inside A or starting in B have no routes. $\square$

The forward model is one of three ways to orient a Berge edge. In general a directed r -uniform hyperedge is a split of its r vertices into a non-empty *tail set* T and a non-empty *head set* H , a route entering at a tail and leaving at a head. The *forward* model is $|T| = 1$, the *backward* model is $|H| = 1$, and the *general* model allows any split (genuinely mixed splits, both parts at least two, exist from $r \geq 4$). The next two results place all three. The first is that backward needs no separate study, and the second is that the general model, despite admitting more hyperedges on a single vertex set, obeys the very same first-order bound as forward.

Lemma A.51 (Backward is the reverse of forward). *Reversing every hyperarc, $(T, H) \mapsto (H, T)$, is an edge-count-preserving bijection between the forward directed r -uniform hypergraphs and the backward ones on the same vertices, and the reverse $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$ and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$. Hence $\lambda^{\max}$ and $\kappa^{\max}$ are invariant under reversal, and the backward edge and vertex extremal numbers equal the forward ones, for every r, m and n .*

Proof. Reversal sends a backward hyperarc ($|T| = r - 1, |H| = 1$) to a forward one ($|T| = 1$) and is its own inverse, so it is an edge-count-preserving bijection of the two families. Let $u = x_0, e_1, x_1, \dots, e_k, x_k = v$ be a directed Berge route in $\mathcal{H}$, where x_{i-1} is a tail and x_i



Figure A.8. How the thesis draws a single hyperedge on its r vertices, in the metro colours of Figure 1.3. *Left:* an undirected hyperedge is one metro line threading all its members, with no arrows. *Right:* a directed hyperedge, a tail with $r - 1$ heads, is drawn as a star, the tail sending one thick arc to each head, so a Berge route enters at the tail and leaves at any head. The hypergraph panels of Figure 1.4 use this same star. Where a directed hyperedge instead has to sit on a metro line it already occupies undirected, as in Figure 2.8, the line is oriented along its own length away from the tail rather than fanned out, which draws the same object with fewer crossings. Either way one rule recovers the tail from the picture: it is the single station of that line into which none of its own arrows points. The shape is a drawing convention, not a path. Each coloured line or star is a *single* edge on several vertices, not a sequence of ordinary edges, so the line on the left is one hyperedge and not three. One colour means one hyperedge, and where two of them meet they cross at a shared station rather than joining into a longer line. What might be mistaken for a hyperedge is a *route*, which does run edge by edge from one vertex to another, but a route is drawn as a thin thread laid over the lines, so the thick coloured hyperedge and the thin route never look alike.

a head of e_i . The reversed edge $e_i^R = (H_i, T_i)$ has x_i as a tail and x_{i-1} as a head, so $v = x_k, e_k^R, x_{k-1}, \dots, e_1^R, x_0 = u$ is a directed Berge route from v to u in $\mathcal{H}^R$ on the same edges and the same internal vertices. This is a bijection between u – v routes in $\mathcal{H}$ and v – u routes in $\mathcal{H}^R$ that preserves which edges and which internal vertices any two routes share, so it carries an edge-disjoint (respectively internally vertex-disjoint) family to one of equal size. Therefore $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$ and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$, and maximising over ordered pairs gives $\lambda^{\max}(\mathcal{H}) = \lambda^{\max}(\mathcal{H}^R)$ and the same for κ . Since reversal is a bijection preserving feasibility and edge count, the two models share every extremal number. $\square$

Proposition A.52 (The general model obeys the forward bound). *Let a directed r -uniform hyperedge be any split into a non-empty tail set T and head set H . If every ordered pair has at most $m - 1$ arc-disjoint, or at most $m - 1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \leq \frac{(m - 1) n(n - 1)}{r - 1},$$

the same bound as the forward model of Proposition A.50. The forward construction of that proposition is itself a set of general hyperedges, so the general value obeys the same upper bound and inherits the same quadratic lower bound, and in particular is $\Theta(n^2)$.

Proof. Fix an ordered pair (u, v) . A hyperedge (T, H) with $u \in T$ and $v \in H$ carries the one-step route $u \rightarrow v$ that enters at the tail u and leaves at the head v . Distinct such hyperedges give routes that share no edge and no internal vertex, so if k hyperedges had u in their tail set and v in their head set then $\lambda(u, v) \geq k$ and $\kappa(u, v) \geq k$, so feasibility forces $k \leq m - 1$. Each hyperedge (T, H) is counted by exactly the $|T| |H|$ ordered pairs (u, v) with $u \in T$ and $v \in H$, so summing

the per-pair bound over all pairs gives $\sum_{(T,H)} |T| |H| \leq (m-1) n(n-1)$. As $|T| + |H| = r$ with both parts at least one, $|T| |H| \geq 1 \cdot (r-1) = r-1$, whence $(r-1)|E| \leq (m-1)n(n-1)$. The forward family of [Proposition A.50](#) consists of general hyperedges, so it witnesses the same $\approx (m-1)n^2/(4(r-1))$ lower bound. $\square$

The two models are therefore trapped between the same pair of bounds, which differ by a factor of 4. That is enough to fix the *order* of both values and not enough to fix either leading constant, so it does not follow that the forward and general models agree asymptotically, only that any disagreement between them lives inside that factor. Determining $\lim_{n \rightarrow \infty} \text{ex}(n)/n^2$ for even one of the two, and then asking whether the two limits coincide, is the open question [Chapter 4](#) records.

It is worth saying which end of that gap is the likely culprit, because the two bounds are not equally crude and the analogy with the simple digraph case is informative. The upper bound charges each hyperedge to the ordered tail-head pairs it occupies and then allows all $n(n-1)$ ordered pairs to be used to capacity. The construction uses only the $\alpha(n-\alpha) \leq n^2/4$ pairs that run from the tail class to the head class, and leaves the pairs inside each class, and those running backwards, entirely idle. The factor of four is exactly that difference, n^2 against $n^2/4$, and it is the same difference that appears in the simple directed problem, where [Construction 1.13](#) also uses only a one-directional bipartite pattern. There, [Theorem A.24](#) now proves that the bipartite pattern is right to first order, that no arrangement recovers the idle pairs. If that phenomenon is a feature of directedness rather than of graphs specifically, then it is the upper bound here that should come down by a factor of four, not the construction that should climb, which suggests the following.

Conjecture A.53 (The directed hypergraph constant). *For fixed $r \geq 3$ and $m \geq 2$, the maximum number of hyperedges of a directed r -uniform hypergraph on n vertices with $\lambda^{\max} \leq m-1$ is $(1+o(1)) (m-1)n^2/(4(r-1))$, so the bipartite construction of [Proposition A.50](#) is asymptotically optimal.*

The proof of [Theorem A.24](#) does not transfer directly, and the place it breaks is worth recording for anyone who takes this up. There, two-step routes through distinct midpoints were automatically arc-disjoint. Here a single hyperedge carries $r-1$ heads at once, so two two-step routes with different midpoints can enter through the very same hyperedge, and the family of two-step routes is no longer automatically edge-disjoint. Recovering a packing of them, or replacing the count by one that tolerates the sharing, is the technical step this conjecture waits on.

A.8 The multigraph vertex problem under the other convention

[Remark 2.1](#) collapses parallel copies in vertex mode, which makes the two multigraph vertex variants restatements of their simple counterparts and is why they carry no theorem of their own. This section takes the other reading seriously, because under it the question is genuinely new,

and records what can be proved and what the machine finds. Throughout, q parallel copies of the edge uv count as q internally vertex-disjoint u - v routes, since a single edge is a path with no interior. Write $K_m^{\text{multi}}(n)$ for the largest total multiplicity of a multigraph on n vertices with $\kappa^{\max} \leq m - 1$ under this reading.

The first thing to notice is that the two kinds of route never interfere, which splits the measure cleanly in two.

Lemma A.54 (Splitting the vertex measure). *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0 . For any two vertices $u \neq v$, write $\pi(u, v)$ for the largest number of pairwise internally disjoint u - v paths of length at least two in G_0 . Then, under the convention above,*

$$\kappa_G(u, v) = \mu(u, v) + \pi(u, v).$$

Proof. A parallel copy of uv is a path with empty interior, so any two copies are internally disjoint from each other and from every longer route. A maximum family may therefore be assumed to contain all $\mu(u, v)$ copies, and what it can add to them is a family of pairwise internally disjoint routes of length at least two, whose interiors are disjoint sets of vertices. Since parallel copies are irrelevant to a route of length at least two, which is determined by its vertex sequence, the largest such family has size $\pi(u, v)$, computed in G_0 . $\square$

So feasibility says exactly that $\mu(u, v) \leq m - 1 - \pi(u, v)$ on every edge, and that $\pi(u, v) \leq m - 1$ on every pair. At $m = 2$ this is already decisive.

Theorem A.55 (The other convention at $m = 2$). $K_2^{\text{multi}}(n) = n - 1$, attained exactly by the spanning trees.

Proof. By Lemma A.54, $\kappa^{\max} \leq 1$ forces $\mu(u, v) + \pi(u, v) \leq 1$ on every edge. If G_0 contained a cycle, an edge uv of that cycle would have $\mu(u, v) \geq 1$ and $\pi(u, v) \geq 1$ from the rest of the cycle, a contradiction, so G_0 is a forest and every multiplicity is one. A forest has at most $n - 1$ edges, and a spanning tree attains it. $\square$

For $m \geq 3$ at least three constructions compete, and which one wins depends on how n compares with m , exactly as in every other variant of this thesis.

★ *The thickened tree.* Take a spanning tree and give every edge multiplicity $m - 1$. Tree edges have $\pi = 0$ and non-adjacent pairs have $\pi = 1$, so this is feasible for every $m \geq 2$, and it carries $(m - 1)(n - 1)$ edges, the multigraph edge value of Theorem 1.3.

★ *The thickened complete graph.* Give every edge of K_n the same multiplicity q . Here $\pi(u, v) = n - 2$, so feasibility asks $q \leq m + 1 - n$, and the count is $\binom{n}{2}(m + 1 - n)$, positive only when $n \leq m + 1$.

★ *The thickened theta.* Choose two *poles* and join each of the $n - 2$ remaining vertices to both of them at multiplicity $m - 2$, then join the poles to each other at multiplicity $m + 1 - n$ when that is positive. A pole and a middle vertex have $\pi = 1$, two middle vertices have $\pi = 2$, and the two poles have $\pi = n - 2$, so this is feasible exactly when $n \leq m + 1$, and it carries $2(n - 2)(m - 2) + \max(0, m + 1 - n)$ edges.

The theta family is the one that makes this problem genuinely different from the edge problem, and it was found by the search rather than by hand. Whenever it is feasible, which is to say for $n \leq m + 1$, it beats the thickened tree for every $m \geq 4$, and it is the extremiser in every such case the exhaustion has settled. Comparing the three counts, the tree leads once $n \geq m + 2$, where the other two die for lack of route budget.

Exhaustive search over all multigraphs with multiplicities in $\{0, \dots, m - 1\}$, run under this convention, gives the values in [Table A.1](#). The routine is `max_multigraph_vertex_standard(n, m)`, which is the same include-or-exclude branch and bound as everywhere else in [Chapter 2](#), with the checker's vertex mode switched to the second convention by one flag, `parallel_routes`, on `_split_capacity_matrix`. Every value below was computed twice, once by that routine and once by a separately written script sharing no code with it, and on the sixteen cells both have finished they agree exactly.

		$n = 2$	3	4	5	6	7
$m = 2$	value	1	2	3	4	5	6
	tree	1	2	3	4	5	6
$m = 3$	value	2	4	6	8	10	12
	tree	2	4	6	8	10	12
$m = 4$	value		6	9	12	15	
	tree		6	9	12	15	
	theta		6	9	12		
$m = 5$	value			14	19		
	tree			12	16		
	theta			14	19		
$m = 6$	value			19			
	tree			15			
	theta			19			

Table A.1. The multigraph vertex problem under the convention that parallel copies are distinct routes, computed exhaustively by `max_multigraph_vertex_standard`. Entries in bold are the cases where the value exceeds the multigraph edge value $(m - 1)(n - 1)$, and there the thickened theta is the extremiser. A blank cell is a size not computed, or one where the construction in that row does not apply. Cells with $n \leq m + 1$ are the regime in which the theta and complete constructions are feasible at all.

Two things stand out. First, the problem is *not* the edge problem in disguise: at $m = 5, n = 5$ the value is 19 against the edge value 16, so the separation axis genuinely buys something for multigraphs under this reading. Second, the small- n regime $n \leq m + 1$, where the complete

graph is feasible for the simple problems too, is tempting to read as the *only* place a cycle pays for itself, with the thickened tree optimal once $n \geq m + 2$. It is not: a single triangle grafted anywhere onto the thickened tree already beats it, for every $m \geq 5$ and every such n , which the next theorem makes precise.

By [Lemma A.54](#), a feasible multigraph on top of a fixed connected simple graph G_0 maximizes its total multiplicity by setting $\mu(u, v) = m - 1 - \pi(u, v)$ on every edge, giving total $(m - 1)|E(G_0)| - \sum_{uv \in E(G_0)} \pi(u, v)$. Writing $|E(G_0)| = n - 1 + \text{rank}(G_0)$, the thickened tree is beaten by any G_0 with

$$\sum_{uv \in E(G_0)} \pi(u, v) < (m - 1) \text{rank}(G_0).$$

The obvious guess is that cycles must always pay for themselves against this bound, each independent cycle costing $m - 1$ units of route capacity. The next theorem shows a single triangle already fails to pay, for $m \geq 5$, and packing many of them compounds the failure into a gap that grows with n .

Theorem A.56 (A bouquet of thickened cliques, and how it beats the thickened tree). *Fix $3 \leq r \leq m$ and set $q = m + 1 - r \geq 1$, the multiplicity of a thickened K_r that saturates the vertex ceiling on r vertices (the “thickened complete graph” construction above, restricted to a block of size r rather than all of n). Let $1 \leq k \leq \lfloor (n - 1)/(r - 1) \rfloor$. Take k copies of K_r that all share one common vertex and are otherwise disjoint, thicken every edge of every copy to multiplicity q , and attach any leftover vertices as a pendant path at multiplicity $m - 1$. This multigraph is feasible, $\kappa^{\max} \leq m - 1$, and its total multiplicity is exactly*

$$(m - 1)(n - 1) + k \cdot \text{gain}(r, m), \quad \text{gain}(r, m) := \frac{(r - 1)(r - 2)(m - 1 - r)}{2}.$$

The count therefore exceeds the thickened tree’s $(m - 1)(n - 1)$ exactly when $\text{gain}(r, m) > 0$, that is when $r \leq m - 2$, which is possible for some r in range precisely when $m \geq 5$.

Proof. Write G_0 for the underlying simple graph.

Feasibility. The shared vertex is a cut vertex of G_0 , and so is every vertex along the pendant path, since the blocks meet only there. So a path of length ≥ 2 between two vertices of the same block cannot leave that block and come back: to do so it would have to pass through the shared vertex twice. Hence for an edge uv inside a block, every route counted by $\pi(u, v)$ stays inside its K_r , where deleting uv leaves u, v joined by exactly $r - 2$ pairwise internally-disjoint two-step routes, one through each of the other block vertices, and by Menger no more, since those same $r - 2$ vertices are a cut of that size. So $\pi(u, v) = r - 2$ and, by [Lemma A.54](#), $\kappa_G(u, v) = q + (r - 2) = m - 1$ exactly, never over. For u, v in different blocks, or with either endpoint on the pendant path, a single vertex separates them, the shared vertex in the first case and a path vertex in the second, so $\kappa_G(u, v) \leq 1$.

Count. Since $q \geq 1$, every block edge is genuinely present, so G_0 is connected and $|E(G_0)| = n - 1 + \text{rank}(G_0)$. This is where the hypothesis $r \leq m$ earns its keep: at $r = m + 1$ the multiplicity q would drop to zero, the blocks would carry no edges at all, and G_0 would fall apart into isolated vertices. Only the k blocks contribute to $\text{rank}(G_0)$ or to $\sum \pi$, since the pendant edges have $\pi = 0$ and lie in no cycle. Each K_r has rank $(r - 1)(r - 2)/2$ and contributes $\binom{r}{2}(r - 2)$ to the sum, since each of its $\binom{r}{2}$ edges has $\pi = r - 2$. Substituting into $(m - 1)(n - 1 + \text{rank}(G_0)) - \sum \pi$,

$$(m - 1) \left[n - 1 + k \frac{(r-1)(r-2)}{2} \right] - k \frac{r(r-1)(r-2)}{2} = (m - 1)(n - 1) + k \text{gain}(r, m),$$

$$\text{using } (m - 1) \frac{(r-1)(r-2)}{2} - \frac{r(r-1)(r-2)}{2} = \frac{(r-1)(r-2)[(m-1)-r]}{2} = \text{gain}(r, m). \quad \square$$

At $r = 3$, $\text{gain}(3, m) = m - 4$: zero at $m = 4$, matching the exhaustive finding that $m \leq 4$ has no counterexample, and strictly positive for every $m \geq 5$. So the guess that the thickened tree is optimal once $n \geq m + 2$ is false for every $m \geq 5$: it held only within the reach of the exhaustive table, $m \leq 4$. The smallest witness is small enough to check by hand. Take $m = 5$ and $n = 7$, the first size the guess covers. Put a triangle on $\{u_1, u_2, u_3\}$ at multiplicity 3 and hang a path of four further vertices off u_1 at multiplicity 4. Two triangle vertices have three parallel copies plus one route through the third vertex, so $\kappa = 4$, and every other pair is separated by a single cut vertex or joined only by its own parallel copies, so nothing exceeds $4 = m - 1$. The total multiplicity is $3 \cdot 3 + 4 \cdot 4 = 25$, one more than the $(m - 1)(n - 1) = 24$ the guess allows. Packing k blocks instead of one multiplies the excess by k , and k may be taken as large as $\lfloor (n - 1)/(r - 1) \rfloor$, so the gap over the thickened tree grows linearly in n .

Sharing a vertex is what buys that count. Disjoint blocks joined by bridges would fit only $\lfloor n/r \rfloor$ of them, since each bridge spends a vertex on nothing but the join, and the difference is real rather than cosmetic: at $m = 10$, $r = 6$ and $n = 11$ the shared-vertex form carries 150 against the bridged form's 120.

Optimising r widens the gap further. Treated as continuous, the gain per vertex $\text{gain}(r, m)/(r - 1)$ peaks near $r \sim (m + 1)/2$, where it is $\Theta(m^2)$, so $K_m^{\text{multi}}(n)/n \rightarrow (m - 1) + \Theta(m^2)$ as $n \rightarrow \infty$ at fixed large m , a different order in m from the guess above.

Even this is not the truth. At $m = 5$, $n = 7$ the construction above gives 27, and an exhaustive branch and bound, stopped by its time limit before it could finish, had already found a feasible multigraph with 28. So the value of $K_m^{\text{multi}}(n)$ is genuinely open rather than reduced to a single inequality about cycle rank, and what these theorems establish is a lower bound of the right order in both n and m , not the extremal number.

A.9 The random threshold

The probabilistic ingredients are two Chernoff bounds, quoted in the form we use them.

Primer: what a Chernoff bound says

Both proofs below need to say that a sum of many independent yes-or-no trials almost never lands far from its average. That is the content of a *Chernoff bound*, and the intuition is worth stating even though we only quote the result. Let $X = X_1 + \cdots + X_N$ count the successes among independent trials, with mean $\mu = \mathbb{E}[X]$. A Chernoff bound says the chance that X overshoots μ by a factor $1 + \delta$, or undershoots it by a factor $1 - \delta$, decays *exponentially* in μ , not merely polynomially as Chebyshev's inequality would give. The mechanism is a single trick. Rather than bound $\Pr[X \geq a]$ directly, bound $\Pr[e^{tX} \geq e^{ta}]$ by Markov's inequality for a parameter $t > 0$, use independence to factor $\mathbb{E}[e^{tX}] = \prod_i \mathbb{E}[e^{tX_i}]$ into a product over the trials, and then choose t to make the resulting bound as small as possible. The exponential in the answer is exactly the e^{tX} that was fed through Markov. We quote the two tails in the shape the threshold proof consumes.

Theorem A.57 (Chernoff bounds, see [21, Cor. 4.4, Thm. 4.4]). *Let X be a sum of independent $\{0, 1\}$ -valued random variables with mean μ . For $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \exp(-\frac{\min(\delta, \delta^2)\mu}{4})$, and for $\delta \in (0, 1)$, $\Pr[X \leq (1 - \delta)\mu] \leq \exp(-\frac{\delta^2\mu}{2})$.*

Corollary A.58. *Let $X \sim \text{Binomial}(n - 1, p)$ with $p = cm/n$ for a constant $c \in (0, 1)$. Then for all sufficiently large m , $\Pr[X \geq m] \leq e^{-\alpha m}$ with $\alpha = \alpha(c) > 0$.*

Proof. The mean is $\mu = (n - 1)p \leq cm$, so $m \geq (1 + \delta)\mu$ with $\delta \geq \frac{1-c}{c} > 0$, and Theorem A.57 applies with $\mu \geq cm/2$ for $n \geq 2$, giving $\alpha = \min(\frac{1-c}{c}, (\frac{1-c}{c})^2) \frac{c}{8}$. $\square$

Proof of Theorem 1.16. Recall the hypotheses: $m = m(n) \geq 2$ with $m/\ln n \rightarrow \infty$ and $m = o(n)$, and $p = cm/n$ for a constant $c > 0$.

Case $c > 1$. The edge count $|E|$ of $G(n, p)$ is binomial with mean $\mu = \binom{n}{2}p = \frac{cm(n-1)}{2}$. With $\delta = 1 - \frac{1}{c}$ the lower-tail bound of Theorem A.57 gives

$$\Pr\left[|E| \leq \frac{m(n-1)}{2}\right] \leq \exp\left(-\frac{(c-1)^2 m(n-1)}{4c}\right) \rightarrow 0,$$

so with high probability $|E| > \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and the contrapositive of Mader's theorem (Theorem 1.2) forces a pair with $\lambda \geq m$. Hence $\Pr[\lambda^{\max} \geq m] \rightarrow 1$.

Case $c < 1$. Each degree is $\text{Binomial}(n - 1, p)$, so by Corollary A.58 and a union bound over the n vertices,

$$\Pr[\Delta(G) \geq m] \leq n e^{-\alpha m} = e^{\ln n - \alpha m} \rightarrow 0,$$

since $m/\ln n \rightarrow \infty$. By the degree bound (Lemma A.3) $\lambda^{\max} \leq \Delta(G)$, so $\Pr[\lambda^{\max} \geq m] \rightarrow 0$. $\square$

Remark A.59 (The vertex and directed analogues). Below the threshold, $\kappa^{\max} \leq \lambda^{\max}$ gives the vertex statement for free, and in the random digraph $D(n, p)$, where each ordered pair's arc is tossed independently with probability p , the out-degree version of the same union bound

applies. Above the threshold, the minimum degree exceeds m with high probability by the lower-tail bound. Mader's theorem alone does not carry this to the vertex problem, since $\kappa^{\max} \leq \lambda^{\max}$ runs the wrong way to deduce $\kappa^{\max} \geq m$ from the edge bound. What supplies it instead are the theorems of Bollobás [22, Ch. 7] for $G(n, p)$, in hitting-time form due to Bollobás and Thomason [23], and its directed analogue [24], which state that in this regime the global connectivities equal the minimum degree (respectively the minimum semi-degree, the smaller of a vertex's in- and out-degree) with high probability, so $\kappa^{\max}$ and the directed $\lambda^{\max}$ also cross m at $p^* = m/n$. The growth hypothesis $m/\ln n \rightarrow \infty$ is what lets the union bound beat the factor n . For fixed m , the bound is vacuous and finer tools (Poisson approximation of low-degree vertex counts) would be needed.

Remark A.60 (How far the threshold reaches, and where it stops). [Theorem 1.16](#) is a statement about $G(n, p)$, and [Remark A.59](#) carries it to the vertex separation and to the directed model $D(n, p)$. It does not reach the hypergraph models, and the reason is a change of scale rather than a gap in the argument. The threshold sits where a vertex's expected degree reaches m , since below that the degree bound caps every local connectivity and above it the extremal bound forces a high pair. In $G(n, p)$ a vertex has expected degree $p(n-1)$, so the balance point is $p^* = m/n$ as stated. In the binomial r -uniform hypergraph, where each of the $\binom{n}{r}$ possible hyperedges is included independently with probability p , a vertex lies in an expected $p\binom{n-1}{r-1}$ hyperedges, so the balance point is

$$p^* = \frac{m}{\binom{n-1}{r-1}} = \Theta\left(\frac{m}{n^{r-1}}\right),$$

which for $r \geq 3$ is smaller than m/n by a factor of order n^{r-2} . Reading m/n as the hypergraph threshold would therefore place the transition far above where it happens: at $r = 3$ and $n = 8$, for instance, m/n is $3/8$ while $m/\binom{7}{2}$ is $3/21$, and by the former density the sampled hypergraph is long past the transition. [Figure B.8](#) accordingly sweeps each model in units of its own p^* rather than a shared p . A second limit of scope is worth repeating here: the theorem assumes $m/\ln n \rightarrow \infty$, so any picture at a fixed small m illustrates the shape of the transition without being an instance of the theorem.

A.10 Computational transcripts

The cut-counting optimisation formulation that certifies [Theorem 2.2](#) is given in [Chapter 2](#). Here are the solver transcripts that back it, together with the exhaustive-search output that settles the directed base cases. The complete commented source accompanies the document as a single self-contained file, `erdos915_unified.py` in the `program/` directory. Its core runs on `numpy` and `scipy` alone, the certifier adds `pulp`, and `matplotlib` and `networkx` are optional and used only for the figures. An optional compiled helper accelerates selected small hot paths but is not needed for correctness.

```
1 | Cut-MILP certification of the directed multigraph problem
2 | M*(n) is the scaled optimum; L_m^dir(n) = (m-1) M*(n).
```

```

3
4 n=3: status=OPTIMAL, M*=4.000, target=2(n-1)=4, proof=True, time=0.0s, L_3^
    ↳ dir(n)=8
5 n=4: status=OPTIMAL, M*=6.000, target=2(n-1)=6, proof=True, time=0.3s, L_3^
    ↳ dir(n)=12
6 n=5: status=OPTIMAL, M*=8.000, target=2(n-1)=8, proof=True, time=5.9s, L_3^
    ↳ dir(n)=16
7 n=6: status=OPTIMAL, M*=10.000, target=2(n-1)=10, proof=True, time=1259.9s,
    ↳ L_3^dir(n)=20

```

Listing A.1. Solver output: $M^*(n) = 2(n - 1)$, optimal with zero gap, for $n = 3, 4, 5, 6$.

The cut-counting optimisation closes the gap up to $n = 5$ inside a short budget and reaches $n = 6$ in about twenty minutes. Past that it stalls, because a solver handles the yes-or-no cut labels by case-splitting, trying a label both ways and solving each half, and the number of cases it must open grows too fast at $n = 7$. The base cases of the directed induction ([Theorem 1.10](#)) run to $n = 7$, and those are settled instead by the pruned exhaustive search over all simple digraphs with $\lambda^{\max} \leq 1$. That search is exact and complete at these sizes, and it agrees with $\ell_2^{\text{dir}}(n) = M(n)$ throughout.

```

1 Exhaustive search over all simple digraphs with lambda^max <= 1 (directed,
    ↳ m=2).
2 Branch and bound; reported max arc count = ell_2^dir(n) = max(2(n-1), floor
    ↳ (n^2/4)).
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s
5 n=3: max= 4, formula= 4, proved (complete), nodes=22, 0.0s
6 n=4: max= 6, formula= 6, proved (complete), nodes=537, 0.0s
7 n=5: max= 8, formula= 8, proved (complete), nodes=17823, 0.1s
8 n=6: max=10, formula=10, proved (complete), nodes=788101, 9.1s
9 n=7: max=12, formula=12, proved (complete), nodes=45122129, 718.1s
10
11 The cut-MILP certifier of the multigraph problem closes n <= 6 (n = 6 in
    ↳ about
12 21 minutes); the directed m = 2 base cases through n = 7 that the induction
13 requires are settled by this exhaustive search instead.

```

Listing A.2. Exhaustive-search output for the base cases $2 \leq n \leq 7$: the maximum arc count equals $M(n) = 2, 4, 6, 8, 10, 12$, proved complete at each n .

The vertex problem needs its own base cases, since [Remark A.16](#) shows they cannot be inherited from the arc ones, and the same driver supplies them with only the inner feasibility test swapped. The two separations agree at every size, which is what closes [Theorem 1.11](#).

```

1 Exhaustive search over all simple digraphs with kappa^max <= 1 (directed, m
    ↳ =2).
2 Branch and bound; reported max arc count = k_2^dir(n) = max(2(n-1), floor(n
    ↳ ^2/4)).
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s

```

```

5  n=3:  max= 4,  formula= 4,  proved (complete),  nodes=22,          0.0s
6  n=4:  max= 6,  formula= 6,  proved (complete),  nodes=537,        0.0s
7  n=5:  max= 8,  formula= 8,  proved (complete),  nodes=17823,       1.1s
8  n=6:  max=10,  formula=10,  proved (complete),  nodes=788101,      77.0s
9  n=7:  max=12,  formula=12,  proved (complete),  nodes=45122129,   3124.8s
10
11 Same driver, same prunes, same sizes as the arc run above. Only the
12 inner test changes, from arc-disjoint to internally vertex-disjoint.
13 The two separations agree at every base case, which is what the
14 vertex induction needs and what Whitney's inequality cannot supply.

```

Listing A.3. The same search run with the internally vertex-disjoint feasibility test, settling the base cases of [Theorem 1.11](#): the maximum arc count is again 2, 4, 6, 8, 10, 12, proved complete at each n .

A.11 How the program checks itself

The invariants are checked by the self-test at the foot of `erdos915_unified.py`, run simply with `python erdos915_unified.py`. Every check passes, grouped here by what they cover.

- ★ *Base values.* The checker against the proven $m = 2$ values 2, 4, 6, 8, the first three vertex base cases of [Theorem 1.11](#), and the fast vertex-flow test against the exact checker it stands in for.
- ★ *Sampled inequalities.* The counting step and both bounds of [Proposition A.22](#) and [Theorem A.24](#) on sampled feasible digraphs, including the high-degree case of the latter's proof, and Whitney's inequality on every sample.
- ★ *Constructions.* Two values of the alternative-convention multigraph vertex problem of [Section A.8](#), including the one that separates it from the edge problem, the named constructions against their predicted sizes and connectivities, and the Gomory–Hu shortcut against brute-force max-flow.
- ★ *The checker and driver.* The Berge connectivities of the hypergraph checker, the Monte Carlo appearance threshold, the cut-counting optima for small n , the search reaching the certified optimum, and the driver returning the proven value or a feasible witness in each mode.

A single companion script, `make_figures.py`, regenerates every figure in the thesis from the same file, with the mapping listed in `program/README.md`.

Appendix B

Supplementary Figures

This appendix collects the supplementary figures that the program produces but that the chapters reference rather than reproduce in full. Each one is named where it belongs in the body, and the file name printed in its caption matches the output of `make_figures.py` and the figure table in `program/README.md`. Figures already shown in the running text are not repeated here.

B.1 A gallery of extremal graphs

The extremal graphs are not abstractions. [Figure B.1](#) draws nine of the named families the body argues about, each at a deliberately large representative size rather than at the smallest case, so the structure is visible. These are the special, structured extremisers, not the trivial small trees: the bidirected star (the simple hub, arcs both ways to every leaf), the directed double star (its multigraph analogue, hub arcs at full multiplicity $m - 1$, which ties $2 \cdot B_{3,4}$ at the crossover), the doubled bipartite wall $2 \cdot B_{3,4}$ itself, the dense bidirected complete graph, the multigraph star (the *undirected* multigraph analogue, hub edges at full multiplicity $m - 1$), and the star hypertrees, the last drawn as metro maps in which each hyperedge is one coloured line through its members. The program builds every one of these directly and, at the small sizes the enumeration reaches, also proves them optimal rather than merely feasible, so the gallery is the analysis made concrete at a size a reader can take in.

B.2 Search traces across the variants

[Figure 3.2](#) in [Chapter 3](#) follows one cooling run in detail. The five runs below repeat that experiment across the family. The first four cover every combination of the two matrix models, simple and multigraph, with the two directions, and the fifth changes the separation, so that the behaviour can be seen not to be special to one case. Each plots every visited graph by its binding connectivity (horizontal) against its size (vertical), colouring each point by the step at which it was visited, from the hot early exploration in warm tones to the converged tail in blue. In every one the walk strays past the feasibility boundary while hot and is driven back onto the densest feasible graph as the temperature falls, exactly as in the body figure. The seeds are fixed, so each picture reproduces verbatim.

The last of the five is the one worth comparing against the others. It runs the same directed multigraph problem under the *vertex* separation rather than the arc one, and the stricter test is visible in the trace: the walk spends a markedly larger share of its steps on the infeasible side of

Extremal graphs of the named families, drawn large

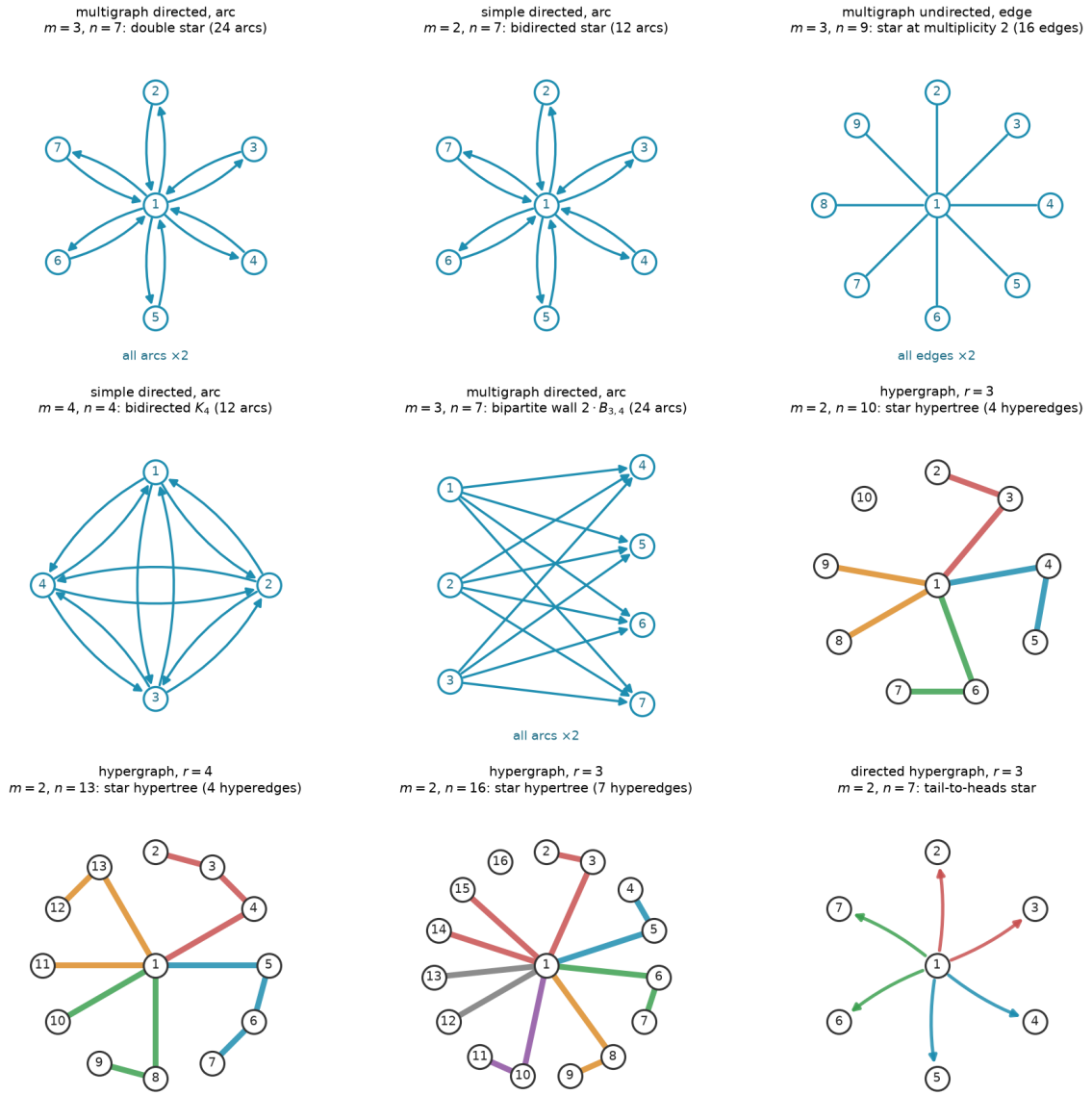


Figure B.1. Extremal graphs of the named families, drawn at a large representative size. *Top and middle rows:* matrix variants, with a central hub where the family has one. Every family here carries one uniform multiplicity, stated once per panel as “all arcs $\times k$ ” rather than repeated on each arc or drawn as parallel curves, and directed arcs carry arrowheads. *Hypergraph panels:* each hyperedge is a metro line through its member vertices, in its own colour, with the hub at the centre, and the directed hypergraph fans coloured arrows from each tail to its heads. Every graph shown is feasible at the stated m . The matrix and undirected-hypergraph families are extremal for their variant, the two directed stars drawn at the crossover $n = 7$, the largest size at which the linear star still ties the one-directional wall before the wall overtakes it. The bipartite wall $2 \cdot B_{3,4}$ is drawn at that same size to show the other side of the tie. It also carries 24 arcs, matching the double star’s count, and it is one of the three families the $n = 7$ classification of [Remark A.39](#) identifies, together with $2 \cdot B_{4,3}$ and the doubled bidirected path P_7 (the double star itself is not among them, since its hub degree exceeds that classification’s degree cap). The final panel is the directed-hypergraph analogue, a tail-to-heads star, shown for the one variant whose exact value is still open ([Proposition A.50](#)). File `extremal_graphs_gallery.png`.

the boundary before the cooling pulls it back, which is what a harder constraint looks like from inside the search.

Search trace: simple undirected, $n = 7$, $m = 3$

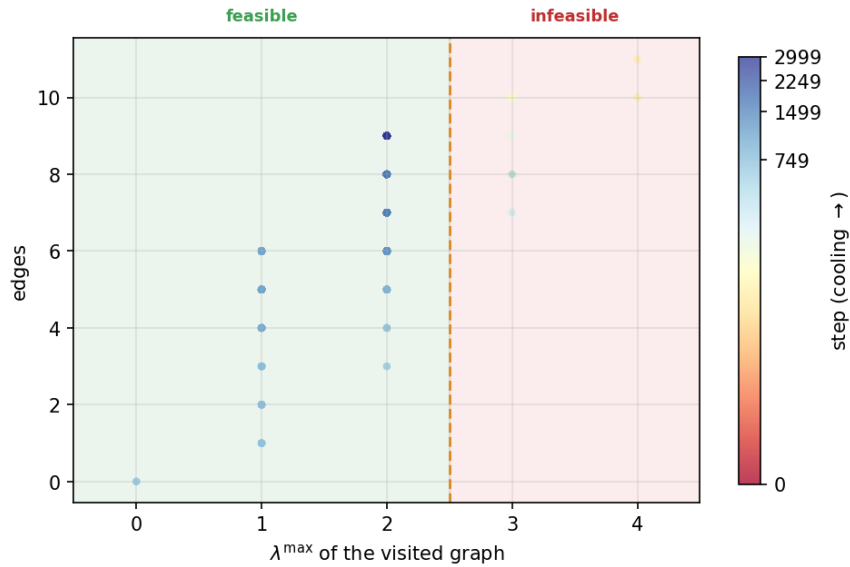


Figure B.2. Search trace for the simple undirected edge problem at $n = 7$, $m = 3$. File `trace_simple_undirected_n7_m3.png`.

B.3 The variant landscape at $m = 6$ and in three dimensions

This appendix shows the per-graph connectivity distribution at the fixed forbidden threshold $m = 6$, alongside a three-dimensional view of the appearance threshold (Figure B.8). They confirm that raising the forbidden threshold relocates the feasibility boundary without changing the qualitative picture. The pooled pair-connectivity and edge-count grids of Chapter 4 (Figures 4.4 and 4.5) already split each variant at its own mid-range connectivity, so they carry no separate $m = 6$ companion.

Search trace: multi directed, $n = 5, m = 3$

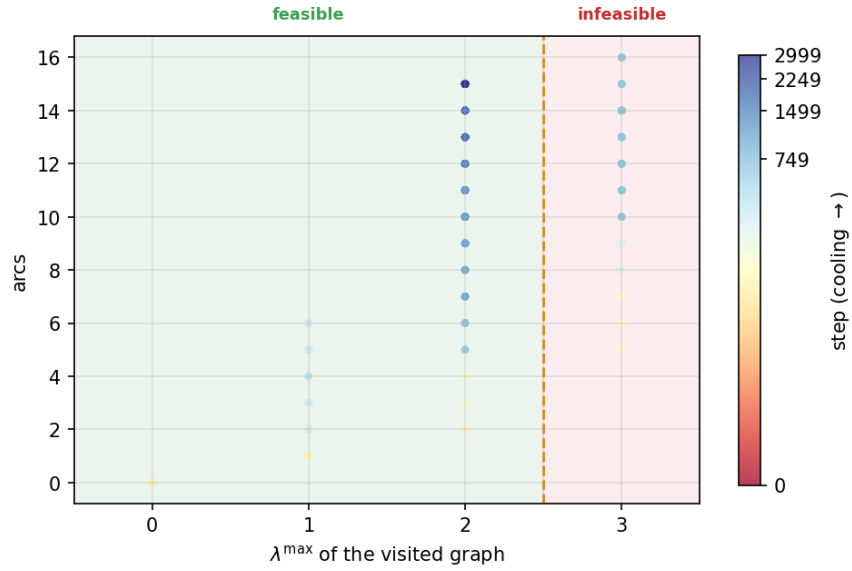


Figure B.3. Search trace for the multigraph directed arc problem at $n = 5, m = 3$. File `trace_multi_directed_n5_m3.png`.

Search trace: simple directed, $n = 5, m = 3$

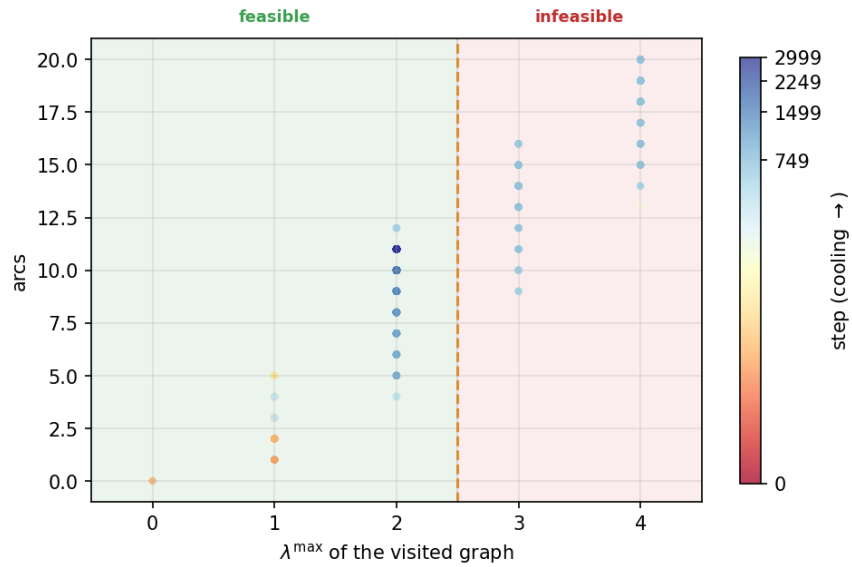


Figure B.4. Search trace for the simple directed arc problem at $n = 5, m = 3$. File `trace_simple_directed_n5_m3.png`.

Search trace: multi undirected, $n = 5, m = 3$

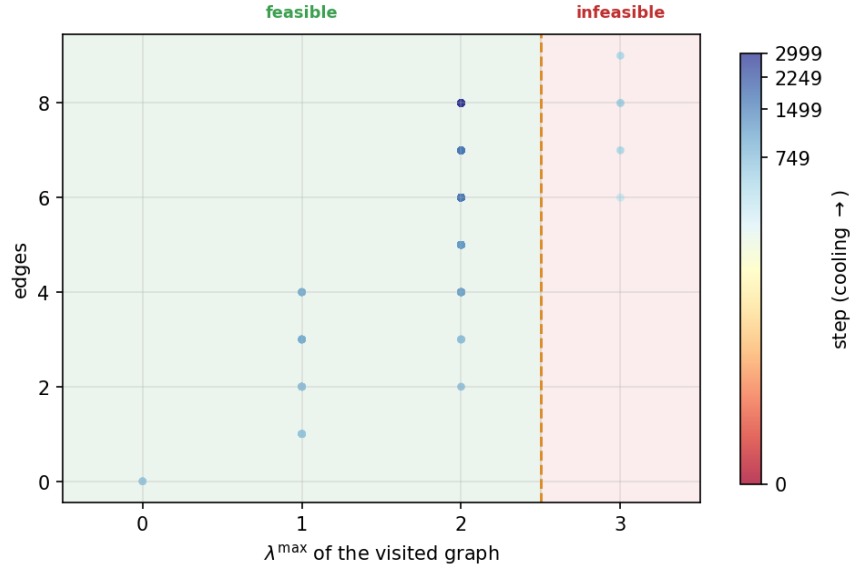


Figure B.5. Search trace for the multigraph undirected edge problem at $n = 5, m = 3$. File `trace_multi_undirected_n5_m3.png`.

Search trace: multi directed vertex-sep, $n = 4, m = 3$

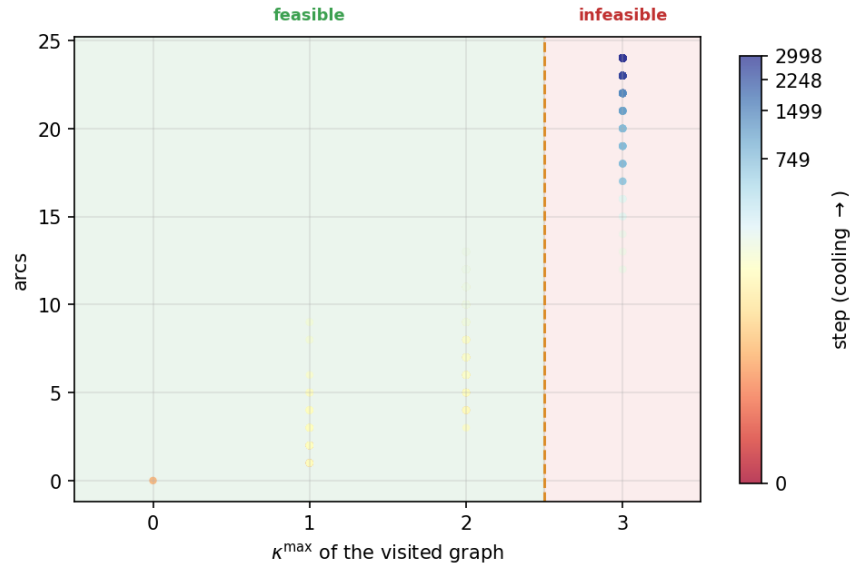


Figure B.6. Search trace for the multigraph directed problem at $n = 4, m = 3$ under the *vertex* separation, so the horizontal axis is $\kappa^{\max}$ rather than $\lambda^{\max}$. Compared with the arc runs above, the walk lingers much longer in the infeasible region before the cooling drives it back, the stricter separation seen from inside the search. File `trace_multi_directed_n4_m3_vertex.png`.

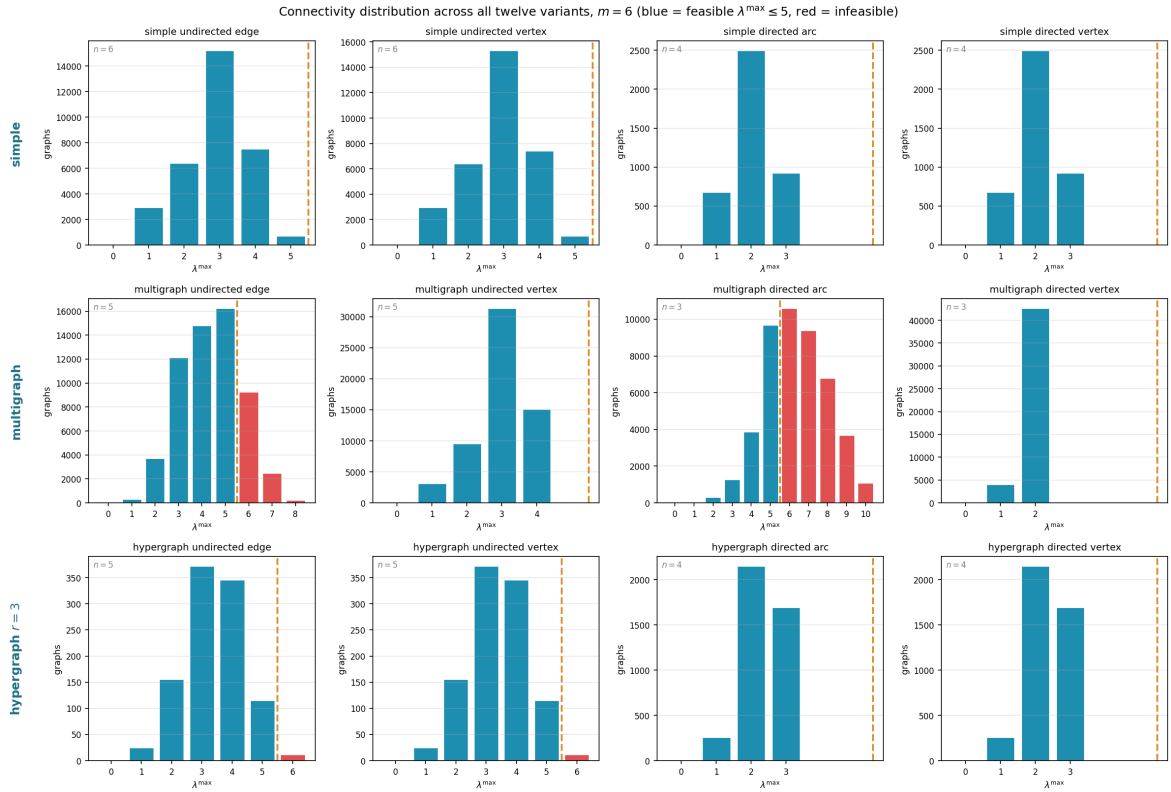


Figure B.7. How the full small- n enumeration distributes by binding connectivity $\lambda^{\max}$ across all twelve variants at the fixed forbidden threshold $m = 6$. Blue bars are feasible graphs ($\lambda^{\max} \leq 5$), red bars infeasible ones, and the dashed line marks the feasibility boundary. At this higher threshold most of the small graphs the enumeration reaches are feasible, so the blue mass dominates: raising m moves the boundary out past the bulk of the distribution. File `conn_dist_m6.png`.

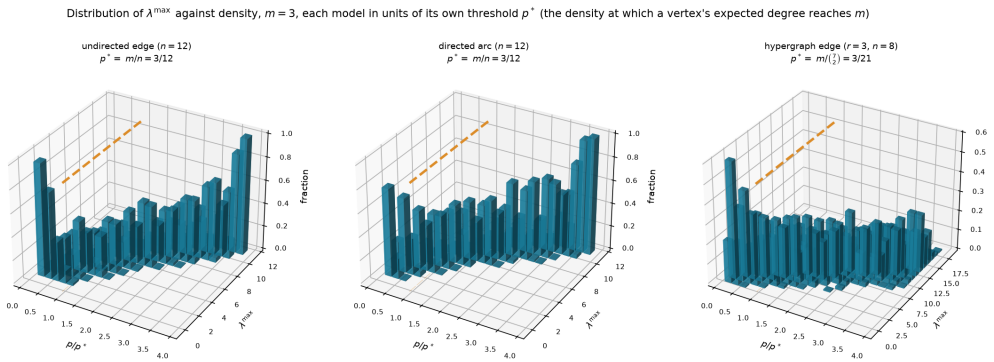


Figure B.8. The appearance threshold as a three-dimensional bar chart, for three representative variants (simple undirected edge and simple directed arc at $n = 12$, three-uniform hypergraph edge at $n = 8$), all at $m = 3$. Each panel plots the fraction of samples (height) reaching each binding connectivity $\lambda^{\max}$ (depth) against density (width), and the mass migrates from low to high connectivity as the density crosses the dashed line. The three models do *not* share a density scale, so each panel is swept in units of its *own* threshold p^* , the density at which a vertex's expected degree reaches m : that is m/n for the two graph models and $m/\binom{n-1}{r-1}$ for the hypergraph, smaller by a factor of order n^{r-2} (Remark A.60). Each panel title gives its own p^* . Only the two graph panels fall under Theorem 1.16, and even they sit outside its asymptotic regime $m/\ln n \rightarrow \infty$ at this fixed $m = 3$, so the figure illustrates the shape of the transition rather than instantiating the theorem. File `threshold_3d.png`.

Appendix C

The Complete Source Code

Every number in this thesis that was not proved by hand came out of the program printed here, and this appendix prints all of it, with nothing summarised away. The point is that a reader who doubts a computed value can find the exact lines that produced it, rather than taking a description of them on trust.

There is no need to read this front to back. It is laid out so that you can jump to whichever part you want to check. The listing is broken into the same four movements as the thesis itself, because the program is organised that way on purpose: it defines the objects, then measures them, then proves things about them, then searches for examples and draws the results. Each part below opens with a short account of what it does and which chapter it belongs to, and every listing carries line numbers so that a reference like “line 1412” has one unambiguous meaning. Where a line of code was too long for the page it is wrapped, and the continuation is marked with a $\hookrightarrow$ arrow rather than being silently cut.

Four files make up the whole of it. The main program, `erdos915_unified.py`, is the one the chapters discuss throughout and the only one that computes a reported result. `make_figures.py` redraws every picture in the thesis by calling into that program, so a figure can never drift away from the numbers behind it. `_erdos_fast.c` is a small optional accelerator: it speeds up two inner loops and changes no answer, and the program runs correctly without it. The `tests/` directory holds the suite that checks all of the above.

C.1 The main program

C.1.1 How the file opens

The program begins by saying what it is for, then imports what it needs and fixes a few constants. Two decisions made here shape everything after. The first is that the heavy scientific libraries are optional: only `numpy` and `scipy` are required, and `matplotlib`, `networkx` and `pulp` are each wrapped so that the file still loads without them, with the features that need them disabled rather than the whole program failing. The second is that every import is hoisted to the top, so there is one place to look to see what the program depends on.

```
1 """
2 erdos915_unified: the complete unified program for Erdos Problem 915, in one file.
3
4 This single module is the whole program, kept in one namespace so it can be read
```

```

5 top to bottom as a continuous story. It is meant to be read, not just run: every
6 non-trivial step carries an inline comment explaining *why* it is there, not
7 merely what it does. Running it ('python erdos915_unified.py') executes the
8 invariant self-check at the foot of the file; 'make_figures.py' next to it
9 regenerates the thesis figures.
10
11 -----
12 The problem, in one breath
13 -----
14 Erdos Problem 915 asks, in its many guises, for the densest graph on 'n'
15 vertices in which no pair of vertices has 'm' independent routes between them.
16 "Independent" can mean edge-disjoint or internally vertex-disjoint; the object
17 can be a simple graph, a multigraph, a digraph, or a hypergraph; and forbidding
18 'm' routes is exactly the constraint that the largest *local* connectivity
19 stays at or below 'm - 1'. Twelve concrete variants fall out of these
20 choices, and this program treats all of them with one representation and one
21 search.
22
23 -----
24 The epistemic spine: measure / prove / discover
25 -----
26 The whole architecture is organised around three different *kinds of claim* a
27 program can make about extremal values, and it keeps them scrupulously apart so
28 they can never silently masquerade as one another:
29
30 * MEASURE. The checker computes exact local connectivities by max-flow
31 (Menger's theorem). Whatever it reports is a fact about a concrete graph,
32 not an estimate. This is the trusted core; everything else is checked
33 against it.
34
35 * PROVE. The cut-counting method proves *upper* bounds for a fixed number of
36 vertices: it shows that no graph on 'n' vertices can be denser, with a
37 zero optimality gap, so an optimal solution is a genuine proof.
38
39 * DISCOVER. The random search finds concrete dense graphs, hence
40 *lower* bounds. A construction it returns is a witness that the extremal
41 value is at least so large; it never proves that no denser graph exists.
42
43 The closed-form 'bounds' library sits to the side as the theory's predictions
44 (proved, cited, or merely conjectured, each labelled honestly), so that "what
45 the program found" can always be held against "what the theory predicts" without
46 the two drifting apart. The Monte Carlo module sits in the DISCOVER/observe
47 corner: a sampled estimate is an estimate, used to make a proved threshold
48 *visible*, never to prove it.
49
50 -----
51 Chapter map (sections are grouped below by the thesis chapter they support)
52 -----
53 CHAPTER 1 The problem and its base cases
54     VARIANT, BOUNDS, HYPERGRAPH, CONSTRUCTIONS
55 CHAPTER 2 Proving bounds by machine
56     CHECKER, GOMORY-HU, PROVE (the brute-force and exhaustive provers
57     live inside the SOLVE driver, in Chapter 3)
58 CHAPTER 3 Discovering bounds by search
59     SENSITIVITY, SEARCH, SOLVE (the one driver: prove or discover)
60 CHAPTER 4 Synthesis and results
61     MONTE CARLO, FIGURES, __main__ (the invariant self-check suite)
62

```

```

63 A bold CHAPTER divider precedes each group, so as you scroll you always know
64 which chapter's machinery you are reading. Each section keeps its own banner.
65 """
66
67 # =====
68 # Hoisted imports
69 # -----
70 # All external dependencies, gathered into a single block.  “from __future__
71 import annotations” must be (and is) the very first statement so that
72 # string-form type hints work uniformly. Everything in this program lives in one
73 # namespace, so there are no intra-module imports to follow.
74 # =====
75 from __future__ import annotations
76
77 import copy
78 import json
79 import math
80 import pickle
81 import random
82 import shutil
83 import statistics
84 import subprocess
85 import time
86 import warnings
87 from collections import deque
88 from collections.abc import Iterable
89 from dataclasses import dataclass, field
90 from itertools import combinations, combinations_with_replacement, permutations,
    ↪ product
91 from pathlib import Path
92 from typing import Callable, Iterator
93
94 import concurrent.futures
95
96 import numpy as np
97 from scipy.sparse import csr_matrix as _csr
98 from scipy.sparse.csgraph import maximum_flow as _csgraph_maxflow
99
100 # networkx is OPTIONAL. Every connectivity measure the thesis reports is computed
101 # through scipy's integer max-flow on a capacity matrix, so the core checker,
102 # search, provers, and enumeration run on numpy + scipy alone. networkx is needed
103 # only for the Gomory-Hu tree (one figure/analysis helper), which raises a clear
104 # message if called without it installed.
105 try:
106     import networkx as nx
107     NETWORKX_AVAILABLE = True
108 except ImportError:
109     nx = None
110     NETWORKX_AVAILABLE = False
111
112
113 def _require_networkx(feature: str):
114     """Raise a clear error when an optional networkx-only feature is used headless
    ↪ """
115     if not NETWORKX_AVAILABLE:
116         raise ImportError(
117             f"{feature} needs networkx (pip install networkx). The core
    ↪ connectivity "

```



```

118         f"measures do not: they run on numpy + scipy alone."
119     )
120
121     # pulp is OPTIONAL. It backs the MILP certifier (prove_directed_multigraph and
122     # prove_integral_arc_bound): one solver-agnostic cut-counting model that CBC (pulp
123     # ↪ 's
124     # bundled solver) runs by default and Gurobi runs by a one-line switch. The model
125     # ↪ ,
126     # checker, search, enumeration, and sampling do not need it.
127     try:
128         import pulp
129         # pulp 3.x emits migration notices for its 4.0 API (LpVariable construction,
130         # the PULP_CBC_CMD name). We use the stable 3.x API on purpose and pin it in
131         # requirements.txt, so silence that forward-looking noise.
132         import warnings as _warnings
133         _warnings.filterwarnings("ignore", message=r".*PuLP 4\.0.*",
134                                 category=DeprecationWarning)
135         PULP_AVAILABLE = True
136     except ImportError:
137         pulp = None
138         PULP_AVAILABLE = False
139
140     def _require_pulp():
141         """Raise a clear error when the MILP certifier is used without pulp installed.
142         ↪ """
143         if not PULP_AVAILABLE:
144             raise ImportError(
145                 "the MILP certifier needs pulp (pip install pulp). The checker, search
146                 ↪ , "
147                 "and enumeration do not: they run on numpy + scipy alone."
148             )
149
150     import ctypes as _ct
151     _C = None
152     _so_path = Path(__file__).with_name("_erdos_fast.so")
153     if _so_path.exists():
154         _C = _ct.CDLL(str(_so_path))
155         _C.tiny_maxflow.restype = _ct.c_int
156         _C.tiny_maxflow.argtypes = [
157             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.c_int, _ct.c_int, _ct.c_int,
158         ]
159         _C.max_connectivity_exceeds.restype = _ct.c_int
160         _C.max_connectivity_exceeds.argtypes = [
161             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.c_int, _ct.c_int,
162         ]
163         _C.canonical_form_min.restype = None
164         _C.canonical_form_min.argtypes = [
165             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.POINTER(_ct.c_int),
166         ]
167     C_EXTENSION_LOADED = _C is not None
168
169     # matplotlib is OPTIONAL: it is needed only to render the figures (the plot_*
170     # functions in the FIGURES section). It needs its backend chosen before pyplot is
171     # imported, so that the figure code runs head-less (writes PNG files, never opens
172     # ↪ a
173     # window). When it is absent the model, checker, provers, search, and enumeration
174     # all still run; only figure generation is unavailable.

```

```

171 try:
172     import matplotlib
173     matplotlib.use("Agg")
174     import matplotlib.pyplot as plt # noqa: E402 (after backend selection)
175     from matplotlib.ticker import MaxNLocator # noqa: E402 (integer-only axis
        ↪ ticks)
176     from mpl_toolkits.mplot3d import Axes3D # noqa: F401 (registers the '3d'
        ↪ projection)
177     MATPLOTLIB_AVAILABLE = True
178 except ImportError:
179     plt = None
180     MaxNLocator = None
181     MATPLOTLIB_AVAILABLE = False

```

C.1.2 The objects: graphs, hypergraphs, and the values we already know

This part corresponds to [Chapter 1](#). It defines what a graph is for the purposes of this thesis, which is a single square table of numbers: entry (u, v) records how many parallel connections run from u to v . That one table covers all eight of the matrix variants at once, because a Variant record says whether the graph is directed and whether it is simple, and those two switches are the only difference between them. Hypergraphs need a different shape, so they are stored as a list of the sets of vertices each hyperedge joins.

After the objects come the closed-form values the thesis proves or conjectures, each written as a small function whose docstring records whether it is proved, conjectured, or a bound only. Then come the named constructions, the specific graphs that achieve those values, so that a claimed bound can always be met with a concrete example rather than an existence argument.

```

184 #####
185 ##
186 ## CHAPTER 1 -- THE PROBLEM AND ITS BASE CASES
187 ## Objects, variants, and the closed-form values we already know.
188 ##
189 #####
190
191 # --- VARIANT: mu[u,v] = multiplicity from u to v; simple = {0,1}; undirected =
    ↪ symmetric ---
192
193
194 @dataclass(frozen=True)
195 class Variant:
196     """Which kind of graph object we are working with.
197
198     Attributes:
199         directed: ‘True’ for digraphs (arcs), ‘False’ for graphs (edges).
200         simple: ‘True’ caps every multiplicity at one. ‘False’ allows
201             arbitrarily many parallel edges or arcs (a multigraph).
202         name: a short human-readable label, used in figures and reports.
203     """
204
205     directed: bool
206     simple: bool
207     # loops: bool
208     name: str = ""

```

```

209
210     def describe(self) -> str:
211         """Return a readable description such as "simple directed"."""
212         # Compose the label from the two boolean axes; used in plot titles.
213         words = ["simple" if self.simple else "multi",
214                 "directed" if self.directed else "undirected"]
215         return " ".join(words)
216
217
218     # The four matrix-representable variants, named for convenience. Random graphs
219     # reuse these (a sampled simple undirected graph is SIMPLE_UNDIRECTED).
220     SIMPLE_UNDIRECTED = Variant(directed=False, simple=True, name="simple undirected")
221     MULTI_UNDIRECTED = Variant(directed=False, simple=False, name="multi undirected")
222     SIMPLE_DIRECTED = Variant(directed=True, simple=True, name="simple directed")
223     MULTI_DIRECTED = Variant(directed=True, simple=False, name="multi directed")
224
225
226     class Graph:
227         """A graph or digraph stored as an integer multiplicity matrix.
228
229         The single source of truth is 'self.mu', an 'n by n' array of
230         non-negative integers with a zero diagonal (no self-loops). For an
231         undirected variant the matrix is kept symmetric at all times, so the
232         invariant 'mu[u, v] == mu[v, u]' may be relied upon everywhere.
233
234         All mutating methods respect the variant: they keep an undirected graph
235         symmetric and never let a simple graph exceed multiplicity one.
236         """
237
238         def __init__(self, num_vertices: int, variant: Variant):
239             if num_vertices < 0:
240                 raise ValueError("number of vertices must be non-negative")
241             self.variant = variant
242             # The whole graph is this one matrix; integer dtype keeps it exact.
243             self.mu = np.zeros((num_vertices, num_vertices), dtype=int)
244
245             # -- basic queries -----
246
247             @property
248             def num_vertices(self) -> int:
249                 """The number of vertices 'n'."""
250                 return self.mu.shape[0]
251
252             def vertices(self) -> range:
253                 """The vertex set '{0, 1, ..., n - 1}' as a range."""
254                 return range(self.num_vertices)
255
256             def multiplicity(self, u: int, v: int) -> int:
257                 """Number of parallel edges or arcs from 'u' to 'v'."""
258                 return int(self.mu[u, v])
259
260             def has_edge(self, u: int, v: int) -> bool:
261                 """Whether at least one edge or arc joins 'u' to 'v'."""
262                 return self.mu[u, v] > 0
263
264             def edge_count(self) -> int:
265                 """Total number of edges or arcs, counted with multiplicity.
266

```

```

267     For an undirected graph each edge is stored twice in the matrix, so we
268     count the strict upper triangle. For a digraph every ordered pair
269     counts once.
270     """
271     if self.variant.directed:
272         return int(self.mu.sum())
273     # Strict upper triangle (k=1) avoids double-counting the symmetric pair.
274     return int(np.triu(self.mu, k=1).sum())
275
276 def out_degree(self, v: int) -> int:
277     """Number of edges or arcs leaving ‘v’ (with multiplicity)."""
278     return int(self.mu[v, :].sum())
279
280 def in_degree(self, v: int) -> int:
281     """Number of edges or arcs entering ‘v’ (with multiplicity)."""
282     return int(self.mu[:, v].sum())
283
284 def degree(self, v: int) -> int:
285     """Total degree of ‘v’.”
286
287     For a digraph this is the sum of in- and out-degree. For an undirected
288     graph in- and out-degree coincide, so we report the out-degree.
289     """
290     if self.variant.directed:
291         return self.out_degree(v) + self.in_degree(v)
292     return self.out_degree(v)
293
294 def edges(self) -> Iterator[tuple[int, int, int]]:
295     """Yield ‘(u, v, multiplicity)’ for every present edge or arc.
296
297     For an undirected graph each edge is yielded once, with ‘u < v’.
298     """
299     n = self.num_vertices
300     for u in range(n):
301         # Undirected: only scan v > u so each edge is emitted once.
302         # Directed: scan all v so both arc directions are emitted.
303         start = u + 1 if not self.variant.directed else 0
304         for v in range(start, n):
305             if u != v and self.mu[u, v] > 0:
306                 yield u, v, int(self.mu[u, v])
307
308 # -- mutation -----
309
310 def add_edge(self, u: int, v: int, count: int = 1) -> None:
311     """Add ‘count’ parallel edges or arcs from ‘u’ to ‘v’.”
312
313     A simple graph silently saturates at multiplicity one. An undirected
314     graph updates both matrix entries to stay symmetric.
315     """
316     self._require_distinct(u, v)
317     new_value = self.mu[u, v] + count
318     if self.variant.simple:
319         new_value = min(new_value, 1) # enforce the 0/1 simple-graph cap
320     self._assign(u, v, new_value)
321
322 def remove_edge(self, u: int, v: int, count: int = 1) -> None:
323     """Remove ‘count’ parallel edges or arcs (never below zero)."""
324     self._require_distinct(u, v)

```

```

325         new_value = max(0, self.mu[u, v] - count)
326         self._assign(u, v, new_value)
327
328     def set_multiplicity(self, u: int, v: int, value: int) -> None:
329         """Set the multiplicity from 'u' to 'v' directly."""
330         self._require_distinct(u, v)
331         if value < 0:
332             raise ValueError("multiplicity must be non-negative")
333         if self.variant.simple and value > 1:
334             raise ValueError("a simple graph cannot have multiplicity above one")
335         self._assign(u, v, value)
336
337     def copy(self) -> "Graph":
338         """Return an independent copy with the same variant and edges."""
339         clone = Graph(self.num_vertices, self.variant)
340         clone.mu = self.mu.copy() # deep-copy the matrix so edits don't alias
341         return clone
342
343     # -- internal helpers -----
344
345     def _assign(self, u: int, v: int, value: int) -> None:
346         """Write 'value' into the matrix, mirroring it when undirected."""
347         self.mu[u, v] = value
348         # Maintain the symmetry invariant for undirected graphs in one place.
349         if not self.variant.directed:
350             self.mu[v, u] = value
351
352     def _require_distinct(self, u: int, v: int) -> None:
353         if u == v:
354             raise ValueError("self-loops are not allowed in this model")
355
356     def __repr__(self) -> str:
357         return (f"Graph(n={self.num_vertices}, variant={self.variant.describe()}!r
358                 ↪ }, "
359                 f"edges={self.edge_count()}")
360
361     # --- BOUNDS: closed-form extremal values; docstrings record proved/conjectured
362     ↪ status ---
363
364     def simple_undirected_edge(n: int, m: int) -> int:
365         """'ell_m(n) = floor(m (n-1) / 2)' . Proved (Mader, 1973)."""
366         return (m * (n - 1)) // 2
367
368
369     def multigraph_undirected_edge(n: int, m: int) -> int:
370         """'L_m(n) = (m-1)(n-1)' . Proved (multi-tree construction and bound)."""
371         return (m - 1) * (n - 1)
372
373
374     def simple_undirected_vertex_le4(n: int, m: int) -> int:
375         """'k_m(n) = floor(m (n-1) / 2)' for 'm <= 4' . Cited (Leonard, 1972)."""
376         if m > 4:
377             raise ValueError("this equality is only known for m <= 4")
378         # For m <= 4 the vertex problem coincides with the edge problem.
379         return simple_undirected_edge(n, m)
380

```

```

381
382 def simple_undirected_vertex_m5(n: int) -> int:
383     """'k5(n) = floor(8n/3) - 3' for large 'n'. Cited (Sorensen-Thomassen).
        ↪ ""
384     return (8 * n) // 3 - 3
385
386
387 def directed_arc_m2(n: int) -> int:
388     """'ell2dir(n) = max(2(n-1), floor(n2/4))'. Proved (induction on n)."""
389     # The max picks whichever of the two competing constructions wins at this n:
390     # the linear hub (double star) versus the quadratic one-directional wall.
391     return max(2 * (n - 1), (n * n) // 4)
392
393
394 def directed_arc_lower_bound(n: int, m: int) -> int:
395     """Lower bound and conjectured value for 'ellmdir(n), 'm ≥ 2.".
396
397     'max(m(n-1), floor((n+m-2)2/4))'. The two branches are the hub
398     construction ('const:directed-hub') and the shifted-partition augmented
399     bipartite construction ('const:augmented-bipartite'). Proved as a lower
400     bound for all 'm'; conjectured to be tight for 'm ≥ 3
401     ('conj:dir-arc'), proved tight for 'm = 2.".
402     """
403     hub_branch = m * (n - 1)
404     bipartite_branch = (n + m - 2) ** 2 // 4
405     return max(hub_branch, bipartite_branch)
406
407
408 def hypergraph_edge(n: int, m: int, r: int) -> int:
409     """'floor((m-1)(n-1)/(r-1))' hyperedges for the 'r'-uniform edge problem.
410
411     Proved modulo the hypergraph Gomory-Hu theorem; the star hypertree attains
412     the 'm = 2' case and a sparse hypergraph attains the general bound.
413     """
414     return ((m - 1) * (n - 1)) // (r - 1)
415
416
417 def directed_multigraph_arc(n: int, m: int) -> int:
418     """'Lmdir(n) = 2(n-1)(m-1). Proved for 'n ≤ 6' (by counting cuts).
419
420     Conjectured to continue to hold while the double star dominates the
421     one-directional bipartite branch (small 'n'); for larger 'n' the
422     quadratic branch takes over.
423     """
424     double_star_branch = 2 * (n - 1) * (m - 1)
425     bipartite_branch = (m - 1) * ((n * n) // 4)
426     return max(double_star_branch, bipartite_branch)
427
428
429 # --- HYPERGRAPH: stored as incidence list; Berge-path connectivity via helper-
    ↪ node max-flow ---
430
431
432 class Hypergraph:
433     """A hypergraph stored by its incidence (a list of hyperedges).
434
435     An undirected hyperedge is a set of vertices, crossed in either direction. A
436     directed hyperedge ('directed=True') splits an 'r'-set into a non-empty

```

```

437 tail set 'T' and a non-empty head set 'H': a Berge route may enter it only
438 at a tail and leave only at a head. Two storage forms are accepted, and the
439 checker treats them identically:
440
441 * the legacy *forward* form '(tail, frozenset(heads))' with a single integer
442   tail and 'r - 1' heads ('|T| = 1'), the one-tail / many-head model the
443   rest of the thesis uses; and
444 * the *general* form '(frozenset(tails), frozenset(heads))' for any split,
445   which subsumes forward ('|T| = 1'), backward ('|H| = 1') and, from
446   'r >= 4', genuinely mixed ('|T|, |H| >= 2') hyperarcs.
447
448 For 'r = 2' every form collapses to a single arc, so a directed graph is the
449 two-vertex special case.
450 """
451
452 def __init__(self, num_vertices: int, hyperedges: Iterable = (),
453              *, directed: bool = False):
454     self.num_vertices = num_vertices
455     self.directed = directed
456     # Each entry is a frozenset, or a '(tail, frozenset(heads))' pair.
457     self.hyperedges: list = []
458     for edge in hyperedges:
459         self.add_hyperedge(edge)
460
461 def add_hyperedge(self, edge) -> None:
462     """Add an undirected vertex set, or a directed '(tail, heads)' pair.
463
464     A directed edge may be the forward form '(tail:int, heads)' or the
465     general form '(tails:frozenset, heads:frozenset)'; the general form is
466     kept verbatim, the forward form is kept verbatim too (no normalisation),
467     so existing forward results are unchanged byte-for-byte.
468     """
469     if self.directed:
470         first, raw_heads = edge
471         if isinstance(first, (set, frozenset)): # general (tails, heads)
472             tails, heads = frozenset(first), frozenset(raw_heads)
473             members = tails | heads
474             if (tails & heads) or not tails or not heads or len(members) < 2:
475                 raise ValueError("a directed hyperedge needs disjoint, non-
476                                     ↪ empty tail and head sets")
477             if any(v < 0 or v >= self.num_vertices for v in members):
478                 raise ValueError("hyperedge refers to a vertex outside the
479                                     ↪ graph")
480             self.hyperedges.append((tails, heads))
481             return
482         tail, heads = first, frozenset(raw_heads) # legacy forward (tail,
483                                     ↪ heads)
484         members = heads | {tail}
485         if tail in heads or len(members) < 2:
486             raise ValueError("a directed hyperedge needs a tail and a distinct
487                                     ↪ head")
488         if any(v < 0 or v >= self.num_vertices for v in members):
489             raise ValueError("hyperedge refers to a vertex outside the graph")
490         self.hyperedges.append((tail, heads))
491     else:
492         vertices = frozenset(edge) # a hyperedge is a set of distinct
493                                     ↪ vertices
494         if len(vertices) < 2:

```

```

490         raise ValueError("a hyperedge must contain at least two vertices")
491     if any(v < 0 or v >= self.num_vertices for v in vertices):
492         raise ValueError("hyperedge refers to a vertex outside the graph")
493     self.hyperedges.append(vertices)
494
495     def members(self, edge) -> frozenset:
496         """The vertex set of a stored hyperedge, directed or not."""
497         if self.directed:
498             tails, heads = _dir_tails_heads(edge)
499             return tails | heads
500         return edge
501
502     def vertices(self) -> range:
503         return range(self.num_vertices)
504
505     def edge_count(self) -> int:
506         """Number of hyperedges."""
507         return len(self.hyperedges)
508
509     def incident_hyperedges(self, v: int) -> list[int]:
510         """Indices of the hyperedges containing vertex 'v'."""
511         return [i for i, edge in enumerate(self.hyperedges) if v in self.members(
512             ↪ edge)]
513
514     def _dir_tails_heads(edge) -> tuple[frozenset, frozenset]:
515         """Tail and head sets of a stored directed hyperedge, as frozensets.
516
517         Accepts the legacy forward form '(tail:int, heads)' and the general form
518         '(tails:frozenset, heads:frozenset)', so every consumer of a directed
519         hyperedge can read one '(tails, heads)' shape regardless of how it was
520         built.
521         """
522         first, heads = edge
523         if isinstance(first, (set, frozenset)):
524             return frozenset(first), frozenset(heads)
525         return frozenset((first,)), frozenset(heads)
526
527     def _hyper_capacity_matrix(hypergraph: Hypergraph, *, vertex_split: bool = False):
528         """Integer capacity matrix of the hypergraph flow network (one per variant).
529
530         Each hyperedge becomes an in/out gate joined by a capacity-1 arc, so at most
531         one disjoint Berge route may traverse it: this is the hyperedge-as-node trick.
532         With 'vertex_split' each ORIGINAL vertex is *also* split into an in/out pair
533         of capacity one, so the flow counts internally vertex-disjoint Berge routes
534         instead of edge-disjoint ones. For a directed hypergraph the gate is entered
535         only from a tail and left only toward a head (one tail for the forward model,
536         several for the general one); for an undirected one every member links to the
537         gate both ways. The hypergraph variants are thus the same construction with a
538         boolean or two flipped, not separate measures.
539
540         Vertices index '0..base-1' ('base = 2n' split, else 'n'); hyperedge gate
541         'i' indexes 'base+2i' (in) and 'base+2i+1' (out). 'leave(v)'/enter(
542             ↪ v)'
543         return the index a route leaves/enters vertex 'v' through, so the same scipy
544         max-flow that serves plain graphs serves hypergraphs too.
545         """

```



```

546     n = hypergraph.num_vertices
547     base = 2 * n if vertex_split else n
548     size = base + 2 * len(hypergraph.hyperedges)
549     cap = np.zeros((size, size), dtype=int)
550     leave = (lambda v: 2 * v + 1) if vertex_split else (lambda v: v)
551     enter = (lambda v: 2 * v) if vertex_split else (lambda v: v)
552     if vertex_split:
553         for v in range(n):
554             cap[2 * v, 2 * v + 1] = 1           # one route through each vertex
555     for index, edge in enumerate(hypergraph.hyperedges):
556         gate_in, gate_out = base + 2 * index, base + 2 * index + 1
557         cap[gate_in, gate_out] = 1           # one route through each
558             ↪ hyperedge
559     if hypergraph.directed:
560         tails, heads = _dir_tails_heads(edge)
561         for tail in tails:           # enter the gate from any tail
562             cap[leave(tail), gate_in] = _UNBOUNDED
563         for head in heads:           # leave the gate toward any head
564             cap[gate_out, enter(head)] = _UNBOUNDED
565     else:
566         for vertex in edge:
567             cap[leave(vertex), gate_in] = _UNBOUNDED
568             cap[gate_out, enter(vertex)] = _UNBOUNDED
569     return cap, size, leave, enter
570
571 def hyper_connectivity(hypergraph: Hypergraph, source: int, target: int,
572                        *, vertex_split: bool = False) -> int:
573     """Edge- or vertex-disjoint Berge routes from ‘source’ to ‘target’.”
574
575     A single scipy integer max-flow on the matrix above. An isolated endpoint
576     (in no hyperedge) has no incident arcs, so the flow is then zero.
577     """
578     cap, _, leave, enter = _hyper_capacity_matrix(hypergraph, vertex_split=
579         ↪ vertex_split)
580     if vertex_split:
581         # Endpoints must not be the bottleneck: only interior vertices are capped.
582         cap[2 * source, 2 * source + 1] = _UNBOUNDED
583         cap[2 * target, 2 * target + 1] = _UNBOUNDED
584     return int(_csgraph_maxflow(_csr(cap, dtype=int), leave(source), enter(target)
585         ↪ ).flow_value)
586
587 def max_hyper_connectivity(hypergraph: Hypergraph, *, vertex_split: bool = False)
588     ↪ -> int:
589     """‘‘lambda^max’’ (edge) or ‘‘kappa^max’’ (vertex) over all pairs.””
590     return max((hyper_connectivity(hypergraph, source, target, vertex_split=
591         ↪ vertex_split)
592         for source, target in _pairs(hypergraph)), default=0)
593
594 def hyperedge_connectivity(hypergraph: Hypergraph, source: int, target: int) ->
595     ↪ int:
596     """Maximum number of edge-disjoint Berge ‘source’-‘target’ paths.””
597     return hyper_connectivity(hypergraph, source, target, vertex_split=False)
598
599 def max_hyperedge_connectivity(hypergraph: Hypergraph) -> int:

```

```

598     """ $\lambda^{\max}$  for the undirected edge hypergraph problem."""
599     return max_hyper_connectivity(hypergraph, vertex_split=False)
600
601
602 # --- CONSTRUCTIONS: named extremal graphs; each states its edge count and  $\lambda^{\max}$ 
603      $\hookrightarrow$  max ---
604
605 def double_star(n: int, m: int, directed: bool = True) -> Graph:
606     """The double star: one hub joined to every leaf with multiplicity  $m-1$ .
607
608     For a digraph the hub carries  $m-1$  arcs in both directions to each
609     leaf, attaining  $2(n-1)(m-1)$  arcs with  $\lambda^{\max} = m-1$ . For an
610     undirected multigraph it attains  $(m-1)(n-1)$  edges. This is the
611     small- $n$  extremiser of the multigraph problems.
612     """
613     variant = MULTI_DIRECTED if directed else MULTI_UNDIRECTED
614     graph = Graph(n, variant)
615     hub = 0 # vertex 0 is the central hub; everything else is a leaf
616     for leaf in range(1, n):
617         graph.set_multiplicity(hub, leaf, m - 1)
618         if directed:
619             graph.set_multiplicity(leaf, hub, m - 1) # arcs in both directions
620     return graph
621
622
623 def one_directional_bipartite(n: int) -> Graph:
624     """All arcs from one part to the other:  $\lfloor n^2/4 \rfloor$  arcs,  $\lambda^{\max} =$ 
625          $\hookrightarrow 1$ .
626
627     The vertices split into a smaller part  $A$  and a larger part  $B$  and
628     every arc runs  $A \rightarrow B$ . No two vertices have two arc-disjoint routes, so
629     this single-direction wall of arcs is feasible for  $m = 2$  while holding
630     quadratically many arcs: the phenomenon that has no undirected analogue.
631     """
632     graph = Graph(n, SIMPLE_DIRECTED)
633     size_a = n // 2 # smaller part; the  $n^2/4$  maximum is at the balanced split
634     part_a = range(size_a)
635     part_b = range(size_a, n)
636     for a in part_a:
637         for b in part_b:
638             graph.add_edge(a, b) # every arc runs  $A \rightarrow B$  only (one direction)
639     return graph
640
641 def augmented_bipartite(n: int, m: int) -> Graph:
642     """The conjectured directed extremiser for  $m \geq 2$ .
643
644     Set  $|B| = \lceil (n+m-2)/2 \rceil$  and  $|A| = n - |B|$ . Fill every arc
645      $A \rightarrow B$  (the one-directional wall), then give each  $B$ -vertex
646      $m - 2$  extra in-arcs from its circulant predecessors inside  $B$ .
647     Because no arc leaves  $B$  toward  $A$ , a route between two  $B$ 
648     vertices never escapes  $B$ , and a route from  $A$  to  $b$  uses the
649     direct arc plus one short detour through each of  $b$ 's  $m - 2$ 
650     in-arcs, giving  $\lambda^{\max} = m - 1$ . The arc count is
651      $\lfloor (n+m-2)^2/4 \rfloor$  (const:augmented-bipartite in the thesis).
652
653     Requires  $n \geq m$  so that  $|A| \geq 1$  and the circulant offsets stay

```

```

654     distinct. At ' $m = 2$ ' or ' $m = 3$ ' the partition equals the balanced
655     ' $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$ ' split and the formula reduces to
656     ' $\lfloor n^2/4 \rfloor + (m-2)\lceil n/2 \rceil$ '. At ' $m \geq 4$ ' the shifted partition
657     gives strictly more arcs. For ' $m = 3, n = 10$ ' this is the 30-arc
658     counterexample.
659     """
660     size_b = (n + m - 2 + 1) // 2 #  $\lceil (n+m-2)/2 \rceil$ 
661     size_a = n - size_b
662     if size_a < 1:
663         raise ValueError("augmented_bipartite needs  $n \geq m$ ")
664
665     graph = Graph(n, SIMPLE_DIRECTED)
666     part_a = range(size_a)
667     part_b = list(range(size_a, n))
668
669     # Complete wall of arcs from the small part to the large part.
670     for a in part_a:
671         for b in part_b:
672             graph.add_edge(a, b)
673
674     # Thicken B by a circulant giving every B-vertex in-degree exactly  $m - 2$ .
675     # Each offset adds a full cycle of arcs within B;  $m - 2$  offsets give each
676     # B-vertex  $m - 2$  extra in-arcs from other B-vertices (the short detours).
677     for offset in range(1, m - 1):
678         for i in range(size_b):
679             source = part_b[i]
680             target = part_b[(i + offset) % size_b]
681             graph.add_edge(source, target)
682     return graph
683
684
685 def clique_tree(clique_size: int, blocks_added: int) -> Graph:
686     """A tree of cliques (a ' $k$ '-tree): the chordal scaffold of the vertex case.
687
688     Begin with a complete graph ' $K_r$ ' on ' $r = \text{clique\_size}$ ' vertices, then
689     repeatedly attach a new vertex to the previous ' $r - 1$ ' vertices, which
690     always form a clique. The result is a maximal chordal graph of treewidth
691     ' $r - 1$ ' and *global* vertex connectivity ' $r - 1$ '.
692
693     A caution that the thesis makes explicit: the controlled quantity here is the
694     global connectivity, not ' $\kappa^{\max}$ '. Attaching a vertex to a triangle
695     raises the local connectivity of that triangle's vertices above ' $r - 1$ ', so
696     a ' $k$ '-tree is not itself feasible for the ' $\kappa^{\max}$ ' problem. It appears
697     in the thesis to illustrate the chordal structure underlying the
698     Sorensen-Thomassen analysis of the ' $m = 5$ ' divergence, not as a feasible
699     extremiser.
700     """
701     if clique_size < 2:
702         raise ValueError("clique_size must be at least two")
703     n = clique_size + blocks_added
704     graph = Graph(n, SIMPLE_UNDIRECTED)
705
706     # The seed clique on the first  $r$  vertices.
707     for u in range(clique_size):
708         for v in range(u + 1, clique_size):
709             graph.add_edge(u, v)
710
711     # Each new vertex joins the  $r - 1$  most recently added vertices.

```

```

712     # Those  $r - 1$  always form a clique, which keeps the graph chordal (a  $k$ -tree).
713     for new_vertex in range(clique_size, n):
714         attach_to = range(new_vertex - (clique_size - 1), new_vertex)
715         for old_vertex in attach_to:
716             graph.add_edge(new_vertex, old_vertex)
717     return graph
718
719
720 def complete_uniform_hypergraph(n: int, r: int) -> Hypergraph:
721     """The complete ' $r$ '-uniform hypergraph: every ' $r$ '-subset is a hyperedge.
722
723     It is dense and highly connected; for example  $K_4^{\{3\}}$  has Berge
724     edge-connectivity three between every pair, exceeding the two-hyperedge count
725     of the pairs because longer Berge routes also contribute.
726     """
727     # One hyperedge per  $r$ -subset of the vertex set.
728     return Hypergraph(n, (set(subset) for subset in combinations(range(n), r)))
729
730
731 def star_hypertree(n: int, r: int) -> Hypergraph:
732     """A hub joined to disjoint blocks: the feasible extremiser at ' $m = 2$ '.
733
734     Vertex ' $0$ ' is the hub, and the remaining ' $n - 1$ ' vertices are split into
735     blocks of size ' $r - 1$ '; each block together with the hub forms one
736     hyperedge. No two vertices share more than one hyperedge and the only routes
737     between blocks pass through the single hub, so the Berge edge-connectivity is
738     one everywhere, and the construction holds  $\lfloor (n-1)/(r-1) \rfloor$ 
739     hyperedges, the ' $m = 2$ ' case of the hypergraph edge bound.
740     """
741     if r < 2:
742         raise ValueError("hyperedge size r must be at least two")
743     hub = 0
744     others = list(range(1, n))
745     hypergraph = Hypergraph(n)
746     # Walk the non-hub vertices in blocks of  $r - 1$ ; each block plus the hub is
747     # one hyperedge (a "star" of  $r$ -sets all sharing the single hub).
748     for start in range(0, len(others) - (r - 2), r - 1):
749         block = others[start:start + (r - 1)]
750         if len(block) == r - 1: # drop a trailing partial block
751             hypergraph.add_hyperedge([hub, *block])
752     return hypergraph

```

C.1.3 Measuring, and proving

This part corresponds to [Chapter 2](#). Measuring comes first. The question “how many independent routes join these two vertices” is answered exactly, by turning it into a maximum-flow problem and solving that, which is Menger’s theorem used as an algorithm. The same routine answers all four versions of the question, because whether routes must avoid shared edges or shared intermediate vertices is controlled by one flag that decides how the flow network is built, and hypergraphs are handled by giving each hyperedge a gate of its own.

Proving comes second and is a different kind of work. To show that no graph of a given size can beat a bound, it is not enough to fail to find one. The program instead writes the whole question as a single optimisation problem whose variables describe an arbitrary graph and whose

constraints say that every pair of vertices admits a small cut. If that problem has no solution, no such graph exists, and that is a proof rather than an absence of evidence.

```

755 #####
756 ##
757 CHAPTER 2 -- PROVING BOUNDS BY MACHINE
758 ## Measure connectivity exactly, then prove small-case upper bounds.
759 ##
760 #####
761
762 # --- CHECKER: exact max-flow connectivity; vertex_split=True gives kappa^max ---
763
764 # A capacity large enough to never be the bottleneck of any cut we build.
765 _UNBOUNDED = 10 ** 9
766
767
768 # -----
769 # One flow network and one measure, parameterised by edge vs vertex
770 # -----
771 # Every measure here is a single scipy integer max-flow on a capacity matrix
772 # (Menger). EDGE mode runs directly on the multiplicity matrix mu: each unit of
773 # capacity is one parallel edge, so a max-flow of value f is f edge-disjoint
774 # routes. VERTEX mode runs on the 2n x 2n split matrix (see
775     ↪ _split_capacity_matrix)
776 # where each vertex becomes a capacity-one in/out gate, so a route uses a vertex
777     ↪ at
778 # most once. Edge vs vertex is one boolean, not a second code path.
779
780 def local_connectivity(graph: Graph, source: int, target: int,
781                        *, vertex_split: bool = False) -> int:
782     """Disjoint 'source'-'target' routes by a single max-flow (Menger).
783
784     Edge-disjoint with 'vertex_split=False'; internally vertex-disjoint with
785     'vertex_split=True'. An isolated endpoint yields zero (no augmenting path).
786     """
787     if not vertex_split:
788         # scipy.sparse.csgraph.maximum_flow is a C implementation; faster than
789         # NetworkX for the small integer capacity matrices we use.
790         return int(_csgraph_maxflow(_csr(graph.mu, dtype=int), source, target).
791                    ↪ flow_value)
792
793     # Vertex mode: the same scipy max-flow on the split matrix. Uncapping the two
794     # endpoints' own in->out gates makes Menger count internally vertex-disjoint
795     # routes (only interior vertices stay capacity-capped at one).
796     cap, _ = _split_capacity_matrix(graph)
797     cap[2 * source, 2 * source + 1] = _UNBOUNDED
798     cap[2 * target, 2 * target + 1] = _UNBOUNDED
799     return int(_csgraph_maxflow(_csr(cap, dtype=int), 2 * source + 1, 2 * target).
800               ↪ flow_value)
801
802
803 def max_connectivity(graph: Graph, *, vertex_split: bool = False) -> int:
804     """The largest local connectivity over all pairs: 'lambda^max' (edge) or
805     'kappa^max' (vertex). For a digraph every ordered pair is examined, for an
806     undirected graph each unordered pair once.
807     """
808     if not vertex_split:
809         # Build the CSR matrix once and reuse it for every pair.
810         csr = _csr(graph.mu, dtype=int)

```

```

806         return max((int(_csggraph_maxflow(csr, s, t).flow_value)
807                     for s, t in _pairs(graph)), default=0)
808     return max((local_connectivity(graph, s, t, vertex_split=True)
809               for s, t in _pairs(graph)), default=0)
810
811
812 # -----
813 # Named measures: thin views on the single path above (so the rest of the
814 # program can pass ‘‘max_edge_connectivity’’ / ‘‘max_vertex_connectivity’’ around
815 # by name -- just as ‘‘hyperedge_connectivity’’ names one branch of
816 # ‘‘hyper_connectivity’’ in the hypergraph section).
817 # -----
818
819 def local_edge_connectivity(graph: Graph, source: int, target: int) -> int:
820     """Maximum number of edge-disjoint ‘‘source’’-‘‘target’’ paths."""
821     return local_connectivity(graph, source, target, vertex_split=False)
822
823
824 def max_edge_connectivity(graph: Graph) -> int:
825     """‘‘lambda^max(G)’’: the largest local edge-connectivity over all pairs."""
826     return max_connectivity(graph, vertex_split=False)
827
828
829 def local_vertex_connectivity(graph: Graph, source: int, target: int) -> int:
830     """Maximum number of internally vertex-disjoint ‘‘source’’-‘‘target’’ paths."""
831     ↪ "
832     return local_connectivity(graph, source, target, vertex_split=True)
833
834
835 def max_vertex_connectivity(graph: Graph) -> int:
836     """‘‘kappa^max(G)’’: the largest local vertex-connectivity over all pairs."""
837     return max_connectivity(graph, vertex_split=True)
838
839
840 def min_vertex_connectivity(graph: Graph) -> int:
841     """The global vertex connectivity ‘‘kappa(G)’’: the smallest over all pairs.
842
843     This is the classical connectivity of a graph (the least number of vertices
844     whose removal disconnects it). It is distinct from ‘‘kappa^max’’: a ‘‘k’’-
845     ↪ tree,
846     for instance, is globally ‘‘k’’-connected yet has pairs with higher local
847     connectivity.
848     """
849     pairs = list(_pairs(graph))
850     if not pairs:
851         return 0
852     # Global connectivity is the MINIMUM local connectivity over all pairs.
853     return min(local_vertex_connectivity(graph, source, target)
854               for source, target in pairs)
855
856 # -----
857 # Shared helpers
858 # -----
859
860 def _pairs(obj):
861     """Yield the vertex pairs to test: ordered if directed, unordered if not.

```

```

862     Works for both a ‘‘Graph’’ (directedness on ‘‘variant’’) and a ‘‘Hypergraph’’
863     (directedness on the object), so one pair sweep serves every variant.
864     """
865     directed = obj.directed if hasattr(obj, "directed") else obj.variant.directed
866     n = obj.num_vertices
867     for source in range(n):
868         # Directed objects need both (s, t) and (t, s); undirected need each
869         # unordered pair once, so we start the inner loop above the source.
870         start = 0 if directed else source + 1
871         for target in range(start, n):
872             if source != target:
873                 yield source, target
874
875
876 # -----
877 # Capped feasibility predicate (used inside the search, Section 7)
878 # -----
879 # The exact measures above answer "what is  $\lambda^{\max} / \kappa^{\max}$ ". The search
880 # only ever needs the cheaper YES/NO question "is it greater than k?", because
881 # its energy depends on the value solely through the excess  $\max(0, \text{value} - (m-1))$ .
882 # A capped flow answers exactly that and stops the moment the answer is known, so
883 # this predicate is equivalent to ‘‘ $\max\_connectivity(\text{graph}, \dots) > k$ ’’ but usually
884 # far cheaper: each pair’s flow halts after k+1 augmenting paths, and the pair
885 # loop halts at the first violating pair. The exact value is recomputed only on
886 # the rare step where the search genuinely needs it (an infeasible proposal).
887 # DEPENDENCIES: _pairs (above), _tiny_maxflow (Section ENUMERATE), _UNBOUNDED.
888
889 def _split_capacity_matrix(graph: Graph,
890                           parallel_routes: bool = False) -> tuple[np.ndarray, int
891                               ↪ ]:
892
893     """The ‘‘2n x 2n’’ capacity matrix of the vertex-split network.
894
895     The vertex-mode network as a plain matrix, consumed by both the exact measure
896     (:func:‘local_connectivity’ via scipy max-flow) and the capped search
897     ↪ predicate
898     (:func:‘exceeds_bound’ via :func:‘_tiny_maxflow’). Vertex ‘‘v’’ becomes an
899     in-copy ‘‘2v’’ and an out-copy ‘‘2v+1’’ joined by an internal arc of capacity
900     one (so any route uses ‘‘v’’ at most once), and each adjacency ‘‘u -> v’’
901     becomes an arc ‘‘(2u+1) -> (2v)’’. The caller uncaps the two endpoints’ own
902     in->out gates so Menger counts internally disjoint routes (see
903     :func:‘exceeds_bound’).
904
905     ‘‘parallel_routes’’ selects the counting convention for parallel copies, the
906     choice discussed at length in the thesis (sec:parallel-convention). The
907     default False caps every adjacency at one, so a bundle of parallel edges is a
908     single direct route and a multigraph measures as its underlying simple graph:
909     that is the convention the twelve variants use. Setting it True gives the
910     adjacency capacity ‘‘ $\mu(u,v)$ ’’, so ‘‘q’’ parallel copies count as ‘‘q’’
911     internally disjoint routes, which is the alternative convention explored in
912     sec:multi-vertex-standard.
913     """
914     n = graph.num_vertices
915     size = 2 * n
916     cap = np.zeros((size, size), dtype=int)
917     for v in range(n):
918         cap[2 * v, 2 * v + 1] = 1 # internal gate: one route through v
919     for u in range(n):
920         for v in range(n):

```

```

918         if u != v and graph.mu[u, v] > 0:
919             cap[2 * u + 1, 2 * v] = (int(graph.mu[u, v]) if parallel_routes
920                                     else 1)
921     return cap, size
922
923
924 def max_multigraph_vertex_standard(n: int, m: int,
925                                   deadline: float | None = None) -> tuple[int,
926                                   ↪ Graph | None, bool]:
927
928     """Exhaustive maximum for the multigraph vertex problem, OTHER convention.
929
930     Maximises the total multiplicity of an undirected multigraph on ‘n’
931     vertices subject to ‘ $\kappa^{\max} \leq m - 1$ ’ when parallel copies are counted
932     as distinct internally vertex-disjoint routes. Every multiplicity is capped
933     at ‘m - 1’ because ‘m’ parallel copies already exceed the ceiling on
934     their own. Branch and bound over the pairs in a fixed order, trying the
935     largest multiplicity first, pruning on feasibility (which is monotone) and
936     on the incumbent. Returns ‘(best_total, witness, completed)’.”
937
938     pairs = list(combinations(range(n), 2))
939     cap = m - 1
940     graph = Graph(n, MULTI_UNDIRECTED)
941     best = [0, None]
942     timed_out = [False]
943
944     def feasible() -> bool:
945         return not exceeds_bound(graph, m - 1, separation="vertex",
946                                 parallel_routes=True)
947
948     def recurse(i: int, total: int) -> None:
949         if deadline is not None and time.time() > deadline:
950             timed_out[0] = True
951             return
952         if total + cap * (len(pairs) - i) <= best[0]:
953             return
954         if i == len(pairs):
955             if total > best[0]:
956                 best[0] = total
957                 best[1] = graph.copy()
958             return
959         u, v = pairs[i]
960         for q in range(cap, -1, -1):
961             graph.set_multiplicity(u, v, q)
962             if q == 0 or feasible():
963                 recurse(i + 1, total + q)
964             if timed_out[0]:
965                 break
966         graph.set_multiplicity(u, v, 0)
967
968     recurse(0, 0)
969     return best[0], best[1], not timed_out[0]
970
971 def exceeds_bound(graph: Graph, k: int, *, separation: str = "edge",
972                  parallel_routes: bool = False) -> bool:
973     """‘True’ iff ‘ $\lambda^{\max}(G) > k$ ’ (edge) or ‘ $\kappa^{\max}(G) > k$ ’ (vertex).
974
975     Equivalent to ‘max_connectivity(graph, vertex_split=(separation=='vertex')) >

```



```

    ↪ k''
975 but with two early exits: each pair's capped flow stops after 'k+1'
    ↪ augmenting
976 paths, and the pair loop stops at the first violating pair. On an infeasible
977 graph this typically returns after a single pair. The pair set is exactly the
978 one :func:'max_connectivity' iterates, so the predicate matches it pair for
    ↪ pair.

979
980 'parallel_routes' is passed through to :func:'_split_capacity_matrix' and
981 only affects vertex mode: it selects the convention in which parallel copies
982 are distinct routes, used by :func:'max_multigraph_vertex_standard'.
983 """
984 n = graph.num_vertices
985 if separation == "edge":
986     mu = graph.mu
987     if _C is not None and n <= 16:
988         flat, ptr = _c_flat(mu)
989         directed = int(graph.variant.directed)
990         return bool(_C.max_connectivity_exceeds(ptr, n, k, directed))
991     for s, t in _pairs(graph):
992         if _tiny_maxflow(mu, n, s, t, k):
993             return True
994     return False
995 if separation == "vertex":
996     cap, size = _split_capacity_matrix(graph, parallel_routes)
997     for s, t in _pairs(graph):
998         # Uncap the endpoints' own in->out gates so Menger counts INTERNAL
999         # routes (mirrors local_connectivity's vertex mode); flow leaves s's
1000         # out-copy (2s+1) and arrives at t's in-copy (2t).
1001         cap[2 * s, 2 * s + 1] = _UNBOUNDED
1002         cap[2 * t, 2 * t + 1] = _UNBOUNDED
1003         hit = _tiny_maxflow(cap, size, 2 * s + 1, 2 * t, k)
1004         cap[2 * s, 2 * s + 1] = 1 # restore the gates for the next
            ↪ pair
1005         cap[2 * t, 2 * t + 1] = 1
1006         if hit:
1007             return True
1008     return False
1009 raise ValueError("separation must be 'edge' or 'vertex'")
1010
1011
1012 # --- GOMORY-HU: encode all pairwise edge-connectivities in n-1 flows (undirected
    ↪ only) ---
1013
1014
1015 def _undirected_capacity_graph(graph: Graph) -> nx.Graph:
1016     """Build the undirected capacitated graph that Gomory-Hu consumes."""
1017     if graph.variant.directed:
1018         raise ValueError("Gomory-Hu trees are defined for undirected graphs only")
1019     capacity_graph = nx.Graph()
1020     capacity_graph.add_nodes_from(graph.vertices())
1021     # One undirected edge per adjacency, capacity = multiplicity (its edge count).
1022     for u, v, multiplicity in graph.edges():
1023         capacity_graph.add_edge(u, v, capacity=float(multiplicity))
1024     return capacity_graph
1025
1026
1027 def gomory_hu_tree(graph: Graph) -> nx.Graph:

```

```

1028     """Return a Gomory-Hu tree of ‘graph’ as a weighted ‘networkx’ tree.
1029
1030     Each tree edge carries a ‘weight’ equal to the capacity of the minimum cut
1031     it represents. This is the one place that genuinely needs networkx’s
1032     Gomory-Hu routine; it backs a figure, not any reported bound.
1033     """
1034     _require_networkx("gomory_hu_tree")
1035     return nx.gomory_hu_tree(_undirected_capacity_graph(graph), capacity="capacity
        ↳ ")
1036
1037
1038 def max_edge_connectivity_via_tree(graph: Graph) -> int:
1039     """‘lambda^max(G)’ read off the Gomory-Hu tree as its heaviest edge.
1040
1041     Returns ‘0’ for graphs with fewer than two vertices or no edges.
1042     """
1043     if graph.num_vertices < 2:
1044         return 0
1045     tree = gomory_hu_tree(graph)
1046     # READ-OFF: lambda(u, v) = lightest tree edge on the u-v path, so the LARGEST
1047     # such bottleneck over all pairs is simply the heaviest edge in the tree.
1048     weights = [data["weight"] for _, _, data in tree.edges(data=True)]
1049     if not weights:
1050         return 0
1051     return int(round(max(weights)))
1052
1053
1054 # --- PROVE: MILP for M*(n) via cut-counting (scipy/HiGHS or Gurobi backend) ---
1055 # L_m^dir(n) = (m-1)*M*(n) where M*(n) = max sum(w) s.t. maxflow(s,t;w)<=1 all
        ↳ pairs.
1056 # Flow constraint encoded exactly: choose a cut x in {0,1}^n per pair, cap
        ↳ crossing
1057 # weight via p; zero MIP gap = proof. Shared helpers build constraints for both
        ↳ backends.
1058
1059
1060 @dataclass
1061 class ProofResult:
1062     """The outcome of a proof run."""
1063
1064     n: int
1065     status: str # OPTIMAL, LIMIT, INFEASIBLE, UNBOUNDED, ...
1066     scaled_optimum: float # M*(n) = max total weight, exact if status OPTIMAL
1067     relative_gap_zero: bool # whether the solver closed the gap to zero
1068     solve_seconds: float
1069     weight_matrix: np.ndarray | None # a witnessing matrix w, if one was found
1070
1071     def value_for(self, m: int) -> int:
1072         """The proved directed multigraph value ‘L_m^dir(n) = (m-1) M*(n)’.”””
1073         # Undo the (m-1) scaling: one proved M*(n) yields every m.
1074         return (m - 1) * int(round(self.scaled_optimum))
1075
1076     def is_proof(self) -> bool:
1077         """Whether this run is a genuine upper-bound proof (optimal, gap zero)."""
1078         # Only an OPTIMAL solve with a closed gap counts as a proof; a LIMIT
1079         # (timed out) result is merely a feasible lower bound, never a proof.
1080         return self.status == "OPTIMAL" and self.relative_gap_zero
1081

```

```

1082
1083 def _ordered_pairs(n: int) -> list[tuple[int, int]]:
1084     return [(u, v) for u in range(n) for v in range(n) if u != v]
1085
1086
1087 def _two_hop_triples(n: int) -> list[tuple[int, int, int]]:
1088     # All (source, middle, target) with three distinct vertices: the s -> x -> t
1089     # detours that the two-hop inequality reasons about.
1090     return [(s, x, t)
1091             for s in range(n) for t in range(n) if s != t
1092             for x in range(n) if x != s and x != t]
1093
1094
1095 # Proved values of M*(k) for the deletion cuts below (proved: M*(k) =
1096 # 2(k-1) for k <= 6, proved by this very optimisation with zero gap).
1097 _PROVEN_MSTAR = {2: 2, 3: 4, 4: 6, 5: 8, 6: 10}
1098
1099
1100 # -----
1101 # The cut-counting MILP: one PuLP model for both provers, any solver
1102 # -----
1103 # The two provers below are the SAME cut-counting optimisation over two weight
1104 # domains: the fractional one (prove_directed_multigraph, weights w in [0, 1], cut
1105 # cap 1) maximises the total weight M*(n); the integral one (prove_integral_arc_
1106 # bound, multiplicities mu in {0..m-1}, cut cap m-1) decides feasibility at a
1107 ↳ fixed
1108 # arc target. One builder serves both, and any MILP solver runs it: CBC (bundled
1109 # with pulp) by default, Gurobi by a one-line switch. Because the cut formulation
1110 ↳ proof.
1111
1112 _PULP_STATUS = {"Optimal": "OPTIMAL", "Infeasible": "INFEASIBLE",
1113                 "Unbounded": "UNBOUNDED"}
1114
1115 def _pick_solver(time_limit: float, show_log: bool, use_gurobi: bool | None):
1116     """Choose the MILP solver. 'gapRel=0' demands a closed gap, which is what
1117     makes an OPTIMAL verdict a proof rather than a heuristic.
1118
1119     'use_gurobi': 'True' forces Gurobi (error if it is not usable), 'False'
1120     forces CBC, 'None' uses Gurobi when its licence is active and CBC otherwise.
1121     Switching to Gurobi is the whole change needed to attack the open n=7 facts.
1122     """
1123     if use_gurobi is not False:
1124         gurobi = pulp.GUROBI_CMD(msg=show_log, timeLimit=time_limit,
1125                                  options=[("MIPGap", 0)])
1126         if gurobi.available():
1127             return gurobi
1128         if use_gurobi is True:
1129             raise RuntimeError(
1130                 "use_gurobi=True but no usable Gurobi was found. This path runs "
1131                 "Gurobi through PuLP's GUROBI_CMD, which calls the gurobi_cl "
1132                 "command line, so gurobipy is not required: check that "
1133                 "'gurobi_cl --version' runs and that the licence is active."
1134             )
1135     return pulp.PULP_CBC_CMD(msg=show_log, timeLimit=time_limit, gapRel=0.0)
1136
1137

```

```

1138 def _cut_counting_model(n: int, *, cap: float, integer: bool, two_hop: bool,
1139                          symmetry: bool, deletion: bool, degree_pair: bool):
1140     """Build the shared cut-counting MILP as a PuLP model (no objective set yet).
1141
1142     For every ordered pair (s, t) the model CHOOSES one cut: the side indicators
1143     fix s on the source side (constant 1) and t on the sink side (constant 0), and
1144     every other vertex takes a binary side. The helper p picks up each crossing
1145     arc's weight through 'p >= w + cap*(x_u - x_v) - cap', which forces 'p >= w
1146     ↪ exactly on a crossing arc (x_u = 1, x_v = 0) and leaves p free at 0 otherwise.
1147     Capping the crossing total at 'cap' says maxflow(s, t) <= cap, and letting
1148     ↪ the
1149     optimisation pick the cut means it picks the MINIMUM cut, so this is exactly
1150     maxflow(s, t) <= cap, the true feasible region. 'cap=1' with continuous
1151     weights is the scaled prover; 'cap=m-1' with integer weights the arc one.
1152
1153     The optional families are all proved-valid inequalities that tighten the
1154     relaxation without ever moving the optimum: 'two_hop' (a two-arc-disjoint
1155     detour bound via z = min of the two hops), 'degree_pair' (a flow lower bound
1156     on each pair), 'deletion' (an induced-subgraph bound from proved smaller
1157     M*(k)), and 'symmetry' (a degree ordering that prunes relabelled duplicates)
1158     ↪ .
1159
1160     Returns '(prob, w)' with 'w' the weight variables keyed by ordered pair.
1161     """
1162     pairs = _ordered_pairs(n)
1163     prob = pulp.LpProblem("erdos915", pulp.LpMaximize)
1164     category = "Integer" if integer else "Continuous"
1165     w = {(u, v): pulp.LpVariable(f"w_{u}_{v}", 0, cap, cat=category)
1166          for (u, v) in pairs}
1167
1168     for (s, t) in pairs:
1169         # s is always on the source side, t always on the sink side: constants,
1170         ↪ not
1171         # variables (which also avoids pulp resetting a Binary's bounds to [0, 1])
1172         ↪ .
1173         side = {u: 1 if u == s else 0 if u == t
1174                 else pulp.LpVariable(f"x_{s}_{t}_{u}", cat="Binary")
1175                 for u in range(n)}
1176         crossing = []
1177         for (u, v) in pairs:
1178             p_uv = pulp.LpVariable(f"p_{s}_{t}_{u}_{v}", 0, cap)
1179             prob += p_uv >= w[(u, v)] + cap * (side[u] - side[v]) - cap
1180             crossing.append(p_uv)
1181         prob += pulp.lpSum(crossing) <= cap # crossing weight <= cap
1182
1183     if two_hop:
1184         z = {}
1185         for (s, x, t) in _two_hop_triples(n):
1186             z[(s, x, t)] = pulp.LpVariable(f"z_{s}_{x}_{t}", 0, cap)
1187             selector = pulp.LpVariable(f"b_{s}_{x}_{t}", cat="Binary")
1188             # z = min(w[s,x], w[x,t]): two upper bounds, and the selector forces z
1189             # up to whichever argument is the minimum (big-M with M = cap).
1190             prob += z[(s, x, t)] <= w[(s, x)]
1191             prob += z[(s, x, t)] <= w[(x, t)]
1192             prob += z[(s, x, t)] >= w[(s, x)] - cap * (1 - selector)
1193             prob += z[(s, x, t)] >= w[(x, t)] - cap * selector
1194         for (s, t) in pairs:
1195             prob += w[(s, t)] + pulp.lpSum(

```

```

1191         z[(s, x, t)] for x in range(n) if x != s and x != t) <= cap
1192
1193     if degree_pair:
1194         #  $d+(s) + d-(t) - w[s,t] \leq (n-1)*cap$ . This is the two-hop bound
1195         # aggregated:  $w[s,t] + \sum_x \min(w[s,x], w[x,t]) \leq cap$ , plus each middle
1196         #  $\max(w[s,x], w[x,t]) \leq cap$ , sums to  $(n-1)*cap$ . It is a genuine cut, NOT
1197         # trivially satisfied: the box bound alone allows the LHS up to  $(2n-3)*cap$ 
1198         #  $\hookrightarrow$ ,
1199         # and a two-route witness reaches  $2(n-1)*cap$  (e.g.  $n=4$ :  $s \rightarrow x_1 \rightarrow t$ ,  $s \rightarrow x_2 \rightarrow t$ 
1200         # gives LHS 4 > 3). Dominated by two_hop when that family is on, but
1201         # standalone-useful when it is off; valid in both weight domains.
1202         for (s, t) in pairs:
1203             prob += (pulp.lpSum(w[(s, x)] for x in range(n) if x != s)
1204                     + pulp.lpSum(w[(x, t)] for x in range(n) if x != t)
1205                     - w[(s, t)]) <= (n - 1) * cap
1206
1207     if deletion:
1208         for k, holes in ((n - 1, 1), (n - 2, 2)):
1209             if k not in _PROVEN_MSTAR:
1210                 continue
1211             bound = cap * _PROVEN_MSTAR[k] # induced subgraph on k vertices
1212             for gone in combinations(range(n), holes):
1213                 prob += pulp.lpSum(w[(a, b)] for (a, b) in pairs
1214                                     if a not in gone and b not in gone) <= bound
1215
1216     if symmetry:
1217         degree = [pulp.lpSum(w[(u, v)] for (u, v) in pairs if v0 in (u, v))
1218                  for v0 in range(n)]
1219         for v0 in range(n - 1):
1220             prob += degree[v0] <= degree[v0 + 1]
1221
1222     return prob, w
1223
1224 def prove_directed_multigraph(
1225     n: int,
1226     *,
1227     time_limit: float = 1500.0,
1228     use_gurobi: bool | None = None,
1229     use_two_hop: bool = True,
1230     use_symmetry_breaking: bool = True,
1231     use_deletion_cuts: bool = False,
1232     show_solver_log: bool = False,
1233 ) -> ProofResult:
1234     """Prove ' $M*(n)$ ' for the directed multigraph arc problem.
1235
1236     Returns a :class:'ProofResult'. When its :meth:'is_proof' is true the value
1237     is a proven upper bound, and ' $value\_for(m)$ ' gives ' $L_m^{dir}(n)$ ' for every
1238     ' $m$ '. ' $use\_gurobi$ ' picks the solver (see :func:'_pick_solver'); the
1239      $\hookrightarrow$  default
1240     uses Gurobi when its licence is active and CBC otherwise.
1241     """
1242     _require_pulp()
1243     prob, w = _cut_counting_model(
1244         n, cap=1.0, integer=False, two_hop=use_two_hop,
1245         symmetry=use_symmetry_breaking, deletion=use_deletion_cuts,
1246         degree_pair=use_deletion_cuts,
1247     )

```

```

1247     prob += pulp.lpSum(w.values())                                # maximise total weight = M*(n
        ↪ )
1248
1249     start = time.time()
1250     prob.solve(_pick_solver(time_limit, show_solver_log, use_gurobi))
1251     elapsed = time.time() - start
1252     status = _PULP_STATUS.get(pulp.LpStatus[prob.status], "LIMIT")
1253
1254     weight_matrix = None
1255     if status == "OPTIMAL":
1256         weight_matrix = np.zeros((n, n))
1257         for (u, v), variable in w.items():
1258             weight_matrix[u, v] = variable.value() or 0.0
1259
1260     return ProofResult(
1261         n=n, status=status,
1262         scaled_optimum=(pulp.value(prob.objective) if status == "OPTIMAL"
1263                        else float("nan")),
1264         relative_gap_zero=(status == "OPTIMAL"),
1265         solve_seconds=elapsed, weight_matrix=weight_matrix,
1266     )

```

C.1.4 Searching

This part corresponds to [Chapter 3](#). Where proving is out of reach, the program looks for good examples instead. It starts from a graph, repeatedly makes a small random change, and keeps the change if it helps. Keeping only improvements gets stuck in the first decent shape it finds, so early on it also accepts some changes that make things worse, and it does this less and less as it goes, in the manner of hot metal cooling into an ordered state. A second engine, tabu search, walks the same landscape with a different rule: it always takes the best available move but refuses to revisit recent positions, which is what stops it circling.

Two ideas make this fast enough to be useful. The search is told which edges are load-bearing, so that it changes those rather than wasting moves on edges that carry nothing. And it almost never measures connectivity fully: adding one connection can raise the count by at most one and removing one can never raise it at all, so most steps can be settled by arithmetic on a bound the search already carries, with an exact measurement only when the answer is genuinely in doubt. This part ends with `solve`, the single entry point that either proves or searches depending on one argument.

```

1269 #####
1270 ##
1271 ## CHAPTER 3 -- DISCOVERING BOUNDS BY SEARCH
1272 ## The temperature search, and the one driver that proves or discovers.
1273 ##
1274 #####
1275
1276 # --- SENSITIVITY: sigma(e) = lambda^max(G) - lambda^max(G-e); dial for the
        ↪ cooling schedule ---
1277
1278 # A connectivity measure maps a graph to its lambda^max or kappa^max.
1279 ConnectivityMeasure = Callable[[Graph], int]

```

```

1280
1281
1282 def edge_sensitivity(
1283     graph: Graph,
1284     u: int,
1285     v: int,
1286     connectivity: ConnectivityMeasure = max_edge_connectivity,
1287 ) -> int:
1288     """Return 'sigma(e)' for the edge or arc 'e = (u, v)'.
1289
1290     The edge is deleted in full (all parallel copies) before remeasuring, so the
1291     result reflects the structural role of the adjacency rather than of a single
1292     parallel copy. If the edge is absent the sensitivity is zero by convention.
1293     """
1294     if not graph.has_edge(u, v):
1295         return 0 # absent edge: nothing to remove, sensitivity is zero
1296     before = connectivity(graph)
1297     reduced = graph.copy()
1298     reduced.set_multiplicity(u, v, 0) # delete the WHOLE adjacency (all copies)
1299     after = connectivity(reduced)
1300     # How much  $\lambda^{\max}$  dropped: positive means e was load-bearing.
1301     return before - after
1302
1303
1304 def sensitivity_map(
1305     graph: Graph,
1306     connectivity: ConnectivityMeasure = max_edge_connectivity,
1307 ) -> dict[tuple[int, int], int]:
1308     """Return '{(u, v): sigma((u, v))}' over all present edges or arcs.
1309
1310     Useful for visualising an extremiser: high-sensitivity edges are the load
1311     bearers, low-sensitivity edges are slack.
1312
1313     The baseline connectivity 'before = connectivity(graph)' is the SAME for
1314     every edge, so it is measured once here rather than re-measured inside each
1315     :func:'edge_sensitivity' call. This changes no value (the result equals
1316     '{e: edge_sensitivity(graph, *e) for e in graph.edges()}'); it only avoids
1317     recomputing the shared baseline once per edge.
1318     """
1319     before = connectivity(graph)
1320     result: dict[tuple[int, int], int] = {}
1321     for u, v, _ in graph.edges():
1322         reduced = graph.copy()
1323         reduced.set_multiplicity(u, v, 0) # delete the WHOLE adjacency (all
            ↪ copies)
1324         result[(u, v)] = before - connectivity(reduced)
1325     return result
1326
1327
1328 # --- SEARCH: simulated annealing;  $E = -|E(G)| + \text{penalty} \cdot \max(0, \lambda^{\max} - (m-1))$ 
            ↪ ---
1329 # Metropolis rule with geometric cooling; removal proposals biased by edge
            ↪ sensitivity.
1330
1331
1332 @dataclass
1333 class SearchStep:
1334     """One step of the search, recorded for plotting the temperature trace."""

```

```

1335
1336     step: int
1337     temperature: float
1338     energy: float
1339     edge_count: int
1340     connectivity: int
1341     accepted: bool
1342     seconds: float = 0.0          # wall-clock since the search began (for timed
        ↳ traces)
1343     best_feasible: int = 0        # densest feasible graph seen so far (for
        ↳ convergence)
1344
1345
1346 @dataclass
1347 class SearchResult:
1348     """The outcome of a search: the best feasible graph and the full trace."""
1349
1350     best_graph: Graph
1351     best_edge_count: int
1352     feasible_found: bool
1353     history: list[SearchStep] = field(default_factory=list)
1354
1355     def acceptance_rate(self) -> float:
1356         """Fraction of proposed moves that were accepted."""
1357         if not self.history:
1358             return 0.0
1359         return sum(record.accepted for record in self.history) / len(self.history)
1360
1361
1362     def _connectivity_measure(separation: str):
1363         """Return the connectivity function named by 'separation'."""
1364         if separation == "edge":
1365             return max_edge_connectivity
1366         if separation == "vertex":
1367             return max_vertex_connectivity
1368         raise ValueError("separation must be 'edge' or 'vertex'")
1369
1370
1371     def _energy(graph: Graph, m: int, measure, penalty: float) -> float:
1372         """The energy '-|E| + penalty * excess connectivity'."""
1373         # Excess is how far connectivity exceeds the allowed m-1; zero when feasible.
1374         excess = max(0, measure(graph) - (m - 1))
1375         # Lower energy is better: more edges (negative term) and no excess.
1376         return -graph.edge_count() + penalty * excess
1377
1378
1379     def _proposal_energy(
1380         proposal: Graph,
1381         *,
1382         is_add: bool,
1383         feasible_current: bool,
1384         lam_ub: int,
1385         m: int,
1386         penalty: float,
1387         measure_full,
1388         separation: str,
1389     ) -> tuple[float, bool]:
1390         """Return '(_energy(proposal), proposal_is_feasible)' doing the least work.

```



```

1391
1392 The energy '-|E| + penalty * max(0, connectivity - (m-1))' depends on the
1393 connectivity only through the excess, and by monotonicity (Facts M1/M2 in the
1394 thesis) that excess is provably zero on most steps, so no flow is needed:
1395
1396 - a removal from a feasible graph stays feasible (M1), excess 0;
1397 - an addition while 'lam_ub <= m-2' stays feasible (M2), excess 0.
1398
1399 Only when neither shortcut applies do we run one capped predicate, and only
    ↳ when
1400 that reports an infeasible proposal do we spend the exact 'measure_full' to
    ↳ get
1401 the penalty term right. The float returned is byte-for-byte the one '_energy
    ↳ '
1402 would return (the same arithmetic on the same excess), so the search
    ↳ trajectory
1403 is unchanged. The boolean is the EXACT feasibility of 'proposal'.
1404 """
1405 edges = proposal.edge_count()
1406 # Decide excess = max(0, measure(proposal) - (m-1)) with the least work.
1407 if (is_add and lam_ub <= m - 2) or (not is_add and feasible_current):
1408     excess = 0 # M2 (add) or M1 (remove): provably
    ↳ feasible
1409 elif not exceeds_bound(proposal, m - 1, separation=separation):
1410     excess = 0 # at the boundary but the capped flow
    ↳ clears it
1411 else:
1412     excess = measure_full(proposal) - (m - 1) # infeasible: exact penalty
    ↳ term
1413 # Identical arithmetic to _energy, so the returned float matches it exactly.
1414 return -edges + penalty * excess, excess == 0
1415
1416
1417 def _candidate_pairs(graph: Graph) -> list[tuple[int, int]]:
1418     """All ordered pairs (directed) or unordered pairs (undirected)."""
1419     n = graph.num_vertices
1420     pairs = []
1421     for u in range(n):
1422         start = 0 if graph.variant.directed else u + 1
1423         for v in range(start, n):
1424             if u != v:
1425                 pairs.append((u, v))
1426     return pairs
1427
1428
1429 def _propose_removal(
1430     graph: Graph,
1431     temperature: float,
1432     rng: random.Random,
1433     cached_sensitivity: dict[tuple[int, int], int] | None,
1434 ) -> tuple[int, int]:
1435     """Choose a present edge to thin out, biased toward free (low sigma) edges.
1436
1437     Without a sensitivity cache the choice is uniform. With one, edge 'e' is
1438     drawn with weight 'exp(-sigma(e) / T)': at high 'T' the weights flatten
1439     (uniform exploration), at low 'T' they concentrate on 'sigma = 0' edges.
1440     """
1441     present = [(u, v) for u, v, _ in graph.edges()]

```

```

1442     if cached_sensitivity is None:
1443         return rng.choice(present) # no guidance: uniform removal
1444     # exp(-sigma/T): big sigma (load-bearing) is suppressed, and the suppression
1445     # sharpens as T -> 0. max(T, eps) guards against division by zero.
1446     weights = [
1447         math.exp(-cached_sensitivity.get((u, v), 0) / max(temperature, 1e-9))
1448         for (u, v) in present
1449     ]
1450     return rng.choices(present, weights=weights, k=1)[0]
1451
1452
1453 def search_for_dense_graph(
1454     variant: Variant,
1455     n: int,
1456     m: int,
1457     *,
1458     separation: str = "edge",
1459     steps: int = 4000,
1460     initial_temperature: float = 3.0,
1461     cooling: float = 0.9985,
1462     penalty: float = 6.0,
1463     max_multiplicity: int | None = None,
1464     sensitivity_guided: bool = True,
1465     bias_refresh: int = 200,
1466     seed: int = 0,
1467     deadline: float | None = None,
1468     record_exact_connectivity: bool = False,
1469     reference_mode: bool = False,
1470 ) -> SearchResult:
1471     """Find the densest feasible graph by a guided random search (simulated
1472         ↪ annealing).
1473
1474     Args:
1475         variant: which graph model to search in.
1476         n: number of vertices.
1477         m: forbidden value. Feasibility means ‘connectivity <= m - 1’.
1478         separation: ‘“edge”’ for arc/edge-disjoint, ‘“vertex”’ for
1479             internally vertex-disjoint paths.
1480         steps: number of proposed moves.
1481         initial_temperature: starting temperature ‘T_0’.
1482         cooling: factor (< 1) the temperature is multiplied by each step.
1483         penalty: weight of the connectivity violation in the energy.
1484         max_multiplicity: cap on parallel edges (defaults to ‘m - 1’, a simple
1485             variant is always capped at one regardless).
1486         sensitivity_guided: bias removals toward free edges (the temperature
1487             dial described in the module docstring).
1488         bias_refresh: recompute the sensitivity cache every this many steps.
1489         seed: random seed, fixed so figures are reproducible.
1490         record_exact_connectivity: log the EXACT ‘lambda^max’ per step in
1491             ‘SearchStep.connectivity’ (the figure-trace needs it). When False,
1492             the cheaper maintained upper bound ‘lam_ub’ is logged instead, which
1493             does not affect the search at all (the energy never reads this field).
1494         reference_mode: INTERNAL, test only. Force the slow, exact per-step
1495             energy (call :func:‘_energy’ and :func:‘max_connectivity’ directly)
1496             instead of the capped/monotone fast path. Used by the equivalence
1497             test to prove the fast path reproduces this reference step for step.
1498
1499     Returns:

```

```

1499         An :class:'SearchResult' with the best feasible graph found and the
1500         full step-by-step history.
1501
1502     The fast path is *trajectory-preserving*: it computes the identical per-step
1503     energy as 'reference_mode' (proven by 'test_search.
1504         ↪ test_fast_path_matches_exact'),
1505     so every accept/reject decision, the random-number draw order, and the
1506         ↪ returned
1507     witness are unchanged. It only avoids flows whose result the energy cannot
1508     depend on (Facts M1/M2: a removal from a feasible graph stays feasible, an
1509     addition while 'lam_ub <= m-2' stays feasible), and resyncs the upper bound
1510     'lam_ub' to the exact value at the 'bias_refresh' cadence.
1511     """
1512     rng = random.Random(seed)
1513     measure = _connectivity_measure(separation)
1514     # Multiplicity cap: simple graphs are always 0/1; multigraphs default to m-1.
1515     cap = 1 if variant.simple else (max_multiplicity if max_multiplicity else m -
1516         ↪ 1)
1517     pairs = None # built lazily once we know we need it
1518
1519     current = Graph(n, variant) # start from the empty graph: feasible, energy 0
1520     current_energy = _energy(current, m, measure, penalty)
1521
1522     best_graph = current.copy()
1523     best_edges = 0
1524     feasible_found = True # the empty graph is feasible
1525
1526     # State carried across steps so the energy can avoid recomputing connectivity
1527     # from scratch (see the trajectory-preserving note in the docstring):
1528     #   feasible -- is 'current' feasible (lambda^max <= m-1)? Exact at all times
1529     ↪ .
1530     #   lam_ub -- a SOUND upper bound on lambda^max(current). Facts M1/M2 keep
1531     ↪ it
1532
1533     #           sound: +1 on an accepted addition, unchanged on a removal.
1534     feasible = True # the empty graph is feasible
1535     lam_ub = 0 # lambda^max(empty graph) = 0
1536
1537     temperature = initial_temperature
1538     cached_sensitivity = None
1539     history: list[SearchStep] = []
1540     clock_start = time.perf_counter() # for the timed convergence trace only
1541
1542     for step in range(steps):
1543         # Periodically recompute the sensitivity map that guides removals. It
1544         # is expensive, so we cache it and refresh only every bias_refresh steps.
1545         if sensitivity_guided and step % bias_refresh == 0 and current.edge_count
1546             ↪ () > 0:
1547             cached_sensitivity = sensitivity_map(current, measure)
1548         # Resync the upper bound to the truth at the same cadence (the bound can
1549         # only drift upward over a run of removals, which costs extra capped
1550         ↪ checks
1551         # but never correctness). This step already pays for flows above, and the
1552         # resync touches neither the energy nor the RNG, so the trajectory is safe
1553         ↪ .
1554         if step % bias_refresh == 0:
1555             lam_ub = measure(current)
1556             feasible = lam_ub <= m - 1
1557

```

```

1549 proposal = current.copy()
1550 # Flip a coin: remove an edge (only if some exist) or add one.
1551 removing = current.edge_count() > 0 and rng.random() < 0.5
1552 if removing:
1553     # Sensitivity-guided removal: prefers free edges, especially cold.
1554     u, v = _propose_removal(current, temperature, rng, cached_sensitivity)
1555     proposal.remove_edge(u, v)
1556 else:
1557     if pairs is None:
1558         pairs = _candidate_pairs(current)
1559     # Only pairs that have room under the multiplicity cap can grow.
1560     addable = [(u, v) for (u, v) in pairs if proposal.multiplicity(u, v) <
1561                 ↪ cap]
1562     if not addable:
1563         continue # nothing to add; skip this step
1564     u, v = rng.choice(addable)
1565     proposal.add_edge(u, v)
1566 is_add = not removing
1567
1568 # The proposal's energy and exact feasibility. reference_mode is the slow
1569 ↪ ,
1570 # exact path the equivalence test compares against; the fast path returns
1571 # the IDENTICAL energy by Facts M1/M2 (see _proposal_energy).
1572 if reference_mode:
1573     proposal_energy = _energy(proposal, m, measure, penalty)
1574     proposal_feasible = measure(proposal) <= m - 1
1575 else:
1576     proposal_energy, proposal_feasible = _proposal_energy(
1577         proposal, is_add=is_add, feasible_current=feasible, lam_ub=lam_ub,
1578         m=m, penalty=penalty, measure_full=measure, separation=separation)
1579 change = proposal_energy - current_energy
1580 # ACCEPT OR NOT: always take an improvement (change <= 0); take a
1581 ↪ worsening
1582 # move only with probability exp(-change / T). As T falls this
1583 ↪ probability
1584 # shrinks, so the walk increasingly refuses to go uphill.
1585 accept = change <= 0 or rng.random() < math.exp(-change / max(temperature,
1586 ↪ 1e-9))
1587
1588 if accept:
1589     current, current_energy = proposal, proposal_energy
1590     # Carry the exact feasibility and update the bound by M2 (an accepted
1591     # addition can raise lambda^max by at most 1; a removal cannot raise
1592     ↪ it).
1593     feasible = proposal_feasible
1594     if is_add:
1595         lam_ub += 1
1596     # Track the best FEASIBLE graph found (within bound and denser than
1597     ↪ any
1598     # seen so far): this is the lower-bound witness. 'feasible' is exact
1599     # here, so the witness is certified with no extra flow.
1600     if feasible and current.edge_count() > best_edges:
1601         best_graph, best_edges = current.copy(), current.edge_count()
1602         feasible_found = True
1603
1604 # The connectivity field feeds fig:trace only. Log the exact value when
1605 ↪ the
1606 # figure asks for it (or in reference_mode), else the maintained upper

```

```

1599         ↪ bound.
1600     if record_exact_connectivity or reference_mode:
1601         recorded_connectivity = measure(current)
1602         # Cheap audit in the exact paths: the maintained bound must never
1603         # underestimate, and the feasibility flag must match the truth.
1604         assert lam_ub >= recorded_connectivity, (lam_ub, recorded_connectivity
1605         ↪ )
1606         assert feasible == (recorded_connectivity <= m - 1)
1607     else:
1608         recorded_connectivity = lam_ub
1609     history.append(SearchStep(
1610         step=step,
1611         temperature=temperature,
1612         energy=current_energy,
1613         edge_count=current.edge_count(),
1614         connectivity=recorded_connectivity,
1615         accepted=accept,
1616         seconds=time.perf_counter() - clock_start,
1617         best_feasible=best_edges,
1618     ))
1619     temperature *= cooling # cool down: T <- cooling * T each step
1620
1621     # Respect the deadline when supplied; check every 100 steps to amortise
1622     # the time.time() call.
1623     if deadline is not None and step % 100 == 99 and time.time() > deadline:
1624         break
1625
1626     return SearchResult(best_graph, best_edges, feasible_found, history)
1627
1628 def _neighbour_moves(graph: Graph, cap: int) -> list[tuple[int, int, int]]:
1629     """Every single add ('+1') or remove ('-1') of one edge/arc under the cap.
1630
1631     The shared neighbourhood of both discoverers: one ordered pair for a digraph,
1632     one unordered pair for an undirected graph, addable while below the
1633     multiplicity 'cap' and removable while above zero.
1634     """
1635     n = graph.num_vertices
1636     directed = graph.variant.directed
1637     moves: list[tuple[int, int, int]] = []
1638     for u in range(n):
1639         for v in range(n):
1640             if u == v or (not directed and v < u):
1641                 continue
1642             mult = graph.multiplicity(u, v)
1643             if mult < cap:
1644                 moves.append((u, v, +1))
1645             if mult > 0:
1646                 moves.append((u, v, -1))
1647     return moves
1648
1649 def tabu_search_for_dense_graph(
1650     variant: Variant,
1651     n: int,
1652     m: int,
1653     *,
1654     separation: str = "edge",

```

```

1655     steps: int = 4000,
1656     penalty: float = 6.0,
1657     tenure: int = 7,
1658     stall_limit: int = 40,
1659     max_multiplicity: int | None = None,
1660     seed: int = 0,
1661     deadline: float | None = None,
1662     record_exact_connectivity: bool = True,
1663 ) -> SearchResult:
1664     """Find the densest feasible graph by tabu search, the deterministic twin of
1665     :func:'search_for_dense_graph'.
1666
1667     Same energy '-|E| + penalty * max(0, connectivity - (m-1))' and the same
1668     neighbourhood (:func:'_neighbour_moves') as the annealer, but a different
1669     engine. Each step scans the whole neighbourhood and takes the best move whose
1670     pair is not on the tabu list, forbids that pair for 'tenure' steps, and
1671     perturbs from the incumbent once it has stalled for 'stall_limit' steps.
1672     Aspiration overrides the tabu flag for any move that beats the global best.
1673     There is no temperature and no accept-worse coin: the only randomness is the
1674     seed driving the perturbation kicks, so two runs from one seed are identical.
1675
1676     Returns the same :class:'SearchResult' as the annealer (the best feasible
1677     ↪ graph
1678     and the full timed history), so the two engines are interchangeable through
1679     ↪ the
1680     'method' argument of :func:'best_of_searches' and :func:'solve'.
1681     """
1682     rng = random.Random(seed)
1683     measure = _connectivity_measure(separation)
1684     cap = 1 if variant.simple else (max_multiplicity if max_multiplicity else m -
1685     ↪ 1)
1686
1687     current = Graph(n, variant) # start empty: feasible, energy 0
1688     best_graph, best_edges = current.copy(), 0
1689     tabu: dict[tuple[int, int], int] = {}
1690     stall = 0
1691     history: list[SearchStep] = []
1692     clock_start = time.perf_counter()
1693
1694     def trial_energy_and_feasibility(graph: Graph) -> tuple[float, bool]:
1695         """The annealer's energy and exact feasibility, by the same capped trick.
1696
1697         'exceeds_bound' decides feasibility with an early-exit capped flow, and
1698         the full 'measure' runs only on the rare infeasible trial that needs its
1699         penalty term, so most moves cost one cheap predicate instead of a full
1700         all-pairs flow. The returned float equals '_energy' exactly.
1701         """
1702         edges = graph.edge_count()
1703         if not exceeds_bound(graph, m - 1, separation=separation):
1704             return float(-edges), True # feasible: excess is zero
1705         return -edges + penalty * (measure(graph) - (m - 1)), False
1706
1707     for step in range(steps):
1708         if deadline is not None and time.time() > deadline:
1709             break
1710
1711         best_move: tuple[int, int, int] | None = None
1712         best_move_energy: float | None = None
1713         best_move_global = False

```

```

1710     for (u, v, d) in _neighbour_moves(current, cap):
1711         trial = current.copy()
1712         trial.add_edge(u, v) if d > 0 else trial.remove_edge(u, v)
1713         trial_energy, trial_feasible = trial_energy_and_feasibility(trial)
1714         improves_global = trial_feasible and trial.edge_count() > best_edges
1715         if tabu.get((u, v), 0) > step and not improves_global:
1716             continue # tabu, with no aspiration to override
1717         if (best_move_energy is None or trial_energy < best_move_energy
1718             or (improves_global and not best_move_global)):
1719             best_move, best_move_energy = (u, v, d), trial_energy
1720             best_move_global = improves_global
1721     if best_move is None: # every move tabu: clear the list,
1722         ↪ retry
1723         tabu.clear()
1724         continue
1725
1726     u, v, d = best_move
1727     current.add_edge(u, v) if d > 0 else current.remove_edge(u, v)
1728     tabu[(u, v)] = step + tenure
1729
1730     connectivity = measure(current) # computed anyway for best-tracking
1731     if connectivity <= m - 1 and current.edge_count() > best_edges:
1732         best_graph, best_edges = current.copy(), current.edge_count()
1733         stall = 0
1734     else:
1735         stall += 1
1736     history.append(SearchStep(
1737         step=step, temperature=0.0, energy=best_move_energy,
1738         edge_count=current.edge_count(),
1739         connectivity=connectivity if record_exact_connectivity else -1,
1740         accepted=True, # tabu always takes its best legal move
1741         seconds=time.perf_counter() - clock_start,
1742         best_feasible=best_edges,
1743     ))
1744
1745     if stall >= stall_limit: # perturb from the incumbent and reset
1746         current = best_graph.copy()
1747         for _ in range(rng.randint(2, 4)):
1748             kicks = _neighbour_moves(current, cap)
1749             if kicks:
1750                 a, b, dd = rng.choice(kicks)
1751                 current.add_edge(a, b) if dd > 0 else current.remove_edge(a, b)
1752                 ↪ )
1753             tabu.clear()
1754             stall = 0
1755
1756     return SearchResult(best_graph, best_edges, feasible_found=True, history=
1757         ↪ history)
1758
1759 def _run_one_search(args: tuple) -> SearchResult:
1760     """Module-level worker for parallel restarts (must be picklable for
1761         ↪ ProcessPoolExecutor)."""
1762     variant, n, m, seed, method, search_options = args
1763     engine = tabu_search_for_dense_graph if method == "tabu" else
1764         ↪ search_for_dense_graph
1765     return engine(variant, n, m, seed=seed, **search_options)
1766

```

```

1763
1764 def best_of_searches(
1765     variant: Variant,
1766     n: int,
1767     m: int,
1768     *,
1769     restarts: int = 6,
1770     seed: int = 0,
1771     parallel: bool = True,
1772     method: str = "sa",
1773     **search_options,
1774 ) -> SearchResult:
1775     """Run several independent discovery schedules and keep the densest result.
1776
1777     A single search can settle into a local optimum, so for reliable
1778     rediscovery we restart from a fresh random seed a handful of times and return
1779     the run that found the most edges. Any keyword understood by the chosen
1780     engine ('steps', 'penalty', 'separation', and so on) is forwarded
1781     unchanged.
1782
1783     Args:
1784         parallel: when 'True' (default), run the restarts concurrently using
1785             :class:`~concurrent.futures.ProcessPoolExecutor`. Each restart has
1786             a distinct seed so the runs are genuinely independent. Falls back
1787             to sequential execution if multiprocessing is unavailable.
1788         method: '"sa"' (default, simulated annealing,
1789             :func:`search_for_dense_graph`) or '"tabu"'
1790             (:func:`tabu_search_for_dense_graph`). Both optimise the same energy
1791             over the same neighbourhood and return the same result type.
1792     """
1793     if method not in ("sa", "tabu"):
1794         raise ValueError("method must be 'sa' or 'tabu'")
1795     task_args = [(variant, n, m, seed + r, method, search_options) for r in range(
1796         ↳ restarts)]
1797
1798     if parallel and restarts > 1:
1799         try:
1800             with concurrent.futures.ProcessPoolExecutor() as executor:
1801                 results = list(executor.map(_run_one_search, task_args))
1802             except (OSError, concurrent.futures.BrokenExecutor) as exc:
1803                 # Multiprocessing is genuinely unavailable here (a sandbox with no
1804                 # fork/spawn, or a worker that could not start). Fall back to a
1805                 # sequential run, but say so rather than masking the loss of cores.
1806                 # A bug inside the worker is NOT caught here, so it fails loudly.
1807                 warnings.warn(
1808                     f"parallel restarts unavailable ({exc!r}); running sequentially",
1809                     RuntimeWarning, stacklevel=2,
1810                 )
1811                 results = [_run_one_search(a) for a in task_args]
1812             else:
1813                 results = [_run_one_search(a) for a in task_args]
1814
1815     return max(results, key=lambda r: r.best_edge_count)
1816
1817 # --- SOLVE: unified driver; exhaustive=True proves, exhaustive=False discovers
1818     ↳ ---

```



```

1819
1820 @dataclass
1821 class SolveResult:
1822     """The outcome of a solve, carrying an honest bound label."""
1823
1824     n: int
1825     m: int
1826     variant: str          # human label, e.g. "simple directed"
1827     separation: str       # "edge" or "vertex"
1828     value: int            # the extremal count we report
1829     bound: str            # "exact" / "lower" / "upper"
1830     method: str           # how the value was obtained
1831     seconds: float
1832     complete: bool        # an exhaustive run that actually finished
1833     witness: object | None # a Graph/Hypergraph attaining 'value', if any
1834     note: str = ""
1835
1836     @property
1837     def proven(self) -> bool:
1838         """Whether 'value' is a proved exact extremal value."""
1839         return self.bound == "exact"
1840
1841     def describe(self) -> str:
1842         """One readable line summarising the result."""
1843         relation = {"exact": "=", "lower": ">=", "upper": "<="}[self.bound]
1844         line = (f"{self.variant} ({self.separation}), n={self.n}, m={self.m}: "
1845                f"value {relation}{self.value} "
1846                f"[{self.bound}, {self.method}, {self.seconds:.1f}s]")
1847         return line if not self.note else f"{line}\n    note: {self.note}"
1848
1849
1850 def _variant_for(directed: bool, simple: bool) -> Variant:
1851     """Pick the matching named Variant constant from the two booleans."""
1852     if directed:
1853         return SIMPLE_DIRECTED if simple else MULTI_DIRECTED
1854     return SIMPLE_UNDIRECTED if simple else MULTI_UNDIRECTED
1855
1856
1857 def _matrix_cells(n: int, directed: bool) -> list[tuple[int, int]]:
1858     """The off-diagonal matrix cells a graph of this kind may fill.
1859
1860     Directed graphs vary every ordered pair; undirected graphs vary only the
1861     upper triangle, since 'set_multiplicity' mirrors the lower half for us.
1862     """
1863     if directed:
1864         return [(u, v) for u in range(n) for v in range(n) if u != v]
1865     return [(u, v) for u in range(n) for v in range(n) if u < v]
1866
1867
1868 def _directed_witness(n: int, m: int, simple: bool) -> Graph | None:
1869     """The densest named directed construction that is feasible for '(n, m)'.
1870
1871     Used to exhibit a concrete graph attaining the proved value: the better
1872     of the double star and the (augmented) one-directional bipartite, or
1873     'None' if neither applies to this case.
1874     """
1875     candidates: list[Graph] = []
1876     if not (simple and m > 2):
1877         # a simple star needs m == 2

```

```

1877         candidates.append(double_star(n, m, directed=True))    #  $2(n-1)(m-1)$  arcs
1878     if m == 2:
1879         candidates.append(one_directional_bipartite(n))          #  $\text{floor}(n^2/4)$  arcs
1880     elif (m - 2) < (n - n // 2):                                # augmented needs  $m-2 < \text{ceil}(n/2)$ 
1881         candidates.append(augmented_bipartite(n, m))
1882     feasible = [g for g in candidates if max_edge_connectivity(g) <= m - 1]
1883     return max(feasible, key=lambda g: g.edge_count()) if feasible else None
1884
1885
1886 def _brute_force_matrix(
1887     variant: Variant, n: int, m: int, separation: str, deadline: float,
1888 ) -> tuple[int, Graph | None, bool]:
1889     """Exhaustively maximise edges over every graph of ‘variant’ on ‘n’.
1890
1891     Returns ‘(best_edge_count, witness, completed)’. ‘completed’ is False if
1892     the budget ran out first, in which case the value is only a lower bound.
1893     A simple cell ranges over ‘{0, 1}’; a multigraph cell over ‘{0, ..., m-1}’
1894     because multiplicity ‘m’ already gives ‘m’ parallel disjoint routes.
1895     """
1896     measure = _connectivity_measure(separation)
1897     cells = _matrix_cells(n, variant.directed)
1898     span = 2 if variant.simple else m          # cell value lives in range(span)
1899     best_count, best_graph, completed = 0, None, True
1900     base = Graph(n, variant)
1901     for tick, values in enumerate(product(range(span), repeat=len(cells))):
1902         if tick % 256 == 0 and time.time() > deadline:
1903             completed = False                  # ran out before exhausting the space
1904             break
1905         candidate = base.copy()
1906         for (u, v), value in zip(cells, values):
1907             if value:
1908                 candidate.set_multiplicity(u, v, value)
1909             if measure(candidate) <= m - 1 and candidate.edge_count() > best_count:
1910                 best_count, best_graph = candidate.edge_count(), candidate.copy()
1911     return best_count, best_graph, completed
1912
1913
1914 def _search_within_budget(
1915     variant: Variant, n: int, m: int, separation: str,
1916     deadline: float, seed: int, method: str = "sa",
1917 ) -> SearchResult:
1918     """Repeated search restarts until the deadline; keep the densest run."""
1919     engine = tabu_search_for_dense_graph if method == "tabu" else
1920         ↪ search_for_dense_graph
1921     best: SearchResult | None = None
1922     restart = 0
1923     while True:
1924         # Pass the deadline through so a single long restart can be cut short.
1925         outcome = engine(variant, n, m, separation=separation,
1926                         steps=4000, seed=seed + restart, deadline=deadline)
1927         if best is None or outcome.best_edge_count > best.best_edge_count:
1928             best = outcome
1929         restart += 1
1930         if time.time() > deadline or restart >= 200:
1931             break
1932     assert best is not None
1933     return best

```

```

1934
1935 def _hyperedge_candidates(n: int, r: int, directed: bool, *, kind: str = "forward"
↳ ) -> list:
1936     """Every possible 'r'-uniform hyperedge.
1937
1938     Undirected: every 'r'-subset. Directed: the 'kind' selects the
1939     orientation model.
1940
1941     * "forward" (default): one tail, 'r - 1' heads, stored in the legacy
1942       '(tail:int, heads)' form so existing forward results are untouched.
1943     * "backward": 'r - 1' tails, one head (the arc-reversal dual of
1944       forward), stored in the general '(tails, heads)' form.
1945     * "general": every split of an 'r'-subset into a non-empty tail set and
1946       a non-empty head set, i.e. forward, backward, and (from 'r >= 4') mixed.
1947     """
1948     if not directed:
1949         return [frozenset(s) for s in combinations(range(n), r)]
1950     if kind == "forward":
1951         # One tail and r-1 heads chosen from the rest (legacy form).
1952         return [(tail, frozenset(heads))
1953                 for tail in range(n)
1954                 for heads in combinations([v for v in range(n) if v != tail], r -
↳ 1)]
1955     if kind == "backward":
1956         # r-1 tails and one head: the reverse of every forward arc.
1957         return [(frozenset(tails), frozenset((head,)))
1958                 for head in range(n)
1959                 for tails in combinations([v for v in range(n) if v != head], r -
↳ 1)]
1960     if kind == "general":
1961         # Every ordered (tails, heads) split of each r-subset, both sides non-
↳ empty.
1962         out: list = []
1963         for subset in combinations(range(n), r):
1964             for mask in range(1, (1 << r) - 1): # exclude all-heads and all-
↳ tails
1965                 tails = frozenset(subset[i] for i in range(r) if mask & (1 << i))
1966                 heads = frozenset(subset[i] for i in range(r) if not (mask & (1 <<
↳ i)))
1967                 out.append((tails, heads))
1968         return out
1969     raise ValueError(f"unknown directed hyperedge kind {kind!r}")
1970
1971
1972 def _brute_force_hypergraph(
1973     n: int, r: int, m: int, deadline: float,
1974     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
1975 ) -> tuple[int, Hypergraph | None, bool]:
1976     """Exhaustively maximise hyperedges over all 'r'-uniform hypergraphs.
1977
1978     Each possible hyperedge is present or absent, so the search visits
1979     '2 ** (#candidates)' hypergraphs; only tiny '(n, r)' finish. The
1980     'directed' and 'vertex_split' flags select which of the four hypergraph
1981     measures decides feasibility, exactly as 'solve' passes them through, and
1982     'kind' selects the directed orientation model (forward/backward/general).
1983     """
1984     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
1985     best_count, best_h, completed = 0, None, True

```

```

1986     for tick, mask in enumerate(product((0, 1), repeat=len(candidates))):
1987         if tick % 256 == 0 and time.time() > deadline:
1988             completed = False
1989             break
1990         chosen = [candidates[i] for i, on in enumerate(mask) if on]
1991         hypergraph = Hypergraph(n, chosen, directed=directed)
1992         if (max_hyper_connectivity(hypergraph, vertex_split=vertex_split) <= m - 1
1993             and hypergraph.edge_count() > best_count):
1994             best_count, best_h = hypergraph.edge_count(), hypergraph
1995     return best_count, best_h, completed
1996
1997
1998 def _random_hypergraph_search(
1999     n: int, r: int, m: int, deadline: float, seed: int,
2000     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
2001 ) -> tuple[int, Hypergraph | None]:
2002     """Greedy randomised growth: add random hyperedges while feasible, restart.
2003
2004     A discovery heuristic for the hypergraph model (search is matrix-only):
2005     each pass shuffles the candidate hyperedges and adds each one that keeps the
2006     Berge connectivity within ‘m - 1’; the densest pass within budget wins. The
2007     feasible hypergraph it returns is the easy construction behind a lower bound.
2008     ‘kind’ selects the directed orientation model.
2009     """
2010     rng = random.Random(seed)
2011     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
2012     best_count, best_h = 0, None
2013     while time.time() < deadline:
2014         order = candidates[:]
2015         rng.shuffle(order)
2016         hypergraph = Hypergraph(n, directed=directed)
2017         for edge in order:
2018             hypergraph.add_hyperedge(edge)
2019             if max_hyper_connectivity(hypergraph, vertex_split=vertex_split) > m -
2020                 ↪ 1:
2021                 hypergraph.hyperedges.pop() # this one broke feasibility; undo
2022             if hypergraph.edge_count() > best_count:
2023                 best_count, best_h = hypergraph.edge_count(), hypergraph
2024     return best_count, (best_h if best_h is not None else Hypergraph(n, directed=
2025         ↪ directed))
2026
2027
2028 def _arc_flow_at_least(out_adj: list[set[int]], n: int, s: int, t: int,
2029     k: int) -> bool:
2030     """True if there are at least ‘k’ arc-disjoint ‘s’-‘t’ paths.
2031
2032     A plain Ford--Fulkerson with unit capacities: find an augmenting path in the
2033     residual (forward arcs and the reverse arcs left by earlier augmentations),
2034     augment, and repeat at most ‘k’ times. Used as the inner feasibility test
2035     of the exhaustive digraph search, where the question is only whether the
2036     local connectivity has reached the forbidden value ‘m’ (so ‘k = m’).
2037     """
2038     cap: dict[tuple[int, int], int] = {}
2039     for a in range(n):
2040         for b in out_adj[a]:
2041             cap[(a, b)] = 1
2042     for _ in range(k):
2043         prev = {s: s}

```

```

2042     stack = [s]                                # DFS for any residual augmenting path
2043     while stack:
2044         x = stack.pop()
2045         if x == t:
2046             break
2047         for y in range(n):
2048             if y not in prev and cap.get((x, y), 0) > 0:
2049                 prev[y] = x
2050                 stack.append(y)
2051     if t not in prev:
2052         return False                                # no further path: fewer than k exist
2053     node = t                                        # walk back, pushing one unit of flow
2054     while node != s:
2055         p = prev[node]
2056         cap[(p, node)] -= 1
2057         cap[(node, p)] = cap.get((node, p), 0) + 1
2058         node = p
2059     return True
2060
2061
2062 def _vertex_flow_at_least(out_adj: list[set[int]], n: int, s: int, t: int,
2063                           k: int) -> bool:
2064     """True if there are at least 'k' internally vertex-disjoint 's'-'t'
2065         ⇨ paths.
2066
2067     The vertex twin of :func:`_arc_flow_at_least`, and the same Ford--Fulkerson
2068     loop, run on the split network instead of the plain one: every vertex 'x'
2069     becomes an entry copy '2x' and an exit copy '2x+1' joined by a single
2070     unit-capacity arc, so a path may pass through 'x' at most once. The two
2071     endpoints keep an uncapped gate, since a route is allowed to start at 's'
2072     and finish at 't'. A direct arc 's -> t' therefore counts as one route
2073     with no interior, exactly as the thesis's separation axis intends.
2074
2075     """
2076     cap: dict[tuple[int, int], int] = {}
2077     for x in range(n):
2078         cap[(2 * x, 2 * x + 1)] = k if (x == s or x == t) else 1
2079     for a in range(n):
2080         for b in out_adj[a]:
2081             cap[(2 * a + 1, 2 * b)] = 1
2082     source, sink = 2 * s + 1, 2 * t
2083     for _ in range(k):
2084         prev = {source: source}
2085         stack = [source]
2086         while stack:
2087             x = stack.pop()
2088             if x == sink:
2089                 break
2090             for y in range(2 * n):
2091                 if y not in prev and cap.get((x, y), 0) > 0:
2092                     prev[y] = x
2093                     stack.append(y)
2094         if sink not in prev:
2095             return False
2096     node = sink
2097     while node != source:
2098         p = prev[node]
2099         cap[(p, node)] -= 1
2100         cap[(node, p)] = cap.get((node, p), 0) + 1

```

```

2099         node = p
2100     return True
2101
2102
2103 def _exhaustive_directed(
2104     n: int, m: int, separation: str, deadline: float,
2105     stats: dict[str, int] | None = None,
2106 ) -> tuple[int, Graph | None, bool]:
2107     """Prove the simple directed maximum by a pruned exhaustive search.
2108
2109     This is the honest version of the thesis's base-case search: branch and bound
2110     over every simple digraph on ' $n$ ' vertices, keeping the densest one whose
2111     connectivity stays within ' $m - 1$ '. Two prunes make it finish where naive
2112     enumeration cannot. Feasibility is monotone, so the include branch adds an
2113     arc only when the digraph is still feasible (an infeasible prefix can never be
2114     rescued by adding more arcs), and the bound prune drops a subtree once even
2115     taking every remaining arc could not beat the best found. Both separations
2116     use the same incremental feasibility test: a new arc can only lift the
2117     connectivity of a pair whose source reaches its tail and whose sink is
2118     reachable from its head, so only those pairs are re-measured, with
2119     :func: '_arc_flow_at_least' or :func: '_vertex_flow_at_least' according to the
2120     separation. Returns '(max_count, witness, completed)'; 'completed' is
2121     False if the time budget ran out first, leaving the value a lower bound.
2122     Pass 'stats' to receive the number of search-tree nodes visited.
2123     """
2124     pairs = [(u, v) for u in range(n) for v in range(n) if u != v]
2125     total = len(pairs)
2126     out: list[set[int]] = [set() for _ in range(n)]
2127     inc: list[set[int]] = [set() for _ in range(n)]
2128     # Seed the best at one below a known construction so the bound prune bites
2129     # immediately, and the search still records an actual witness when it ties it.
2130     # The same seed is valid in both separations: by Whitney's inequality
2131     #  $\kappa \leq \lambda$ , a digraph feasible for the arc problem is feasible for the
2132     # vertex problem too, so an arc construction is an honest vertex lower bound.
2133     seed = _directed_witness(n, m, simple=True)
2134     seed_value = seed.edge_count() if seed is not None else 0
2135     best_count = [max(0, seed_value - 1)]
2136     best_arcs: list[tuple[int, int]] = []
2137     timed_out = [False]
2138
2139     def reaches_to(target: int) -> set[int]:
2140         seen, stack = {target}, [target] # reverse reachability (can reach t)
2141         while stack:
2142             x = stack.pop()
2143             for y in inc[x]:
2144                 if y not in seen:
2145                     seen.add(y); stack.append(y)
2146         return seen
2147
2148     def reaches_from(source: int) -> set[int]:
2149         seen, stack = {source}, [source] # forward reachability (s can reach)
2150         while stack:
2151             x = stack.pop()
2152             for y in out[x]:
2153                 if y not in seen:
2154                     seen.add(y); stack.append(y)
2155         return seen
2156

```

```

2157     flow_at_least = (_arc_flow_at_least if separation == "edge"
2158                     else _vertex_flow_at_least)
2159
2160     def feasible_after(u: int, v: int) -> bool:
2161         for s in reaches_to(u):
2162             for t in reaches_from(v):
2163                 if s != t and flow_at_least(out, n, s, t, m):
2164                     return False # this pair reached m disjoint paths
2165             return True
2166
2167     def recurse(idx: int, count: int) -> None:
2168         if stats is not None:
2169             stats["nodes"] = stats.get("nodes", 0) + 1
2170         if time.time() > deadline:
2171             timed_out[0] = True
2172             return
2173         if count + (total - idx) <= best_count[0]:
2174             return # cannot beat the incumbent: prune
2175         if idx == total:
2176             if count > best_count[0]:
2177                 best_count[0] = count
2178                 best_arcs[:] = [(a, b) for a in range(n) for b in out[a]]
2179             return
2180         u, v = pairs[idx]
2181         out[u].add(v); inc[v].add(u) # include the arc, if still feasible
2182         if feasible_after(u, v):
2183             recurse(idx + 1, count + 1)
2184         out[u].discard(v); inc[v].discard(u)
2185         if timed_out[0]:
2186             return
2187         recurse(idx + 1, count) # exclude the arc
2188
2189     recurse(0, 0)
2190     witness = None
2191     if best_arcs:
2192         witness = Graph(n, SIMPLE_DIRECTED)
2193         for (a, b) in best_arcs:
2194             witness.mu[a, b] = 1
2195     elif seed is not None:
2196         witness = seed # max equalled the seed construction
2197     # A completed search always reaches at least the seed's count, so the max
2198     # below only matters on a timeout before any recorded leaf: there it lifts
2199     # the reported lower bound from seed_value - 1 (the pruning incumbent) to
2200     # the seed witness's own arc count, keeping value and witness consistent.
2201     return max(best_count[0], seed_value), witness, not timed_out[0]
2202
2203
2204 def solve(
2205     n: int, m: int, *,
2206     directed: bool = False, simple: bool = True,
2207     hypergraph: bool = False, r: int = 3,
2208     exhaustive: bool = False, separation: str = "edge",
2209     max_seconds: float = 60.0, seed: int = 0, method: str = "tabu",
2210 ) -> SolveResult:
2211     """The single driver: prove the exact value, or discover a dense example.
2212
2213     Discovery defaults to "method="tabu": tabu search is the stronger engine on
2214     the harder directed problems and the one used to solve the problems in this

```

```

2215 work. Simulated annealing ('method="sa"' is kept for the self-check and
2216 verification, where the lighter engine is enough.
2217
2218 Args:
2219     n, m: the vertices and the forbidden number of independent routes.
2220     directed, simple: the two matrix axes (ignored when 'hypergraph').
2221     hypergraph, r: switch to the 'r'-uniform hypergraph model instead.
2222     exhaustive: 'True' to PROVE the optimum, 'False' to DISCOVER one.
2223     separation: '"edge"' or '"vertex"' disjointness (matrix models).
2224     max_seconds: wall-clock budget; always respected.
2225     seed: random seed for the discovery searches.
2226
2227 Returns:
2228     A :class:'SolveResult' whose 'bound' is '"exact"' (proved),
2229         '"lower"' (a witness), or '"upper"' (proved, no matching witness).
2230     ""
2231     start = time.time()
2232     deadline = start + max_seconds
2233
2234     # ----- the hypergraph model -----
2235     # Same driver, a different internal measure: the edge/vertex separation and
2236     # the direction pick which of the four Berge measures decides feasibility.
2237     if hypergraph:
2238         vertex_split = (separation == "vertex")
2239         label = f"{r}-uniform {'directed ' if directed else ''}hypergraph"
2240         if exhaustive:
2241             value, witness, done = _brute_force_hypergraph(
2242                 n, r, m, deadline, directed=directed, vertex_split=vertex_split)
2243             bound = "exact" if done else "lower"
2244             method = "brute-force enumeration"
2245             note = "" if done else "budget ran out; value is only a lower bound"
2246         else:
2247             value, witness = _random_hypergraph_search(
2248                 n, r, m, deadline, seed, directed=directed, vertex_split=
2249                     ↪ vertex_split)
2250             bound, done, method = "lower", True, "randomised greedy search"
2251             note = "discovery only ever yields a lower bound"
2252         return SolveResult(n, m, label, separation, value, bound, method,
2253                             time.time() - start, done, witness, note)
2254
2255     # ----- the matrix models -----
2256     variant = _variant_for(directed, simple)
2257     label = variant.describe()
2258
2259     # DISCOVER: search within the budget; the witness found is a lower bound.
2260     if not exhaustive:
2261         if method not in ("sa", "tabu"):
2262             raise ValueError("method must be 'sa' or 'tabu'")
2263         result = _search_within_budget(variant, n, m, separation, deadline, seed,
2264             ↪ method)
2265         method_label = "tabu search" if method == "tabu" else "random search"
2266         return SolveResult(
2267             n, m, label, separation, result.best_edge_count, "lower",
2268             method_label, time.time() - start, True,
2269             result.best_graph, "discovery only ever yields a lower bound")
2270
2271     # EXHAUSTIVE, simple directed: a pruned exhaustive digraph search is exact.
2272     # This is the prover for the m=2 base cases of the directed theorem, and it

```



```

2271     # reaches  $n = 6, 7$  where the cut-counting cannot close the gap in any sane
        ↳ time.
2272     if directed and simple:
2273         value, witness, done = _exhaustive_directed(n, m, separation, deadline)
2274         bound = "exact" if done else "lower"
2275         method = "exhaustive digraph search (branch and bound)"
2276         note = "" if done else "budget ran out; value is only a lower bound"
2277         return SolveResult(n, m, label, separation, value, bound, method,
2278                             time.time() - start, done, witness, note)
2279
2280     # EXHAUSTIVE, directed MULTIGRAPH arc problem: the cut-counting is the prover.
2281     if directed and separation == "edge":
2282         budget = max(1.0, deadline - time.time())
2283         proof = prove_directed_multigraph(n, time_limit=budget)
2284         if proof.is_proof():
2285             # We only reach here for a multigraph, where  $(m-1) M*(n)$  IS the value.
2286             value = proof.value_for(m) #  $(m-1) * M*(n)$ 
2287             witness = _directed_witness(n, m, simple)
2288             return SolveResult(n, m, label, separation, value, "exact",
2289                               "cut-counting (gap zero)", time.time() - start,
2290                               True, witness, "")
2291         # Timed out: fall back to the best construction as a lower bound.
2292         witness = _directed_witness(n, m, simple)
2293         low = witness.edge_count() if witness else 0
2294         return SolveResult(
2295             n, m, label, separation, low, "lower",
2296             "cut-counting hit the time limit; reporting a construction",
2297             time.time() - start, False, witness,
2298             "raise max_seconds to let the cut-counting close the gap")
2299
2300     # EXHAUSTIVE otherwise (undirected, or vertex separation): brute force.
2301     value, witness, done = _brute_force_matrix(
2302         variant, n, m, separation, deadline)
2303     bound = "exact" if done else "lower"
2304     method = "brute-force enumeration"
2305     note = (" if done else "budget ran out; value is only a lower bound "
2306            "(no cut-counting exists for this case)")
2307     return SolveResult(n, m, label, separation, value, bound, method,
2308                       time.time() - start, done, witness, note)

```

C.1.5 The random model

Here begins the material behind [Chapter 4](#). This first piece builds random graphs and measures them, which is how the thesis looks at the typical case rather than the extreme one. Each of the six models gets its own sampler, and the multigraph samplers use a two-stage rule: decide whether a pair is connected at all, then decide how many parallel copies it gets, which recovers the ordinary random graph when the second stage is switched off.

```

2311 #####
2312 ##
2313 ## CHAPTER 4 -- SYNTHESIS AND RESULTS
2314 ## Sampling the random model, the thesis figures, and the self-check.
2315 ##
2316 #####
2317

```

```

2318 # --- MONTE CARLO: sample  $G(n,p)$  graphs, measure with exact checker, average ---
2319
2320
2321 def sample_random_graph(n: int, p: float, directed: bool, rng: random.Random) ->
    ↳ Graph:
2322     """Draw one sample of  $G(n, p)$  (or the random digraph  $D(n, p)$ ).
2323
2324     Each possible edge is included independently with probability ‘p’: unordered
2325     pairs for an undirected graph, ordered pairs for a digraph.
2326     """
2327     variant = SIMPLE_DIRECTED if directed else SIMPLE_UNDIRECTED
2328     graph = Graph(n, variant)
2329     for u in range(n):
2330         # Undirected: scan  $v > u$  (each pair once). Directed: scan all  $v$ .
2331         start = 0 if directed else u + 1
2332         for v in range(start, n):
2333             if u != v and rng.random() < p: # include this edge with prob p
2334                 graph.add_edge(u, v)
2335     return graph
2336
2337
2338 def estimate_appearance_probability(
2339     n: int, p: float, m: int, *, trials: int = 200,
2340     separation: str = "edge", directed: bool = False, seed: int = 0,
2341 ) -> float:
2342     """Estimate the probability that a sample has  $\lambda^{\max} \geq m$ ."""
2343     rng = random.Random(seed)
2344     measure = _connectivity_measure(separation)
2345     # Count the fraction of samples that already exhibit  $m$  independent routes
2346     # somewhere ( $\lambda^{\max} \geq m$ ): the empirical appearance probability.
2347     appearances = sum(
2348         measure(sample_random_graph(n, p, directed, rng)) >= m
2349         for _ in range(trials)
2350     )
2351     return appearances / trials
2352
2353
2354 def connectivity_distribution(
2355     n: int, p: float, *, trials: int = 200,
2356     separation: str = "edge", directed: bool = False, seed: int = 0,
2357 ) -> list[int]:
2358     """Return the list of  $\lambda^{\max}$  (or  $\kappa^{\max}$ ) over samples.
2359
2360     Same arguments as the estimators above. The returned list has one entry per
2361     sample, so its histogram shows how the binding connectivity is distributed
2362     for that kind of graph. Calling this with the same ‘seed’ and ‘(n, p)’
2363     but different ‘separation’ reuses the very same random graphs, which is why
2364     the edge and vertex distributions can be compared sample by sample.
2365     """
2366     rng = random.Random(seed)
2367     measure = _connectivity_measure(separation)
2368     # Same seed + same (n, p) reproduces the identical sequence of graphs, so an
2369     # edge run and a vertex run can be compared sample by sample (Whitney check).
2370     return [measure(sample_random_graph(n, p, directed, rng)) for _ in range(
        ↳ trials)]
2371
2372
2373 @dataclass

```

```

2374 class ThresholdCurve:
2375     """The appearance probability sampled across a range of densities."""
2376
2377     n: int
2378     m: int
2379     probabilities: list[tuple[float, float]] # (p, estimated probability)
2380
2381     @property
2382     def predicted_threshold(self) -> float:
2383         """The proved location  $p^{\{*\}} = m/n$ ."""
2384         return self.m / self.n
2385
2386
2387 def threshold_curve(
2388     n: int, m: int, p_values: list[float], *, trials: int = 200,
2389     separation: str = "edge", directed: bool = False, seed: int = 0,
2390 ) -> ThresholdCurve:
2391     """Sweep 'p' and return the appearance probability at each value."""
2392     points = [
2393         (p, estimate_appearance_probability(
2394             n, p, m, trials=trials, separation=separation, directed=directed, seed
2395             ↪ =seed))
2396         for p in p_values
2397     ]
2398     return ThresholdCurve(n=n, m=m, probabilities=points)
2399
2400 # --- edge against vertex disjointness in the random model -----
2401 #
2402 # Whitney's inequality  $\kappa^{\max} \leq \lambda^{\max}$  leaves open whether the two
2403 # binding connectivities of a graph actually differ. For the EXTREMAL problem
2404 # they first part company at  $m = 5$  (Sorensen-Thomassen). The function below lets
2405 # the same split be watched in the random model: it draws samples of  $G(n, p)$  and
2406 # measures both maxima on each one. On small graphs, where the binding
2407 # connectivity stays at or below four, the two coincide on every sample; on
2408 # larger, denser graphs, where it climbs past five, a pair can carry more
2409 # edge-disjoint routes than internally vertex-disjoint ones, and  $\kappa^{\max}$  drops
2410 # below  $\lambda^{\max}$  on a growing fraction of samples.
2411
2412
2413 def edge_vertex_distribution(
2414     n: int, p: float, *, trials: int = 300, seed: int = 0,
2415 ) -> tuple[list[int], list[int]]:
2416     """Return paired  $(\lambda^{\max}, \kappa^{\max})$  over the same samples of  $G(n, p)$ .
2417
2418     Both lists are measured on the identical sequence of random graphs, because
2419     connectivity_distribution reuses the seed, so the two can be compared sample
2420     by sample: the fraction of entries with  $\kappa^{\max} < \lambda^{\max}$  is exactly how
2421     often vertex disjointness is strictly more demanding than edge disjointness
2422     at that size and density.
2423     """
2424     lambda_max = connectivity_distribution(
2425         n, p, trials=trials, separation="edge", seed=seed)
2426     kappa_max = connectivity_distribution(
2427         n, p, trials=trials, separation="vertex", seed=seed)
2428     return lambda_max, kappa_max
2429
2430

```

```

2431 # --- SAMPLING ALL VARIANTS: Bernoulli for simple/hypergraph, hurdle-geometric for
      ↪ multigraph ---
2432
2433
2434 def sample_random_multigraph(
2435     n: int, p: float, alpha: float, directed: bool, rng: random.Random,
2436     *, cap: int | None = None,
2437 ) -> Graph:
2438     """Random multigraph with hurdle-geometric edge multiplicities.
2439
2440     Each cell is empty with probability '1 - p'; otherwise it carries one copy
2441     plus a Geometric('alpha') number of further copies, so the multiplicity
2442     decays exponentially with rate 'alpha' and 'alpha = 0' recovers simple
2443     'G(n, p)'. With 'cap' set (use 'm - 1') a single fat edge cannot
2444     trivially break feasibility, so the panel stays a structural question.
2445     """
2446     variant = MULTI_DIRECTED if directed else MULTI_UNDIRECTED
2447     graph = Graph(n, variant)
2448     for u in range(n):
2449         start = 0 if directed else u + 1
2450         for v in range(start, n):
2451             if u == v or rng.random() >= p:
2452                 continue
2453             k = 1
2454             while rng.random() < alpha:
2455                 k += 1
2456             graph.set_multiplicity(u, v, k if cap is None else min(k, cap))
2457     return graph
2458
2459
2460 def sample_random_hypergraph(
2461     n: int, p: float, r: int, directed: bool, rng: random.Random,
2462 ) -> Hypergraph:
2463     """Random 'r'-uniform hypergraph: each candidate hyperedge included w.p. 'p'
2464     ↪ 'p'"""
2465     chosen = [c for c in _hyperedge_candidates(n, r, directed) if rng.random() < p
2466               ↪ ]
2467     return Hypergraph(n, chosen, directed=directed)
2468
2469
2470 def _sample_variant(n, p, *, directed, simple, hypergraph, r, alpha, cap, rng):
2471     """Draw one sample of whichever generative model the flags select."""
2472     if hypergraph:
2473         return sample_random_hypergraph(n, p, r, directed, rng)
2474     if simple:
2475         return sample_random_graph(n, p, directed, rng)
2476     return sample_random_multigraph(n, p, alpha, directed, rng, cap=cap)
2477
2478
2479 def _measure_variant(obj, *, separation, hypergraph):
2480     """Binding connectivity of a sampled object under the chosen separation."""
2481     if hypergraph:
2482         return max_hyper_connectivity(obj, vertex_split=(separation == "vertex"))
2483     return _connectivity_measure(separation)(obj)
2484
2485
2486 def _mean_binding_degree(obj, *, directed, hypergraph):

```

```

2485     """The degree that the degree bound caps connectivity by, averaged over
        ↪ vertices.
2486
2487     Undirected: total incident multiplicity. Directed: ‘‘min(d+, d-)’’ (the
2488     quantity bounding an arc-disjoint count). Hypergraph: the Berge degree, the
2489     number of hyperedges through a vertex.
2490     """
2491     n = obj.num_vertices
2492     if hypergraph:
2493         deg = [0] * n
2494         for edge in obj.hyperedges:
2495             for v in obj.members(edge):
2496                 deg[v] += 1
2497         return sum(deg) / n
2498     if directed:
2499         return sum(min(obj.out_degree(v), obj.in_degree(v)) for v in range(n)) / n
2500     return sum(obj.degree(v) for v in range(n)) / n
2501
2502
2503 def _p_for_target_degree(target_deg, n, *, directed, simple, hypergraph, r, alpha)
        ↪ :
2504     """Presence probability ‘‘p’’ that puts the expected degree near ‘‘target_deg
        ↪ ‘‘.
2505
2506     Only an aiming heuristic -- the figures plot the *measured* mean degree -- so
2507     the exact value of the cap or of the geometric tail does not need inverting.
2508     """
2509     if hypergraph:
2510         per_vertex = math.comb(n - 1, r - 1)
2511         if directed:
2512             per_vertex += (n - 1) * math.comb(n - 2, r - 2)
2513         return min(1.0, target_deg / max(per_vertex, 1))
2514     copies = 1.0 if simple else 1.0 / (1.0 - alpha)
2515     return min(1.0, target_deg / ((n - 1) * copies))
2516
2517
2518 # Twelve sampled panels, mirroring _VARIANT_ENUM_CONFIGS but at sampling sizes
2519 # well past the enumeration wall (n=6). ‘‘sample_n’’ is chosen per variant so
2520 # the binding-connectivity sweep stays a few minutes overall.
2521 _VARIANT_SAMPLE_CONFIGS: list[dict] = [
2522     dict(key="simple_undirected_edge", title="simple undirected edge",
2523         directed=False, simple=True, hypergraph=False, r=3, separation="edge",
2524         ↪ sample_n=26),
2525     dict(key="simple_undirected_vertex", title="simple undirected vertex",
2526         directed=False, simple=True, hypergraph=False, r=3, separation="vertex",
2527         ↪ sample_n=18),
2528     dict(key="simple_directed_edge", title="simple directed arc",
2529         directed=True, simple=True, hypergraph=False, r=3, separation="edge",
2530         ↪ sample_n=14),
2531     dict(key="simple_directed_vertex", title="simple directed vertex",
2532         directed=True, simple=True, hypergraph=False, r=3, separation="vertex",
2533         ↪ sample_n=14),
2534     dict(key="multi_undirected_edge", title="multigraph undirected edge",
2535         directed=False, simple=False, hypergraph=False, r=3, separation="edge",
2536         ↪ sample_n=22),
2537     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
2538         directed=False, simple=False, hypergraph=False, r=3, separation="vertex",
2539         ↪ sample_n=16),

```

```

2534     dict(key="multi_directed_edge",      title="multigraph directed arc",
2535           directed=True, simple=False, hypergraph=False, r=3, separation="edge",
           ↪ sample_n=12),
2536     dict(key="multi_directed_vertex",    title="multigraph directed vertex",
2537           directed=True, simple=False, hypergraph=False, r=3, separation="vertex",
           ↪ sample_n=12),
2538     dict(key="hyper_undirected_edge",    title="hypergraph undirected edge",
2539           directed=False, simple=True, hypergraph=True, r=3, separation="edge",
           ↪ sample_n=12),
2540     dict(key="hyper_undirected_vertex",  title="hypergraph undirected vertex",
2541           directed=False, simple=True, hypergraph=True, r=3, separation="vertex",
           ↪ sample_n=12),
2542     dict(key="hyper_directed_edge",      title="hypergraph directed arc",
2543           directed=True, simple=True, hypergraph=True, r=3, separation="edge",
           ↪ sample_n=9),
2544     dict(key="hyper_directed_vertex",    title="hypergraph directed vertex",
2545           directed=True, simple=True, hypergraph=True, r=3, separation="vertex",
           ↪ sample_n=9),
2546 ]

```

C.1.6 Drawing the figures

Every picture in the thesis is produced by the routines below, from the same data structures the rest of the program uses. They are collected here rather than in the figure script next door so that a plot and the computation behind it stay in one file.

```

2549 # --- FIGURES: headless matplotlib PNGs; each function takes a path and writes a
           ↪ file ---
2550
2551 _KUL_BLUE = "#1D8DB0"      # feasible / proved
2552 _KUL_DARK = "#1E6E87"     # row labels, known-max lines
2553 _KUL_LIGHT = "#52BDEC"    # named branches
2554 _WARM = "#DC8C28"         # feasibility boundary, certain interval
2555 _RED = "#E05050"          # infeasible bars
2556 _GREEN = "#3CA050"        # machine-checked exact markers
2557 _VIOLET = "#9B5DE5"       # search lower-bound markers
2558 _GUESS = "#E6B800"        # "guess" interpolation: a clear yellow/gold, kept well
2559                             # apart from the red conjecture curve (proved=blue,
2560                             # conjectured=red, guess=yellow)
2561
2562 # Four-level sensitivity palette: cool (sigma=0) -> hot (sigma=max).
2563 _SIGMA_PALETTE = ["#52BDEC", "#F9C74F", "#F4914B", "#DC8C28"] # 0,1,2,3+
2564
2565
2566 def plot_directed_crossover(m: int, max_n: int, path: str | Path) -> None:
2567     """Plot the two competing directed branches and their maximum versus 'n'".
2568
2569     Shows how the linear hub branch 'm(n-1)' is overtaken by the quadratic
2570     augmented-bipartite branch near 'n ~ 2m': the heart of the directed story.
2571     """
2572     ns = list(range(2, max_n + 1))
2573     hub = [m * (n - 1) for n in ns] # linear hub branch
2574     # Quadratic branch: the shifted-partition augmented bipartite count
2575     # floor((n+m-2)^2/4) of const: augmented-bipartite (at m <= 3 it equals the
2576     # balanced-partition count floor(n^2/4) + (m-2)ceil(n/2)).
2577     bipartite = [(n + m - 2) ** 2 // 4 for n in ns]

```

```

2578 envelope = [directed_arc_lower_bound(n, m) for n in ns] # their pointwise max
2579
2580 plt.figure(figsize=(7, 4.3))
2581 plt.plot(ns, hub, "--", color=_KUL_DARK, label=r"hub branch $m(n-1)$")
2582 plt.plot(ns, bipartite, "--", color=_WARM,
2583          label=r"bipartite branch $\lfloor (n+m-2)^2/4 \rfloor$")
2584 plt.plot(ns, envelope, "-", color=_KUL_BLUE, linewidth=2.4, label="their
    ↳ maximum")
2585
2586 # Mark the crossover: the first n at which the quadratic branch overtakes the
2587 # linear one, the order (linear in m) where the directed story changes hands.
2588 cross_n = next((n for n, h, b in zip(ns, hub, bipartite) if b > h), None)
2589 if cross_n is not None:
2590     y_cross = directed_arc_lower_bound(cross_n, m)
2591     plt.axvline(cross_n, color="gray", linestyle=":", linewidth=1.3, alpha
    ↳ =0.8)
2592     plt.scatter([cross_n], [y_cross], color=_KUL_BLUE, zorder=5,
2593                s=45, edgecolor="white", linewidth=0.8)
2594     plt.annotate(f"crossover at $n = {cross_n}$",
2595                xy=(cross_n, y_cross),
2596                xytext=(cross_n - 2.8, max(envelope) * 0.48),
2597                fontsize=9.5, color="black", ha="center",
2598                arrowprops=dict(arrowstyle="->", color="gray", lw=1.1))
2599
2600 plt.xlabel("number of vertices $n$")
2601 plt.ylabel("arcs")
2602 plt.title(f"Directed lower bound at $m = {m}$")
2603 plt.legend(loc="upper left")
2604 plt.grid(True, alpha=0.3)
2605 _save(path)
2606
2607
2608 def plot_edge_vertex_divergence(max_n: int, path: str | Path) -> None:
2609     """Show agreement through  $m \leq 4$  and divergence at  $m=5$ , all starting at the same
    ↳  $n$ .
2610
2611     All curves begin at  $n_{\text{start}}$  so the reader can compare directly.
2612     For  $m=2,3,4$  the two problems coincide (one line each, blue shades).
2613     At  $m=5$  they part: edge below, vertex above, with the gap shaded.
2614     """
2615     n_start = 4 # common start for all curves
2616     ns = list(range(n_start, max_n + 1))
2617
2618     plt.figure(figsize=(8, 5))
2619
2620     #  $m=2,3,4$ : edge == vertex -- one curve per  $m$ 
2621     agree_palette = ["#AED9EE", "#5CB4D9", _KUL_DARK]
2622     for m_val, color in zip([2, 3, 4], agree_palette):
2623         vals = [simple_undirected_edge(n, m_val) for n in ns]
2624         plt.plot(ns, vals, "-", color=color, linewidth=1.8, alpha=0.9,
2625                label=f"$m={m\_val}$: edge $=$ vertex")
2626
2627     #  $m=5$ : edge and vertex (same starting point, diverge from  $n=6$  onwards)
2628     edge5 = [simple_undirected_edge(n, 5) for n in ns]
2629     vert5 = [simple_undirected_vertex_m5(n) for n in ns]
2630     plt.plot(ns, edge5, "--", color=_KUL_BLUE, linewidth=2.4,
2631            label=r"$m=5$: edge $\ell_5(n)=\lfloor 5(n-1)/2 \rfloor$")
2632     plt.plot(ns, vert5, "-", color=_WARM, linewidth=2.4,

```

```

2633         label=r"$m=5$: vertex $k_5(n)=\lfloor 8n/3 \rfloor - 3$"
2634     # shade only where the gap is positive
2635     plt.fill_between(ns, edge5, vert5, where=[v > e for v, e in zip(vert5, edge5)
        ↪ ],
2636                     alpha=0.15, color=_WARM)
2637
2638     plt.xlabel("number of vertices $n$")
2639     plt.ylabel("maximum edges")
2640     plt.title("Agreement through $m \leq 4$, first divergence at $m = 5$")
2641     plt.legend(fontsize=9, loc="upper left")
2642     plt.grid(True, alpha=0.3)
2643     _save(path)
2644
2645
2646     # NOTE: plot_degree_threshold generated the 2-D random-sampling threshold figure
2647     # (the old Figure 4.2) removed from the thesis on 2026-06-20. Kept here so the
2648     # threshold phenomenon can be restored; see
2649     # research_notes/removed_threshold_phenomenon.md for the recipe. (Its 3-D
2650     # companion plot_conn_threshold_3d is still used, for the appendix.)
2651     def plot_degree_threshold(
2652         path: str | Path, *, n: int = 12, m: int = 4, alpha: float = 0.5,
2653         trials: int = 70, seed: int = 7,
2654         targets=(0.3, 0.5, 0.65, 0.8, 0.95, 1.1, 1.3, 1.6, 2.0),
2655     ) -> None:
2656         """Appearance probability against mean binding-degree, across every model.
2657
2658         Six random models -- simple, multigraph and $3$-uniform hypergraph, each
2659         undirected and directed -- are swept in density and plotted by their
2660         *measured* mean binding degree (in units of 'm') against the probability
2661         that a sample already contains a pair of (Berge) edge-connectivity at least
2662         'm'. The vertical line at degree '= m' is the asymptotic threshold the
2663         degree argument of thm:gnp-threshold predicts; at the fixed small 'm' a
2664         computation can reach the curves are ORDERED rather than coincident -- a
2665         directed multigraph forces the pair at the least degree (heavy parallel edges
2666         concentrate connectivity), a directed hypergraph at the most (a hyperedge
2667         serves only one Berge route), with the simple cases between. Universality is
2668         the 'm/ln n -> infinity' limit; the spread here is the finite-m gap.
2669         """
2670         models = [
2671             ("simple, undirected", dict(directed=False, simple=True,
        ↪ hypergraph=False, r=3), _KUL_LIGHT, "o", "-"),
2672             ("simple, directed", dict(directed=True, simple=True,
        ↪ hypergraph=False, r=3), _KUL_BLUE, "s", "-"),
2673             ("multigraph, undirected", dict(directed=False, simple=False,
        ↪ hypergraph=False, r=3), _WARM, "o", "--"),
2674             ("multigraph, directed", dict(directed=True, simple=False,
        ↪ hypergraph=False, r=3), "#C0392B", "s", "--"),
2675             ("3-uniform hyper, undir.", dict(directed=False, simple=True,
        ↪ hypergraph=True, r=3), _VIOLET, "^", ":"),
2676             ("3-uniform hyper, dir.", dict(directed=True, simple=True,
        ↪ hypergraph=True, r=3), _GREEN, "^", ":"),
2677         ]
2678     plt.figure(figsize=(7.8, 5.0))
2679     for label, spec, colour, mk, ls in models:
2680         xs, ys = [], []
2681         for t in targets:
2682             p = _p_for_target_degree(t * m, n, alpha=alpha, **spec)
2683             rng = random.Random(seed)

```



```

2684         degs, hits = [], 0
2685         for _ in range(trials):
2686             obj = _sample_variant(n, p, alpha=alpha, cap=m - 1, rng=rng, **
                ↪ spec)
2687             degs.append(_mean_binding_degree(
2688                 obj, directed=spec["directed"], hypergraph=spec["hypergraph"])
                ↪ )
2689             if _measure_variant(obj, separation="edge",
2690                                 hypergraph=spec["hypergraph"]) >= m:
2691                 hits += 1
2692             xs.append(sum(degs) / len(degs) / m)
2693             ys.append(hits / trials)
2694             plt.plot(xs, ys, ls, marker=m, color=colour, markersize=4.5,
2695                     linewidth=1.8, label=label)
2696             plt.axvline(1.0, color="black", linestyle="--", linewidth=1.6,
2697                         label=r"asymptotic threshold (degree $= m$)")
2698             plt.axhline(0.5, color="gray", linestyle=":", linewidth=1.0, alpha=0.6)
2699             plt.xlabel(r"mean binding degree, in units of $m$")
2700             plt.ylabel(r"$\hat{P}\backslash, \backslash, \text{binding connectivity} \geq m\backslash, \backslash$")
2701             plt.title(f"Appearance vs expected degree, by model "
2702                      f"$n = {n}$, $m = {m}$, multiplicities capped at $m-1$")
2703             plt.ylim(-0.03, 1.03)
2704             plt.legend(fontsize=8.5, loc="lower right")
2705             plt.grid(True, alpha=0.3)
2706             _save(path)
2707
2708
2709 def plot_connectivity_distribution(series: dict, n: int, p: float, path: str |
    ↪ Path) -> None:
2710     """Plot overlaid histograms of the binding connectivity for several models.
2711
2712     ‘series’ maps a label to a list of connectivity values (one per random
2713     sample), as produced by :func:‘montecarlo.connectivity_distribution’. The
2714     integer-valued histograms are drawn on shared integer bins so the shapes can
2715     be compared directly across the kinds of graph.
2716
2717     # Shared integer bins across every series so the shapes line up exactly.
2718     all_values = [value for values in series.values() for value in values]
2719     low, high = min(all_values), max(all_values)
2720     levels = list(range(low, high + 1))
2721     colours = [_KUL_BLUE, _WARM, _KUL_DARK, _KUL_LIGHT]
2722     group_count = len(series)
2723     bar_width = 0.8 / group_count # split each integer slot among the series
2724
2725     plt.figure(figsize=(7, 4.3))
2726     for index, ((label, values), colour) in enumerate(zip(series.items(), colours)
    ↪ ):
2727         counts = [values.count(level) for level in levels]
2728         # Offset each series' bars so grouped bars sit side by side per level.
2729         offset = (index - (group_count - 1) / 2) * bar_width
2730         positions = [level + offset for level in levels]
2731         plt.bar(positions, counts, width=bar_width, color=colour, label=label,
2732                 edgecolor="white", linewidth=0.4)
2733     plt.xticks(levels)
2734     plt.xlabel("binding connectivity")
2735     plt.ylabel("number of samples")
2736     plt.title(f"Connectivity over random graphs, $n = {n}$, $p = {p}$")
2737     plt.legend()

```

```

2738     plt.grid(True, axis="y", alpha=0.3)
2739     _save(path)
2740
2741
2742 _FRACTION_LABEL = {0.25: "1/4", 0.5: "1/2", 0.75: "3/4"}
2743
2744
2745 def plot_edge_vertex_histograms(
2746     data: dict[float, tuple[list[int], list[int]]],
2747     n: int,
2748     p_values: list[float],
2749     path: str | Path,
2750 ) -> None:
2751     """Six histograms of the binding connectivity in  $G(n, p)$ , at one size  $n$ .
2752
2753     'data' maps each density 'p' to a pair '(lambda_max_values,
2754     kappa_max_values)' measured on the same samples, from
2755     :func:'montecarlo.edge_vertex_distribution'. The grid is two rows by three
2756     columns: the top row is the edge-connectivity ' $\lambda^{\max}$ ' and the bottom
2757     row the vertex-connectivity ' $\kappa^{\max}$ ', one column per density. Reading a
2758     column top to bottom compares the two distributions at a fixed density, so
2759     Whitney's inequality shows up as the vertex histogram sitting at or to the
2760     left of the edge one. The fraction of samples with ' $\kappa^{\max} <
2761     \lambda^{\max}$ ' is printed on each vertex panel. All six share one axis range.
2762     """
2763     rows = [(r"edge  $\lambda^{\max}$ ", 0, _KUL_BLUE),
2764             (r"vertex  $\kappa^{\max}$ ", 1, _WARM)]
2765     all_values = [v for p in p_values for series in data[p] for v in series]
2766     lo, hi = min(all_values), max(all_values)
2767     bins = range(lo, hi + 2) # one integer-wide bin per connectivity value
2768     trials = len(data[p_values[0]][0])
2769
2770     fig, axes = plt.subplots(2, len(p_values), figsize=(11, 6),
2771                             sharex=True, sharey=True)
2772     for col, p in enumerate(p_values):
2773         lam, kap = data[p]
2774         gap = 100 * sum(k < l for k, l in zip(kap, lam)) / len(lam)
2775         plabel = _FRACTION_LABEL.get(p, f"{p:.2f}")
2776         for label, ridx, colour in rows:
2777             axis = axes[ridx][col]
2778             values = lam if ridx == 0 else kap
2779             axis.hist(values, bins=bins, align="left", color=colour,
2780                     edgecolor="white", linewidth=0.4, rwidth=0.9)
2781             axis.axvline(statistics.mean(values), color="black", linestyle="--",
2782                         linewidth=1.2)
2783             if ridx == 0:
2784                 axis.set_title(f"$p = {plabel}$")
2785             else:
2786                 axis.set_xlabel("binding connectivity")
2787                 axis.text(0.03, 0.92, fr"$\kappa \!< \! \lambda$: {gap:.0f}%",
2788                         transform=axis.transAxes, ha="left", va="top",
2789                         fontsize=9, color=_WARM)
2790             if col == 0:
2791                 axis.set_ylabel(f"{label}\nsamples")
2792             axis.grid(True, axis="y", alpha=0.3)
2793
2794     fig.suptitle(fr"Distribution of the binding connectivity in  $G(\{n\}, p)$  "
2795                fr"over {trials} samples", fontsize=12)

```

```

2796     _save(path)
2797
2798
2799 def plot_search_trace(result: SearchResult, path: str | Path,
2800                      optimum: int | None = None,
2801                      ceiling: int | None = None,
2802                      *,
2803                      show_cooling: bool = True,
2804                      show_histogram: bool = True,
2805                      connectivity_label: str = r"$\lambda^{\max}$",
2806                      edge_label: str = "arcs",
2807                      title: str = "Cooling toward an extremal graph") -> None:
2808     """Plot the search trajectory; optionally include the cooling schedule.
2809
2810     Layout: [cooling schedule (optional)] | [scatter] | [visit histogram].
2811
2812     The scatter colours points by step number with a heavily nonlinear
2813     PowerNorm (gamma=0.2, RdYlBu colourmap) so that only the brief hot
2814     exploration phase (early steps) stands out in warm reds/yellows, while
2815     the long converged tail is uniformly blue. When 'ceiling' is given the
2816     feasibility boundary is marked and "feasible"/"infeasible" labels are
2817     placed just above the axes box using a blended data/axes transform, so
2818     they sit centred over their respective shaded regions regardless of zoom.
2819
2820     The rightmost panel is an optional horizontal density histogram showing how
2821     many search steps landed at each edge-count level; it shares the y-axis with
2822     the scatter so the bars align with the dots.
2823
2824     New keyword arguments (all optional, all have sensible defaults):
2825     - 'show_cooling': set False to omit the left cooling-schedule panel.
2826     - 'show_histogram': set False to omit the right visit-density panel.
2827     - 'connectivity_label': x-axis label for the scatter (default  $\lambda^{\max}$ ).
2828     - 'edge_label': y-axis label (default "arcs"; use "edges" for undirected).
2829     - 'title': figure supptitle.
2830     """
2831     from collections import Counter
2832     from matplotlib.colors import PowerNorm
2833     from matplotlib.transforms import blended_transform_factory
2834     from matplotlib.gridspec import GridSpec
2835
2836     steps = [record.step for record in result.history]
2837     temperature = [record.temperature for record in result.history]
2838     edges = [record.edge_count for record in result.history]
2839     connectivity = [record.connectivity for record in result.history]
2840
2841     # Build layout: cooling (optional) | scatter | histogram (optional).
2842     # Histogram axes use no sharey -- that avoids a tight_layout warning, so the
2843     # y-limits are synced to the scatter manually after it is fully drawn.
2844     if show_cooling and show_histogram:
2845         fig = plt.figure(figsize=(13.5, 4.8))
2846         gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.6, 0.28],
2847                      wspace=0.40)
2848         ax_cool = fig.add_subplot(gs[0])
2849         ax_scatter = fig.add_subplot(gs[1])
2850         ax_hist = fig.add_subplot(gs[2])
2851     elif show_cooling and not show_histogram:
2852         fig = plt.figure(figsize=(11.5, 4.8))
2853         gs = GridSpec(1, 2, figure=fig, width_ratios=[1.0, 1.6], wspace=0.40)

```

```

2854     ax_cool = fig.add_subplot(gs[0])
2855     ax_scatter = fig.add_subplot(gs[1])
2856     ax_hist = None
2857     elif (not show_cooling) and show_histogram:
2858         fig = plt.figure(figsize=(8.5, 4.8))
2859         gs = GridSpec(1, 2, figure=fig, width_ratios=[1.6, 0.28], wspace=0.12)
2860         ax_cool = None
2861         ax_scatter = fig.add_subplot(gs[0])
2862         ax_hist = fig.add_subplot(gs[1])
2863     else:
2864         fig = plt.figure(figsize=(6.6, 4.8))
2865         ax_cool = None
2866         ax_scatter = fig.add_subplot(1, 1, 1)
2867         ax_hist = None
2868
2869     # Left: cooling schedule.
2870     if ax_cool is not None:
2871         ax_cool.plot(steps, temperature, color=_WARM, linewidth=1.8)
2872         ax_cool.set_xlabel("step")
2873         ax_cool.set_ylabel("temperature")
2874         ax_cool.set_title("Cooling schedule")
2875         ax_cool.grid(True, alpha=0.3)
2876
2877     # Middle: scatter of (connectivity, edge_count), coloured by step.
2878     ax = ax_scatter
2879     lo = min(connectivity) - 0.5
2880     hi = max(connectivity) + 0.5
2881     if ceiling is not None:
2882         boundary = ceiling + 0.5
2883         ax.axvspan(lo, boundary, color=_GREEN, alpha=0.10)
2884         ax.axvspan(boundary, hi, color=_RED, alpha=0.10)
2885         ax.axvline(boundary, color=_WARM, linestyle="--", linewidth=1.4)
2886         ax.set_xlim(lo, hi)
2887         # Labels centred over each shaded region, placed just above the axes box.
2888         # blended_transform mixes data-x (tracks the region) with axes-fraction-y
2889         # (1.03 is always just above the top of the box regardless of zoom level).
2890         xt = blended_transform_factory(ax.transData, ax.transAxes)
2891         ax.text((lo + boundary) / 2, 1.03, "feasible",
2892                transform=xt, ha="center", va="bottom",
2893                fontsize=8.5, color=_GREEN, fontweight="bold", clip_on=False)
2894         ax.text((boundary + hi) / 2, 1.03, "infeasible",
2895                transform=xt, ha="center", va="bottom",
2896                fontsize=8.5, color="#C03030", fontweight="bold", clip_on=False)
2897
2898     # Nonlinear colour norm: gamma < 1 squashes large step values toward 1,
2899     # mapping the long converged tail (most points) to the BLUE end of RdYlBu
2900     # while only the first few hot steps appear in warm reds and yellows.
2901     max_step = max(steps) if steps else 1
2902     norm = PowerNorm(gamma=0.2, vmin=0, vmax=max_step)
2903     sc = ax.scatter(connectivity, edges, c=steps, cmap="RdYlBu",
2904                    norm=norm, s=16, alpha=0.75, linewidths=0)
2905
2906     if optimum is not None:
2907         ax.axhline(optimum, color=_KUL_DARK, linestyle="--", linewidth=1.4,
2908                   label=f"certified optimum = {optimum}")
2909     if optimum is not None and ceiling is not None:
2910         ax.plot([ceiling], [optimum], "*", color="#D11A2A", markersize=20,
2911                markeredgewidth=0.9, zorder=5,

```

```

2912         label="densest feasible graph")
2913     ax.set_xlabel(connectivity_label + " of the visited graph")
2914     ax.set_ylabel(edge_label)
2915     # No panel title: the supitle and the feasible/infeasible labels describe
2916     # the panel; a panel title would overlap with the labels above the axes box.
2917     ax.xaxis.set_major_locator(MaxNLocator(integer=True))
2918     ax.grid(True, alpha=0.3)
2919     handles, labels = ax.get_legend_handles_labels()
2920     if labels:
2921         ax.legend(loc="upper left", fontsize=8, framealpha=0.9)
2922     cbar = fig.colorbar(sc, ax=ax, shrink=0.92)
2923     cbar.set_label(r"step (cooling $\rightarrow$)")
2924     cbar.set_ticks([0, max_step // 4, max_step // 2,
2925                    3 * max_step // 4, max_step])
2926
2927     # Right: horizontal histogram -- visit density per edge-count level.
2928     # Sync y-limits with the scatter manually (avoids tight_layout warning
2929     # that sharey axes cause inside _save's plt.tight_layout() call).
2930     if ax_hist is not None:
2931         visit_counts = Counter(edges)
2932         arc_vals = sorted(visit_counts)
2933         ax_hist.barh(arc_vals, [visit_counts[a] for a in arc_vals],
2934                     height=0.72, color=_KUL_BLUE, alpha=0.75, linewidth=0)
2935         ax_hist.set_ylim(ax.get_ylim())
2936         ax_hist.set_xlabel("visits", fontsize=8)
2937         ax_hist.set_title("Distribution", fontsize=9)
2938         ax_hist.tick_params(axis="y", left=False, labelleft=False)
2939         ax_hist.xaxis.set_major_locator(MaxNLocator(integer=True, nbins=3))
2940         ax_hist.tick_params(labelsize=7)
2941         ax_hist.grid(True, axis="x", alpha=0.3)
2942
2943     fig.suptitle(title, fontsize=12, y=1.04)
2944     _save(path)
2945
2946
2947 def draw_graph_with_sensitivity(
2948     graph: Graph,
2949     path: str | Path,
2950     layout: dict | None = None,
2951     show_sigma: bool = False,
2952     node_labels: dict | None = None,
2953 ) -> None:
2954     """Draw a directed multigraph with every parallel arc drawn, coloured by sigma
2955     → .
2956
2957     Each parallel copy of an adjacency is rendered as its own curved arc, so the
2958     multiplicity ‘‘ $\mu(u, v)$ ’’ is read by *counting* arcs rather than from a label,
2959     and the whole adjacency is coloured by its sensitivity ‘‘ $\sigma(e)$ ’’ -- how far
2960     ‘‘ $\lambda^{\max}$ ’’ falls when the adjacency is deleted -- on a four-level
2961     cool-to-hot palette. An over-provisioned adjacency (several parallel copies
2962     but a downstream bottleneck) therefore shows as a bundle of arcs in a cool
2963     colour: the visual point that multiplicity and load-bearing role differ.
2964
2965     Pass a ‘‘layout’’ dict mapping vertex id to (x, y) and a ‘‘node_labels’’ dict
2966     to override the numeric labels. ‘‘show_sigma’’ is accepted for backward
2967     compatibility and ignored (sigma is always shown).
2968     """
2969     from matplotlib.patches import FancyArrowPatch

```

```

2969
2970 sensitivities = sensitivity_map(graph)
2971 if layout is None:
2972     # Evenly spaced on the unit circle (the same picture nx.circular_layout
2973     # draws), computed directly so drawing needs no networkx.
2974     verts = list(graph.vertices())
2975     layout = {v: (math.cos(2 * math.pi * i / len(verts)),
2976                  math.sin(2 * math.pi * i / len(verts)))
2977              for i, v in enumerate(verts)}
2978 labels = (node_labels if node_labels is not None
2979          else {v: str(v) for v in graph.vertices()})
2980
2981 fig, ax = plt.subplots(figsize=(8.6, 6.2))
2982
2983 for (u, v), sigma in sensitivities.items():
2984     mu = graph.multiplicity(u, v)
2985     colour = _SIGMA_PALETTE[min(sigma, len(_SIGMA_PALETTE) - 1)]
2986     # Symmetric fan of curvatures, one per parallel copy.
2987     if mu <= 1:
2988         rads = [0.0]
2989     else:
2990         spread = 0.13 * (mu - 1)
2991         rads = [-spread + 2 * spread * k / (mu - 1) for k in range(mu)]
2992     for rad in rads:
2993         ax.add_patch(FancyArrowPatch(
2994             layout[u], layout[v], connectionstyle=f"arc3,rad={rad}",
2995             arrowstyle="-|>", mutation_scale=13, lw=2.3, color=colour,
2996             shrinkA=13, shrinkB=13, zorder=1))
2997         # One sigma label per adjacency, at the chord midpoint.
2998         (x0, y0), (x1, y1) = layout[u], layout[v]
2999         ax.text((x0 + x1) / 2, (y0 + y1) / 2, f"$\\sigma={sigma}$",
3000              fontsize=9, ha="center", va="center", zorder=3,
3001              bbox={"boxstyle": "round,pad=0.18", "fc": "white",
3002                  "ec": colour, "alpha": 0.9})
3003
3004 xs = [p[0] for p in layout.values()]
3005 ys = [p[1] for p in layout.values()]
3006 ax.scatter(xs, ys, s=620, c=_KUL_DARK, edgecolors="white", linewidths=1.2,
3007           zorder=2)
3008 for v, (x, y) in layout.items():
3009     ax.text(x, y, str(labels.get(v, v)), color="white", fontweight="bold",
3010           ha="center", va="center", zorder=4)
3011
3012 mx = (max(xs) - min(xs)) * 0.12 + 0.3
3013 my = (max(ys) - min(ys)) * 0.12 + 0.3
3014 ax.set_xlim(min(xs) - mx, max(xs) + mx)
3015 ax.set_ylim(min(ys) - my, max(ys) + my)
3016 ax.set_aspect("equal")
3017 ax.axis("off")
3018 ax.set_title("Parallel arcs drawn explicitly, coloured by sensitivity "
3019             "$\\sigma$: \ncool ($\\sigma=0$, free) $\\to$ warm (load-bearing)"
3020             "\n↔",
3021             fontsize=12)
3022 _save(path)
3023
3024 def _running_best_trace(result: SearchResult) -> tuple[list[float], list[int]]:
3025     """Best feasible edge count so far against wall-clock seconds, from a history.

```

```

3026
3027 Reads the timed :class:'SearchStep' log (its 'best_feasible' field already
3028 carries the densest feasible graph seen up to each step) and returns the timed
3029 convergence curve both engines record in their native fast modes.
3030 """
3031 xs = [record.seconds for record in result.history]
3032 ys = [record.best_feasible for record in result.history]
3033 return xs, ys
3034
3035
3036 def _settle_time(xs: list[float], ys: list[int]) -> float:
3037     """Wall-clock second at which a trace first reached its own final best value.
3038
3039     The convergence curves plateau long before the time budget runs out, so this
3040     is what the x-axis should track: cropping to the budget would leave a long
3041     flat tail and the cropped view would barely change when the budget does.
3042     """
3043     if not xs:
3044         return 0.0
3045     final = ys[-1]
3046     return next((x for x, y in zip(xs, ys) if y == final), xs[-1])
3047
3048
3049 def plot_sa_vs_tabu_convergence(
3050     path: str | Path, *,
3051     cases: tuple[tuple[int, int], ...] = ((5, 3), (7, 3)),
3052     budget: float = 8.0,
3053     seed: int = 0,
3054     xmaxs: tuple[float, ...] | None = None,
3055     settle_margin: float = 1.3,
3056 ) -> None:
3057     """Best-feasible-value against wall-clock for annealing vs tabu search.
3058
3059     For each '(n, m)' it runs one simulated-annealing schedule
3060     (:func:'search_for_dense_graph') and one tabu schedule
3061     (:func:'tabu_search_for_dense_graph') on the directed multigraph at an equal
3062     wall-clock 'budget', and overlays their timed best-so-far traces against the
3063     conjectured optimum. The chosen cases are where the two engines diverge: tabu
3064     assembles the bipartite extremiser and reaches the optimum while annealing
3065     plateaus a few arcs short. The time axis is wall-clock, so this one figure is
3066     a representative timed run on the reference machine rather than a seed-exact
3067     reproduction like the others.
3068
3069     By default each panel's x-axis stops shortly after the slower engine settles
3070     ('settle_margin' times the later of the two plateau times), so the view
3071     tracks the actual run instead of a fixed window: cropping to a hard-coded
3072     second count leaves a long flat tail because both engines converge well
3073     inside one second. Pass 'xmaxs' to force fixed per-panel limits instead.
3074     """
3075     if not MATPLOTLIB_AVAILABLE:
3076         raise RuntimeError("matplotlib is required for figures")
3077     fig, axes = plt.subplots(1, len(cases), figsize=(11, 4.3))
3078     if len(cases) == 1:
3079         axes = [axes]
3080     for i, (ax, (n, m)) in enumerate(zip(axes, cases)):
3081         sa = search_for_dense_graph(MULTI_DIRECTED, n, m, seed=seed, steps=10**7,
3082                                     deadline=time.time() + budget)
3083         tb = tabu_search_for_dense_graph(MULTI_DIRECTED, n, m, seed=seed,

```

```

3084         steps=10**7, deadline=time.time() +
           ↪ budget)

3085     settle = 0.0
3086     for res, label, colour in ((sa, "simulated annealing", _KUL_BLUE),
3087                               (tb, "tabu search", _WARM)):
3088         xs, ys = _running_best_trace(res)
3089         ax.step(xs, ys, where="post", label=label, color=colour, linewidth
           ↪ =2.0)
3090         settle = max(settle, _settle_time(xs, ys))
3091     opt = directed_multigraph_arc(n, m)
3092     ax.axhline(opt, linestyle=":", color=_KUL_DARK, linewidth=1.6,
3093               label=f"optimum  $L_3^{\{\mathrm{dir}\}}(n) = \{opt\}$ ")
3094     ax.set_title(f"directed multigraph,  $n = \{n\}$ ,  $m = \{m\}$ ", fontsize=11)
3095     ax.set_xlabel("wall-clock seconds", fontsize=9.5)
3096     ax.set_ylabel("densest feasible arc count", fontsize=9.5)
3097     # Crop the long flat tail: stop just after the slower engine settles, so
3098     # the rise fills the panel instead of being squeezed against the y-axis.
3099     # A fixed second count cannot do this, since the settle time shifts run to
3100     # run; an explicit amaxes overrides for a reproducible fixed window.
3101     if xaxes is not None and i < len(xaxes):
3102         ax.set_xlim(0, xaxes[i])
3103     else:
3104         ax.set_xlim(0, max(settle * settle_margin, 0.5))
3105     ax.grid(True, alpha=0.3)
3106     ax.legend(fontsize=8.5, loc="lower right")
3107     # Fewer x-ticks declutters the time axis: the action concentrates early
3108     # and dense ticks add noise.
3109     if MaxNLocator is not None:
3110         ax.xaxis.set_major_locator(MaxNLocator(nbins=5))
3111     _save(path)
3112
3113
3114 def _save(path: str | Path, *, tight: bool = True) -> None:
3115     """Tighten the layout, write the file, and close the figure.
3116
3117     ‘tight=False’ skips ‘tight_layout’ for the 3-D figures, where it
3118     misbehaves with the projected axes (they manage their own spacing).
3119     """
3120     import warnings
3121     Path(path).parent.mkdir(parents=True, exist_ok=True)
3122     if tight:
3123         # tight_layout can warn when figures contain colorbars or shared axes;
3124         # bbox_inches="tight" in savefig handles the actual bounding box so the
3125         # layout warning is cosmetic and safe to suppress.
3126         with warnings.catch_warnings():
3127             warnings.simplefilter("ignore", UserWarning)
3128             plt.tight_layout()
3129     plt.savefig(path, dpi=150, bbox_inches="tight")
3130     plt.close() # release the figure so head-less runs don't leak memory
3131
3132
3133 def plot_complexity_growth(path: str | Path) -> None:
3134     """The combinatorial explosion: number of graphs to enumerate, by direction.
3135
3136     Two panels, directed (left) and undirected (right), sharing one axis range.
3137     Each plots, on a log scale, how many graphs blind enumeration must visit
3138     against the vertex count ‘n’, for the simple model, two multigraph
3139     connectivity caps, and (dotted) the 3-uniform hypergraph, with a dashed line

```



```

3140 at a generous budget of  $10^9$  candidates. The count does not depend on the
3141 edge-versus-vertex separation, because both search the very same set of
3142 graphs, so one pair of panels covers all of those variants at once.
3143 Direction doubles the number of cells to fill, which is the whole difference
3144 between the two panels.
3145 """
3146 ns = list(range(2, 13))
3147
3148 def log10_pow(base: float, exponent: float) -> float:
3149     return exponent * math.log10(base)
3150
3151 def curves(directed: bool):
3152     # Each model fills a number of off-diagonal cells; the candidate count is
3153     # (values per cell) ^ (number of cells). Labels stay plain -- the precise
3154     # exponential forms live in the caption, not the legend. Direction
3155     # multiplies the cell count: by 2 for graphs (ordered vs unordered pairs,
3156     #  $n(n-1)$  vs  $C(n,2)$ ), but by 3 for 3-uniform hypergraphs, since a directed
3157     # hyperedge is a tail plus 2 heads ( $nC(n-1,2) = 3nC(n,3)$  candidates).
3158     cells = (lambda n: n * (n - 1)) if directed else (lambda n: n * (n - 1) /
3159         ↪ 2)
3160     hyper_cells = ((lambda n: n * (n - 1) * (n - 2) / 2) if directed
3161         else (lambda n: n * (n - 1) * (n - 2) / 6))
3162     return [
3163         ("simple / random", _KUL_BLUE, "-",
3164         [log10_pow(2, cells(n)) for n in ns]),
3165         ("multigraph  $m=3$ ", _KUL_DARK, "-",
3166         [log10_pow(3, cells(n)) for n in ns]),
3167         ("multigraph  $m=4$ ", _WARM, "-",
3168         [log10_pow(4, cells(n)) for n in ns]),
3169         (" $3$ -uniform hypergraph", _VIOLET, ":",
3170         [log10_pow(2, hyper_cells(n)) for n in ns]),
3171     ]
3172
3173 fig, axes = plt.subplots(1, 2, figsize=(11, 4.6), sharey=True)
3174 for ax, directed, title in [(axes[0], True, "directed graphs"),
3175     (axes[1], False, "undirected graphs")]:
3176     for label, colour, style, vals in curves(directed):
3177         ax.plot(ns, vals, style, color=colour, marker="o", markersize=3.5,
3178             label=label)
3179     ax.axhline(9, color="gray", linestyle="--", linewidth=1,
3180         label=r"$10^{\{9\}}$ budget")
3181     ax.set_title(title)
3182     ax.set_xlabel("vertices  $n$ ")
3183     ax.grid(True, alpha=0.3)
3184     # Cap the shared vertical axis at 100 so the slowest curve (the directed
3185     # 3-uniform hypergraph) runs off the top instead of compressing every other
3186     # curve into a thin band near the bottom. sharey=True propagates the limit.
3187     axes[0].set_ylim(0, 100)
3188     axes[0].annotate(r"$3$-uniform directed" + "\n" + r"continues off the top",
3189         xy=(9.55, 99.3), xytext=(7.5, 92), fontsize=8, color=_VIOLET,
3190         ha="center", va="center",
3191         arrowprops=dict(arrowstyle="->", color=_VIOLET, lw=1.0))
3192     axes[0].set_ylabel(r"$\log_{10}$ (number of graphs)")
3193     axes[0].legend(fontsize=8.5, loc="upper left", title="model")
3194     fig.suptitle("Number of graphs blind enumeration must visit",
3195         fontsize=12)
3196     _save(path)

```

```

3197
3198 def _circle_layout(n, hub=None):
3199     """Vertex positions for a panel: all on a unit circle, or the ‘hub‘ vertex
3200     at the centre and the rest on the circle, which draws stars and hub graphs
3201     far more clearly than a plain circle does."""
3202     if hub is None:
3203         ang = [math.pi / 2 - 2 * math.pi * k / n for k in range(n)]
3204         return [(math.cos(a), math.sin(a)) for a in ang]
3205     pos = [(0.0, 0.0) for _ in range(n)]
3206     others = [k for k in range(n) if k != hub]
3207     for idx, k in enumerate(others):
3208         a = math.pi / 2 - 2 * math.pi * idx / len(others)
3209         pos[k] = (math.cos(a), math.sin(a))
3210     return pos
3211
3212
3213 def _draw_extremal_panel(ax, matrix, *, directed: bool, hub=None) -> None:
3214     """Draw one extremal multiplicity matrix on ‘ax‘ as a graph.
3215
3216     Vertices sit on a circle (or, when ‘hub‘ is given, that vertex sits at the
3217     centre and the rest on the circle). An adjacency of multiplicity one is a
3218     single line (an arrowed line when ‘directed‘), and a higher multiplicity is
3219     shown by a small count label rather than parallel curves, so the picture
3220     stays legible. Opposite arcs of a bidirected pair are bent apart.
3221
3222     Every named family the gallery draws is *uniform*: all its adjacencies carry
3223     one and the same multiplicity. Repeating that count on every arc buries the
3224     structure under labels (worst on the bidirected ‘K4‘, twelve labels over
3225     six crossing arcs), so a uniform multiplicity above one is stated once in a
3226     corner note, and the per-arc labels are kept only for the genuinely mixed
3227     case. ‘hub‘ may also be an explicit list of ‘(x, y)’ positions, used
3228     for the bipartite wall panel, where two columns read far better than a circle.
3229     """
3230     from matplotlib.patches import FancyArrowPatch
3231     n = len(matrix)
3232     pos = hub if isinstance(hub, list) else _circle_layout(n, hub)
3233     mults = {int(matrix[i][j]) for i in range(n) for j in range(n)
3234              if i != j and matrix[i][j] > 0}
3235     uniform = mults.pop() if len(mults) == 1 else None
3236     per_arc_labels = uniform is None
3237
3238     def edge(i, j, rad, arrow):
3239         ax.add_patch(FancyArrowPatch(
3240             pos[i], pos[j], connectionstyle=f"arc3,rad={rad}",
3241             arrowstyle="->" if arrow else "-"), mutation_scale=12,
3242             lw=1.7, color=_KUL_BLUE, shrinkA=10, shrinkB=10, zorder=1))
3243
3244     def mlabel(i, j, rad, mu):
3245         if mu <= 1:
3246             return
3247         mx, my = (pos[i][0] + pos[j][0]) / 2, (pos[i][1] + pos[j][1]) / 2
3248         dx, dy = pos[j][0] - pos[i][0], pos[j][1] - pos[i][1]
3249         length = math.hypot(dx, dy) or 1.0
3250         mx += (rad * 0.9 + 0.06) * (-dy) / length
3251         my += (rad * 0.9 + 0.06) * (dx) / length
3252         ax.text(mx, my, f"$\\times{mu}$", fontsize=8, ha="center", va="center",
3253                 color=_KUL_DARK, zorder=3,
3254                 bbox=dict(boxstyle="round,pad=0.08", fc="white", ec="none"))

```

```

3255
3256     if directed:
3257         for i in range(n):
3258             for j in range(n):
3259                 if i == j or matrix[i][j] == 0:
3260                     continue
3261                 rad = 0.16 if matrix[j][i] else 0.0
3262                 edge(i, j, rad, True)
3263                 if per_arc_labels:
3264                     mlabel(i, j, rad, matrix[i][j])
3265     else:
3266         for i in range(n):
3267             for j in range(i + 1, n):
3268                 if matrix[i][j] == 0:
3269                     continue
3270                 edge(i, j, 0.0, False)
3271                 if per_arc_labels:
3272                     mlabel(i, j, 0.0, matrix[i][j])
3273     for k, (x, y) in enumerate(pos):
3274         ax.scatter([x], [y], s=300, c="white", edgecolors=_KUL_BLUE,
3275                   linewidths=1.6, zorder=2)
3276         ax.text(x, y, str(k + 1), ha="center", va="center", fontsize=9,
3277               zorder=3, color=_KUL_DARK)
3278     if uniform is not None and uniform > 1:
3279         word = "arcs" if directed else "edges"
3280         ax.text(0.0, -1.34, f"all {word}  $\times$ {uniform}", ha="center",
3281               va="center", fontsize=9, color=_KUL_DARK)
3282     ax.set_xlim(-1.45, 1.45)
3283     ax.set_ylim(-1.45, 1.45)
3284
3285
3286     # metro-line palette for hypergraph panels: red, blue, green, orange, purple, grey
3287     _GALLERY_METRO = ["#C85050", "#1D8DB0", "#3CA050", "#DC8C28", "#8C50A0", "#787878"
3288                      ↪ ]
3289
3290 def _draw_hyper_panel(ax, hyperedges, n, *, directed: bool, hub=None) -> None:
3291     """Draw a hypergraph extremiser as a metro map: vertices on a circle (or a
3292     hub at the centre), each hyperedge a thick coloured line through its members
3293     when undirected, or a fan of coloured arrows from its tail when directed."""
3294     from matplotlib.patches import FancyArrowPatch
3295     pos = _circle_layout(n, hub)
3296     for idx, he in enumerate(hyperedges):
3297         col = _GALLERY_METRO[idx % len(_GALLERY_METRO)]
3298         if directed:
3299             tail, heads = he
3300             tx, ty = pos[tail]
3301             for h in heads:
3302                 hx, hy = pos[h]
3303                 ax.add_patch(FancyArrowPatch(
3304                     (tx, ty), (hx, hy), arrowstyle="->", mutation_scale=11,
3305                     lw=2.4, color=col, alpha=0.85, shrinkA=11, shrinkB=11,
3306                     zorder=1, connectionstyle="arc3,rad=0.10"))
3307         else:
3308             pts = [pos[v] for v in sorted(he)]
3309             cx = sum(p[0] for p in pts) / len(pts)
3310             cy = sum(p[1] for p in pts) / len(pts)
3311             pts.sort(key=lambda p: math.atan2(p[1] - cy, p[0] - cx))

```

```

3312         ax.plot([p[0] for p in pts], [p[1] for p in pts], "-", color=col,
3313                 lw=4.2, solid_capstyle="round", solid_joinstyle="round",
3314                 alpha=0.85, zorder=1)
3315     for k, (x, y) in enumerate(pos):
3316         ax.scatter([x], [y], s=300, c="white", edgecolors="#333333",
3317                 linewidths=1.5, zorder=2)
3318         ax.text(x, y, str(k + 1), ha="center", va="center", fontsize=9,
3319                 zorder=3, color="#222222")
3320     ax.set_xlim(-1.45, 1.45)
3321     ax.set_ylim(-1.45, 1.45)
3322
3323
3324 def plot_extremal_gallery(path: str | Path, *,
3325                          gallery_json: str | Path | None = None) -> None:
3326     """A gallery of the named extremal graphs, drawn at a large representative
3327         ↪ size.
3328
3329     These are the structured extremisers the analysis identifies, not the
3330     trivial small trees: the directed double and bidirected stars, the doubled
3331     bipartite wall, the dense bidirected complete graph, the multigraph star at
3332     full multiplicity, and the star hypertrees (drawn as metro maps). The program
3333     ↪ both builds these and,
3334     at small 'n', proves them optimal by the enumeration of
3335     :func:'gallery_extremal_graphs'. The 'gallery_json' argument is accepted for
3336     backward compatibility and ignored.
3337     """
3338     if not MATPLOTLIB_AVAILABLE:
3339         raise RuntimeError("matplotlib is required for figures")
3340     # dense bidirected complete graphs, built directly
3341     bidir_k4 = Graph(4, SIMPLE_DIRECTED)
3342     for i in range(4):
3343         for j in range(4):
3344             if i != j:
3345                 bidir_k4.set_multiplicity(i, j, 1)
3346     # 2.B_{3,4}: the doubled one-directional bipartite wall, one of the three
3347     # proved 24-arc extremisers at n = 7, m = 3 (rem:odd-step-roadmap, fact (b)).
3348     # It ties the double star at the crossover, so the gallery shows both branches
3349     ↪ .
3350     bip_2b34 = Graph(7, MULTI_DIRECTED)
3351     for a in range(3):
3352         for b in range(3, 7):
3353             bip_2b34.set_multiplicity(a, b, 2)
3354     bip_pos = [(-0.85, 0.85), (-0.85, 0.0), (-0.85, -0.85),
3355               (0.85, 1.05), (0.85, 0.35), (0.85, -0.35), (0.85, -1.05)]
3356     multi_star = Graph(9, MULTI_UNDIRECTED) # undirected star, mult 2
3357     for i in range(1, 9):
3358         multi_star.set_multiplicity(0, i, 2)
3359     dir_hypertree = [(0, frozenset({1, 2})), (0, frozenset({3, 4})),
3360                     (0, frozenset({5, 6}))] # directed star hypertree on 7
3361     ↪ vertices
3362
3363     # ("g"/"h"/"hd", object, directed, hub, title)
3364     panels = [
3365         ("g", double_star(7, 3, directed=True), True, 0,
3366          "multigraph directed, arc\n$m=3$, $n=7$: double star ($24$ arcs)"),
3367         ("g", double_star(7, 2, directed=True), True, 0,
3368          "simple directed, arc\n$m=2$, $n=7$: bidirected star ($12$ arcs)"),
3369         ("g", multi_star, False, 0,

```

```

3366         "multigraph undirected, edge\n$m=3$, $n=9$: star at multiplicity $2$ (
           ↪ $16$ edges)"),
3367     ("g", bidir_k4, True, None,
3368         "simple directed, arc\n$m=4$, $n=4$: bidirected $K_4$ ($12$ arcs)"),
3369     ("g", bip_2b34, True, bip_pos,
3370         "multigraph directed, arc\n$m=3$, $n=7$: bipartite wall $2 \cdot B_{\{3,4\}}$
           ↪ $24$ arcs)"),
3371     ("h", star_hypertree(10, 3), False, 0,
3372         "hypergraph, $r=3$\n$m=2$, $n=10$: star hypertree ($4$ hyperedges)"),
3373     ("h", star_hypertree(13, 4), False, 0,
3374         "hypergraph, $r=4$\n$m=2$, $n=13$: star hypertree ($4$ hyperedges)"),
3375     ("h", star_hypertree(16, 3), False, 0,
3376         "hypergraph, $r=3$\n$m=2$, $n=16$: star hypertree ($7$ hyperedges)"),
3377     ("hd", (dir_hypertree, 7), True, 0,
3378         "directed hypergraph, $r=3$\n$m=2$, $n=7$: tail-to-heads star"),
3379 ]
3380 fig, axes = plt.subplots(3, 3, figsize=(12, 12))
3381 for ax, (kind, obj, directed, hub, title) in zip(axes.flat, panels):
3382     if kind == "g":
3383         _draw_extremal_panel(ax, obj.mu.tolist(), directed=directed, hub=hub)
3384     elif kind == "h":
3385         _draw_hyper_panel(ax, list(obj.hyperedges), obj.num_vertices,
3386                         directed=False, hub=hub)
3387     else: # "hd": (hyperedges, n)
3388         hes, nn = obj
3389         _draw_hyper_panel(ax, hes, nn, directed=True, hub=hub)
3390     ax.set_title(title, fontsize=9.5)
3391     ax.set_aspect("equal")
3392     ax.axis("off")
3393 fig.suptitle("Extremal graphs of the named families, drawn large", fontsize
           ↪ =14)
3394 _save(path)
3395
3396
3397 _GUESS_STYLE = (0, (1, 1)) # dense dots: "we are very much guessing"
3398
3399
3400 def plot_variant_grid(panels: list[dict], path: str | Path,
3401                      m: int | None = None) -> None:
3402     """All twelve variants on one grid, in the proved/conjectured/guessed language
           ↪ .
3403
3404     ‘panels’ is a row-major list of twelve dictionaries (rows = model simple,
3405     multi, hyper; columns = undirected edge, undirected vertex, directed arc,
3406     directed vertex). Each panel may carry any of:
3407
3408     ‘proved’ ‘(xs, ys)’ solid blue line, a theorem holding for all ‘n’;
3409     ‘conj’ ‘(xs, ys)’ solid red line, a conjecture formula;
3410     ‘guess’ ‘(xs, ys)’ solid yellow line, a bare extrapolation of the
3411     machine points where no formula is known;
3412     ‘band’ ‘(xs, lo, hi)’ the certain interval, an easy construction below
3413     and the trivial maximum edge count above;
3414     ‘exact’ ‘(xs, ys)’ filled squares, sizes the machine proved;
3415     ‘search’ ‘(xs, ys)’ open circles, the search lower bounds.
3416
3417     The point is one honest picture, distinguished by colour rather than dash
3418     pattern: a blue line is settled, a red line is conjectured, a yellow line is
3419     a guess, and the shaded band is the interval we are certain the truth lies in.

```

```

3420 """
3421 def draw_panel(ax, panel):
3422     band = panel.get("band")
3423     if band is not None:
3424         xs, lo, hi = band
3425         ax.fill_between(xs, lo, hi, color=_WARM, alpha=0.12,
3426                         label="certain interval")
3427         ax.plot(xs, lo, "-", color=_WARM, linewidth=0.8, alpha=0.6)
3428         ax.plot(xs, hi, "-", color=_WARM, linewidth=0.8, alpha=0.6)
3429     for branch in panel.get("branches", []):
3430         bxs, bys, blabel = branch
3431         ax.plot(bxs, bys, ":", color=_KUL_LIGHT, linewidth=1.3, label=blabel)
3432     if panel.get("proved") is not None:
3433         xs, ys = panel["proved"]
3434         ax.plot(xs, ys, "-", color=_KUL_BLUE, linewidth=2.3, label="proved")
3435     if panel.get("conj") is not None:
3436         xs, ys = panel["conj"]
3437         ax.plot(xs, ys, "-", color=_RED, linewidth=2.0, label="conjectured")
3438     if panel.get("guess") is not None:
3439         xs, ys = panel["guess"]
3440         ax.plot(xs, ys, "-", color=_GUESS, linewidth=2.0,
3441                 label="guess (interpolated)")
3442     if panel.get("exact") is not None:
3443         xs, ys = panel["exact"]
3444         if len(xs):
3445             ax.plot(xs, ys, "s", color=_GREEN, markersize=7,
3446                     label="machine-checked (exact)")
3447     if panel.get("search") is not None:
3448         xs, ys = panel["search"]
3449         if len(xs):
3450             ax.plot(xs, ys, "o", mfc="none", mec=_VIOLET, mew=1.8,
3451                     markersize=8, label="search (lower bound)")
3452     ax.set_title(panel["title"], fontsize=9.5)
3453     ax.set_xlabel("vertices $n$", fontsize=8.5)
3454     ax.set_ylabel(panel.get("ylabel", "edges"), fontsize=8.5)
3455     ax.tick_params(labelsize=8)
3456     ax.grid(True, alpha=0.3)
3457     ax.legend(fontsize=6.8, loc="upper left", framealpha=0.85)
3458
3459     # The line styles (solid / dashed / dotted) are named in the per-panel
3460     # legends and spelled out in the caption, so the supitle stays short and
3461     # just records the variant family and the threshold m.
3462     supitle = "Erdős 915 across the twelve variants"
3463     if m is not None:
3464         supitle += fr"    $m = {m}$"
3465     _variant_panel_grid(draw_panel, configs=panels, supitle=supitle, path=path,
3466                         supitle_fontsize=13)

```

C.1.7 Walking the whole space

At small sizes it is possible to visit every graph there is and record what it looks like, which is what these routines do. The result is the distribution behind the landscape figures, and, more importantly, a way to confirm a claimed maximum by exhaustion rather than by search. The three-dimensional surfaces are built on the same idea, run across a grid of sizes and route limits and cached so that the sweep is paid for once.

```

3469 # --- ENUMERATION LANDSCAPES: visit every labeled graph, collect (edges, lambda^
    ↪ max) ---
3470
3471 # Row-major table of all twelve variants, matching gather_variant_grid.
3472 # "enum_n" is the vertex count used for full enumeration.
3473 # "max_mult" caps per-cell multiplicity for multigraph variants.
3474 _VARIANT_ENUM_CONFIGS: list[dict] = [
3475     # row 1 -- simple
3476     dict(key="simple_undirected_edge", title="simple undirected edge",
3477          directed=False, simple=True, hypergraph=False, r=3, separation="edge",
3478          enum_n=6, max_mult=1),
3479     dict(key="simple_undirected_vertex", title="simple undirected vertex",
3480          directed=False, simple=True, hypergraph=False, r=3, separation="vertex",
3481          enum_n=6, max_mult=1),
3482     dict(key="simple_directed_edge", title="simple directed arc",
3483          directed=True, simple=True, hypergraph=False, r=3, separation="edge",
3484          enum_n=4, max_mult=1),
3485     dict(key="simple_directed_vertex", title="simple directed vertex",
3486          directed=True, simple=True, hypergraph=False, r=3, separation="vertex",
3487          enum_n=4, max_mult=1),
3488     # row 2 -- multigraph
3489     dict(key="multi_undirected_edge", title="multigraph undirected edge",
3490          directed=False, simple=False, hypergraph=False, r=3, separation="edge",
3491          enum_n=5, max_mult=2),
3492     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
3493          directed=False, simple=False, hypergraph=False, r=3, separation="vertex",
3494          enum_n=5, max_mult=2),
3495     dict(key="multi_directed_edge", title="multigraph directed arc",
3496          directed=True, simple=False, hypergraph=False, r=3, separation="edge",
3497          enum_n=3, max_mult=5),
3498     dict(key="multi_directed_vertex", title="multigraph directed vertex",
3499          directed=True, simple=False, hypergraph=False, r=3, separation="vertex",
3500          enum_n=3, max_mult=5),
3501     # row 3 -- hypergraph r=3
3502     dict(key="hyper_undirected_edge", title="hypergraph undirected edge",
3503          directed=False, simple=True, hypergraph=True, r=3, separation="edge",
3504          enum_n=5, max_mult=1),
3505     dict(key="hyper_undirected_vertex", title="hypergraph undirected vertex",
3506          directed=False, simple=True, hypergraph=True, r=3, separation="vertex",
3507          enum_n=5, max_mult=1),
3508     dict(key="hyper_directed_edge", title="hypergraph directed arc",
3509          directed=True, simple=True, hypergraph=True, r=3, separation="edge",
3510          enum_n=4, max_mult=1),
3511     dict(key="hyper_directed_vertex", title="hypergraph directed vertex",
3512          directed=True, simple=True, hypergraph=True, r=3, separation="vertex",
3513          enum_n=4, max_mult=1),
3514 ]
3515
3516
3517 def enumerate_all_graphs(
3518     n: int, *,
3519     directed: bool,
3520     simple: bool,
3521     hypergraph: bool = False,
3522     r: int = 3,
3523     max_mult: int = 3,
3524     separation: str = "edge",
3525 ) -> list[tuple[int, int]]:

```

```

3526     """Return '(edge_count, lambda_max)' for every labeled graph of this type on
        ↳ 'n' vertices.
3527
3528     For simple graphs each cell is 0/1; for multigraphs each cell ranges over
3529     '{0,...,max_mult}'. For hypergraphs each candidate hyperedge is present or
3530     absent. The list covers the full labeled graph space (isomorphic copies
3531     included), which is what we want for the distribution over the search space.
3532     """
3533     results: list[tuple[int, int]] = []
3534
3535     if hypergraph:
3536         vertex_split = (separation == "vertex")
3537         candidates = _hyperedge_candidates(n, r, directed)
3538         for mask in product((0, 1), repeat=len(candidates)):
3539             chosen = [candidates[i] for i, flag in enumerate(mask) if flag]
3540             h = Hypergraph(n, chosen, directed=directed)
3541             conn = max_hyper_connectivity(h, vertex_split=vertex_split)
3542             results.append((h.edge_count(), conn))
3543         return results
3544
3545     variant = _variant_for(directed, simple)
3546     measure = _connectivity_measure(separation)
3547     cells = _matrix_cells(n, directed)
3548     span = 2 if simple else (max_mult + 1)
3549     base = Graph(n, variant)
3550     for values in product(range(span), repeat=len(cells)):
3551         candidate = base.copy()
3552         for (u, v), value in zip(cells, values):
3553             if value:
3554                 candidate.set_multiplicity(u, v, value)
3555             conn = measure(candidate)
3556             results.append((candidate.edge_count(), conn))
3557     return results
3558
3559
3560 def _pair_connectivities(obj, *, separation: str, hypergraph: bool) -> list[int]:
3561     """Local connectivity of every vertex pair of one graph or hypergraph.
3562
3563     The same pair sweep 'max_connectivity' runs, but keeping each value rather
3564     than only the maximum. For the edge case the capacity matrix is built once
3565     and reused across pairs (as 'max_connectivity' does); vertex and hypergraph
3566     cases reuse their named per-pair routines.
3567     """
3568     vsplit = (separation == "vertex")
3569     if hypergraph:
3570         return [hyper_connectivity(obj, s, t, vertex_split=vsplit)
3571                 for s, t in _pairs(obj)]
3572     if not vsplit:
3573         csr = _csr(obj.mu, dtype=int)
3574         return [int(_csgraph_maxflow(csr, s, t).flow_value) for s, t in _pairs(obj)
3575                 ↳ ]
3576     return [local_connectivity(obj, s, t, vertex_split=True) for s, t in _pairs(
3577         ↳ obj)]
3578
3579 def enumerate_pair_connectivities(
3580     n: int, *,
3581     directed: bool,

```



```

3581     simple: bool,
3582     hypergraph: bool = False,
3583     r: int = 3,
3584     max_mult: int = 3,
3585     separation: str = "edge",
3586 ) -> dict[tuple[int, int], int]:
3587     """Pooled 2-D histogram '{(lambda_max, pair_conn): count}' over all graphs.
3588
3589     For every labeled graph on 'n' vertices and every vertex pair, record the
3590     pair's local connectivity tagged with the graph's own 'lambda^max'. The
3591     result is a small threshold-independent table: at plot time a feasibility
3592     threshold 'T' splits each pair-connectivity column into observations from
3593     graphs that stay at 'lambda^max <= T' and those that exceed it. Much
3594     smaller than the raw observation list (a few dozen entries per variant), so
3595     it pickles compactly even though the sweep itself enumerates every graph.
3596     """
3597     from collections import Counter
3598     table: Counter = Counter()
3599
3600     def record(obj):
3601         pcs = _pair_connectivities(obj, separation=separation, hypergraph=
3602             ↪ hypergraph)
3603         lmax = max(pcs, default=0)
3604         for c in pcs:
3605             table[(lmax, c)] += 1
3606
3607     if hypergraph:
3608         candidates = _hyperedge_candidates(n, r, directed)
3609         for mask in product((0, 1), repeat=len(candidates)):
3610             chosen = [candidates[i] for i, flag in enumerate(mask) if flag]
3611             record(Hypergraph(n, chosen, directed=directed))
3612         return dict(table)
3613
3614     variant = _variant_for(directed, simple)
3615     cells = _matrix_cells(n, directed)
3616     span = 2 if simple else (max_mult + 1)
3617     base = Graph(n, variant)
3618     for values in product(range(span), repeat=len(cells)):
3619         candidate = base.copy()
3620         for (u, v), value in zip(cells, values):
3621             if value:
3622                 candidate.set_multiplicity(u, v, value)
3623         record(candidate)
3624     return dict(table)
3625
3626 def compute_pair_enumeration_cache(
3627     cache_path: str | Path = "figures/pair_enumeration_cache.pkl",
3628     configs: list[dict] | None = None,
3629 ) -> dict[str, dict[tuple[int, int], int]]:
3630     """Per-variant pooled pair-connectivity tables, loading from cache if present.
3631
3632     The pooled analogue of :func:'compute_enumeration_cache': one
3633     '{(lambda_max, pair_conn): count}' table per variant. Slow on first run
3634     (it re-sweeps the enumeration measuring every pair), cached thereafter.
3635     """
3636     cache_path = Path(cache_path)
3637     if configs is None:

```

```

3638         configs = _VARIANT_ENUM_CONFIGS
3639
3640     if cache_path.exists():
3641         with open(cache_path, "rb") as f:
3642             cached = pickle.load(f)
3643     else:
3644         cached = {}
3645
3646     changed = False
3647     for cfg in configs:
3648         key = cfg["key"]
3649         if key in cached:
3650             continue
3651         print(f"    pooling pairs for {cfg['title']} (n={cfg['enum_n']})...", flush=
↳ True)
3652         cached[key] = enumerate_pair_connectivities(
3653             cfg["enum_n"],
3654             directed=cfg["directed"], simple=cfg["simple"],
3655             hypergraph=cfg["hypergraph"], r=cfg["r"],
3656             max_mult=cfg["max_mult"], separation=cfg["separation"])
3657         changed = True
3658
3659     if changed:
3660         cache_path.parent.mkdir(parents=True, exist_ok=True)
3661         with open(cache_path, "wb") as f:
3662             pickle.dump(cached, f)
3663
3664     return cached
3665
3666
3667 def compute_enumeration_cache(
3668     cache_path: str | Path = "figures/enumeration_cache.pkl",
3669     configs: list[dict] | None = None,
3670 ) -> dict[str, list[tuple[int, int]]]:
3671     """Return the full enumeration for every variant, loading from cache if
↳ available.
3672
3673     On first run this takes several minutes; subsequent runs load the pickle
3674     instantly. The cache lives at 'cache_path' relative to the caller's
3675     working directory.
3676     """
3677     cache_path = Path(cache_path)
3678     if configs is None:
3679         configs = _VARIANT_ENUM_CONFIGS
3680
3681     if cache_path.exists():
3682         with open(cache_path, "rb") as f:
3683             cached = pickle.load(f)
3684     else:
3685         cached = {}
3686
3687     changed = False
3688     for cfg in configs:
3689         key = cfg["key"]
3690         if key in cached:
3691             continue
3692         print(f"    enumerating {cfg['title']} (n={cfg['enum_n']})...", flush=True)
3693         data = enumerate_all_graphs(

```

```

3694         cfg["enum_n"],
3695         directed=cfg["directed"],
3696         simple=cfg["simple"],
3697         hypergraph=cfg["hypergraph"],
3698         r=cfg["r"],
3699         max_mult=cfg["max_mult"],
3700         separation=cfg["separation"],
3701     )
3702     cached[key] = data
3703     changed = True
3704
3705     if changed:
3706         cache_path.parent.mkdir(parents=True, exist_ok=True)
3707         with open(cache_path, "wb") as f:
3708             pickle.dump(cached, f)
3709
3710     return cached
3711
3712
3713 def plot_scatter_lambda_edges(
3714     enum_data: dict[str, list[tuple[int, int]]],
3715     path: str | Path,
3716 ) -> None:
3717     """Extremal envelope of edge count against binding connectivity, all twelve
3718         ↪ variants.
3719
3720     For each variant the full enumeration is drawn as a faint grey cloud (one
3721     point per labeled graph) with the binding connectivity ' $\lambda^{\max}$ ' on the
3722     horizontal axis and the edge count on the vertical, matching the
3723     edges-against-parameter layout of the all-variant grid. The extremal
3724     envelope -- the densest feasible graph available at each connectivity level
3725     -- is overlaid as an integer staircase with one dot per level, so every
3726     connectivity value carries an exact extremal edge count and the frontier is
3727     read off without any interpolation between levels.
3728
3729     """
3730     def draw_panel(ax, cfg):
3731         key = cfg["key"]
3732         data = enum_data.get(key, [])
3733         if not data:
3734             ax.set_title(cfg["title"], fontsize=9.5)
3735             return
3736
3737         edges_arr = [e for e, _ in data]
3738         conn_arr = [c for _, c in data]
3739
3740         # Faint cloud of every graph: connectivity (x) against edge count (y).
3741         ax.scatter(conn_arr, edges_arr, s=1.5, c="grey", alpha=0.25,
3742             linewidths=0, rasterized=True)
3743
3744         # Extremal envelope  $e(\lambda)$  = densest feasible graph with  $\lambda^{\max}$ 
3745         # at most this level: a non-decreasing integer staircase. Reading it at
3746         #  $\lambda = m-1$  gives the extremal value of Problem 915 at this n.
3747         levels = sorted(set(conn_arr))
3748         running, env_edges = 0, []
3749         for lv in levels:
3750             running = max(running, max(e for e, c in data if c == lv))
3751             env_edges.append(running)
3752         ax.step(levels, env_edges, where="mid", color=_KUL_BLUE, linewidth=1.8,

```

```

3751         zorder=3)
3752     ax.plot(levels, env_edges, "o", color=_KUL_BLUE, markersize=6,
3753             zorder=4, label="extremal envelope")
3754
3755     ax.set_title(cfg["title"], fontsize=9.5)
3756     ax.set_xlabel(r"$\lambda^{\{\max\}}$", fontsize=8.5)
3757     ax.set_ylabel("edge count", fontsize=8.5)
3758     ax.tick_params(labels=8)
3759     # Both axes count discrete objects (independent routes, edges).
3760     ax.xaxis.set_major_locator(MaxNLocator(integer=True))
3761     ax.yaxis.set_major_locator(MaxNLocator(integer=True))
3762     ax.grid(True, alpha=0.3)
3763     ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3764            ha="left", va="top", fontsize=8, color="grey")
3765
3766     _variant_panel_grid(
3767         draw_panel, path=path,
3768         suptitle=("Extremal envelope: edge count against binding connectivity "
3769                 "across all twelve variants (full enumeration)"),
3770         suptitle_fontsize=13)
3771
3772
3773 def _variant_panel_grid(draw_panel, *, suptitle: str, path: str | Path,
3774                        configs=None, suptitle_fontsize: float = 12.0,
3775                        row_label_fontsize: float = 12.0) -> None:
3776     """Shared scaffold for every twelve-panel variant grid (three model rows by
3777     four columns): the distribution grids, the proved/conjectured bound grid, the
3778     sampled grid, and the extremal-envelope scatter. Builds the axes, calls
3779     ‘draw_panel(ax, cfg)’ on each panel config in turn, adds the per-row model
3780     labels and the suptitle, and saves. Only the per-panel drawing and the panel
3781     configs differ between the grids, so the caller supplies both; everything else
3782     (layout, row labels, save) lives here once. ‘configs’ defaults to the
3783     twelve enumeration variants but may be any 12-item list (e.g. the sampled
3784     configs or a precomputed ‘panels’ list).
3785     """
3786     if configs is None:
3787         configs = _VARIANT_ENUM_CONFIGS
3788     fig, axes = plt.subplots(3, 4, figsize=(16, 11))
3789     for cfg, ax in zip(configs, axes.flat):
3790         draw_panel(ax, cfg)
3791     for row, name in enumerate(("simple", "multigraph", "hypergraph $r=3$")):
3792         axes[row, 0].annotate(name, xy=(-0.32, 0.5), xycoords="axes fraction",
3793                              rotation=90, ha="center", va="center",
3794                              fontsize=row_label_fontsize, fontweight="bold",
3795                              color=_KUL_DARK)
3796     fig.suptitle(suptitle, fontsize=suptitle_fontsize)
3797     fig.tight_layout(rect=(0.02, 0.0, 1.0, 0.97))
3798     _save(path)
3799
3800
3801 def plot_conn_dist_grid(
3802     enum_data: dict[str, list[tuple[int, int]]],
3803     m: int,
3804     path: str | Path,
3805     known_maxima: dict[str, int] | None = None,
3806 ) -> None:
3807     """12-panel histogram of lambda_max distribution, all variants, fixed m.
3808

```

```

3809     Bars to the left of the feasibility boundary  $m-1$  are coloured blue (the
3810     graphs we study); bars at or above  $m$  are coloured red (infeasible for this
3811      $m$ ). A vertical dashed line marks the boundary, and a dotted line marks
3812     the known or conjectured maximum if supplied in ‘‘known_maxima‘‘.
3813     """
3814     threshold = m - 1 # feasible means  $\lambda_{\max} \leq \text{threshold}$ 
3815
3816     def draw_panel(ax, cfg):
3817         key = cfg["key"]
3818         data = enum_data.get(key, [])
3819         if not data:
3820             ax.set_title(cfg["title"], fontsize=9)
3821             return
3822
3823         conn_vals = [c for _, c in data]
3824         lo, hi = min(conn_vals), max(conn_vals)
3825         levels = list(range(lo, hi + 1))
3826         counts = [conn_vals.count(lv) for lv in levels]
3827         bar_colours = [_KUL_BLUE if lv <= threshold else _RED for lv in levels]
3828         ax.bar(levels, counts, color=bar_colours, edgecolor="white", linewidth
3829             ↪ =0.4)
3830
3831         ax.axvline(threshold + 0.5, color=_WARM, linestyle="--", linewidth=1.8)
3832         if known_maxima and key in known_maxima:
3833             ax.axvline(known_maxima[key], color=_KUL_DARK, linestyle=":",
3834                 ↪ linewidth=1.6, label=f"known max = {known_maxima[key]}")
3835
3836         ax.set_title(cfg["title"], fontsize=9.5)
3837         ax.set_xlabel(r"$\lambda^{\{\max\}}$", fontsize=8.5)
3838         ax.set_ylabel("graphs", fontsize=8.5)
3839         ax.tick_params(labelsize=8)
3840         # Connectivity is integer-valued: force integer ticks so the directed
3841         # panels stop showing spurious half-integer labels (0.5, 1.5, ...).
3842         ax.set_xticks(levels)
3843         ax.grid(True, axis="y", alpha=0.3)
3844         # The feasibility boundary is identical in all twelve panels, so it is
3845         # explained once in the title and caption rather than repeated as a
3846         # legend in every panel. A legend is drawn only when a per-panel known
3847         # maximum line is present (a value that genuinely differs by panel).
3848         if known_maxima and key in known_maxima:
3849             ax.legend(fontsize=6.8, loc="upper right", framealpha=0.85)
3850             ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3851                 ↪ ha="left", va="top", fontsize=8, color="grey")
3852
3853     suptitle = (fr"Connectivity distribution across all twelve variants,  $m = \{m\}$ 
3854         ↪ "
3855         fr"(blue = feasible  $\lambda^{\{\max\}} \leq \{\text{threshold}\}$ , red =
3856             ↪ infeasible)")
3857     _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
3858         ↪ row_label_fontsize=12)
3859
3860     def _midrange_lambda_threshold(hi: int) -> int:
3861         """A per-variant connectivity boundary at the midpoint of the achievable range
3862             ↪ .

```

```

3862     a red (high) population. A single global boundary collapses some panels to
3863         ↳ one
3864     colour (every graph below it, or almost none), because the twelve variants
3865     reach very different connectivity ranges at the sizes enumeration allows.
3866     Splitting instead at ‘‘round(hi/2)’’ -- the middle of what each enumeration
3867         ↳ can
3868     reach -- keeps both populations visible in every panel, scaled to what is
3869     possible there. Floored at one so the boundary is never zero, which would
3870     push every graph above it and leave the blue (low) side empty.
3871     """
3872     return max(1, round(hi / 2))
3873
3874 def plot_pair_conn_dist_grid(
3875     pair_data: dict[str, dict[tuple[int, int], int]],
3876     path: str | Path,
3877 ) -> None:
3878     """12-panel histogram of per-PAIR connectivity, pooled over the full
3879         ↳ enumeration.
3880
3881     Where :func:‘plot_conn_dist_grid‘ plots one ‘‘lambda^max’’ per graph, this
3882     pools every vertex pair of every graph: one observation per (graph, pair).
3883     Each bar is split and STACKED by the connectivity of the graph the pair came
3884     from -- blue (bottom) for pairs from graphs whose ‘‘lambda^max’’ stays at or
3885     below the per-panel boundary ‘‘T’’, red (top) for pairs from graphs above it
3886     -- so the bar’s full height is the true number of observations at that level.
3887
3888     ‘‘T’’ is set per variant by :func:‘_midrange_lambda_threshold‘, not by a
3889         ↳ single
3890     global ‘‘m’’: at a fixed boundary some panels are entirely one colour, while
3891         ↳ the
3892     mid-range split keeps both populations visible everywhere. Pairs whose own
3893     connectivity exceeds ‘‘T’’ can only come from graphs above ‘‘T’’, so those
3894         ↳ bars
3895     are wholly red; below ‘‘T’’ the colours mix, since a high-connectivity graph
3896     still has many low-connectivity pairs.
3897     """
3898     def draw_panel(ax, cfg):
3899         table = pair_data.get(cfg["key"], {})
3900         if not table:
3901             ax.set_title(cfg["title"], fontsize=9)
3902             return
3903
3904         hi = max(lmax for (lmax, _pc) in table)
3905         T = _midrange_lambda_threshold(hi)
3906         levels = sorted({pc for (_lmax, pc) in table})
3907         blue = [sum(c for (lmax, pc), c in table.items()
3908                     if pc == lv and lmax <= T) for lv in levels]
3909         red = [sum(c for (lmax, pc), c in table.items()
3910                    if pc == lv and lmax > T) for lv in levels]
3911
3912         ax.bar(levels, blue, color=_KUL_BLUE, edgecolor="white", linewidth=0.4,
3913               label=fr"from $\lambda^{\{\max\}}\!\!\leq\!\{T\}$")
3914         ax.bar(levels, red, bottom=blue, color=_RED, edgecolor="white",
3915               linewidth=0.4, label=fr"from $\lambda^{\{\max\}}\!\!>\!\{T\}$")
3916         ax.axvline(T + 0.5, color=_WARM, linestyle="--", linewidth=1.8)
3917
3918     ax.set_title(cfg["title"], fontsize=9.5)

```

```

3914     ax.set_xlabel("pair connectivity", fontsize=8.5)
3915     ax.set_ylabel("pair observations", fontsize=8.5)
3916     ax.tick_params(labelsize=8)
3917     ax.set_xticks(levels)
3918     ax.grid(True, axis="y", alpha=0.3)
3919     ax.legend(fontsize=6.8, loc="upper right", framealpha=0.85)
3920     ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3921            ha="left", va="top", fontsize=8, color="grey")
3922
3923     suptitle = ("Pair-connectivity distribution across all twelve variants "
3924               "(every vertex pair of every graph pooled, split at each variant's
3925               ↪ "
3926               r"mid-range  $\lambda^{\max}$ : blue from low-connectivity graphs,
3927               ↪ red from high)")
3928     _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
3929                        row_label_fontsize=12)
3930
3931 def plot_edge_dist_grid(
3932     enum_data: dict[str, list[tuple[int, int]]],
3933     path: str | Path,
3934 ) -> None:
3935     """12-panel histogram of edge count distribution, all variants.
3936
3937     Each bar is split and STACKED by graph connectivity: blue (bottom) for graphs
3938     whose ' $\lambda^{\max}$ ' stays at or below the per-panel boundary ' $T$ ', red (top)
3939     for graphs above it, so the full bar height is the true number of graphs at
3940     that edge count. ' $T$ ' is the midpoint of each variant's achievable
3941     connectivity range (:func: '_midrange_lambda_threshold'); a single global
3942     boundary leaves some panels all one colour, the mid-range split keeps both
3943     populations visible. A dotted line marks the densest graph at or below ' $T$ '.
3944
3945     """
3946     def draw_panel(ax, cfg):
3947         key = cfg["key"]
3948         data = enum_data.get(key, [])
3949         if not data:
3950             ax.set_title(cfg["title"], fontsize=9)
3951             return
3952
3953         hi = max(c for _e, c in data)
3954         T = _midrange_lambda_threshold(hi)
3955         max_e = max(e for e, _ in data)
3956         levels = list(range(0, max_e + 1))
3957         blue = [0] * (max_e + 1)
3958         red = [0] * (max_e + 1)
3959         for e, c in data:
3960             (blue if c <= T else red)[e] += 1
3961
3962         # Stacked, not overlaid: low-connectivity graphs (blue) on the bottom,
3963         # high-connectivity (red) on top, so the full height is the true count.
3964         ax.bar(levels, blue, color=_KUL_BLUE, edgecolor="white", linewidth=0.3,
3965              label=fr"$\lambda^{\{\max\}}\!\!\leq\!\{T\}$")
3966         ax.bar(levels, red, bottom=blue, color=_RED, edgecolor="white",
3967              linewidth=0.3, label=fr"$\lambda^{\{\max\}}\!\!>\!\{T\}$")
3968
3969         # The densest graph in the lower-connectivity population: read straight
3970         # off the data, so it lands on the right edge of the blue mass.
3971         if any(blue):

```

```

3970         kmax = max(e for e in levels if blue[e])
3971         ax.axvline(kmax, color=_KUL_DARK, linestyle=":", linewidth=1.8,
3972                 label=fr"densest  $\leq \{T\}$ : {kmax}")
3973
3974         ax.set_title(cfg["title"], fontsize=8.5)
3975         ax.set_xlabel("edge count", fontsize=7.5)
3976         ax.set_ylabel("graphs", fontsize=7.5)
3977         ax.tick_params(labels=7)
3978         # Edge counts are integers: keep the tick labels integer-valued.
3979         ax.xaxis.set_major_locator(MaxNLocator(integer=True))
3980         ax.grid(True, axis="y", alpha=0.25)
3981         ax.legend(fontsize=6.2, loc="upper right", framealpha=0.8)
3982         ax.text(0.98, 0.02, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3983                 ha="right", va="bottom", fontsize=7, color="grey")
3984
3985         suptitle = ("Edge-count distribution across all twelve variants "
3986                     r"(stacked by connectivity, split at each variant's mid-range "
3987                     r"$\lambda^{\max}$: blue low-connectivity graphs, red high)")
3988         _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
3989                             row_label_fontsize=11)
3990
3991
3992 # --- 3-D BOUND SURFACE: cache solve() over (variant, n, m) grid;
3993     ↳ plot_variant_3d_surfaces draws it ---
3994
3995 # The two multigraph vertex problems reduce exactly to the simple ones: vertex
3996 # mode is blind to parallel copies (see _split_capacity_matrix), so a multigraph
3997 # and its underlying simple graph have the same vertex connectivity. We mirror
3998 # the simple-vertex surface into these keys rather than searching multigraphs,
3999 # whose degenerate free-parallel fills would report a lower bound far above the
4000 # true (simple) value and spike the plot.
4001 _SURFACE_ALIASED_VERTEX = {
4002     "multi_undirected_vertex": "simple_undirected_vertex",
4003     "multi_directed_vertex": "simple_directed_vertex",
4004 }
4005
4006 # 12 variant configs for the surface computation.
4007 _SURFACE_VARIANT_CONFIGS: list[dict] = [
4008     dict(key="simple_undirected_edge", title="simple undirected edge",
4009          directed=False, simple=True, hypergraph=False, r=3, separation="edge"),
4009     dict(key="simple_undirected_vertex", title="simple undirected vertex",
4010          directed=False, simple=True, hypergraph=False, r=3, separation="vertex")
4011     ↳ ,
4012     dict(key="simple_directed_edge", title="simple directed arc",
4013          directed=True, simple=True, hypergraph=False, r=3, separation="edge"),
4014     dict(key="simple_directed_vertex", title="simple directed vertex",
4015          directed=True, simple=True, hypergraph=False, r=3, separation="vertex")
4016     ↳ ,
4017     dict(key="multi_undirected_edge", title="multigraph undirected edge",
4018          directed=False, simple=False, hypergraph=False, r=3, separation="edge"),
4019     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
4020          directed=False, simple=False, hypergraph=False, r=3, separation="vertex")
4021     ↳ ,
4022     dict(key="multi_directed_edge", title="multigraph directed arc",
4023          directed=True, simple=False, hypergraph=False, r=3, separation="edge"),
4024     dict(key="multi_directed_vertex", title="multigraph directed vertex",
4025          directed=True, simple=False, hypergraph=False, r=3, separation="vertex")
4026     ↳ ,

```



```

4023     dict(key="hyper_undirected_edge",    title="hypergraph undirected edge",
4024           directed=False, simple=True, hypergraph=True, r=3, separation="edge"),
4025     dict(key="hyper_undirected_vertex",  title="hypergraph undirected vertex",
4026           directed=False, simple=True, hypergraph=True, r=3, separation="vertex")
4027     ↪ ,
4028     dict(key="hyper_directed_edge",      title="hypergraph directed arc",
4029           directed=True, simple=True, hypergraph=True, r=3, separation="edge"),
4030     dict(key="hyper_directed_vertex",    title="hypergraph directed vertex",
4031           directed=True, simple=True, hypergraph=True, r=3, separation="vertex")
4032     ↪ ,
4033 ]
4034
4035 def _surface_known_value(vkey: str, n: int, m: int) -> int | None:
4036     """The proved exact value at '(n, m)' for 'vkey', or 'None' if open here
4037     ↪ .
4038
4039     Returns the closed-form extremal value precisely in the regime where the
4040     thesis proves it, so the surface can colour those bars as exact (blue)
4041     rather than leaving every bar a search lower bound. Outside the proved
4042     regime it returns 'None' and the caller falls back to a discovery search
4043     (a purple lower bound). Each value is capped at the trivial maximum for
4044     its model, since the closed form overshoots in the small-'n' large-'m'
4045     corner where no graph that dense exists.
4046     """
4047     tri_simple = n * (n - 1) // 2
4048     tri_dir = n * (n - 1)
4049     tri_hyp = math.comb(n, 3)
4050     if vkey == "simple_undirected_edge":      # Mader, all m
4051         return min(simple_undirected_edge(n, m), tri_simple)
4052     if vkey == "simple_undirected_vertex":    # Leonard, m<=4
4053         return min(simple_undirected_edge(n, m), tri_simple) if m <= 4 else None
4054     if vkey == "simple_directed_edge":        # exact only at m=2
4055         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4056     if vkey == "simple_directed_vertex":      # exact only at m=2
4057         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4058     if vkey == "multi_undirected_edge":      # multi-tree bound, all m
4059         return min(multigraph_undirected_edge(n, m), (m - 1) * tri_simple)
4060     if vkey == "multi_undirected_vertex":    # = simple vertex, m<=4
4061         return min(simple_undirected_edge(n, m), tri_simple) if m <= 4 else None
4062     if vkey == "multi_directed_edge":        # cut-counting proves n<=6
4063         return min(directed_multigraph_arc(n, m), (m - 1) * tri_dir) if n <= 6
4064         ↪ else None
4065     if vkey == "multi_directed_vertex":      # = simple digraph, exact m=2
4066         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4067     if vkey == "hyper_undirected_edge":      # incidence-rank, all m
4068         return min(hypergraph_edge(n, m, 3), tri_hyp)
4069     if vkey == "hyper_undirected_vertex":    # incidence-rank lemma, m<=3
4070         return min(hypergraph_edge(n, m, 3), tri_hyp) if m <= 3 else None
4071     # hyper_directed_edge, hyper_directed_vertex: open (new model), no formula.
4072     return None
4073
4074 def compute_surface_cache(
4075     n_range: tuple[int, int] = (3, 9),
4076     m_range: tuple[int, int] = (2, 6),
4077     max_seconds: float = 20.0,
4078     cache_path: str | Path = "figures/surface_cache.json",

```

```

4077 ) -> dict:
4078     """Compute the optimal bound at every (variant, n, m) triple, save to a JSON
        ↳ cache.

4079
4080     Returns the cache dict: '{variant_key: {str(n): {str(m): {value, bound}}}}'.
4081     Where the value is proved (see :func: '_surface_known_value') the bound is
4082     recorded as '"exact"' using the closed form; otherwise a discovery search
4083     supplies a '"lower"' bound. Missing entries are skipped on later runs.
4084     """
4085     cache_path = Path(cache_path)
4086     if cache_path.exists():
4087         with open(cache_path) as f:
4088             cache = json.load(f)
4089     else:
4090         cache = {}
4091
4092     changed = False
4093     ns = list(range(n_range[0], n_range[1] + 1))
4094     ms = list(range(m_range[0], m_range[1] + 1))
4095
4096     for cfg in _SURFACE_VARIANT_CONFIGS:
4097         vkey = cfg["key"]
4098         if vkey in _SURFACE_ALIASED_VERTEX:
4099             continue # filled by mirroring its simple counterpart, below
4100         if vkey not in cache:
4101             cache[vkey] = {}
4102         for n in ns:
4103             sn = str(n)
4104             if sn not in cache[vkey]:
4105                 cache[vkey][sn] = {}
4106             for m in ms:
4107                 sm = str(m)
4108                 known = _surface_known_value(vkey, n, m)
4109                 if known is not None:
4110                     # Proved regime: use the closed form, mark it exact (blue).
4111                     # Always (re)write, so an older cache whose every cell was
4112                     # tagged "lower" is corrected here without recomputation.
4113                     entry = {"value": known, "bound": "exact"}
4114                     if cache[vkey][sn].get(sm) != entry:
4115                         cache[vkey][sn][sm] = entry
4116                         changed = True
4117                     continue
4118                 if sm in cache[vkey][sn]:
4119                     continue # open cell already searched; keep the lower bound
4120                 print(f" surface: {cfg['title']} n={n} m={m}", flush=True)
4121                 res = solve(
4122                     n, m,
4123                     directed=cfg["directed"],
4124                     simple=cfg["simple"],
4125                     hypergraph=cfg["hypergraph"],
4126                     r=cfg["r"],
4127                     exhaustive=False,
4128                     separation=cfg["separation"],
4129                     max_seconds=max_seconds,
4130                     seed=0,
4131                 )
4132                 cache[vkey][sn][sm] = {"value": res.value, "bound": res.bound}
4133                 changed = True

```

```

4134
4135     # Mirror the simple-vertex surfaces into their multigraph aliases, so the two
4136     # provably-equal problems show identical bars instead of two independent
4137     # (and possibly degenerate) searches.
4138     for multi_key, simple_key in _SURFACE_ALIASED_VERTEX.items():
4139         if simple_key in cache and cache.get(multi_key) != cache[simple_key]:
4140             cache[multi_key] = copy.deepcopy(cache[simple_key])
4141             changed = True
4142
4143     if changed:
4144         cache_path.parent.mkdir(parents=True, exist_ok=True)
4145         with open(cache_path, "w") as f:
4146             json.dump(cache, f)
4147
4148     return cache
4149
4150
4151 def plot_variant_3d_surfaces(
4152     cache_path: str | Path,
4153     path: str | Path,
4154 ) -> None:
4155     """3-D bar chart of the optimal bound over the (n, m) grid, all twelve
4156     → variants.
4157
4158     Each bar's height is the bound value; exact points are blue, lower-bound
4159     points are purple. The 3x4 layout matches the flat grid figure.
4160     """
4161     with open(cache_path) as f:
4162         cache = json.load(f)
4163
4164     fig = plt.figure(figsize=(20, 13))
4165     row_names = ("simple", "multigraph", "hypergraph $r=3$")
4166
4167     for idx, cfg in enumerate(_SURFACE_VARIANT_CONFIGS):
4168         vkey = cfg["key"]
4169         ax = fig.add_subplot(3, 4, idx + 1, projection="3d")
4170
4171         variant_data = cache.get(vkey, {})
4172         ns_exact, ms_exact, vs_exact = [], [], []
4173         ns_lower, ms_lower, vs_lower = [], [], []
4174
4175         for sn, m_dict in sorted(variant_data.items(), key=lambda x: int(x[0])):
4176             n = int(sn)
4177             for sm, entry in sorted(m_dict.items(), key=lambda x: int(x[0])):
4178                 m_val = int(sm)
4179                 v = entry["value"]
4180                 if entry["bound"] == "exact":
4181                     ns_exact.append(n)
4182                     ms_exact.append(m_val)
4183                     vs_exact.append(v)
4184                 else:
4185                     ns_lower.append(n)
4186                     ms_lower.append(m_val)
4187                     vs_lower.append(v)
4188
4189         dx = dy = 0.6
4190         if ns_lower:
4191             ax.bar3d(

```

```

4191         [x - dx / 2 for x in ns_lower],
4192         [y - dy / 2 for y in ms_lower],
4193         [0] * len(vs_lower),
4194         dx, dy, vs_lower,
4195         color=_VIOLET, alpha=0.7, shade=True,
4196         edgecolor="white", linewidth=0.25,
4197     )
4198     if ns_exact:
4199         ax.bar3d(
4200             [x - dx / 2 for x in ns_exact],
4201             [y - dy / 2 for y in ms_exact],
4202             [0] * len(vs_exact),
4203             dx, dy, vs_exact,
4204             color=_KUL_BLUE, alpha=0.9, shade=True,
4205             edgecolor="white", linewidth=0.25,
4206         )
4207
4208     # Same camera for every panel so the twelve are read side by side, and a
4209     # z-axis anchored at 0 so bar heights are comparable within a panel.
4210     ax.view_init(elev=24, azim=-58)
4211     peak = max(vs_exact + vs_lower, default=1)
4212     ax.set_zlim(0, peak * 1.05)
4213     for pane in (ax.xaxis, ax.yaxis, ax.zaxis):
4214         pane.pane.set_alpha(0.04)
4215
4216     ax.set_title(cfg["title"], fontsize=7.5, pad=2)
4217     ax.set_xlabel("$n$", fontsize=7, labelpad=2)
4218     ax.set_ylabel("$m$", fontsize=7, labelpad=2)
4219     ax.set_zlabel("bound", fontsize=7, labelpad=2)
4220     ax.tick_params(labelsize=6)
4221
4222     # Row labels
4223     for row, name in enumerate(row_names):
4224         row_ax = fig.axes[row * 4]
4225         row_ax.text2D(-0.28, 0.5, name, transform=row_ax.transAxes,
4226                     rotation=90, ha="center", va="center",
4227                     fontsize=11, fontweight="bold", color=_KUL_DARK)
4228
4229     # Manual legend patches
4230     from matplotlib.patches import Patch
4231     legend_elems = [
4232         Patch(facecolor=_KUL_BLUE, alpha=0.85, label="exact (proved)"),
4233         Patch(facecolor=_VIOLET, alpha=0.65, label="lower bound (search)"),
4234     ]
4235     fig.legend(handles=legend_elems, loc="lower center", ncol=2,
4236              fontsize=10, framealpha=0.9)
4237
4238     fig.suptitle("Erdős 915: optimal bound over $(n, m)$, "
4239                "blue where proved, purple where only a search lower bound is  

4240                ↔ known",
4241                fontsize=13, y=0.99)
4242     _save(path, tight=False)
4243
4244     # --- 3-D THRESHOLD HISTOGRAM: lambda^max distribution vs p for three variants ---
4245
4246     def plot_conn_threshold_3d(

```

```

4248     path: str | Path,
4249     n: int = 30,
4250     m: int = 3,
4251     p_values: list[float] | None = None,
4252     samples: int = 400,
4253     seed: int = 7,
4254 ) -> None:
4255     """3-D bar chart of the lambda_max distribution over density, for three
        ↳ variants.

4256
4257     x = density in units of that model's OWN threshold, y = lambda_max value,
4258     z = fraction of samples. Each panel is swept in units of its own p*, and the
4259     threshold line sits at 1 in all three, because the models do NOT share a
4260     density scale. The threshold is where a vertex's expected degree reaches m,
4261     which for a graph is  $p*(n-1) = m$ , giving  $p* = m/n$ , but for an r-uniform
4262     binomial hypergraph is  $p*C(n-1, r-1) = m$ , giving  $p* = m/C(n-1, r-1) =$ 
4263      $\Theta(m/n^{(r-1)})$ . Plotting the hypergraph panel against m/n would put the
4264     line in the wrong place by a factor of order  $n^{(r-2)}$ . Only the graph panels
4265     are covered by thm:gnp-threshold; the hypergraph panel is an observation.
4266
4267     Three panels: undirected edge (uses 'n'), directed arc (uses 'n'),
4268     hypergraph edge (uses min(n, 8) to keep flow computation fast).
4269     """
4270     # Sweep in units of the model's own threshold, so all three panels are
4271     # directly comparable and each transition is visible where it actually is.
4272     ratios = ([i / 5.0 for i in range(1, 20)] if p_values is None else None)
4273
4274     rng = random.Random(seed)
4275     # Hypergraph flow computation scales poorly: cap n at 8 for that panel.
4276     n_hyper = min(n, 8)
4277
4278     # The hypergraph threshold is  $m / C(n-1, r-1)$  with  $r = 3$ . Write the binomial
4279     # with its arguments already evaluated: matplotlib's mathtext lays out
4280     # "\binom{8-1}{2}" so that the "-1" reads as a superscript on the 8, which
4281     # renders as a garbled binomial rather than "7 choose 2".
4282     hyper_top = n_hyper - 1
4283     hyper_pstar = math.comb(hyper_top, 2)
4284     variants_3d = [
4285         dict(label=f"undirected edge ( $n={n}$ )", directed=False, separation="edge"
        ↳ ,
4286             hypergraph=False, panel_n=n, p_star=m / n,
4287             star_text=f" $m/n = \{m\}/\{n\}$ "),
4288         dict(label=f"directed arc ( $n={n}$ )", directed=True, separation="edge"
        ↳ ,
4289             hypergraph=False, panel_n=n, p_star=m / n,
4290             star_text=f" $m/n = \{m\}/\{n\}$ "),
4291         dict(label=f"hypergraph edge ( $r=3$ ,  $n={n\_hyper}$ )", directed=False,
4292             separation="edge", hypergraph=True, panel_n=n_hyper,
4293             p_star=m / hyper_pstar,
4294             star_text=(fr" $m/\binom{{{hyper\_top}}}{{2}} = "$ 
4295                       fr" $\{m\}/\{hyper\_pstar\}$ ")),
4296     ]
4297
4298     fig = plt.figure(figsize=(18, 6))
4299
4300     for panel_idx, vdesc in enumerate(variants_3d):
4301         ax = fig.add_subplot(1, 3, panel_idx + 1, projection="3d")
4302         panel_n = vdesc["panel_n"]

```

```

4303 p_star = vdesc["p_star"]
4304 panel_ps = (p_values if p_values is not None
4305             else [min(1.0, r * p_star) for r in ratios])
4306
4307 all_conn_vals: list[int] = []
4308 dist_by_p: list[tuple[float, list[int]]] = []
4309
4310 for p in panel_ps:
4311     conn_list: list[int] = []
4312     for _ in range(samples):
4313         if vdesc["hypergraph"]:
4314             cands = _hyperedge_candidates(panel_n, 3, vdesc["directed"])
4315             chosen = [e for e in cands if rng.random() < p]
4316             h = Hypergraph(panel_n, chosen, directed=vdesc["directed"])
4317             c = max_hyper_connectivity(h, vertex_split=False)
4318         else:
4319             g = sample_random_graph(panel_n, p, vdesc["directed"], rng)
4320             if vdesc["separation"] == "edge":
4321                 c = max_edge_connectivity(g)
4322             else:
4323                 c = max_vertex_connectivity(g)
4324             conn_list.append(c)
4325         dist_by_p.append((p, conn_list))
4326     all_conn_vals.extend(conn_list)
4327
4328 if not all_conn_vals:
4329     continue
4330
4331 lo, hi = 0, max(all_conn_vals)
4332 levels = list(range(lo, hi + 1))
4333
4334 # x is the density in units of this panel's own threshold.
4335 xs = [p / p_star for p, _ in dist_by_p]
4336 width = 0.9 * min((b - a) for a, b in zip(xs, xs[1:])) if len(xs) > 1 else
4337     ↪ 0.1
4338 for x, (_, conn_list) in zip(xs, dist_by_p):
4339     total = len(conn_list)
4340     for lv in levels:
4341         frac = conn_list.count(lv) / total
4342         if frac > 0:
4343             ax.bar3d(x - width / 2, lv - 0.4, 0, width, 0.8, frac,
4344                     color=_KUL_BLUE, alpha=0.7, shade=True)
4345
4346 # The threshold sits at 1 in these units, by construction.
4347 for lv in levels:
4348     ax.plot([1.0, 1.0], [lv - 0.4, lv + 0.4], [0, 0],
4349             color=_WARM, linewidth=0.5, alpha=0.4)
4350 zmax = max(conn_list.count(lv) / len(conn_list)
4351            for _, conn_list in dist_by_p
4352            for lv in levels
4353            if conn_list.count(lv) > 0)
4354 ax.plot([1.0, 1.0], [lo, hi], [zmax * 0.9, zmax * 0.9],
4355         color=_WARM, linewidth=2.5, linestyle="--", label="$p / p^* = 1$")
4356
4357 ax.set_title(f"{vdesc['label']}\n$p^* = $ {vdesc['star_text']}", fontsize
4358             ↪ =9)
4359 ax.set_xlabel("$p / p^*$", fontsize=9)
4360 ax.set_ylabel(r"$\lambda^{\{\max\}}$", fontsize=9)

```

```

4359         ax.set_zlabel("fraction", fontsize=9)
4360         ax.tick_params(labelsize=7)
4361
4362     fig.suptitle(
4363         fr"Distribution of  $\lambda^{\{\max\}}$  against density,  $m = \{m\}$ , each
            ↳ model "
4364         fr"in units of its own threshold  $p^*_{\{m\}}$  (the density at which a vertex's "
4365         fr"expected degree reaches  $m_{\{m\}}$ ",
4366         fontsize=11,
4367     )
4368     _save(path, tight=False)

```

C.1.8 The open variants

These routines exist to attack the questions the thesis leaves open. They measure the hypergraph vertex problem, work with fractional weightings where the integer problem is out of reach, and provide the exhaustive maximisers used to test conjectures at the sizes where testing is still possible. Several results in [Appendix A](#) were found, or in two cases refuted, by running exactly these functions.

```

4371 # --- OPEN-VARIANT EXPLORATION: hypergraph vertex connectivity, fractional search
            ↳ tools ---
4372
4373
4374 def max_hyper_vertex_connectivity(hypergraph: Hypergraph) -> int:
4375     """ $\kappa^{\max}$  for the hypergraph vertex problem (readability wrapper)."""
4376     return max_hyper_connectivity(hypergraph, vertex_split=True)
4377
4378
4379 def _hyper_codegree_ok(edges, bound: int) -> bool:
4380     """Necessary feasibility filter: two vertices in  $> \text{bound}$  common
4381     hyperedges already carry that many one-step disjoint Berge routes."""
4382     from collections import Counter
4383
4384     codegree: Counter = Counter()
4385     for edge in edges:
4386         for pair in combinations(sorted(edge), 2):
4387             codegree[pair] += 1
4388             if codegree[pair] > bound:
4389                 return False
4390     return True
4391
4392
4393 def hyper_vertex_feasible_exists(n: int, r: int, m: int, q: int,
4394                                 multi: bool = True) -> bool:
4395     """Exhaustively decide whether some  $r$ -uniform (multi-)hypergraph on
4396      $n$  vertices with  $q$  hyperedges has  $\kappa^{\max} \leq m - 1$ .
4397
4398     Repeated hyperedges are allowed when  $\text{multi}$  is set (they are forced
4399     infeasible at  $m = 2$  by the codegree filter, so  $m = 2$  needs no repeats).
4400     Practical up to roughly  $C(C(n,r), q) \sim$  a few hundred thousand candidates.
4401     """
4402     all_edges = list(combinations(range(n), r))
4403     chooser = combinations_with_replacement if multi else combinations
4404     for sub in chooser(all_edges, q):

```

```

4405         if not _hyper_codegree_ok(sub, m - 1):
4406             continue
4407         candidate = Hypergraph(n, [frozenset(edge) for edge in sub])
4408         feasible = True
4409         for s in range(n):
4410             for t in range(s + 1, n):
4411                 if hyper_connectivity(candidate, s, t, vertex_split=True) > m - 1:
4412                     feasible = False
4413                     break
4414             if not feasible:
4415                 break
4416         if feasible:
4417             return True
4418     return False
4419
4420
4421 def verify_hyper_vertex_value(n: int, r: int, m: int) -> bool:
4422     """Confirm exhaustively that the ‘r’-uniform vertex value at ‘(n, m)’
4423     equals ‘floor((m-1)(n-1)/(r-1))’: the bound’s witness is feasible
4424     (kappa_max <= m-1) and one more hyperedge never is."""
4425     target = ((m - 1) * (n - 1)) // (r - 1)
4426     if m == 2:
4427         witness = star_hypertree(n, r)
4428         attained = (len(witness.hyperedges) == target
4429                     and max_hyper_vertex_connectivity(witness) <= 1)
4430     else:
4431         attained = hyper_vertex_feasible_exists(n, r, m, target)
4432     return attained and not hyper_vertex_feasible_exists(n, r, m, target + 1)
4433
4434
4435 def max_feasible_hyperedges(
4436     n: int, r: int, m: int,
4437     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
4438     time_limit: float = 20.0, seed_lb: int = 0,
4439 ) -> tuple[int, bool]:
4440     """Largest number of ‘r’-uniform hyperedges with Berge connectivity ‘<= m
4441     ⇔ -1’.”
4442
4443     Returns ‘(value, exact)’. ‘exact’ is ‘True’ when the branch and bound
4444     below proved optimality within ‘time_limit’ (the value is then the true
4445     extremal number for that variant); otherwise ‘value’ is the best feasible
4446     construction found, a lower bound. ‘kind’ selects the directed orientation
4447     model (forward / backward / general); for the undirected case it is ignored.
4448
4449     The search is a depth-first include/exclude over the candidate hyperedges with
4450     two prunings. Feasibility is monotone (adding a hyperedge never lowers any
4451     Berge connectivity), so once the active set is infeasible no superset can be
4452     feasible and the include branch is dropped; and a branch is cut once
4453     ‘active + remaining’ cannot beat the best feasible count already seen. A
4454     short randomised warm start raises that incumbent first, so the bound prune
4455     bites early. ‘seed_lb’ is an externally known feasible lower bound (e.g.
4456     ⇔ the
4457     forward value when maximising over the general model, which contains every
4458     forward hypergraph) used to start the incumbent; it never changes the proven
4459     optimum, only the pruning.
4460     """
4461     deadline = time.time() + time_limit
4462     # Warm start: a short greedy randomised lower bound sharpens the bound prune;

```



```

4461     # kept brief so the branch and bound, which does the actual proving, keeps
4462     # the budget.
4463     warm_share = min(0.3 * time_limit, 1.5)
4464     lb, _ = _random_hypergraph_search(
4465         n, r, m, time.time() + warm_share, seed=0,
4466         directed=directed, vertex_split=vertex_split, kind=kind)
4467     lb = max(lb, seed_lb)
4468
4469     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
4470     total = len(candidates)
4471     best = [lb]
4472     timed_out = [False]
4473     active = Hypergraph(n, directed=directed)
4474
4475     def dfs(pos: int, count: int) -> None:
4476         if timed_out[0]:
4477             return
4478         if count + (total - pos) <= best[0]:
4479             return # cannot beat the incumbent
4480         if time.time() > deadline:
4481             timed_out[0] = True
4482             return
4483         if pos == total:
4484             if count > best[0]:
4485                 best[0] = count
4486             return
4487         # Include candidates[pos], but only if the set stays feasible.
4488         active.add_hyperedge(candidates[pos])
4489         if max_hyper_connectivity(active, vertex_split=vertex_split) <= m - 1:
4490             dfs(pos + 1, count + 1)
4491         active.hyperedges.pop()
4492         # Exclude candidates[pos].
4493         dfs(pos + 1, count)
4494
4495     dfs(0, 0)
4496     return best[0], (not timed_out[0])
4497
4498
4499 def _c_flat(mu: np.ndarray) -> tuple[np.ndarray, "_ct.POINTER[_ct.c_int]"]:
4500     """Return a contiguous int32 copy of 'mu' and a ctypes c_int pointer into it
4501     ↪ ."""
4502     flat = np.ascontiguousarray(mu, dtype=np.int32)
4503     ptr = flat.ctypes.data_as(_ct.POINTER(_ct.c_int))
4504     return flat, ptr # caller must keep flat alive while ptr is in use
4505
4506 def _tiny_maxflow(mu: np.ndarray, n: int, s: int, t: int, cap: int) -> bool:
4507     """Return True iff the integer max-flow from s to t under capacities 'mu'
4508     exceeds 'cap'. Plain DFS augmentation (Ford-Fulkerson with a stack, not
4509     BFS/Edmonds-Karp); built for n <= 7, where it beats both the networkx call
4510     and scipy's C 'maximum_flow' by orders of magnitude inside the hot
4511     enumeration/search loops (measured ~16x faster than scipy at n=7: the
4512     library calls' per-call sparse-matrix construction and dispatch overhead
4513     dominates at this size, while the 'cap' lets this routine stop after
4514     cap+1 augmenting paths). Correct for integer capacities: each augmentation
4515     increases flow by at least 1, so the loop terminates in at most
4516     max-flow-value iterations."""
4517     if _C is not None and n <= 7:

```

```

4518         flat, ptr = _c_flat(mu)
4519         return bool(_C.tiny_maxflow(ptr, n, s, t, cap))
4520 residual = mu.astype(int)          # astype already returns a fresh, mutable
    ↪ copy
4521 flow = 0
4522 while flow <= cap:
4523     parent = [-1] * n
4524     parent[s] = s
4525     queue = [s]
4526     while queue and parent[t] == -1:
4527         u = queue.pop()
4528         for v in range(n):
4529             if parent[v] == -1 and residual[u, v] > 0:
4530                 parent[v] = u
4531                 queue.append(v)
4532     if parent[t] == -1:
4533         return False # flow is maximal and <= cap
4534     bottleneck = 10 ** 9
4535     v = t
4536     while v != s:
4537         bottleneck = min(bottleneck, residual[parent[v], v])
4538         v = parent[v]
4539     v = t
4540     while v != s:
4541         residual[parent[v], v] -= bottleneck
4542         residual[v, parent[v]] += bottleneck
4543         v = parent[v]
4544     flow += bottleneck
4545 return True
4546
4547
4548 def _canonical_form(mu: np.ndarray) -> bytes:
4549     """A canonical key for the isomorphism class of a directed multigraph.
4550
4551     The key is the lexicographically smallest flattened multiplicity matrix over
4552     all vertex relabellings: two multiplicity matrices have the same key iff one
4553     is a vertex permutation of the other. This is the SOURCE OF TRUTH for
4554     isomorphism here, with no dependency on any external library, so the
4555     enumeration's correctness never rests on an outside binary. For 'n=7' it is
4556     5040 permutations per graph, which is cheap on CPU; storing one key per class
4557     is what bounds the enumeration's memory to the number of classes rather than
4558     the number of labelled copies.
4559     """
4560     n = mu.shape[0]
4561     if _C is not None and n <= 7:
4562         flat, ptr = _c_flat(mu)
4563         out = np.empty(n * n, dtype=np.int32)
4564         out_ptr = out.ctypes.data_as(_ct.POINTER(_ct.c_int))
4565         _C.canonical_form_min(ptr, n, out_ptr)
4566         return out.tobytes()
4567     best: bytes | None = None
4568     for perm in permutations(range(n)):
4569         p = list(perm)
4570         cand = mu[np.ix_(p, p)].tobytes()
4571         if best is None or cand < best:
4572             best = cand
4573     assert best is not None # n >= 1, so at least the identity permutation exists
4574     return best

```

```

4575
4576
4577 def enumerate_extremal_directed_multigraphs(
4578     n: int, m: int, target_arcs: int, max_degree: int | None = None,
4579     up_to_iso: bool = True,
4580 ) -> list[np.ndarray]:
4581     """Exhaustively list the directed multigraphs on 'n' vertices with
4582     multiplicities in {0..m-1}, ' $\lambda^{\max} \leq m-1$ ', exactly 'target_arcs'
4583     arcs and (optionally) maximum total degree at most 'max_degree'.
4584
4585     This is the finite-base tool for the m = 3 chain (see claude.md and the
4586     commented rem:odd-step-roadmap): the characterisations at n = 4, 5 feed
4587     the recursion behind statement (b) at n = 7. DFS in vertex-block order
4588     with three prunings: multiplicity capacity, the induced-subgraph arc bound
4589     on every completed prefix, and exact flow checks on completed prefixes
4590     (induced flows only grow, so a violation is final).
4591
4592     SOUNDNESS: the induced-subgraph bound is a PROVED upper bound for j <= 6
4593     (from the cut-counting MILP; see 'known' dict below). For j >= 7 it
4594     falls back to the CONJECTURED bipartite bound  $\lfloor j^2/4 \rfloor$ , which is not
4595     yet proved. Therefore this function is a sound complete search only for
4596     n <= 6. For n >= 7 it is a heuristic lower-bound search: it may miss
4597     extremal graphs whose j=7 prefix exceeds the bipartite bound (if the
4598     conjecture is false) or whose search branch was pruned by it.
4599     Returns multiplicity matrices, deduplicated up to vertex permutation when
4600     'up_to_iso' is set. Deduplication is STREAMED through a canonical form
4601     (:func: '_canonical_form') as graphs are found, so peak memory is bounded by
4602     the number of isomorphism classes rather than the number of labelled copies
4603     (the previous buffer-everything approach exhausted RAM on n=7). The returned
4604     list, and its order, are unchanged by this.
4605     """
4606     # Ordered pairs grouped so that all pairs inside {0..j} precede vertex j+1.
4607     order: list[tuple[int, int]] = []
4608     for j in range(1, n):
4609         for i in range(j):
4610             order.append((i, j))
4611             order.append((j, i))
4612     block_end = {j: 2 * ((j + 1) * j // 2) for j in range(1, n)} # prefix length
4613     # SOUNDNESS NOTE: for j <= 6 the prefix cap (m-1)*M*(j) is a PROVED upper
4614     # bound, so pruning at that cap is safe. For j >= 7, _PROVEN_MSTAR.get(j, 0)
4615     #  $\hookrightarrow$  == 0
4616     # and we fall back to (m-1)*floor(j^2/4), which equals the conjectured
4617     # bipartite bound and is NOT yet proved. Any call with n >= 7 therefore
4618     # uses an unverified pruning at the j=7 boundary: the enumeration is sound
4619     # as a lower-bound search but cannot certify completeness for n >= 7.
4620     # To produce a proof for n=7 the j=7 pruning must be disabled or replaced
4621     # by a proved bound (e.g. from the MILP once M*(7) is confirmed).
4622     prefix_cap = {j: (m - 1) * max(_PROVEN_MSTAR.get(j, 0), (j * j) // 4)
4623                   for j in range(2, n + 1)}
4624
4625     mu = np.zeros((n, n), dtype=int)
4626     total_pairs = len(order)
4627     # Stream results straight into the deduplicated output instead of buffering
4628     # every labelled feasible matrix first. Memory is then bounded by the number
4629     # of ISOMORPHISM CLASSES (one canonical key in 'seen', one representative in
4630     # the list), not by the number of labelled copies, which is what previously
4631     # blew up to tens of gigabytes on n=7. The set of matrices returned, and
4632     # their order, are identical to the old buffer-then-dedup code.

```

```

4632 seen: set[bytes] = set()
4633 representatives: list[np.ndarray] = []
4634
4635 def _record(matrix: np.ndarray) -> None:
4636     # The DFS only checked prefixes {0..j} for j < n, so confirm the full
4637     # n-vertex graph is feasible before keeping it (cheap, and done first so a
4638     # rejected graph never pays for the canonical form).
4639     if any(_tiny_maxflow(matrix, n, s, t, m - 1)
4640           for s in range(n) for t in range(n) if s != t):
4641         return
4642     if up_to_iso:
4643         canon = _canonical_form(matrix)
4644         if canon in seen:
4645             return
4646         seen.add(canon)
4647     representatives.append(matrix.copy())
4648
4649 def dfs(pos: int, arcs: int) -> None:
4650     if arcs + (m - 1) * (total_pairs - pos) < target_arcs or arcs >
4651         ↪ target_arcs:
4652         return
4653     if pos == total_pairs:
4654         if arcs == target_arcs:
4655             _record(mu)
4656             return
4657     completed = next((j for j in range(1, n) if block_end[j] == pos), None)
4658     if completed is not None:
4659         j = completed + 1 # prefix {0..completed} has j vertices
4660         if arcs > prefix_cap.get(j, 10 ** 9):
4661             return
4662         for s in range(j):
4663             for t in range(j):
4664                 if s != t and _tiny_maxflow(mu, n, s, t, m - 1):
4665                     return
4666     u, v = order[pos]
4667     for value in range(m):
4668         mu[u, v] = value
4669         if max_degree is None or (mu[u, :].sum() + mu[:, u].sum() <=
4670             ↪ max_degree
4671             and mu[v, :].sum() + mu[:, v].sum() <=
4672             ↪ max_degree):
4673             dfs(pos + 1, arcs + value)
4674         mu[u, v] = 0
4675
4676 dfs(0, 0)
4677 return representatives
4678
4679 def _geng_support_graphs(
4680     n: int, min_edges: int, max_edges: int, geng_path: str = "geng",
4681 ) -> Iterator[list[tuple[int, int]]]:
4682     """Yield one edge list per isomorphism class of simple graph on 'n'
4683     vertices with 'min_edges' to 'max_edges' edges, via nauty's 'geng'.
4684
4685     Each edge is an ordered tuple '(u, v)' with 'u < v'; isolated vertices
4686     are kept ('geng' always emits all 'n' vertices). 'geng' writes the
4687     non-isomorphic graphs in graph6 to stdout and we parse each with _graph6_edges
4688     ↪ .

```

```

4686     Raises ‘‘RuntimeError’’ if ‘‘geng’’ is not on PATH.
4687     """
4688     exe = shutil.which(geng_path)
4689     if exe is None:
4690         raise RuntimeError(
4691             f"nauty's '{geng_path}' was not found on PATH. Install nauty "
4692             "(e.g. the 'nauty' package) or pass geng_path=...; the "
4693             "generation enumerator needs it."
4694         )
4695     if max_edges < min_edges:
4696         return
4697     proc = subprocess.run(
4698         [exe, str(n), f"{{min_edges}}:{{max_edges}}"],
4699         capture_output=True, check=True,
4700     )
4701     for line in proc.stdout.splitlines():
4702         line = line.strip()
4703         if not line:
4704             continue
4705         yield _graph6_edges(line)
4706
4707
4708 def _graph6_edges(data: bytes) -> list[tuple[int, int]]:
4709     """Decode one graph6 line (geng's output format) into its ‘‘(i, j)’’ edges, ‘‘
4710         ↳ i < j’’.
4711
4712     graph6 stores ‘‘n’’ in the first byte (offset 63), then the upper-triangle
4713     adjacency bits column by column (for ‘‘j’’ from 1, all ‘‘i < j’’), six bits
4714         ↳ per
4715     subsequent byte, high bit first. We only ever generate ‘‘n <= 12’’ supports,
4716         ↳ so
4717     the single-byte header case is all that is needed and the format needs no
4718     networkx to read.
4719     """
4720     values = [byte - 63 for byte in data]
4721     n = values[0]
4722     bits = [(value >> shift) & 1 for value in values[1:] for shift in range(5, -1,
4723         ↳ -1)]
4724     index = 0
4725     edges: list[tuple[int, int]] = []
4726     for j in range(1, n):
4727         for i in range(j):
4728             if bits[index]:
4729                 edges.append((i, j))
4730             index += 1
4731     return edges
4732
4733
4734 def _decorate_support_worker(task: tuple) -> list[np.ndarray]:
4735     """Every feasible extremal decoration of ONE undirected support graph.
4736
4737     Module-level and picklable, so :func:‘
4738         ↳ enumerate_extremal_directed_multigraphs_via_generation‘
4739     can fan the supports out across processes (each support is independent, and
4740     two non-isomorphic supports can never yield the same multigraph). Returns one
4741     multiplicity matrix per decoration, deduplicated within this support when
4742     ‘‘up_to_iso’’. The decoration logic and its pruning are identical to the
4743     sequential generator; only the ownership of the scratch arrays moves here.

```

```

4739 """
4740 n, m, target_arcs, max_degree, up_to_iso, edges = task
4741 per_edge_max = 2 * (m - 1)
4742 # Per-edge states (a, b) = (mu[u,v], mu[v,u]) excluding (0,0): a support edge
4743 # carries at least one arc in some direction.
4744 states = [(a, b) for a in range(m) for b in range(m) if (a, b) != (0, 0)]
4745 mu = np.zeros((n, n), dtype=int)
4746 deg = np.zeros(n, dtype=int)
4747 out_deg = np.zeros(n, dtype=int)
4748 in_deg = np.zeros(n, dtype=int)
4749 seen: set[bytes] = set()
4750 reps: list[np.ndarray] = []
4751
4752 def feasible_prefix(j: int) -> bool:
4753     cap = m - 1
4754     for s in range(j):
4755         if out_deg[s] <= cap:
4756             continue
4757         for t in range(j):
4758             if s != t and in_deg[t] > cap and _tiny_maxflow(mu, n, s, t, cap):
4759                 return False
4760     return True
4761
4762 # Decorate in vertex-block order: pairs sorted by larger endpoint, then
4763 # smaller, so after the last pair whose larger endpoint is w the induced
4764 # subgraph on {0..w} is complete and the prefix bound for size w+1 applies.
4765 sps = sorted(edges, key=lambda e: (e[1], e[0]))
4766 bound: list[int | None] = [None] * len(sps)
4767 for i, (_, w) in enumerate(sps):
4768     if i + 1 == len(sps) or sps[i + 1][1] > w:
4769         bound[i] = w + 1
4770
4771 def decorate(idx: int, arcs: int) -> None:
4772     remaining = len(sps) - idx
4773     if arcs + remaining > target_arcs:
4774         return
4775     if arcs + remaining * per_edge_max < target_arcs:
4776         return
4777     if idx == len(sps):
4778         # every support edge decorated
4779         if arcs == target_arcs and feasible_prefix(n):
4780             if up_to_iso:
4781                 canon = _canonical_form(mu)
4782                 if canon in seen:
4783                     return
4784                 seen.add(canon)
4785                 reps.append(mu.copy())
4786             return
4787     u, v = sps[idx]
4788     j = bound[idx]
4789     # prefix size to verify after this edge, or None
4790     for a, b in states:
4791         s = a + b
4792         if arcs + s > target_arcs:
4793             continue
4794         if max_degree is not None and (deg[u] + s > max_degree
4795                                         or deg[v] + s > max_degree):
4796             continue
4797         mu[u, v] = a
4798         mu[v, u] = b

```

```

4797         deg[u] += s
4798         deg[v] += s
4799         out_deg[u] += a; in_deg[v] += a
4800         out_deg[v] += b; in_deg[u] += b
4801         ok = True
4802         if j is not None:      # block boundary: prefix {0..j-1} is complete
4803             cap = _PROVEN_MSTAR.get(j)      # j == prefix size
4804             if cap is not None and arcs + s > (m - 1) * cap:
4805                 ok = False                  # PROVED induced-arc bound
4806             elif not feasible_prefix(j):
4807                 ok = False
4808         if ok:
4809             decorate(idx + 1, arcs + s)
4810         deg[u] -= s
4811         deg[v] -= s
4812         out_deg[u] -= a; in_deg[v] -= a
4813         out_deg[v] -= b; in_deg[u] -= b
4814         mu[u, v] = 0
4815         mu[v, u] = 0
4816
4817     decorate(0, 0)
4818     return reps
4819
4820
4821 def enumerate_extremal_directed_multigraphs_via_generation(
4822     n: int, m: int, target_arcs: int, max_degree: int | None = None,
4823     up_to_iso: bool = True, geng_path: str = "geng", parallel: bool = True,
4824 ) -> list[np.ndarray]:
4825     """Sound, geng-seeded twin of :func:'enumerate_extremal_directed_multigraphs'.
4826
4827     Lists every directed multigraph on 'n' vertices with multiplicities in
4828     {0..m-1}, 'lambda^max <= m-1', exactly 'target_arcs' arcs and
4829     (optionally) maximum total degree '<= max_degree', deduplicated up to
4830     isomorphism. Same object and same return format (multiplicity matrices) as
4831     the DFS enumerator.
4832
4833     Following J. Goedgebeur's suggestion, the underlying undirected SUPPORT
4834     graph (the pairs carrying at least one arc) is generated once per
4835     isomorphism class by nauty's 'geng'; each support is then decorated, in
4836     vertex-block order, with a directed multiplicity pair '(mu[u,v], mu[v,u])'
4837     in {0..m-1}^2 minus (0,0) per support edge. Pruning is the running arc
4838     total, the degree cap, and -- at each completed prefix on 'j' vertices --
4839     the PROVED induced-arc bound 'arcs(prefix) <= (m-1) M*(j)' together with a
4840     prefix feasibility check (induced max-flows only grow, so a prefix that
4841     already exceeds the cap is dead).
4842
4843     Soundness for every 'n'. Two directed multigraphs with non-isomorphic
4844     supports are non-isomorphic, and every isomorphism class with a given
4845     support class has a labelling whose support equals geng's representative, so
4846     decorating every representative and canonical-deduplicating returns exactly
4847     one matrix per isomorphism class. Crucially the induced-arc bound is used
4848     ONLY for 'j <= 6', where 'M*(j)' is proved; for 'j >= 7' no arc bound
4849     is applied (only the global 'target_arcs' and feasibility). This is the
4850     difference from :func:'enumerate_extremal_directed_multigraphs', which falls
4851     back to the CONJECTURED 'floor(j^2/4)' at 'j >= 7' and is therefore a
4852     complete search only for 'n <= 6'. This generator is complete for all
4853     'n' (it never assumes an unproved value), so it can certify the finite
4854     'n = 7' classification once it finishes.

```

```

4855
4856 Requires nauty's 'geng' on PATH. For 'n <= 6' it returns the same set
4857 of isomorphism classes as :func:'enumerate_extremal_directed_multigraphs';
4858 'tests/test_solve' checks that equality.
4859
4860 With 'parallel' (default) the supports are decorated concurrently across
4861 processor cores via a :class:'~concurrent.futures.ProcessPoolExecutor', since
4862 each support is independent. This is where the 'n = 7' classification
4863 becomes practical on a multi-core machine: the work scales with the number of
4864 supports, which is exactly what fans out. It falls back to a sequential run
4865 (with a warning) where multiprocessing is unavailable, and the result is
4866 identical either way.
4867 """
4868 if m < 1:
4869     raise ValueError("m must be >= 1")
4870 per_edge_max = 2 * (m - 1)          # most arcs one undirected pair can hold
4871 if per_edge_max == 0:              # m == 1: only the empty graph is
    ↪ feasible
4872     return [np.zeros((n, n), dtype=int)] if target_arcs == 0 else []
4873 min_edges = (target_arcs + per_edge_max - 1) // per_edge_max
4874 max_edges = min(target_arcs, n * (n - 1) // 2)
4875
4876 # geng emits each undirected support iso-class once, and decorating one
4877 # support is independent of the others (two non-isomorphic supports can never
4878 # yield the same multigraph), so the supports fan out across processes, each
4879 # with its own scratch arrays inside _decorate_support_worker. The PROVED
4880 # M*(j) bound is applied there only for j <= 6; j >= 7 gets no arc bound, so
4881 # the search stays sound at every n.
4882 tasks = [(n, m, target_arcs, max_degree, up_to_iso, edges)
4883          for edges in _geng_support_graphs(n, min_edges, max_edges, geng_path)
    ↪ ]
4884
4885 representatives: list[np.ndarray] = []
4886 if parallel and len(tasks) > 1:
4887     try:
4888         with concurrent.futures.ProcessPoolExecutor() as executor:
4889             # chunksize=1: supports vary wildly in cost, so hand them out one
4890             # at a time to keep every core busy to the end.
4891             for reps in executor.map(_decorate_support_worker, tasks,
4892                                     chunksize=1):
4893                 representatives.extend(reps)
4894         except (OSError, concurrent.futures.BrokenExecutor) as exc:
4895             warnings.warn(
4896                 f"parallel support enumeration unavailable ({exc!r}); running "
4897                 "sequentially", RuntimeWarning, stacklevel=2,
4898             )
4899             for task in tasks:
4900                 representatives.extend(_decorate_support_worker(task))
4901     else:
4902         for task in tasks:
4903             representatives.extend(_decorate_support_worker(task))
4904
4905 if up_to_iso:
4906     # Distinct geng supports give non-isomorphic multigraphs, so this final
4907     # canonical pass is only a safety net over the process-local per-support
4908     # dedup; it keeps exactly one representative per isomorphism class.
4909     seen: set[bytes] = set()
4910     unique: list[np.ndarray] = []

```



```

4911         for matrix in representatives:
4912             key = _canonical_form(matrix)
4913             if key not in seen:
4914                 seen.add(key)
4915                 unique.append(matrix)
4916         representatives = unique
4917     return representatives

```

C.1.9 The gallery

Given a size and a route limit, these routines find every extremal graph there is, up to relabelling, and count the symmetries of each. Deciding when two graphs are the same graph drawn differently is done by trying all relabellings and keeping the smallest resulting table, which is slow in principle but perfectly quick at the sizes involved.

```

4920 # --- GALLERY: all non-isomorphic extremal graphs for small (n, m); entry point
4921     ↪ gallery_extremal_graphs ---
4922
4923 def _graph_from_mu(mu: np.ndarray, variant: Variant) -> Graph:
4924     """Wrap a multiplicity matrix in a fresh Graph (zero-copy)."""
4925     g = Graph(mu.shape[0], variant)
4926     g.mu[:] = mu
4927     return g
4928
4929
4930 def _aut_count_matrix(mu: np.ndarray) -> int:
4931     """Count vertex permutations that preserve ‘mu’ (equals |Aut(G)|)."""
4932     n = mu.shape[0]
4933     canon = _canonical_form(mu)
4934     return sum(1 for perm in permutations(range(n))
4935                if mu[np.ix_(list(perm), list(perm))].tobytes() == canon)
4936
4937
4938 def _dir_relabel_key(edge, p: list) -> tuple:
4939     """Permutation-relabelled key of one directed hyperedge.
4940
4941     Keeps the legacy forward shape ‘(int, tuple_of_heads)’ for forward edges
4942     (so their canonical keys and dedup counts are unchanged) and uses
4943     ‘(tuple_of_tails, tuple_of_heads)’ for general edges.
4944     """
4945     first, heads = edge
4946     if isinstance(first, (set, frozenset)):
4947         return (tuple(sorted(p[t] for t in first)), tuple(sorted(p[h] for h in
4948                        ↪ heads)))
4949     return (p[first], tuple(sorted(p[h] for h in heads)))
4950
4951 def _hyper_canonical(hyperedges: list, n: int, directed: bool) -> tuple:
4952     """Canonical key for a hyperedge collection under vertex permutation.
4953
4954     Each edge is a frozenset (undirected) or a directed hyperedge in either the
4955     forward ‘(tail, heads)’ or general ‘(tails, heads)’ form. Returns the
4956     lex-minimum over all n! relabellings.
4957     """
4958     best: tuple | None = None

```

```

4959     for perm in permutations(range(n)):
4960         p = list(perm)
4961         if directed:
4962             relabeled: tuple = tuple(sorted(_dir_relabel_key(edge, p) for edge in
4963                                     ↪ hyperedges))
4964         else:
4965             relabeled = tuple(sorted(
4966                 tuple(sorted(p[v] for v in edge)) for edge in hyperedges
4967             ))
4968             if best is None or relabeled < best:
4969                 best = relabeled
4970     return best # type: ignore[return-value]
4971
4972 def _aut_count_hyper(hyperedges: list, n: int, directed: bool) -> int:
4973     """Count vertex permutations that preserve a hyperedge collection."""
4974     canon = _hyper_canonical(hyperedges, n, directed)
4975
4976     def _relabel(p: list) -> tuple:
4977         if directed:
4978             return tuple(sorted(_dir_relabel_key(edge, p) for edge in hyperedges))
4979         return tuple(sorted(
4980             tuple(sorted(p[v] for v in edge)) for edge in hyperedges
4981         ))
4982
4983     return sum(1 for perm in permutations(range(n))
4984               if _relabel(list(perm)) == canon)
4985
4986
4987 def _hyper_to_lists(hyperedges: list, directed: bool) -> list:
4988     """Convert hyperedges to JSON-serialisable sorted integer lists.
4989
4990     Forward edges keep their ‘[tail, [heads]]’ shape; general edges use
4991     ‘[[tails], [heads]]’.
4992     """
4993     if directed:
4994         out = []
4995         for edge in hyperedges:
4996             first, heads = edge
4997             if isinstance(first, (set, frozenset)):
4998                 out.append([sorted(first), sorted(heads)])
4999             else:
5000                 out.append([first, sorted(heads)])
5001         return out
5002     return [sorted(edge) for edge in hyperedges]
5003
5004
5005 def _enum_matrix_extremals(
5006     variant: Variant,
5007     n: int,
5008     m: int,
5009     separation: str,
5010     target: int,
5011     deadline: float,
5012 ) -> tuple[list[np.ndarray], bool]:
5013     """DFS over all (variant, n, m) multiplicity matrices achieving exactly ‘
5014         ↪ target’ edges.

```

```

5015     Generalises :func:'enumerate_extremal_directed_multigraphs' to all four
5016     matrix variants (simple/multi × directed/undirected) and to vertex as well
5017     as edge separation. Returns '(reps, timed_out)' where 'reps' are
5018     deduplicated representatives (one per isomorphism class).
5019
5020     Pruning strategy: arc-count reachability at every step; prefix edge-
5021     connectivity check at each completed vertex block (edge separation only;
5022     vertex separation checks only the final graph because the per-node cost is
5023     too high for DFS).
5024     """
5025     max_mult = 1 if variant.simple else (m - 1)
5026     is_directed = variant.directed
5027
5028     # Vertex-block order: all pairs involving vertex j before j+1 appears.
5029     order: list[tuple[int, int]] = []
5030     for j in range(1, n):
5031         for i in range(j):
5032             order.append((i, j))
5033             if is_directed:
5034                 order.append((j, i))
5035     total = len(order)
5036
5037     # Position at which the subgraph on {0..j} is fully decided.
5038     # Directed: j*(j+1) positions; undirected: j*(j+1)//2 positions.
5039     step_per_vertex = 2 if is_directed else 1
5040     block_end: dict[int, int] = {
5041         j: step_per_vertex * j * (j + 1) // 2 for j in range(1, n)
5042     }
5043
5044     mu = np.zeros((n, n), dtype=int)
5045     seen: set[bytes] = set()
5046     reps: list[np.ndarray] = []
5047     timed_out = [False]
5048
5049     def _feasible_full() -> bool:
5050         g = _graph_from_mu(mu, variant)
5051         if separation == "edge":
5052             return max_edge_connectivity(g) <= m - 1
5053         return max_vertex_connectivity(g) <= m - 1
5054
5055     def dfs(pos: int, edges: int) -> None:
5056         if timed_out[0]:
5057             return
5058         if time.time() > deadline:
5059             timed_out[0] = True
5060             return
5061         if edges + max_mult * (total - pos) < target or edges > target:
5062             return
5063         if pos == total:
5064             if edges == target and _feasible_full():
5065                 canon = _canonical_form(mu)
5066                 if canon not in seen:
5067                     seen.add(canon)
5068                     reps.append(mu.copy())
5069             return
5070         # At each block boundary: prune if the prefix subgraph is already
5071         # infeasible (edge connectivity only -- cheap with _tiny_maxflow).
5072         completed_j = next(

```

```

5073         (j for j in range(1, n) if block_end.get(j) == pos), None
5074     )
5075     if completed_j is not None and separation == "edge":
5076         sub_n = completed_j + 1
5077         if any(_tiny_maxflow(mu, n, s, t, m - 1)
5078             for s in range(sub_n) for t in range(sub_n) if s != t):
5079             return
5080     u, v = order[pos]
5081     for val in range(max_mult + 1):
5082         mu[u, v] = val
5083         if not is_directed:
5084             mu[v, u] = val
5085         dfs(pos + 1, edges + val)
5086     mu[u, v] = 0
5087     if not is_directed:
5088         mu[v, u] = 0
5089
5090     dfs(0, 0)
5091     return reps, timed_out[0]
5092
5093
5094 def _enum_hyper_extremals(
5095     directed: bool,
5096     vertex_split: bool,
5097     r: int,
5098     n: int,
5099     m: int,
5100     target: int,
5101     deadline: float,
5102     *,
5103     kind: str = "forward",
5104 ) -> tuple[list[list], bool]:
5105     """DFS over all r-uniform hypergraphs on n vertices with exactly ‘target‘
5106         ⇨ edges.
5107
5108     Iterates candidate hyperedges in order (include / exclude) and prunes
5109     immediately when adding a hyperedge pushes the connectivity above m-1
5110     (monotonicity: adding edges never decreases connectivity, so no superset
5111     can be feasible at that branch either). ‘kind‘ selects the directed
5112     orientation model. Returns ‘(reps, timed_out)’ where each representative
5113     in ‘reps‘ is a list of JSON-serialisable sorted vertex lists.
5114
5115     """
5116     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
5117     total = len(candidates)
5118     active: list = []
5119     seen: set[tuple] = set()
5120     reps: list[list] = []
5121     timed_out = [False]
5122
5123     def dfs(pos: int, count: int) -> None:
5124         if timed_out[0]:
5125             return
5126         if time.time() > deadline:
5127             timed_out[0] = True
5128             return
5129         if count + (total - pos) < target or count > target:
5130             return
5131         if pos == total:

```

```

5130         if count == target:
5131             canon = _hyper_canonical(active, n, directed)
5132             if canon not in seen:
5133                 seen.add(canon)
5134                 reps.append(_hyper_to_lists(active, directed))
5135             return
5136         # Include candidates[pos] and recurse only if still feasible.
5137         active.append(candidates[pos])
5138         h = Hypergraph(n, active, directed=directed)
5139         if max_hyper_connectivity(h, vertex_split=vertex_split) <= m - 1:
5140             dfs(pos + 1, count + 1)
5141         active.pop()
5142         # Exclude candidates[pos].
5143         dfs(pos + 1, count)
5144
5145     dfs(0, 0)
5146     return reps, timed_out[0]
5147
5148
5149 def gallery_extremal_graphs(
5150     max_n: int = 7,
5151     max_m: int = 4,
5152     r: int = 3,
5153     time_per_case: float = 3.0,
5154 ) -> dict:
5155     """Collect every non-isomorphic extremal graph for all 12 variants at small n,
5156         ↪ m.
5157
5158     For each (variant, n, m) triple where the enumeration finishes within
5159     'time_per_case' seconds, the function:
5160
5161     1. runs a short repeated search to discover the extremal edge count;
5162     2. exhaustively enumerates every graph achieving that count;
5163     3. deduplicates by isomorphism class and records 'n! / |Aut(G)|' (the
5164         number of labelled copies) per class.
5165
5166     Returns a JSON-serialisable dict with structure::
5167
5168         result[variant_key]["n={n}_m={m}"] = {
5169             "extremal_value": int,
5170             "classes": [{"repr": <matrix or edge list>, "labelled_count": int}],
5171             "total_labelled": int,      # sum of labelled_count over all classes
5172             "complete": bool,          # False when the deadline was hit
5173         }
5174
5175     For matrix variants 'repr' is the multiplicity matrix (list of lists of
5176     int). For hypergraph variants it is a list of hyperedges: each edge is a
5177     sorted list of vertex indices (undirected) or '[tail, [head, ...]]'
5178     (directed).
5179
5180     The twelve variant keys are:
5181     'simple_undirected_{edge,vertex}',
5182     'simple_directed_{edge,vertex}',
5183     'multi_undirected_{edge,vertex}',
5184     'multi_directed_{edge,vertex}',
5185     'hyper_undirected_r{r}_{edge,vertex}',
5186     'hyper_directed_r{r}_{edge,vertex}'.
```

```

5187 The two ‘‘multi*_vertex’’ keys report the reduced SIMPLE problem, because
5188 the thesis’s vertex measure is blind to parallel copies and the variant
5189 collapses onto the simple one (see the config comment below).
5190 """
5191 # The two multi-vertex rows run on the SIMPLE variant: the checker’s vertex
5192 # mode gives every adjacency capacity one, so parallel copies never change
5193 # kappa and a raw multigraph search would just fill every cell to m-1 for
5194 # free (it once reported a "36-arc extremal K_4 at multiplicity 3"). The
5195 # thesis reduces these variants to the simple ones (tab:summary), and the
5196 # gallery must report the reduced problem, exactly as the variant grids do.
5197 matrix_configs: list[tuple[str, Variant, str]] = [
5198     ("simple_undirected_edge", SIMPLE_UNDIRECTED, "edge"),
5199     ("simple_undirected_vertex", SIMPLE_UNDIRECTED, "vertex"),
5200     ("simple_directed_edge", SIMPLE_DIRECTED, "edge"),
5201     ("simple_directed_vertex", SIMPLE_DIRECTED, "vertex"),
5202     ("multi_undirected_edge", MULTI_UNDIRECTED, "edge"),
5203     ("multi_undirected_vertex", SIMPLE_UNDIRECTED, "vertex"),
5204     ("multi_directed_edge", MULTI_DIRECTED, "edge"),
5205     ("multi_directed_vertex", SIMPLE_DIRECTED, "vertex"),
5206 ]
5207 hyper_configs: list[tuple[str, bool, bool]] = [
5208     ("hyper_undirected_r{r}_edge", False, False),
5209     ("hyper_undirected_r{r}_vertex", False, True),
5210     ("hyper_directed_r{r}_edge", True, False),
5211     ("hyper_directed_r{r}_vertex", True, True),
5212 ]
5213
5214 result: dict = {}
5215
5216 for key, variant, separation in matrix_configs:
5217     result[key] = {}
5218     for n in range(2, max_n + 1):
5219         for m in range(2, max_m + 1):
5220             case_deadline = time.time() + time_per_case
5221             # Step 1: discover the extremal value via repeated search.
5222             # 2000 steps/restart is enough to converge for n <= 6;
5223             # we run at least 3 seeds before the deadline to be robust
5224             # against unlucky single-seed runs.
5225             best_val = 0
5226             seed = 0
5227             while time.time() < case_deadline - 2.0 or seed < 3:
5228                 if time.time() > case_deadline - 0.5:
5229                     break
5230                 res = search_for_dense_graph(
5231                     variant, n, m, separation=separation,
5232                     steps=2000, seed=seed)
5233                 best_val = max(best_val, res.best_edge_count)
5234                 seed += 1
5235             # Step 2: enumerate all graphs at that value.
5236             enum_deadline = min(case_deadline, time.time() + 2.0)
5237             reps, timed_out = _enum_matrix_extremals(
5238                 variant, n, m, separation, best_val, enum_deadline)
5239             # Step 3: count automorphisms and labelled copies.
5240             fn = math.factorial(n)
5241             classes = [
5242                 {"repr": mu.tolist(),
5243                  "labelled_count": fn // _aut_count_matrix(mu)}
5244                 for mu in reps

```

```

5245         ]
5246         result[key][f"n={n}_m={m}"] = {
5247             "extremal_value": best_val,
5248             "classes": classes,
5249             "total_labelled": sum(c["labelled_count"] for c in classes),
5250             "complete": not timed_out,
5251         }
5252
5253     for key, directed, vertex_split in hyper_configs:
5254         result[key] = {}
5255         for n in range(r, max_n + 1):
5256             for m in range(2, max_m + 1):
5257                 case_deadline = time.time() + time_per_case
5258                 # Brute-force search (small n): finds the extremal value.
5259                 best_val, _, completed = _brute_force_hypergraph(
5260                     n, r, m, case_deadline,
5261                     directed=directed, vertex_split=vertex_split)
5262                 if not completed:
5263                     result[key][f"n={n}_m={m}"] = {
5264                         "extremal_value": best_val, "classes": [],
5265                         "total_labelled": 0, "complete": False,
5266                     }
5267                     continue
5268                 # Enumerate all achieving best_val.
5269                 enum_deadline = min(case_deadline, time.time() + 2.0)
5270                 hyper_reps, timed_out = _enum_hyper_extremals(
5271                     directed, vertex_split, r, n, m, best_val, enum_deadline)
5272                 fn = math.factorial(n)
5273                 classes = []
5274                 for edge_lists in hyper_reps:
5275                     raw = ((row[0], frozenset(row[1])) for row in edge_lists)
5276                     if directed else [frozenset(e) for e in edge_lists]
5277                     classes.append({
5278                         "repr": edge_lists,
5279                         "labelled_count": fn // _aut_count_hyper(raw, n, directed)
5280                         ↵ ,
5281                     })
5282                 result[key][f"n={n}_m={m}"] = {
5283                     "extremal_value": best_val,
5284                     "classes": classes,
5285                     "total_labelled": sum(c["labelled_count"] for c in classes),
5286                     "complete": not timed_out,
5287                 }
5288
5289     return result
5290
5291 def save_gallery_json(gallery: dict, path: str | Path) -> None:
5292     """Write the gallery dict to a JSON file (pretty-printed)."""
5293     with open(path, "w") as f:
5294         json.dump(gallery, f, indent=2)
5295
5296 def prove_integral_arc_bound(n: int, m: int, target: int, *,
5297                             time_limit: float = 3000.0,
5298                             use_gurobi: bool | None = None,
5299                             show_solver_log: bool = False) -> str:
5300     """Decide by MILP whether some directed multigraph on 'n' vertices with

```

```

5302     multiplicities in {0..m-1} and ‘‘lambda^max <= m-1’’ has >= ‘‘target’’ arcs.
5303
5304     Returns "INFEASIBLE" (proving  $L_m^{\text{dir}}(n) < \text{target}$ ), "FEASIBLE", or
5305     "LIMIT". This is the integral companion of the fractional prover: the
5306     m = 3 chain (rem:odd-step-roadmap) needs only  $L_3^{\text{dir}}(7) = 24$ , i.e.
5307     INFEASIBLE at target 25, which is a far friendlier MILP than  $M^*(7)$  because the
5308     weights themselves are integers. Same exact :func:‘_cut_counting_model’ with
5309     cap = m-1 and integer multiplicities, all its proved-valid families on, plus
5310     ↪ the
5311     one constraint that the multiplicities total at least ‘‘target’’.
5312     """
5313     _require_pulp()
5314     prob, w = _cut_counting_model(
5315         n, cap=float(m - 1), integer=True, two_hop=True,
5316         symmetry=True, deletion=True, degree_pair=True,
5317     )
5318     prob += pulp.lpSum(w.values()) >= target           # the arc target
5319     prob += 0                                           # feasibility, no objective
5320
5321     prob.solve(_pick_solver(time_limit, show_solver_log, use_gurobi))
5322     status = pulp.LpStatus[prob.status]
5323     if status == "Infeasible":
5324         return "INFEASIBLE"
5325     if status == "Optimal":
5326         return "FEASIBLE"
5327     return "LIMIT"
5328
5329 def _float_maxflow_value(capacity: np.ndarray, source: int, target: int,
5330                          eps: float = 1e-12) -> float:
5331     """Max-flow value for FLOAT capacities (Edmonds-Karp, shortest augmenting path
5332     ↪ ).
5333
5334     scipy’s csgraph max-flow is integer-only, so the fractional check below needs
5335     its own tiny routine. BFS augmenting paths give a polynomial bound
5336     ↪ independent
5337     of the capacity magnitudes, so the loop terminates even on irrational-looking
5338     floats; ‘eps’ treats a near-zero residual as saturated.
5339     """
5340     n = capacity.shape[0]
5341     residual = capacity.astype(float).copy()
5342     flow = 0.0
5343     while True:
5344         parent = [-1] * n
5345         parent[source] = source
5346         queue = deque([source])
5347         while queue and parent[target] == -1:
5348             u = queue.popleft()           # FIFO: shortest augmenting
5349             ↪ path
5350             for v in range(n):
5351                 if parent[v] == -1 and residual[u, v] > eps:
5352                     parent[v] = u
5353                     queue.append(v)
5354             if parent[target] == -1:
5355                 return flow
5356         bottleneck = float("inf")
5357         v = target
5358         while v != source:

```



```

5356         bottleneck = min(bottleneck, residual[parent[v], v])
5357         v = parent[v]
5358     v = target
5359     while v != source:
5360         residual[parent[v], v] -= bottleneck
5361         residual[v, parent[v]] += bottleneck
5362         v = parent[v]
5363     flow += bottleneck
5364
5365
5366 def fractional_flows_feasible(weights: np.ndarray, tol: float = 1e-9) -> bool:
5367     """Check the scaled multigraph constraint: all pairwise max-flows <= 1.
5368
5369     ‘‘weights‘‘ is an ‘‘n x n‘‘ matrix with entries in [0,1] (zero diagonal),
5370     the scaled multiplicity matrix  $\mu/(m-1)$  of lem:scaling-reduction.
5371     """
5372     n = weights.shape[0]
5373     capacity = np.where(weights > 1e-12, weights, 0.0).astype(float)
5374     np.fill_diagonal(capacity, 0.0)
5375     for s in range(n):
5376         for t in range(n):
5377             if t != s and _float_maxflow_value(capacity, s, t) > 1 + tol:
5378                 return False
5379     return True
5380
5381
5382 def fractional_search(n: int, objective: str = "min_degree", steps: int = 6000,
5383                     seed: int = 0, start: str = "bipartite") -> tuple[np.ndarray
5384                               ↪ , float]:
5385     """Hunt for a fractional counterexample to the odd-n directed multigraph
5386     statements (conj:min-degree and the total-weight bound).
5387
5388     ‘‘objective‘‘: "min_degree" maximises the minimum weighted total degree
5389     (conjecture: cannot exceed  $k = (n-1)/2$  for odd n); "total" maximises the
5390     total weight (conjecture: cannot exceed  $\max(2(n-1), \text{floor}(n^2/4))$ ).
5391     ‘‘start‘‘: "bipartite" (the conjectured extremiser), "zero", or "random".
5392     Returns the best weighting found and its objective value. A value
5393     exceeding the conjectured ceiling would refute the conjecture for every
5394      $m - 1$  divisible by the weight denominators; none was found at  $n = 7, 9$ .
5395     """
5396     rng = random.Random(seed)
5397     weights = np.zeros((n, n))
5398     if start == "bipartite":
5399         for a in range(n // 2):
5400             for b in range(n // 2, n):
5401                 weights[a, b] = 1.0
5402     elif start == "random":
5403         weights = np.array([[rng.random() if i != j else 0.0 for j in range(n)]
5404                             for i in range(n)])
5405         while not fractional_flows_feasible(weights):
5406             weights *= 0.85
5407
5408 def score(w: np.ndarray) -> float:
5409     if objective == "total":
5410         return float(w.sum())
5411     degrees = w.sum(axis=0) + w.sum(axis=1)
5412     return float(degrees.min()) + 1e-3 * float(w.sum())

```

```

5413     current = score(weights)
5414     best, best_w = current, weights.copy()
5415     for it in range(steps):
5416         temperature = 0.08 * (1 - it / steps) + 1e-4
5417         u, v = rng.randrange(n), rng.randrange(n)
5418         if u == v:
5419             continue
5420         old = weights[u, v]
5421         weights[u, v] = min(1.0, max(0.0, old + (rng.random() * 2 - 1)
5422                                 * max(0.02, temperature)))
5423         if weights[u, v] == old:
5424             continue
5425         value = score(weights)
5426         accept = value > current or rng.random() < math.exp((value - current)
5427                                                             / temperature)
5428         if accept and fractional_flows_feasible(weights):
5429             current = value
5430             if value > best:
5431                 best, best_w = value, weights.copy()
5432         else:
5433             weights[u, v] = old
5434     return best_w, best
5435
5436
5437 # --- SELF-CHECK: python erdos915_unified.py runs every invariant; exits 0 on all-
5438 ↪ PASS ---
5439
5439 _failures = 0
5440
5441
5442 def check(label: str, condition: bool) -> None:
5443     """Print a PASS/FAIL line and tally failures."""
5444     global _failures
5445     print(f" {'PASS' if condition else 'FAIL'} {label}")
5446     if not condition:
5447         _failures += 1
5448
5449
5450 def section(title: str) -> None:
5451     """Print a section header."""
5452     print(f"\n{title}")

```

C.1.10 The self-check

The file ends with a suite of checks that runs when the program is executed directly. It re-derives known values, tests each construction against its predicted size and connectivity, confirms the fast routines agree with the slow ones they stand in for, and verifies the bounds proved in [Appendix A](#) on sampled graphs. It prints a line per check and fails loudly rather than quietly if any of them stops holding, which is what makes it useful as a regression test after an edit.

```

5455 def _run_checks() -> int:
5456     """Run every invariant check; return 1 on any failure, else 0.
5457
5458     This is the program checking itself end to end, using the bare top-level
5459     names defined above in this single file.

```

```

5460 """
5461 section("Checker: the directed m=2 values ell_2^dir(n) = 2, 4, 6, 8")
5462 for n, expected in [(2, 2), (3, 4), (4, 6), (5, 8)]:
5463     graph = double_star(n, m=2, directed=True)
5464     check(f"double star n={n}: {graph.edge_count()} arcs, lambda^max={
5465         ↪ max_edge_connectivity(graph)}",
5466           graph.edge_count() == expected and max_edge_connectivity(graph) ==
5467             ↪ 1)
5468
5469 section("Checker: one-directional bipartite is quadratic and feasible")
5470 for n in range(2, 8):
5471     graph = one_directional_bipartite(n)
5472     check(f"bipartite n={n}: {graph.edge_count()} arcs (floor(n^2/4)={ (n*n)
5473         ↪ //4 })",
5474           graph.edge_count() == (n * n) // 4 and max_edge_connectivity(graph)
5475             ↪ == 1)
5476
5477 section("Checker: the undirected cycle is the infeasible witness for m=2")
5478 cycle = Graph(6, SIMPLE_UNDIRECTED)
5479 for v in range(6):
5480     cycle.add_edge(v, (v + 1) % 6)
5481 check(f"C_6 has lambda^max={max_edge_connectivity(cycle)}",
5482       ↪ max_edge_connectivity(cycle) == 2)
5483
5484 if NETWORKX_AVAILABLE:
5485     section("Gomory-Hu shortcut agrees with brute-force lambda^max")
5486     tree = clique_tree(3, 4)
5487     check(f"GH={max_edge_connectivity_via_tree(tree)} vs direct={
5488         ↪ max_edge_connectivity(tree)}",
5489           max_edge_connectivity_via_tree(tree) == max_edge_connectivity(tree))
5490 else:
5491     section("Gomory-Hu shortcut skipped (optional networkx not installed)")
5492
5493 section("Vertex connectivity: K_4-trees are globally 3-connected")
5494 for blocks in range(0, 5):
5495     ktrees = clique_tree(4, blocks)
5496     check(f"K_4-tree on {ktrees.num_vertices} vertices: kappa={
5497         ↪ min_vertex_connectivity(ktrees)}",
5498           min_vertex_connectivity(ktrees) == 3)
5499
5500 section("Constructions: the augmented bipartite conjecture values")
5501 counterexample = augmented_bipartite(10, 3)
5502 check(f"m=3,n=10: {counterexample.edge_count()} arcs, lambda^max={
5503     ↪ max_edge_connectivity(counterexample)}",
5504       counterexample.edge_count() == 30 and max_edge_connectivity(
5505         ↪ counterexample) == 2)
5506
5507 for n, m in [(8, 3), (10, 4), (12, 4), (10, 5)]:
5508     graph = augmented_bipartite(n, m)
5509     expected = (n + m - 2) ** 2 // 4
5510     check(f"n={n},m={m}: {graph.edge_count()} arcs (expected {expected}),
5511         ↪ lambda^max={max_edge_connectivity(graph)}",
5512           graph.edge_count() == expected and max_edge_connectivity(graph) == m
5513             ↪ - 1)
5514
5515 section("Hypergraphs: Berge connectivity, exact and generalising the graph
5516     ↪ case")
5517 k43 = complete_uniform_hypergraph(4, 3)

```

```

5505     check(f"complete 3-uniform  $K_4^{(3)}$ : Berge  $\lambda^{\max} = \{$ 
5506          $\hookrightarrow$   $\max_{\text{hyperedge\_connectivity}}(k43)\}$  (expect 3)",
5507         max_hyperedge_connectivity(k43) == 3)
5508     for n in [3, 4, 5]:
5509         complete = complete_uniform_hypergraph(n, 2)
5510         check(f"2-uniform  $K_{\{n\}}$  reduces to edge-connectivity {
5511              $\hookrightarrow$   $\max_{\text{hyperedge\_connectivity}}(\text{complete})\}$  (expect {n-1})",
5512             max_hyperedge_connectivity(complete) == n - 1)
5513     for n, r in [(7, 3), (10, 4), (13, 4)]:
5514         star = star_hypertree(n, r)
5515         check(f"star hypertree  $n=\{n\}$ ,  $r=\{r\}$ : {star.edge_count()} edges (expect {(n
5516              $\hookrightarrow$  -1)/(r-1)}), "
5517             f" $\lambda^{\max}=\{\max_{\text{hyperedge\_connectivity}}(\text{star})\}$ ",
5518             star.edge_count() == (n - 1) // (r - 1) and
5519              $\hookrightarrow$  max_hyperedge_connectivity(star) == 1)
5520
5521     section("Hypergraphs: directed orientation models (forward / backward /
5522          $\hookrightarrow$  general)")
5523     # The general gate admits a mixed arc {0,1} -> {2,3}: a route enters at a tail
5524     # and leaves at a head, so 0->2 carries one route and 2->0 carries none.
5525     mixed = Hypergraph(4, [(frozenset({0, 1}), frozenset({2, 3}))], directed=True)
5526     check("mixed arc {0,1}->{2,3}:  $\lambda(0,2)=1$  and  $\lambda(2,0)=0$ ",
5527         hyper_connectivity(mixed, 0, 2) == 1 and hyper_connectivity(mixed, 2, 0)
5528          $\hookrightarrow$  == 0)
5529     # Backward is the arc-reversal dual of forward, so their extremal numbers
5530      $\hookrightarrow$  agree.
5531     fwd, _ = max_feasible_hyperedges(4, 3, 3, directed=True, kind="forward",
5532          $\hookrightarrow$  time_limit=1.5)
5533     bwd, _ = max_feasible_hyperedges(4, 3, 3, directed=True, kind="backward",
5534          $\hookrightarrow$  time_limit=1.5)
5535     check(f"forward == backward at n=4, r=3, m=3 ({fwd} == {bwd})", fwd == bwd)
5536     # General contains forward and strictly beats it where vertices are scarce
5537     # (n = r); both values here are proved exact by the branch and bound.
5538     g3, e3 = max_feasible_hyperedges(3, 3, 3, directed=True, kind="general",
5539          $\hookrightarrow$  time_limit=1.5)
5540     f3, _ = max_feasible_hyperedges(3, 3, 3, directed=True, kind="forward",
5541          $\hookrightarrow$  time_limit=1.5)
5542     check(f"general 4 > forward 3 at n=3, r=3, m=3 (got {g3} vs {f3}, exact={e3})"
5543          $\hookrightarrow$  ,
5544         g3 == 4 and f3 == 3 and e3)
5545     g4, e4 = max_feasible_hyperedges(4, 4, 4, directed=True, kind="general",
5546          $\hookrightarrow$  time_limit=10.0)
5547     f4, _ = max_feasible_hyperedges(4, 4, 4, directed=True, kind="forward",
5548          $\hookrightarrow$  time_limit=3.0)
5549     check(f"general 8 doubles forward 4 at n = r = 4, m = 4 ({g4} vs {f4}, exact={
5550          $\hookrightarrow$  e4})",
5551         g4 == 8 and f4 == 4 and e4)
5552
5553     section("Monte Carlo: the appearance probability crosses the m/n scale")
5554     n, m = 30, 3
5555     below = estimate_appearance_probability(n, 0.3 * m / n, m, trials=120, seed=1)
5556     above = estimate_appearance_probability(n, 1.8 * m / n, m, trials=120, seed=1)
5557     check(f"P[appears] rises from {below:.2f} (well below m/n) to {above:.2f} (
5558          $\hookrightarrow$  well above)",
5559         below < 0.5 < above)
5560
5561     section("Monte Carlo: vertex-connectivity never exceeds edge-connectivity (
5562          $\hookrightarrow$  Whitney)")

```

```

5546     edge_dist = connectivity_distribution(20, 0.25, trials=40, separation="edge",
5547         ↪ seed=3)
5547     vertex_dist = connectivity_distribution(20, 0.25, trials=40, separation="
5548         ↪ vertex", seed=3)
5548     check("kappa^max <= lambda^max on every one of the 40 identical samples",
5549         all(kappa <= lam for kappa, lam in zip(vertex_dist, edge_dist)))
5550
5551     section("Edge vs vertex in G(n,p): they agree at small m, part company at
5552         ↪ large m")
5552     lam_small, kap_small = edge_vertex_distribution(5, 0.5, trials=120, seed=7)
5553     small_gap = sum(k < 1 for k, l in zip(kap_small, lam_small))
5554     check("n=5: kappa^max = lambda^max on every sample (no gap)",
5555         small_gap == 0 and all(k <= 1 for k, l in zip(kap_small, lam_small)))
5556     lam_large, kap_large = edge_vertex_distribution(16, 0.25, trials=200, seed=7)
5557     large_gap = sum(k < 1 for k, l in zip(kap_large, lam_large))
5558     check(f"n=16: Whitney holds and kappa^max < lambda^max on {large_gap} of 200
5559         ↪ samples",
5560         large_gap > 0 and all(k <= 1 for k, l in zip(kap_large, lam_large)))
5561
5561     if not PULP_AVAILABLE:
5562         section("Prover sections skipped (optional pulp not installed)")
5563     else:
5564         section("Prover: cut-counting proves  $M^*(n) = 2(n-1)$  optimal for small n")
5565         # The thesis (ch2) claims  $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$  is proved for ALL  $n \leq$ 
5566         ↪ 6
5567         # via the cut-counting MILP (thm:dir-multi-small).  $n=3,4,5$  all finish in
5568         # well under 60 s and are verified here.  $n=6$  takes ~1315 s (~22 min) on a
5569         # typical laptop and is NOT included in this fast loop -- it must be
5570         # verified separately:
5571         #
5572         #     result = prove_directed_multigraph(6, time_limit=2000.0)
5573         #     assert result.is_proof() and round(result.scaled_optimum) == 10
5574         #
5575         # The confirmed run (2026-06-13) is logged in program/logs/selftest_check.
5576         ↪ log
5577         # (search "n=6 OPTIMAL  $M^*(6)=10$  in 1315s"). Omitting  $n=6$  from this loop
5578         ↪ is
5579         # not an error in the proof -- the proof ran -- but it means this self-
5580         ↪ test
5581         # alone does not reproduce the full  $n \leq 6$  certification.
5582         for n in [3, 4, 5]:
5583             result = prove_directed_multigraph(n, time_limit=300.0)
5584             check(f"n={n}: status={result.status},  $M^*={result.scaled\_optimum:.1f}$ ",
5585                 ↪ proof={result.is_proof()}",
5586                 result.is_proof() and round(result.scaled_optimum) == 2 * (n -
5587                     ↪ 1))
5588
5589         section("Prover: the valid inequalities sharpen but never move the optimum
5590             ↪ ")
5591         # Turning the two-hop and symmetry-breaking rows off leaves the BARE exact
5592         # cut formulation, which must still prove the same  $M^*(3) = 4$ . If any row
5593         ↪ we
5594         # call a "valid inequality" were in fact invalid, it would cut a feasible
5595         # point and shift this optimum -- so this is the regression tripwire that
5596         # guards the prover's soundness, not just a re-run of the value.
5597         bare = prove_directed_multigraph(3, time_limit=120.0,
5598             use_two_hop=False, use_symmetry_breaking=
5599                 ↪ False)

```

```

5591     check(f"M*(3) with the tighteners off = {bare.scaled_optimum:.1f} (expect
        ↪ 4), proof={bare.is_proof()}",
5592         bare.is_proof() and round(bare.scaled_optimum) == 4)
5593
5594     section("Prover (integral): an INFEASIBLE verdict is a genuine upper-bound
        ↪ proof")
5595     # L_3^dir(4) = 12: a 12-arc multigraph exists, no 13-arc one does. This
5596     # exercises the exact mechanism behind the thesis's L_3(7) = 24 result
5597     # (INFEASIBLE one above the optimum) on a value small enough to re-run
        ↪ here,
5598     # so a regression in the integral encoding cannot pass unnoticed.
5599     feasible_at_12 = prove_integral_arc_bound(4, 3, 12, time_limit=120.0)
5600     infeasible_at_13 = prove_integral_arc_bound(4, 3, 13, time_limit=120.0)
5601     check(f"target 12 -> {feasible_at_12} (expect FEASIBLE), "
5602         f"target 13 -> {infeasible_at_13} (expect INFEASIBLE)",
5603         feasible_at_12 == "FEASIBLE" and infeasible_at_13 == "INFEASIBLE")
5604
5605     section("Search: rediscovering ell_2^dir(4) = 6 by random search")
5606     result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000, seed=0)
5607     check(f"best found = {result.best_edge_count} arcs, feasible lambda^max={
        ↪ max_edge_connectivity(result.best_graph)}",
5608         result.best_edge_count == 6 and max_edge_connectivity(result.best_graph)
        ↪ <= 1)
5609
5610     section("Driver: one solve() call handles each kind of question")
5611     # Exhaustive directed: the cut-counting proves ell_2^dir(4) = 6 exactly.
5612     proved = solve(4, 2, directed=True, simple=True, exhaustive=True,
5613         max_seconds=120.0)
5614     check(f"solve exhaustive simple-directed n=4,m=2: {proved.value} ({proved.
        ↪ bound})",
5615         proved.proven and proved.value == 6)
5616     # Discovery finds the same 6 arcs as a lower bound within a short budget.
5617     # The checks use sa (lighter); tabu is the default for actually solving.
5618     found = solve(4, 2, directed=True, simple=True, exhaustive=False,
5619         max_seconds=3.0, method="sa")
5620     check(f"solve discovery simple-directed n=4,m=2: {found.value} ({found.bound})
        ↪ ",
5621         found.bound == "lower" and found.value == 6)
5622     # The VERTEX separation is a separate question, not a corollary of the arc
5623     # one: Whitney's kappa <= lambda makes the vertex-feasible family the LARGER
5624     # of the two, so an arc upper bound does not restrict it. thm:dir-vertex-m2
5625     # needs its own base cases, and these are the first three of them.
5626     for base_n, base_value in ((3, 4), (4, 6), (5, 8)):
5627         v_value, _, v_done = _exhaustive_directed(base_n, 2, "vertex",
5628             time.time() + 120.0)
5629         check(f"exhaustive simple-directed VERTEX n={base_n},m=2: {v_value}",
5630             v_done and v_value == base_value)
5631     # The vertex flow helper must agree with the exact checker it stands in for.
5632     _vg = Graph(4, SIMPLE_DIRECTED)
5633     for _a, _b in ((0, 1), (0, 2), (1, 3), (2, 3)):
5634         _vg.mu[_a, _b] = 1
5635     _vout = [{_b for _b in range(4) if _vg.mu[_a, _b]} for _a in range(4)]
5636     check("vertex flow helper agrees with the exact checker on a theta digraph",
5637         _vertex_flow_at_least(_vout, 4, 0, 3, 2)
5638         and not _vertex_flow_at_least(_vout, 4, 0, 3, 3)
5639         and local_connectivity(_vg, 0, 3, vertex_split=True) == 2)
5640     # prop:dir-arc-stability, the unconditional quadratic bound. Both the
5641     # load-bearing counting step (sum of d+ d- capped by m n(n-1)) and the bound

```

```

5642     # it yields are checked on sampled feasible digraphs.
5643     _rng = random.Random(3)
5644     _worst_slack = None
5645     _count_ok = True
5646     for _ in range(120):
5647         _sn, _sm = _rng.randint(3, 7), _rng.randint(2, 4)
5648         _sg = Graph(_sn, SIMPLE_DIRECTED)
5649         for _a in range(_sn):
5650             for _b in range(_sn):
5651                 if _a != _b and _rng.random() < 0.6:
5652                     _sg.mu[_a, _b] = 1
5653             while max_edge_connectivity(_sg) > _sm - 1: # prune down to feasible
5654                 _present = [(a, b) for a in range(_sn) for b in range(_sn) if _sg.mu[a
5655                     ↪ , b]]
5656                 _a, _b = _rng.choice(_present)
5657                 _sg.mu[_a, _b] = 0
5658             _dout = [int(_sg.mu[x].sum()) for x in range(_sn)]
5659             _din = [int(_sg.mu[:, x].sum()) for x in range(_sn)]
5660             # The sharp form of the counting step: (m-1) n(n-1), not m n(n-1).
5661             if sum(_dout[x] * _din[x] for x in range(_sn)) > (_sm - 1) * _sn * (_sn -
5662                 ↪ 1):
5663                 _count_ok = False
5664             _slack = ((_sn * _sn) // 4 + math.sqrt(_sm) * _sn ** 1.5
5665                 - int(_sg.mu.sum()))
5666             _worst_slack = _slack if _worst_slack is None else min(_worst_slack,
5667                 ↪ _slack)
5668             # thm:dir-arc-linear-error, the 0_m(n) bound.
5669             if int(_sg.mu.sum()) > (_sn * _sn) // 4 + 4 * (_sm - 1) * (_sn - 1):
5670                 _count_ok = False
5671             # Case 2 of its proof: min total degree >= n/2 forces the linear bound
5672             # on the sum of the smaller half-degrees.
5673             if min(_dout[x] + _din[x] for x in range(_sn)) >= _sn / 2:
5674                 if (sum(min(_dout[x], _din[x]) for x in range(_sn))
5675                     > 4 * (_sm - 1) * (_sn - 1)):
5676                     _count_ok = False
5677             check("prop:dir-arc-stability and thm:dir-arc-linear-error hold on 120 "
5678                 f"sampled feasible digraphs (tightest slack {_worst_slack:.1f})",
5679                 _count_ok and _worst_slack >= 0)
5680             # sec:multi-vertex-standard: the OTHER counting convention is a different
5681             # problem, and the theta construction beats the thickened tree at m=5, n=4.
5682             _mv4, _, _mv4_done = max_multigraph_vertex_standard(4, 3)
5683             _mv5, _, _mv5_done = max_multigraph_vertex_standard(4, 5)
5684             check(f"multigraph vertex, other convention: K_3(4) = {_mv4} = (m-1)(n-1)",
5685                 _mv4_done and _mv4 == 6)
5686             check(f"multigraph vertex, other convention: K_5(4) = {_mv5} > 12, the "
5687                 "multigraph edge value, so the two problems differ",
5688                 _mv5_done and _mv5 == 14)
5689             # Exhaustive undirected by brute force: ell_2(5) = n-1 = 4 (a spanning tree).
5690             tree = solve(5, 2, directed=False, simple=True, exhaustive=True,
5691                 max_seconds=30.0)
5692             check(f"solve exhaustive simple-undirected n=5,m=2: {tree.value} ({tree.bound
5693                 ↪ })",
5694                 tree.proven and tree.value == 4)
5695             # Hypergraph discovery returns a feasible lower bound for the r=3, m=2 case.
5696             hyper = solve(7, 2, hypergraph=True, r=3, exhaustive=False, max_seconds=3.0,
5697                 method="sa")
5698             check(f"solve discovery 3-uniform hypergraph n=7,m=2: {hyper.value} ({hyper.
5699                 ↪ bound})",

```

```

5695         hyper.bound == "lower" and hyper.value == 3)
5696
5697     # --- open-variant exploration coverage -----
5698     # These three functions are the program's only handle on the OPEN parts of
5699     # the problem (hypergraph vertex value, the fractional relaxation behind the
5700     # odd-n conjectures). The first author-reviewed draft of this block ran
5701     # exhaustive  $n \geq 7$  sweeps and 6000-step searches, which added minutes; the
5702     # checks below cover the same code paths at  $n \leq 5$  so they finish in ~2s.
5703     section("Open variants: hypergraph vertex value ( $\kappa^{\max}$ ) at small n")
5704     #  $m=2$  uses the incidence-rank/forest witness,  $m=3$  the brute-force
5705     # feasibility sweep; both confirm  $\text{value} = \text{floor}((m-1)(n-1)/(r-1))$ .
5706     for n, r, m in [(4, 3, 2), (5, 3, 2), (4, 3, 3), (5, 3, 3)]:
5707         target = ((m - 1) * (n - 1)) // (r - 1)
5708         check(f"hypergraph vertex value at  $n=\{n\}$ ,  $r=\{r\}$ ,  $m=\{m\} = \{target\}$ ",
5709              verify_hyper_vertex_value(n, r, m))
5710
5711     section("Open variants: the fractional  $[0,1]$ -weight checker is exact")
5712     # A directed path keeps every max-flow at 1 (feasible); two internally
5713     # disjoint  $s \rightarrow t$  routes push the  $s \rightarrow t$  flow to 2 (infeasible). This is the
5714     # relaxation that fractional_search optimises over.
5715     path_w = np.zeros((4, 4))
5716     path_w[0, 1] = path_w[1, 2] = path_w[2, 3] = 1.0
5717     two_route = np.zeros((4, 4))
5718     two_route[0, 1] = two_route[0, 2] = two_route[1, 3] = two_route[2, 3] = 1.0
5719     check("path weighting feasible, two-route weighting infeasible",
5720          fractional_flows_feasible(path_w)
5721          and not fractional_flows_feasible(two_route))
5722
5723     section("Open variants: fractional search respects the odd-n ceilings")
5724     # Short  $n=5$  runs (the conjectured bipartite point is rigid). The
5725     # min_degree score carries a  $1e-3$  * total-weight tie-breaker, so its
5726     # comparison allows that slack; a genuine refutation would clear the
5727     # ceiling by a full unit, far above it.
5728     _, total_best = fractional_search(5, "total", steps=1500, seed=1)
5729     check(f"fractional total weight at  $n=5$  stays  $\leq 8$ : {total_best:.3f}",
5730          total_best <= 8 + 1e-6)
5731     _, deg_best = fractional_search(5, "min_degree", steps=1500, seed=1)
5732     check(f"fractional min degree at  $n=5$  stays  $\leq k=2$  (+tie-break slack): {
5733         ↪ deg_best:.3f}",
5734          deg_best <= 2 + 1e-3 * 5 * 5)
5735
5736     print(f"\n{'ALL CHECKS PASSED' if _failures == 0 else f'{'_failures'} CHECK(S)
5737         ↪ FAILED'}")
5738
5739     return 1 if _failures else 0
5740
5741 if __name__ == "__main__":
5742     print(f"C extension: {'LOADED (_erdos_fast.so)' if C_EXTENSION_LOADED else '
5743         ↪ not found (pure Python)'}")
5744     raise SystemExit(_run_checks())

```

C.2 The figure script

This is the only script that writes image files. It imports the main program and calls its plotting routines, so it holds no mathematics of its own. Keeping it separate means the figures can be

regenerated in one command without running anything else, and every random choice it makes is seeded, so running it twice produces identical files.

```
1 """
2 Regenerate every figure the thesis uses, from the one program next door.
3
4 This is the only figure script. It imports the single program
5 ‘‘erdos915_unified.py’’ and calls its plotting routines, so the figures can never
6 drift from the code that produces the numbers. Run it from this ‘‘program/’’
7 directory:
8
9     python make_figures.py
10
11 Each figure is written into ‘‘../figures/’’. Every randomised search uses a
12 fixed seed, so the output is reproducible.
13 """
14
15 from __future__ import annotations
16
17 import math
18 import sys
19 from pathlib import Path
20
21 import numpy as np
22
23 sys.path.insert(0, str(Path(__file__).resolve().parent))
24
25 from erdos915_unified import ( # noqa: E402
26     MULTI_DIRECTED,
27     MULTI_UNDIRECTED,
28     SIMPLE_DIRECTED,
29     SIMPLE_UNDIRECTED,
30     Graph,
31     _VARIANT_ENUM_CONFIGS,
32     search_for_dense_graph,
33     compute_enumeration_cache,
34     compute_pair_enumeration_cache,
35     compute_surface_cache,
36     connectivity_distribution,
37     gallery_extremal_graphs,
38     directed_arc_lower_bound,
39     directed_multigraph_arc,
40     draw_graph_with_sensitivity,
41     edge_vertex_distribution,
42     hypergraph_edge,
43     multigraph_undirected_edge,
44     plot_complexity_growth,
45     plot_extremal_gallery,
46     plot_conn_dist_grid,
47     plot_conn_threshold_3d,
48     plot_connectivity_distribution,
49     plot_pair_conn_dist_grid,
50     plot_directed_crossover,
51     plot_edge_vertex_divergence,
52     plot_edge_vertex_histograms,
53     plot_edge_dist_grid,
54     plot_scatter_lambda_edges,
55     plot_search_trace,
```

```

56     plot_sa_vs_tabu_convergence,
57     plot_variant_3d_surfaces,
58     plot_variant_grid,
59     save_gallery_json,
60     simple_undirected_edge,
61     solve,
62 )
63
64 FIGURES = Path(__file__).resolve().parent.parent / "figures"
65
66
67 # -----
68 # The all-variant grid: proved / conjectured / guessed, with machine points.
69 # Every number below comes from one driver, 'solve': exhaustive for an exact
70 # point, discovery for a search lower bound. The formula curves come from the
71 # closed forms; the guess is a fit; the band is [construction LB, trivial max].
72 # -----
73
74 def _exact_points(ns, m, budget, **kw):
75     """Machine-proved sizes: increasing n until the exhaustion no longer finishes.
76     ↳ """
77     xs, ys = [], []
78     for n in ns:
79         res = solve(n, m, exhaustive=True, max_seconds=budget, **kw)
80         if res.bound == "exact":
81             xs.append(n)
82             ys.append(res.value)
83         else:
84             break
85     return xs, ys
86
87 def _search_points(ns, m, budget, **kw):
88     """Discovery lower bounds: a concrete feasible graph at each n (the LB
89     ↳ construction)."""
90     xs, ys = [], []
91     for n in ns:
92         res = solve(n, m, exhaustive=False, max_seconds=budget, seed=0, **kw)
93         xs.append(n)
94         ys.append(res.value)
95     return xs, ys
96
97 def _band(search, trivial_max, ns):
98     """The certain interval: an easy construction below, the trivial maximum above
99     ↳ .
100
101     The lower edge is the densest feasible graph the search exhibited (a real
102     lower bound); the upper edge is the most edges any graph on n vertices can
103     hold at all (a loose but certain upper bound). The truth lies in between.
104     """
105     found = dict(zip(search[0], search[1]))
106     lower = [min(found.get(n, 0), trivial_max(n)) for n in ns]
107     upper = [trivial_max(n) for n in ns]
108     return list(ns), lower, upper
109
110 def _extend_lower_bounds(search, lb_fn, ns):

```

```

111     """Raise the search lower bounds to a known verified construction where it
        ↳ wins.
112
113     Past the size the quick search reaches, an explicit construction the checker
114     confirms is a better (still honest) lower bound, so we report the best of the
115     two at each n, exactly as the rediscovery section does by hand.
116     """
117     found = dict(zip(search[0], search[1]))
118     for n in ns:
119         found[n] = max(found.get(n, 0), lb_fn(n))
120     xs = sorted(found)
121     return xs, [found[n] for n in xs]
122
123
124 def _reconcile_panel(panel: dict) -> dict:
125     """Make a panel internally consistent before plotting.
126
127     Three honest invariants, enforced here once for every panel rather than
128     scattered through the construction code:
129
130     1. **Nothing exceeds a proved optimum.** A machine-checked exact value is
131     the true maximum at that 'n'; no proved/conjectured curve, no
132     named-construction lower bound, and no search circle may sit above it.
133     (This is what produced the green square *below* its own curve: a closed
134     form -- e.g. the hypergraph bound capped at the complete hypergraph -- can
135     overshoot what is actually achievable at small 'n'.)
136     2. **Search lower bounds rise with 'n'.

```

```

168         running, mono = -1, []
169         for y in sy:
170             running = max(running, y)
171             mono.append(running)
172         panel["search"] = (sx, mono)
173
174         # invariant 3b: the open-panel guess is the interpolation through those
175         # (monotone, exact-capped) points, so it can never ride above them.
176         if panel.get("guess") == "search":
177             panel["guess"] = panel.get("search")
178
179     return panel
180
181
182 def gather_variant_grid(m=3, exact_budget=4.0, search_budget=0.4,
183     ↪ open_search_budget=4.0):
184     """Build the twelve panels of the all-variant grid (row-major, model x col).
185
186     ‘m’ is the forbidden connectivity value shown in every panel.
187
188     Two uniformity rules make the twelve panels directly comparable:
189
190     1. Every matrix panel plots its search lower bounds over the *same* vertex
191     range ‘matrix_ns’ and every hypergraph panel over ‘hyper_ns’, so each
192     panel carries the same number of data points -- no panel trails off early.
193
194     2. For every panel where a construction is known (proved or conjectured) the
195     search lower bound is raised to that named construction with
196     :func:‘_extend_lower_bounds’, exactly as the caption of the figure and
197     the rediscovery section state: the reported circle at each ‘n’ is the
198     better of the cooled search and the verified construction. This is the
199     honest "lands on the curve" rule, applied to *all* panels rather than
200     only the two directed ones it used to cover, so the proved curves pass
201     through their circles instead of floating above a jagged search tail.
202
203     Open panels also get a verified construction planted, not only the proved
204     ones: the open undirected vertex panels get the proved EDGE value (Whitney,
205     kappa <= lambda, so the edge extremiser is vertex-feasible) and the two
206     directed-hypergraph panels get the proved bipartite family of
207     prop:dir-hyper-first. This matters because at ‘search_budget=0.4’ the
208     vertex-mode search on a large matrix graph barely completes one move, so
209     the RAW result actively *degrades* with growing ‘n’ (measured:
210     n=6/10/14/16 gave 15/12/5/4 at m=6), and the monotone running-max in
211     :func:‘_reconcile_panel’ then freezes the plotted curve at its early peak,
212     a flat line that looks like a finding but is really budget starvation.
213     ‘open_search_budget’ (default 4s, ~10x ‘search_budget’) still gives
214     those panels a stronger walk, since the search can genuinely beat the
215     planted construction where the open problem is richer (it does at m=6,
216     n=9 on the undirected vertex panel).
217     """
218
219     panels = []
220     # Uniform x-ranges for the search / construction curves. The machine-PROVED
221     # (exact) points are genuinely limited by enumeration cost and stop early; the
222     # search lower bounds and the formula curves are cheap, so they run well past
223     # that, to show each variant's trend and let the search discover dense
224     # ("maybe extremal") graphs at sizes exhaustion cannot reach.
225     matrix_ns = list(range(2, 17))    # all matrix panels, n = 2..16
226     hyper_ns = list(range(2, 13))     # all hypergraph panels, n = 2..12

```

```

225 def tri_undirected(n):
226     return n * (n - 1) // 2
227
228 def tri_hyper(n, r=3):
229     return math.comb(n, r)
230
231 def tri_dir_hyper(n, r=3):
232     return n * math.comb(n - 1, r - 1)
233
234 # Construction lower bounds (verified, proved-or-conjectured values), each
235 # capped at the trivial maximum for its model. The cap matters at small n:
236 # a closed form like Mader's  $\text{floor}(m(n-1)/2)$  is the value only for  $n \geq m$ ;
237 # for  $n < m$  the complete graph  $K_n$  already has  $\lambda_{\max} = n-1 \leq m-1$ , so it
238 # is feasible and denser, and the true value is  $\text{comb}(n, 2)$ . Without the cap
239 # the plotted curve and its lower-bound circles float ABOVE the exact squares,
240 # which is impossible (a lower bound can never exceed the proved optimum).
241 def lb_simple_edge(n):
242     return min(simple_undirected_edge(n, m), n * (n - 1) // 2)
243
244 def lb_multi_edge(n):
245     return min(multigraph_undirected_edge(n, m), (m - 1) * (n * (n - 1) // 2))
246
247 def lb_dir(n):
248     return min(directed_arc_lower_bound(n, m), n * (n - 1))
249
250 def lb_multi_dir(n):
251     return min(directed_multigraph_arc(n, m), (m - 1) * n * (n - 1))
252
253 def lb_hyper_edge(n):
254     return min(hypergraph_edge(n, m, 3), math.comb(n, 3))
255
256 def lb_dir_hyper(n):
257     # The PROVED bipartite family of prop:dir-hyper-first at  $r = 3$ :
258     #  $\alpha$  tails, and on the other  $n - \alpha$  vertices a shared near-regular
259     # head graph of max degree  $m - 1$  (realisable iff  $m - 1 \leq n - \alpha - 1$ ),
260     # giving  $\alpha * \text{floor}((m-1)(n-\alpha)/2)$  single-step hyperarcs with
261     #  $\lambda_{\max} = \kappa_{\max} = m - 1$ . Sound for BOTH separations, since
262     # every route in it is a single tail  $\rightarrow$  head step.
263     best = 0
264     for alpha in range(1, n - m + 1):
265         best = max(best, alpha * ((m - 1) * (n - alpha) // 2))
266     return min(best, n * math.comb(n - 1, 2))
267
268 # Undirected vertex is proved for  $m \leq 4$  (Leonard/Whitney), open for  $m \geq 5$ .
269 vert_proved = (m <= 4)
270 # Hypergraph vertex is proved for  $m \leq 3$  (incidence-rank lemma, thm:hyper-vertex
271 #  $\hookrightarrow -m3$ ).
272 hyper_vert_proved = (m <= 3)
273
274 def searched(ns, lb_fn, **solve_kw):
275     """Search over the uniform range, then raise to the known construction."""
276     se = _search_points(ns, m, search_budget, **solve_kw)
277     if lb_fn is not None:
278         se = _extend_lower_bounds(se, lb_fn, ns)
279     return se
280
281 # ----- row 1: simple -----
282 # (1) simple undirected edge -- proved (Mader) for all m.

```

```

282 ex = _exact_points(range(2, 9), m, exact_budget,
283                     directed=False, simple=True, separation="edge")
284 se = searched(matrix_ns, lb_simple_edge,
285               directed=False, simple=True, separation="edge")
286 panels.append(dict(
287     title=f"undirected edge, $m={m}$ (proved)", ylabel="edges",
288     proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
289     exact=ex, search=se))
290
291 # (2) simple undirected vertex -- proved for m<=4, open for m>=5.
292 ex2 = _exact_points(range(2, 8), m, exact_budget,
293                     directed=False, simple=True, separation="vertex")
294 if vert_proved:
295     se2 = searched(matrix_ns, lb_simple_edge,
296                   directed=False, simple=True, separation="vertex")
297     panels.append(dict(
298         title=f"undirected vertex, $m={m}$ (proved)", ylabel="edges",
299         proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
300         exact=ex2, search=se2))
301 else:
302     se2 = _search_points(matrix_ns, m, open_search_budget,
303                           directed=False, simple=True, separation="vertex")
304     # Whitney: kappa <= lambda, so Mader's edge extremiser is vertex-feasible
305     # and the proved edge value is an honest lower bound on the open vertex
306     # panel too. Without it the starved vertex-mode search freezes into a
307     # flat plateau at large n (the "cut off" look); the search may still beat
308     # the edge value where the vertex problem is genuinely richer.
309     se2 = _extend_lower_bounds(se2, lb_simple_edge, matrix_ns)
310     band2 = _band(se2, tri_undirected, matrix_ns)
311     panels.append(dict(
312         title=f"undirected vertex, $m={m}$ (open)", ylabel="edges",
313         guess="search",
314         band=band2, exact=ex2, search=se2))
315
316 # (3) simple directed arc -- conjectured.
317 ex3 = _exact_points(range(2, 8), m, exact_budget,
318                     directed=True, simple=True, separation="edge")
319 se3 = searched(matrix_ns, lb_dir,
320               directed=True, simple=True, separation="edge")
321 panels.append(dict(
322     title=f"directed arc, $m={m}$ (conjectured)", ylabel="arcs",
323     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
324     branches=[(matrix_ns, [min(m * (n - 1), n * (n - 1)) for n in matrix_ns],
325                  "hub $m(n{-}1)$"),
326               # Shifted-partition augmented bipartite floor((n+m-2)^2/4)
327               # (the corrected conj:dir-arc branch; the old balanced count
328               # floor(n^2/4)+(m-2)ceil(n/2) agrees at m<=3 but is smaller
329               # at m>=4 and would sit below the conjecture curve at m=6).
330               (matrix_ns,
331                [min((n + m - 2) ** 2 // 4, n * (n - 1))
332                  for n in matrix_ns], "bipartite")],
333     exact=ex3, search=se3))
334
335 # (4) simple directed vertex -- conjectured (= arc).
336 ex4 = _exact_points(range(2, 8), m, exact_budget,
337                     directed=True, simple=True, separation="vertex")
338 se4 = searched(matrix_ns, lb_dir,
339               directed=True, simple=True, separation="vertex")

```

```

340 panels.append(dict(
341     title=f"directed vertex, $m={m}$ (conjectured)", ylabel="arcs",
342     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
343     exact=ex4, search=se4))
344
345 # ----- row 2: multigraph -----
346 # (5) multigraph undirected edge -- proved for all m.
347 ex5 = _exact_points(range(2, 7), m, exact_budget,
348     directed=False, simple=False, separation="edge")
349 se5 = searched(matrix_ns, lb_multi_edge,
350     directed=False, simple=False, separation="edge")
351 panels.append(dict(
352     title=f"undirected edge, $m={m}$ (proved)", ylabel="edges",
353     proved=(matrix_ns, [lb_multi_edge(n) for n in matrix_ns]),
354     exact=ex5, search=se5))
355
356 # (6) multigraph undirected vertex -- equals simple (parallel edges irrelevant
357     ↪ for vertex cuts).
358 multi_vert_label = ("proved, $=$ simple" if vert_proved else "open, $=$ simple
359     ↪ ")
360 if vert_proved:
361     panels.append(dict(
362         title=f"undirected vertex, $m={m}$ ({multi_vert_label})", ylabel="
363             ↪ edges",
364         proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
365         exact=ex2, search=se2))
366 else:
367     panels.append(dict(
368         title=f"undirected vertex, $m={m}$ ({multi_vert_label})", ylabel="
369             ↪ edges",
370         guess="search",
371         band=_band(se2, tri_undirected, matrix_ns), exact=ex2, search=se2))
372
373 # (7) multigraph directed arc -- certified (cut-counting) then conjectured.
374 ex7 = _exact_points(range(3, 6), m, 10.0,
375     directed=True, simple=False, separation="edge")
376 se7 = searched(matrix_ns, lb_multi_dir,
377     directed=True, simple=False, separation="edge")
378 panels.append(dict(
379     title=f"directed arc, $m={m}$ (multigraph, certified)", ylabel="arcs",
380     conj=(matrix_ns, [lb_multi_dir(n) for n in matrix_ns]),
381     exact=ex7, search=se7))
382
383 # (8) multigraph directed vertex -- conjectured (reduces to simple digraph).
384 panels.append(dict(
385     title=f"directed vertex, $m={m}$ (conjectured, $=$ simple)", ylabel="arcs
386     ↪ ",
387     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
388     exact=ex4, search=se4))
389
390 # ----- row 3: hypergraph (r=3) -----
391 # (9) hypergraph undirected edge -- proved for all m.
392 ex9 = _exact_points(range(2, 8), m, exact_budget,
393     hypergraph=True, r=3, directed=False, separation="edge")
394 se9 = searched(hyper_ns, lb_hyper_edge,
395     hypergraph=True, r=3, directed=False, separation="edge")
396 panels.append(dict(
397     title=f"undirected edge, $m={m}$ (proved)", ylabel="hyperedges",

```

```

393         proved=(hyper_ns, [lb_hyper_edge(n) for n in hyper_ns]),
394         exact=ex9, search=se9))
395
396     # (10) hypergraph undirected vertex -- PROVED for m<=3 (incidence-rank lemma),
397     #         open for m>=4.
398     ex10 = _exact_points(range(2, 7), m, exact_budget,
399                         hypergraph=True, r=3, directed=False, separation="vertex"
400                         ↪ )
401
402     if hyper_vert_proved:
403         se10 = searched(hyper_ns, lb_hyper_edge,
404                        hypergraph=True, r=3, directed=False, separation="vertex")
405         panels.append(dict(
406             title=f"undirected vertex, $m={m}$ (proved)", ylabel="hyperedges",
407             proved=(hyper_ns, [lb_hyper_edge(n) for n in hyper_ns]),
408             exact=ex10, search=se10))
409     else:
410         se10 = _search_points(hyper_ns, m, open_search_budget,
411                              hypergraph=True, r=3, directed=False, separation="
412                              ↪ vertex")
413
414         # Whitney again: the proved hyperedge value is a lower bound for the
415         # vertex separation (its extremiser is vertex-feasible). The exact cap
416         # in _reconcile_panel corrects the small-n corner where the simple
417         # attainment condition m-1 <= n-2 has not kicked in yet.
418         se10 = _extend_lower_bounds(se10, lb_hyper_edge, hyper_ns)
419         panels.append(dict(
420             title=f"undirected vertex, $m={m}$ (open)", ylabel="hyperedges",
421             guess="search",
422             band=_band(se10, tri_hyper, hyper_ns), exact=ex10, search=se10))
423
424     # (11) hypergraph directed arc -- OPEN (new directed Berge model).
425     ex11 = _exact_points(range(2, 6), m, exact_budget,
426                         hypergraph=True, r=3, directed=True, separation="edge")
427     se11 = _search_points(hyper_ns, m, open_search_budget,
428                          hypergraph=True, r=3, directed=True, separation="edge")
429
430     # The proved bipartite construction of prop:dir-hyper-first is the named
431     # lower bound here (the search alone slips below the quadratic at larger n).
432     se11 = _extend_lower_bounds(se11, lb_dir_hyper, hyper_ns)
433     panels.append(dict(
434         title=f"directed arc, $m={m}$ (open, new model)", ylabel="hyperarcs",
435         guess="search",
436         band=_band(se11, tri_dir_hyper, hyper_ns), exact=ex11, search=se11))
437
438     # (12) hypergraph directed vertex -- OPEN (new directed Berge model).
439     ex12 = _exact_points(range(2, 6), m, exact_budget,
440                         hypergraph=True, r=3, directed=True, separation="vertex")
441     se12 = _search_points(hyper_ns, m, open_search_budget,
442                          hypergraph=True, r=3, directed=True, separation="vertex"
443                          ↪ )
444
445     # Same construction: all its routes are single tail -> head steps, so it is
446     # feasible for the vertex separation at the same value.
447     se12 = _extend_lower_bounds(se12, lb_dir_hyper, hyper_ns)
448     panels.append(dict(
449         title=f"directed vertex, $m={m}$ (open, new model)", ylabel="hyperarcs",
450         guess="search",
451         band=_band(se12, tri_dir_hyper, hyper_ns), exact=ex12, search=se12))
452
453     return [_reconcile_panel(p) for p in panels]

```



```

448
449 def main() -> None:
450     FIGURES.mkdir(parents=True, exist_ok=True)
451
452     plot_directed_crossover(m=3, max_n=24, path=FIGURES / "directed_crossover.png"
453         ↪ )
454     print("wrote directed_crossover.png")
455
456     plot_edge_vertex_divergence(max_n=35, path=FIGURES / "edge_vertex_divergence.
457         ↪ png")
458     print("wrote edge_vertex_divergence.png")
459
460     # Figure 3.2: cooling trace for the directed multigraph n=4, m=3 case.
461     # record_exact_connectivity logs the true lambda^max per step (more expensive
462     # but needed for an exact scatter; the search itself uses the cheaper upper
463     ↪ bound).
464     run = search_for_dense_graph(MULTI_DIRECTED, n=4, m=3, steps=3000, seed=1,
465         record_exact_connectivity=True)
466     plot_search_trace(run, path=FIGURES / "temperature_trace.png",
467         optimum=12, ceiling=2, show_histogram=False)
468     print(f"wrote temperature_trace.png (search reached {run.best_edge_count} arcs
469         ↪ of 12)")
470
471     # Additional scatter-only traces for other variants (no cooling panel needed
472     # since the cooling schedule is the same across all cases).
473     trace_cases = [
474         # (variant, n, m, filename_stem, ceiling, connectivity_label, edge_label,
475         ↪ title)
476         (SIMPLE_UNDIRECTED, 7, 3, "trace_simple_undirected_n7_m3", 2,
477         r"$\lambda^{\max}$", "edges",
478         r"Search trace: simple undirected, $n=7$, $m=3$"),
479         (SIMPLE_DIRECTED, 5, 3, "trace_simple_directed_n5_m3", 2,
480         r"$\lambda^{\max}$", "arcs",
481         r"Search trace: simple directed, $n=5$, $m=3$"),
482         (MULTI_UNDIRECTED, 5, 3, "trace_multi_undirected_n5_m3", 2,
483         r"$\lambda^{\max}$", "edges",
484         r"Search trace: multi undirected, $n=5$, $m=3$"),
485         (MULTI_DIRECTED, 5, 3, "trace_multi_directed_n5_m3", 2,
486         r"$\lambda^{\max}$", "arcs",
487         r"Search trace: multi directed, $n=5$, $m=3$"),
488         (MULTI_DIRECTED, 4, 3, "trace_multi_directed_n4_m3_vertex", 2,
489         r"$\kappa^{\max}$", "arcs",
490         r"Search trace: multi directed vertex-sep, $n=4$, $m=3$"),
491     ]
492     for variant, n, m, stem, ceil_val, clabel, elabel, tttitle in trace_cases:
493         sep = "vertex" if "vertex" in stem else "edge"
494         tr = search_for_dense_graph(variant, n=n, m=m, separation=sep,
495             steps=3000, seed=1,
496             record_exact_connectivity=True)
497         plot_search_trace(tr, path=FIGURES / f"{stem}.png",
498             ceiling=ceil_val, show_cooling=False,
499             show_histogram=False,
500             connectivity_label=clabel, edge_label=elabel,
501             title=tttitle)
502         print(f"wrote {stem}.png (best={tr.best_edge_count})")
503
504     # A larger directed multigraph: four s->t routes of differing capacity plus a
505     # dead end. Parallel arcs are drawn explicitly, so multiplicity is read by

```

```

501  # counting. The s->a route is over-provisioned (mu=3 into a but a single arc
502  # a->t out), so it appears as three arcs in a COOL colour (sigma=1): the
503  # visual point that multiplicity and load-bearing role are not the same.
504  sens = Graph(8, MULTI_DIRECTED) # 0=s 1=t 2=a 3=b 4=c 5=d 6=e 7=f
505  sens.set_multiplicity(0, 2, 3); sens.set_multiplicity(2, 1, 1) # s->a->t,
    ↳ over-provisioned in
506  sens.set_multiplicity(0, 3, 3); sens.set_multiplicity(3, 1, 3) # s->b->t, cap
    ↳ 3 (load-bearing)
507  sens.set_multiplicity(0, 4, 2); sens.set_multiplicity(4, 1, 2) # s->c->t, cap
    ↳ 2
508  sens.set_multiplicity(0, 6, 1); sens.set_multiplicity(6, 1, 1) # s->e->t, cap
    ↳ 1
509  sens.set_multiplicity(0, 7, 2); sens.set_multiplicity(7, 1, 2) # s->f->t, cap
    ↳ 2
510  sens.set_multiplicity(0, 5, 2) # s->d, dead
    ↳ end
511  sens_layout = {0: (-3.2, 0.0), 1: (3.2, 0.0),
512                2: (0.0, 2.6), 3: (0.0, 1.3), 4: (0.0, 0.0),
513                6: (0.0, -1.3), 7: (0.0, -2.6), 5: (-3.2, -2.4)}
514  sens_labels = {0: "s", 1: "t", 2: "a", 3: "b", 4: "c", 5: "d", 6: "e", 7: "f"}
515  draw_graph_with_sensitivity(
516      sens, path=FIGURES / "sensitivity_mixed.png",
517      layout=sens_layout, node_labels=sens_labels)
518  print("wrote sensitivity_mixed.png")
519
520  # Annealing vs tabu convergence (Ch.3): wall-clock timed, so a representative
521  # run rather than a seed-exact one (see the figure caption).
522  plot_sa_vs_tabu_convergence(FIGURES / "sa_vs_tabu_convergence.pdf",
523                             cases=((5, 3), (7, 3)), budget=8.0, seed=0)
524  print("wrote sa_vs_tabu_convergence.pdf")
525
526  plot_complexity_growth(path=FIGURES / "complexity_growth.png")
527  print("wrote complexity_growth.png")
528
529  # Gallery of machine-found extremal graphs (reads extremal_gallery.json).
530  plot_extremal_gallery(FIGURES / "extremal_graphs_gallery.png")
531  print("wrote extremal_graphs_gallery.png")
532
533  # Bounds grids: one for m=3 and one for m=6 (more purple dots).
534  for m_val in (3, 6):
535      print(f"gathering all-variant grid m={m_val} (runs solve many times)...")
536      panels = gather_variant_grid(m=m_val)
537      out = FIGURES / f"variant_bounds_m{m_val}.png"
538      plot_variant_grid(panels, path=out, m=m_val)
539      print(f"wrote {out.name}")
540
541  # -----
542  # Full-enumeration scatter and distribution figures.
543  # The enumeration cache is written once to figures/enumeration_cache.pkl.
544  # -----
545  print("building enumeration cache (slow on first run, cached thereafter)...")
546  enum_cache_path = FIGURES / "enumeration_cache.pkl"
547  enum_data = compute_enumeration_cache(cache_path=enum_cache_path)
548  print("enumeration cache ready")
549
550  plot_scatter_lambda_edges(enum_data, path=FIGURES / "scatter_lambda_edges.png"
    ↳ )
551  print("wrote scatter_lambda_edges.png")

```

```

552
553 # Pooled per-pair connectivity: every vertex pair of every enumerated graph,
554 # tagged with its graph's  $\lambda_{\max}$ . Slow first run, cached thereafter.
555 print("building pair-connectivity cache (slow on first run, cached thereafter)
    ↳ ...")
556 pair_cache_path = FIGURES / "pair_enumeration_cache.pkl"
557 pair_data = compute_pair_enumeration_cache(cache_path=pair_cache_path)
558 print("pair-connectivity cache ready")
559
560 # Pooled per-pair connectivity histograms (Ch.4): each panel stacks blue/red
561 # at its own mid-range connectivity boundary, so the split is meaningful in
562 # every variant rather than collapsing to one colour at a fixed m.
563 plot_pair_conn_dist_grid(pair_data, path=FIGURES / "pair_conn_dist.png")
564 print("wrote pair_conn_dist.png")
565
566 # Edge-count histograms (Ch.4): same mid-range stacked colouring.
567 plot_edge_dist_grid(enum_data, path=FIGURES / "edges_dist.png")
568 print("wrote edges_dist.png")
569
570 # Per-graph connectivity distribution at the fixed forbidden threshold m=6
571 # (App. B), showing the m=6 feasibility split; m=3 body version removed
    ↳ 2026-06-20.
572 plot_conn_dist_grid(enum_data, 6, path=FIGURES / "conn_dist_m6.png")
573 print("wrote conn_dist_m6.png")
574
575 # -----
576 # 3-D bound surface over the (n, m) grid.
577 # Uses a JSON cache (slow first run, instant thereafter).
578 # -----
579 surface_cache_path = FIGURES / "surface_cache.json"
580 print("building surface cache (slow on first run, cached thereafter)...")
581 compute_surface_cache(
582     n_range=(3, 9), m_range=(2, 6),
583     max_seconds=20, cache_path=surface_cache_path)
584 plot_variant_3d_surfaces(surface_cache_path, path=FIGURES / "
    ↳ variant_surface_3d.png")
585 print("wrote variant_surface_3d.png")
586
587 # -----
588 # 3-D threshold histogram (appendix): how the  $\lambda_{\max}$  distribution shifts
589 # with density p, for three representative variants. Kept as a standalone
590 # appendix figure (fig:threshold-3d) after the Figure 4.2 threshold story was
591 # removed from the body on 2026-06-20.
592 # -----
593 # No p-values: each panel sweeps in units of its own threshold, since the
594 # graph and hypergraph models do not share a density scale.
595 plot_conn_threshold_3d(
596     path=FIGURES / "threshold_3d.png",
597     n=12, m=3, samples=150, seed=7)
598 print("wrote threshold_3d.png")
599
600 # NOTE: the 2-D random-sampling threshold figures (degree_threshold,
601 # sampled_variant_grid) were removed from the thesis on 2026-06-20. Their
602 # generators are kept in erdos915_unified.py and the restore recipe is in
603 # research_notes/removed_threshold_phenomenon.md.
604
605 # Edge vs vertex disjointness in  $G(16, p)$  at three densities (Chapter 1).
606 p_values = [0.25, 0.5, 0.75]

```

```

607     data = {pp: edge_vertex_distribution(16, pp, trials=400, seed=7)
608             for pp in p_values}
609     plot_edge_vertex_histograms(data, 16, p_values,
610                                FIGURES / "edge_vertex_sampling.png")
611     print("wrote edge_vertex_sampling.png")
612
613     # Gallery of extremal graphs: all 12 variants × small (n, m), ~3s per case.
614     print("building extremal graph gallery (this takes a few minutes)...")
615     gallery_path = FIGURES / "extremal_gallery.json"
616     gal = gallery_extremal_graphs(max_n=7, max_m=4, r=3, time_per_case=3.0)
617     save_gallery_json(gal, gallery_path)
618     total_classes = sum(
619         len(cases.get("classes", []))
620         for variant_data in gal.values()
621         for cases in variant_data.values()
622     )
623     print(f"wrote {gallery_path.name} ({total_classes} iso-classes across all
        ↪ variants)")
624
625
626 if __name__ == "__main__":
627     main()

```

C.3 The C accelerator

Two operations dominate the running time at the sizes the searches reach: deciding whether a pair of vertices already has too many independent routes, and reducing a graph to a standard form so that two drawings of the same graph can be recognised as equal. Both are rewritten here in C, which is compiled separately and loaded if it is present.

Nothing depends on it. Every function below has a pure Python counterpart in the main program, the two are tested against each other, and if the compiled library is missing the program simply uses the Python version. The size limits in the code are deliberate: the fixed-size buffers are safe only up to the stated number of vertices, and above it the program falls back rather than risking a buffer it cannot hold.

```

1  /*
2  * _erdos_fast.c - hot-path C helpers for erdos915_unified.py
3  *
4  * Compile: gcc -O3 -march=native -shared -fPIC -o _erdos_fast.so _erdos_fast.c
5  *
6  * All functions take a flat row-major int32 array 'mu' of size n*n.
7  * Access: mu[u*n + v] is the capacity (multiplicity) of arc u->v.
8  */
9
10 #include <stdint.h>
11 #include <string.h>
12 #include <stdlib.h>
13
14 /* ----- */
15 /* tiny_maxflow */
16 /* DFS Ford-Fulkerson; returns 1 iff max-flow(s,t) > cap. */
17 /* Stops as soon as it finds cap+1 augmenting paths (early exit). */

```

```

18  /* n <= 7 in practice; residual lives on the stack. */
19  /* ----- */
20  int tiny_maxflow(const int *mu, int n, int s, int t, int cap)
21  {
22      /* local mutable residual; sized for n up to 16 (vertex-split at n=8) */
23      int res[256];
24      memcpy(res, mu, (size_t)(n * n) * sizeof(int));
25
26      int flow = 0;
27      while (flow <= cap) {
28          /* DFS to find an augmenting path */
29          int parent[16];
30          for (int i = 0; i < n; i++) parent[i] = -1;
31          parent[s] = s;
32
33          /* stack-based DFS (max depth = n vertices) */
34          int stack[16];
35          int sp = 0;
36          stack[sp++] = s;
37
38          while (sp > 0 && parent[t] == -1) {
39              int u = stack[--sp];
40              for (int v = 0; v < n; v++) {
41                  if (parent[v] == -1 && res[u * n + v] > 0) {
42                      parent[v] = u;
43                      stack[sp++] = v;
44                  }
45              }
46          }
47
48          if (parent[t] == -1)
49              return 0; /* max-flow <= cap */
50
51          /* find bottleneck */
52          int bottleneck = 1 << 30;
53          for (int v = t; v != s; ) {
54              int u = parent[v];
55              int cap_uv = res[u * n + v];
56              if (cap_uv < bottleneck) bottleneck = cap_uv;
57              v = u;
58          }
59
60          /* augment */
61          for (int v = t; v != s; ) {
62              int u = parent[v];
63              res[u * n + v] -= bottleneck;
64              res[v * n + u] += bottleneck;
65              v = u;
66          }
67
68          flow += bottleneck;
69      }
70      return 1; /* max-flow > cap */
71  }
72
73  /* ----- */
74  /* max_connectivity_exceeds */
75  /* Sweeps all pairs; returns 1 iff any pair has flow > k. */

```

```

76 /* directed=1: all ordered pairs; directed=0: unordered pairs. */
77 /* ----- */
78 int max_connectivity_exceeds(const int *mu, int n, int k, int directed)
79 {
80     for (int s = 0; s < n; s++) {
81         int start = directed ? 0 : s + 1;
82         for (int t = start; t < n; t++) {
83             if (s == t) continue;
84             if (tiny_maxflow(mu, n, s, t, k))
85                 return 1;
86         }
87     }
88     return 0;
89 }
90
91 /* ----- */
92 /* canonical_form_min */
93 /* Writes the lex-min row-major permutation of the n×n matrix into */
94 /* out (n*n int32 values). Uses Heap's algorithm to enumerate n! */
95 /* permutations. At n=7: 5040 iterations × 49 ints each. */
96 /* ----- */
97 void canonical_form_min(const int *mu, int n, int *out)
98 {
99     int perm[7], best[49], cand[49];
100     int c[7]; /* Heap's algorithm counters */
101     int nn = n * n;
102
103     for (int i = 0; i < n; i++) { perm[i] = i; c[i] = 0; }
104
105     /* identity permutation is the first candidate */
106     for (int r = 0; r < n; r++)
107         for (int col = 0; col < n; col++)
108             best[r * n + col] = mu[perm[r] * n + perm[col]];
109
110     /* Heap's algorithm */
111     int i = 1;
112     while (i < n) {
113         if (c[i] < i) {
114             /* swap */
115             if (i % 2 == 0) {
116                 int tmp = perm[0]; perm[0] = perm[i]; perm[i] = tmp;
117             } else {
118                 int tmp = perm[c[i]]; perm[c[i]] = perm[i]; perm[i] = tmp;
119             }
120
121             /* build candidate permuted matrix */
122             for (int r = 0; r < n; r++)
123                 for (int col = 0; col < n; col++)
124                     cand[r * n + col] = mu[perm[r] * n + perm[col]];
125
126             /* update best if cand is lex-smaller */
127             if (memcmp(cand, best, (size_t)nn * sizeof(int)) < 0)
128                 memcpy(best, cand, (size_t)nn * sizeof(int));
129
130             c[i]++;
131             i = 1;
132         } else {
133             c[i] = 0;

```

```

134         i++;
135     }
136 }
137
138 memcpy(out, best, (size_t)nn * sizeof(int));
139 }

```

C.4 The test suite

The suite checks the program against facts established independently of it: closed forms proved by hand, constructions whose size and connectivity are known in advance, and small cases where an exhaustive walk gives the answer outright. Several tests compare two routines that are supposed to agree, such as a fast path against the slow one it replaces, which is what catches an optimisation that changes an answer. Tests needing an optional library skip themselves when it is absent, so the suite passes on a minimal installation rather than reporting failures for tools it was never given.

C.4.1 Shared setup

```

1  """Test suite for the unified Erdos-915 program.
2
3  The tests import ‘‘erdos915_unified’’ from the parent directory, so we put that
4  directory on the path here, once, for every test module. The whole suite runs
5  with the standard library alone:
6
7      cd program
8      python -m unittest discover -s tests
9
10 (no pytest required, though pytest will also discover and run these if present).
11 """
12
13 import sys
14 from pathlib import Path
15
16 sys.path.insert(0, str(Path(__file__).resolve().parent.parent))

```

C.4.2 The graph model

```

1  """The graph model: the multiplicity matrix and every operation on it."""
2
3  import unittest
4
5  from erdos915_unified import (
6      Graph,
7      MULTI_DIRECTED,
8      MULTI_UNDIRECTED,
9      SIMPLE_DIRECTED,
10     SIMPLE_UNDIRECTED,
11 )
12
13
14 class VariantBooleans(unittest.TestCase):

```

```

15     def test_the_two_booleans(self):
16         self.assertEqual((SIMPLE_UNDIRECTED.directed, SIMPLE_UNDIRECTED.simple), (
17             ↪ False, True))
18         self.assertEqual((MULTI_UNDIRECTED.directed, MULTI_UNDIRECTED.simple), (
19             ↪ False, False))
20         self.assertEqual((SIMPLE_DIRECTED.directed, SIMPLE_DIRECTED.simple), (True
21             ↪ , True))
22         self.assertEqual((MULTI_DIRECTED.directed, MULTI_DIRECTED.simple), (True,
23             ↪ False))
24
25     def test_describe_is_nonempty(self):
26         for variant in (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED, SIMPLE_DIRECTED,
27             ↪ MULTI_DIRECTED):
28             self.assertTrue(variant.describe())
29
30 class EmptyGraph(unittest.TestCase):
31     def test_fresh_graph_is_empty(self):
32         g = Graph(5, SIMPLE_UNDIRECTED)
33         self.assertEqual(g.num_vertices, 5)
34         self.assertEqual(list(g.vertices()), list(range(5)))
35         self.assertEqual(g.edge_count(), 0)
36         self.assertEqual(list(g.edges()), [])
37         self.assertFalse(g.has_edge(0, 1))
38
39 class UndirectedModel(unittest.TestCase):
40     def test_add_edge_is_symmetric(self):
41         g = Graph(3, SIMPLE_UNDIRECTED)
42         g.add_edge(0, 1)
43         self.assertTrue(g.has_edge(0, 1))
44         self.assertTrue(g.has_edge(1, 0))
45         self.assertEqual(g.multiplicity(0, 1), 1)
46         self.assertEqual(g.multiplicity(1, 0), 1)
47         self.assertEqual(g.edge_count(), 1)
48         self.assertEqual(list(g.edges()), [(0, 1, 1)])
49
50     def test_simple_graph_saturates_at_one(self):
51         g = Graph(3, SIMPLE_UNDIRECTED)
52         g.add_edge(0, 1)
53         g.add_edge(0, 1)
54         self.assertEqual(g.multiplicity(0, 1), 1)
55         self.assertEqual(g.edge_count(), 1)
56
57     def test_multigraph_keeps_parallel_edges(self):
58         g = Graph(3, MULTI_UNDIRECTED)
59         g.add_edge(0, 1, 3)
60         self.assertEqual(g.multiplicity(0, 1), 3)
61         self.assertEqual(g.multiplicity(1, 0), 3)
62         self.assertEqual(g.edge_count(), 3)
63         self.assertEqual(g.degree(0), 3)
64         self.assertEqual(g.degree(1), 3)
65
66     def test_undirected_degree_counts_incident_multiplicity(self):
67         g = Graph(3, MULTI_UNDIRECTED)
68         g.add_edge(0, 1, 2)
69         g.add_edge(0, 2, 1)
70         self.assertEqual(g.degree(0), 3)

```



```

68         self.assertEqual(g.edge_count(), 3)
69
70
71 class DirectedModel(unittest.TestCase):
72     def test_arc_is_one_directional(self):
73         g = Graph(3, SIMPLE_DIRECTED)
74         g.add_edge(0, 1)
75         self.assertTrue(g.has_edge(0, 1))
76         self.assertFalse(g.has_edge(1, 0))
77         self.assertEqual(g.edge_count(), 1)
78         self.assertEqual(g.out_degree(0), 1)
79         self.assertEqual(g.in_degree(1), 1)
80         self.assertEqual(g.in_degree(0), 0)
81         self.assertEqual(g.degree(0), 1)
82         self.assertEqual(g.degree(1), 1)
83
84     def test_directed_multiplicities_are_independent(self):
85         g = Graph(3, MULTI_DIRECTED)
86         g.add_edge(0, 1, 2)
87         g.add_edge(1, 0, 1)
88         self.assertEqual(g.multiplicity(0, 1), 2)
89         self.assertEqual(g.multiplicity(1, 0), 1)
90         self.assertEqual(g.edge_count(), 3)
91         self.assertEqual(g.out_degree(0), 2)
92         self.assertEqual(g.in_degree(0), 1)
93         self.assertEqual(g.degree(0), 3)
94
95     def test_edges_yields_both_arc_directions(self):
96         g = Graph(3, SIMPLE_DIRECTED)
97         g.add_edge(0, 1)
98         g.add_edge(1, 0)
99         self.assertEqual(sorted(g.edges()), [(0, 1, 1), (1, 0, 1)])
100
101
102 class Mutation(unittest.TestCase):
103     def test_remove_never_goes_below_zero(self):
104         g = Graph(3, MULTI_UNDIRECTED)
105         g.add_edge(0, 1, 3)
106         g.remove_edge(0, 1, 5)
107         self.assertEqual(g.multiplicity(0, 1), 0)
108         self.assertFalse(g.has_edge(0, 1))
109
110     def test_set_multiplicity_directly(self):
111         g = Graph(3, MULTI_DIRECTED)
112         g.set_multiplicity(0, 1, 4)
113         self.assertEqual(g.multiplicity(0, 1), 4)
114         self.assertEqual(g.multiplicity(1, 0), 0)
115
116     def test_simple_rejects_multiplicity_above_one(self):
117         g = Graph(3, SIMPLE_UNDIRECTED)
118         with self.assertRaises(ValueError):
119             g.set_multiplicity(0, 1, 2)
120
121     def test_negative_multiplicity_rejected(self):
122         g = Graph(3, MULTI_UNDIRECTED)
123         with self.assertRaises(ValueError):
124             g.set_multiplicity(0, 1, -1)
125

```

```

126     def test_self_loops_rejected(self):
127         g = Graph(3, MULTI_UNDIRECTED)
128         with self.assertRaises(ValueError):
129             g.add_edge(1, 1)
130
131     def test_copy_is_independent(self):
132         g = Graph(3, MULTI_UNDIRECTED)
133         g.add_edge(0, 1, 2)
134         clone = g.copy()
135         clone.add_edge(0, 2, 1)
136         self.assertEqual(g.edge_count(), 2)
137         self.assertEqual(clone.edge_count(), 3)
138
139
140 if __name__ == "__main__":
141     unittest.main()

```

C.4.3 The closed-form bounds

```

1  """The closed-form bounds library: every formula, on a range of inputs."""
2
3  import unittest
4
5  from erdos915_unified import (
6      directed_arc_lower_bound,
7      directed_arc_m2,
8      directed_multigraph_arc,
9      hypergraph_edge,
10     multigraph_undirected_edge,
11     simple_undirected_edge,
12     simple_undirected_vertex_le4,
13     simple_undirected_vertex_m5,
14 )
15
16
17 class Bounds(unittest.TestCase):
18     def test_simple_undirected_edge(self):
19         self.assertEqual(simple_undirected_edge(5, 3), 6)
20         for n in range(2, 12):
21             for m in range(2, 6):
22                 self.assertEqual(simple_undirected_edge(n, m), (m * (n - 1)) // 2)
23
24     def test_multigraph_undirected_edge(self):
25         for n in range(2, 12):
26             for m in range(2, 6):
27                 self.assertEqual(multigraph_undirected_edge(n, m), (m - 1) * (n -
28                                     ↪ 1))
29
30     def test_vertex_le4_matches_edge_and_caps_at_four(self):
31         for m in range(2, 5):
32             self.assertEqual(simple_undirected_vertex_le4(10, m),
33                             ↪ simple_undirected_edge(10, m))
34         with self.assertRaises(ValueError):
35             simple_undirected_vertex_le4(10, 5)
36
37     def test_vertex_m5(self):
38         for n in range(6, 20):

```

```

37         self.assertEqual(simple_undirected_vertex_m5(n), (8 * n) // 3 - 3)
38
39     def test_directed_arc_m2_branches_and_crossover(self):
40         self.assertEqual(directed_arc_m2(4), 6)      # linear hub branch wins
41         self.assertEqual(directed_arc_m2(7), 12)     # the two branches meet
42         self.assertEqual(directed_arc_m2(8), 16)     # quadratic branch wins
43         for n in range(2, 20):
44             self.assertEqual(directed_arc_m2(n), max(2 * (n - 1), (n * n) // 4))
45
46     def test_directed_arc_lower_bound(self):
47         self.assertEqual(directed_arc_lower_bound(10, 3), 30) # the 30-arc graph
48         self.assertEqual(directed_arc_lower_bound(4, 2), 6)
49         for n in range(3, 14):
50             for m in range(2, 5):
51                 expected = max(m * (n - 1), (n + m - 2) ** 2 // 4)
52                 self.assertEqual(directed_arc_lower_bound(n, m), expected)
53
54     def test_hypergraph_edge(self):
55         self.assertEqual(hypergraph_edge(7, 2, 3), 3)
56         self.assertEqual(hypergraph_edge(13, 2, 4), 4)
57         for n in range(3, 14):
58             for m in range(2, 5):
59                 for r in range(2, 5):
60                     self.assertEqual(hypergraph_edge(n, m, r), ((m - 1) * (n - 1))
61                                     ↪ // (r - 1))
62
63     def test_directed_multigraph_arc(self):
64         self.assertEqual(directed_multigraph_arc(4, 3), 12)
65         self.assertEqual(directed_multigraph_arc(6, 2), 10)
66         for n in range(2, 12):
67             for m in range(2, 6):
68                 expected = max(2 * (n - 1) * (m - 1), (m - 1) * ((n * n) // 4))
69                 self.assertEqual(directed_multigraph_arc(n, m), expected)
70
71 if __name__ == "__main__":
72     unittest.main()

```

C.4.4 The named constructions

```

1  """The named extremal constructions: their exact sizes and connectivities."""
2
3  import unittest
4
5  from erdos915_unified import (
6      augmented_bipartite,
7      clique_tree,
8      double_star,
9      max_edge_connectivity,
10     min_vertex_connectivity,
11     one_directional_bipartite,
12 )
13
14
15 class Constructions(unittest.TestCase):
16     def test_double_star_directed(self):
17         for n in range(2, 7):

```

```

18         for m in range(2, 5):
19             g = double_star(n, m, directed=True)
20             self.assertEqual(g.edge_count(), 2 * (n - 1) * (m - 1))
21             self.assertEqual(max_edge_connectivity(g), m - 1)
22
23     def test_double_star_undirected(self):
24         for n in range(2, 7):
25             for m in range(2, 5):
26                 g = double_star(n, m, directed=False)
27                 self.assertEqual(g.edge_count(), (m - 1) * (n - 1))
28                 self.assertEqual(max_edge_connectivity(g), m - 1)
29
30     def test_one_directional_bipartite_is_quadratic_and_feasible(self):
31         for n in range(2, 8):
32             g = one_directional_bipartite(n)
33             self.assertEqual(g.edge_count(), (n * n) // 4)
34             self.assertEqual(max_edge_connectivity(g), 1)
35
36     def test_augmented_bipartite_conjecture_values(self):
37         self.assertEqual(augmented_bipartite(10, 3).edge_count(), 30)
38         for n, m in [(8, 3), (10, 4), (12, 4), (10, 5)]:
39             g = augmented_bipartite(n, m)
40             self.assertEqual(g.edge_count(), (n + m - 2) ** 2 // 4)
41             self.assertEqual(max_edge_connectivity(g), m - 1)
42
43     def test_augmented_bipartite_guards_its_precondition(self):
44         with self.assertRaises(ValueError):
45             augmented_bipartite(3, 5) # needs n >= m
46
47     def test_clique_tree_is_globally_connected(self):
48         for blocks in range(0, 5):
49             self.assertEqual(min_vertex_connectivity(clique_tree(4, blocks)), 3)
50
51
52 if __name__ == "__main__":
53     unittest.main()

```

C.4.5 The connectivity checker

```

1  """The connectivity checker (edge and vertex) and the Gomory-Hu shortcut."""
2
3  import random
4  import unittest
5
6  from erdos915_unified import (
7      Graph,
8      MULTI_DIRECTED,
9      MULTI_UNDIRECTED,
10     NETWORKX_AVAILABLE,
11     SIMPLE_DIRECTED,
12     SIMPLE_UNDIRECTED,
13     _vertex_flow_at_least,
14     exceeds_bound,
15     local_edge_connectivity,
16     local_vertex_connectivity,
17     max_connectivity,
18     max_edge_connectivity,

```

```

19     max_edge_connectivity_via_tree,
20     max_vertex_connectivity,
21     sample_random_graph,
22 )
23
24
25 def random_multigraph(variant, n, rng, max_mult=2):
26     """A random graph of ‘variant’ on ‘n’ vertices for predicate testing.
27
28     Simple variants take multiplicities in ‘{0, 1}’; multigraph variants in
29     ‘{0, ..., max_mult}’. Directed variants fill every ordered pair, undirected
30     variants the upper triangle (‘set_multiplicity’ mirrors the lower half).
31     """
32     g = Graph(n, variant)
33     top = 1 if variant.simple else max_mult
34     for u in range(n):
35         start = 0 if variant.directed else u + 1
36         for v in range(start, n):
37             if u != v:
38                 g.set_multiplicity(u, v, rng.randint(0, top))
39     return g
40
41
42 def complete_graph(n, variant=SIMPLE_UNDIRECTED):
43     g = Graph(n, variant)
44     for u in range(n):
45         for v in range(u + 1, n):
46             g.add_edge(u, v)
47     return g
48
49
50 def star(n):
51     g = Graph(n, SIMPLE_UNDIRECTED)
52     for leaf in range(1, n):
53         g.add_edge(0, leaf)
54     return g
55
56
57 def cycle(n):
58     g = Graph(n, SIMPLE_UNDIRECTED)
59     for v in range(n):
60         g.add_edge(v, (v + 1) % n)
61     return g
62
63
64 class EdgeConnectivity(unittest.TestCase):
65     def test_tree_is_one(self):
66         self.assertEqual(max_edge_connectivity(star(6)), 1)
67
68     def test_cycle_is_two(self):
69         self.assertEqual(max_edge_connectivity(cycle(6)), 2)
70
71     def test_complete_graph(self):
72         for n in (3, 4, 5):
73             self.assertEqual(max_edge_connectivity(complete_graph(n)), n - 1)
74             self.assertEqual(local_edge_connectivity(complete_graph(n), 0, 1), n -
75                 ↪ 1)

```

```

76     def test_parallel_edges_count_as_routes(self):
77         g = Graph(3, MULTI_UNDIRECTED)
78         g.add_edge(0, 1, 4)
79         self.assertEqual(local_edge_connectivity(g, 0, 1), 4)
80         self.assertEqual(max_edge_connectivity(g), 4)
81
82     def test_directed_arc_disjoint_routes(self):
83         g = Graph(3, SIMPLE_DIRECTED)
84         g.add_edge(0, 1) # direct route
85         g.add_edge(0, 2) # detour 0 -> 2 -> 1
86         g.add_edge(2, 1)
87         self.assertEqual(local_edge_connectivity(g, 0, 1), 2)
88
89     def test_removing_an_edge_never_raises_lambda(self):
90         rng = random.Random(11)
91         for _ in range(20):
92             g = sample_random_graph(8, 0.4, False, rng)
93             before = max_edge_connectivity(g)
94             for u, v, _count in list(g.edges()):
95                 reduced = g.copy()
96                 reduced.set_multiplicity(u, v, 0)
97                 self.assertLessEqual(max_edge_connectivity(reduced), before)
98
99
100 class VertexConnectivity(unittest.TestCase):
101     def test_complete_graph(self):
102         for n in (3, 4, 5):
103             self.assertEqual(max_vertex_connectivity(complete_graph(n)), n - 1)
104             self.assertEqual(local_vertex_connectivity(complete_graph(n), 0, 1), n
105                 ↪ - 1)
106
107     def test_whitney_inequality_on_random_graphs(self):
108         rng = random.Random(0)
109         for _ in range(40):
110             g = sample_random_graph(12, 0.3, False, rng)
111             self.assertLessEqual(max_vertex_connectivity(g), max_edge_connectivity
112                 ↪ (g))
113
114 class VertexFlowPredicate(unittest.TestCase):
115     """'_vertex_flow_at_least' is the exhaustive search's inner vertex test.
116
117     It must agree with the exact checker it replaces on every (pair, k), or the
118     base cases of the directed vertex theorem would rest on a faster wrong test.
119     """
120
121     def test_matches_the_exact_checker(self):
122         rng = random.Random(17)
123         for _ in range(150):
124             n = rng.randint(2, 6)
125             g = Graph(n, SIMPLE_DIRECTED)
126             out = [set() for _ in range(n)]
127             for a in range(n):
128                 for b in range(n):
129                     if a != b and rng.random() < 0.35:
130                         g.mu[a, b] = 1
131                         out[a].add(b)
132             for s in range(n):

```

```

132         for t in range(n):
133             if s == t:
134                 continue
135             exact = local_vertex_connectivity(g, s, t)
136             for k in range(1, 5):
137                 self.assertEqual(_vertex_flow_at_least(out, n, s, t, k),
138                                 exact >= k)
139
140     def test_a_direct_arc_is_one_route_with_no_interior(self):
141         # s -> t plus the detour s -> x -> t: two internally disjoint routes.
142         out = [{1, 2}, set(), {1}, set()]
143         self.assertTrue(_vertex_flow_at_least(out, 4, 0, 1, 2))
144         self.assertFalse(_vertex_flow_at_least(out, 4, 0, 1, 3))
145
146
147 @unittest.skipUnless(NETWORKX_AVAILABLE, "Gomory-Hu trees need the optional
    ↳ networkx")
148 class GomoryHu(unittest.TestCase):
149     def test_tree_matches_direct_sweep(self):
150         for g in (cycle(6), complete_graph(5), star(7)):
151             self.assertEqual(max_edge_connectivity_via_tree(g),
    ↳ max_edge_connectivity(g))
152
153     def test_multigraph_matches(self):
154         g = Graph(4, MULTI_UNDIRECTED)
155         g.add_edge(0, 1, 3)
156         g.add_edge(1, 2, 2)
157         g.add_edge(2, 3, 1)
158         g.add_edge(0, 3, 2)
159         self.assertEqual(max_edge_connectivity_via_tree(g), max_edge_connectivity(
    ↳ g))
160
161
162 class CappedPredicate(unittest.TestCase):
163     """'exceeds_bound' must agree with the exact checker, pair set for pair set.
164
165     This pins the cheap capped predicate the search relies on to the trusted
166     exact 'max_connectivity' over every variant, separation and threshold.
167     """
168
169     def test_predicate_matches_exact_value(self):
170         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
171                     SIMPLE_DIRECTED, MULTI_DIRECTED)
172         rng = random.Random(2024)
173         for variant in variants:
174             for n in (3, 4, 5):
175                 for _ in range(8):
176                     g = random_multigraph(variant, n, rng)
177                     for sep in ("edge", "vertex"):
178                         split = sep == "vertex"
179                         exact = max_connectivity(g, vertex_split=split)
180                         for k in range(-1, exact + 2):
181                             self.assertEqual(
182                                 exceeds_bound(g, k, separation=sep),
183                                 exact > k,
184                                 msg=f"{variant.name} n={n} sep={sep} k={k} "
185                                     f"exact={exact}",
186                             )

```

```

187
188     def test_predicate_does_not_mutate_the_graph(self):
189         rng = random.Random(7)
190         g = random_multigraph(MULTI_DIRECTED, 5, rng)
191         before = g.mu.copy()
192         exceeds_bound(g, 1, separation="vertex")
193         exceeds_bound(g, 1, separation="edge")
194         self.assertTrue((g.mu == before).all())
195
196
197 if __name__ == "__main__":
198     unittest.main()

```

C.4.6 Hypergraphs

```

1  """The hypergraph model and Berge connectivity by the helper-point network."""
2
3  import unittest
4
5  from erdos915_unified import (
6      Hypergraph,
7      complete_uniform_hypergraph,
8      hyper_connectivity,
9      hyperedge_connectivity,
10     max_feasible_hyperedges,
11     max_hyperedge_connectivity,
12     star_hypertree,
13     _hyperedge_candidates,
14 )
15
16
17 class BergeConnectivity(unittest.TestCase):
18     def test_single_hyperedge_carries_one_route(self):
19         h = Hypergraph(3, [[0, 1, 2]])
20         self.assertEqual(h.edge_count(), 1)
21         self.assertEqual(hyperedge_connectivity(h, 0, 1), 1)
22         self.assertEqual(hyperedge_connectivity(h, 0, 2), 1)
23         self.assertEqual(hyperedge_connectivity(h, 1, 2), 1)
24
25     def test_two_hyperedges_give_two_routes(self):
26         h = Hypergraph(4, [[0, 1, 2], [0, 1, 3]])
27         self.assertEqual(h.edge_count(), 2)
28         self.assertEqual(hyperedge_connectivity(h, 0, 1), 2)
29         self.assertEqual(len(h.incident_hyperedges(0)), 2)
30         self.assertEqual(len(h.incident_hyperedges(2)), 1)
31
32     def test_complete_three_uniform_k4(self):
33         self.assertEqual(max_hyperedge_connectivity(complete_uniform_hypergraph(4,
34             ↪ 3)), 3)
35
36     def test_two_uniform_reduces_to_edge_connectivity(self):
37         for n in (3, 4, 5):
38             self.assertEqual(max_hyperedge_connectivity(
39                 ↪ complete_uniform_hypergraph(n, 2)), n - 1)
40
41     def test_star_hypertree(self):
42         for n, r in [(7, 3), (10, 4), (13, 4)]:

```



```

41         h = star_hypertree(n, r)
42         self.assertEqual(h.edge_count(), (n - 1) // (r - 1))
43         self.assertEqual(max_hyperedge_connectivity(h), 1)
44
45     def test_bad_hyperedge_size_rejected(self):
46         with self.assertRaises(ValueError):
47             star_hypertree(7, 1)
48
49
50 class DirectedOrientationModels(unittest.TestCase):
51     """Forward / backward / general directed hyperedge models."""
52
53     def test_general_gate_is_one_way(self):
54         # A mixed arc {0,1} -> {2,3}: a route enters at a tail, leaves at a head.
55         h = Hypergraph(4, [(frozenset({0, 1}), frozenset({2, 3}))], directed=True)
56         self.assertEqual(sorted(h.members(h.hyperedges[0])), [0, 1, 2, 3])
57         self.assertEqual(hyper_connectivity(h, 0, 2), 1) # tail -> head
58         self.assertEqual(hyper_connectivity(h, 2, 0), 0) # head -> tail: none
59
60     def test_forward_legacy_equals_general_form(self):
61         legacy = Hypergraph(3, [(0, frozenset({1, 2}))], directed=True)
62         general = Hypergraph(3, [(frozenset({0}), frozenset({1, 2}))], directed=
        ↪ True)
63         for s, t in [(0, 1), (0, 2), (1, 2)]:
64             self.assertEqual(hyper_connectivity(legacy, s, t),
65                             hyper_connectivity(general, s, t))
66
67     def test_candidate_counts_split_at_r3(self):
68         # At r = 3 every split has a singleton side, so general = forward +
69         ↪ backward.
70         fwd = _hyperedge_candidates(4, 3, True, kind="forward")
71         bwd = _hyperedge_candidates(4, 3, True, kind="backward")
72         gen = _hyperedge_candidates(4, 3, True, kind="general")
73         self.assertEqual(len(fwd), 12)
74         self.assertEqual(len(bwd), 12)
75         self.assertEqual(len(gen), 24)
76
77     def test_forward_backward_duality(self):
78         # Arc reversal makes backward the dual of forward: same extremal numbers.
79         for n, m in [(3, 3), (4, 3), (4, 2)]:
80             f, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="forward",
81             ↪ time_limit=2.0)
82             b, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="backward"
83             ↪ , time_limit=2.0)
84             self.assertEqual(f, b)
85
86     def test_general_beats_forward_when_vertices_scarce(self):
87         # n = r is the regime where the richer orientation set packs more arcs.
88         g3, e3 = max_feasible_hyperedges(3, 3, 3, directed=True, kind="general",
89         ↪ time_limit=2.0)
90         f3, _ = max_feasible_hyperedges(3, 3, 3, directed=True, kind="forward",
91         ↪ time_limit=2.0)
92         self.assertTrue(e3)
93         self.assertEqual((g3, f3), (4, 3))
94
95     def test_general_contains_forward(self):
96         # General can never be worse than forward (forward hypergraphs are general
97         ↪ ).

```

```

92         for n, m in [(4, 3), (5, 2)]:
93             f, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="forward",
94                 ↪ time_limit=2.0)
95             g, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="general",
96                 time_limit=3.0, seed_lb=f)
97             self.assertGreaterEqual(g, f)
98
99 if __name__ == "__main__":
100     unittest.main()

```

C.4.7 Edge sensitivity

```

1  """Edge sensitivity: the load-bearing measure that guides the search."""
2
3  import random
4  import unittest
5
6  from erdos915_unified import (
7      Graph,
8      MULTI_DIRECTED,
9      MULTI_UNDIRECTED,
10     SIMPLE_DIRECTED,
11     SIMPLE_UNDIRECTED,
12     double_star,
13     edge_sensitivity,
14     max_edge_connectivity,
15     max_vertex_connectivity,
16     one_directional_bipartite,
17     sensitivity_map,
18 )
19
20
21 def _random_graph(variant, n, rng, top):
22     g = Graph(n, variant)
23     cap = 1 if variant.simple else top
24     for u in range(n):
25         start = 0 if variant.directed else u + 1
26         for v in range(start, n):
27             if u != v:
28                 g.set_multiplicity(u, v, rng.randint(0, cap))
29     return g
30
31
32 class Sensitivity(unittest.TestCase):
33     def test_absent_edge_has_zero_sensitivity(self):
34         g = Graph(3, SIMPLE_UNDIRECTED)
35         self.assertEqual(edge_sensitivity(g, 0, 1), 0)
36
37     def test_sensitivity_matches_its_definition(self):
38         g = one_directional_bipartite(6)
39         before = max_edge_connectivity(g)
40         for (u, v), sigma in sensitivity_map(g).items():
41             reduced = g.copy()
42             reduced.set_multiplicity(u, v, 0)
43             self.assertEqual(sigma, before - max_edge_connectivity(reduced))
44             self.assertGreaterEqual(sigma, 0) # removing an edge never raises

```

```

        ↪ lambda
45
46     def test_map_covers_exactly_the_present_edges(self):
47         g = double_star(5, 3, directed=True)
48         self.assertEqual(len(sensitivity_map(g)), sum(1 for _ in g.edges()))
49
50     def test_map_equals_per_edge_sensitivity(self):
51         # The before-once refactor must return the IDENTICAL dict as the old
52         # per-edge definition, across variants and both connectivity measures.
53         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
54                     SIMPLE_DIRECTED, MULTI_DIRECTED)
55         rng = random.Random(99)
56         for variant in variants:
57             for measure in (max_edge_connectivity, max_vertex_connectivity):
58                 for _ in range(6):
59                     g = _random_graph(variant, 5, rng, top=2)
60                     expected = {(u, v): edge_sensitivity(g, u, v, measure)
61                                 for u, v, _ in g.edges()}
62                     self.assertEqual(sensitivity_map(g, measure), expected)
63
64
65 if __name__ == "__main__":
66     unittest.main()

```

C.4.8 The search

```

1  """The temperature search: it rediscovers the known small extremal values.
2
3  The search is seeded, so these outcomes are deterministic and reproducible.
4  """
5
6  import unittest
7
8  from erdos915_unified import (
9      MULTI_DIRECTED,
10     MULTI_UNDIRECTED,
11     SIMPLE_DIRECTED,
12     SIMPLE_UNDIRECTED,
13     best_of_searches,
14     max_connectivity,
15     max_edge_connectivity,
16     search_for_dense_graph,
17     tabu_search_for_dense_graph,
18 )
19
20
21 class Search(unittest.TestCase):
22     def test_search_rediscovers_directed_m2(self):
23         result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000,
24             ↪ seed=0)
25         self.assertEqual(result.best_edge_count, 6) # ell_2^dir(4) = 6
26         self.assertTrue(result.feasible_found)
27         self.assertLessEqual(max_edge_connectivity(result.best_graph), 1)
28
29     def test_acceptance_rate_is_a_fraction(self):
30         result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000,
31             ↪ seed=0)

```

```

30         self.assertTrue(0.0 <= result.acceptance_rate() <= 1.0)
31         self.assertEqual(len(result.history), 6000)
32
33     def test_search_rediscovered_undirected_m3(self):
34         #  $ell_3(n) = \text{floor}(3(n-1)/2)$ :  $n=4 \rightarrow 4$ ,  $n=5 \rightarrow 6$ .
35         for n, expected in [(4, 4), (5, 6)]:
36             result = best_of_searches(SIMPLE_UNDIRECTED, n, 3, restarts=6, steps
37                                     ↪ =3000, seed=0)
38             self.assertEqual(result.best_edge_count, expected)
39             self.assertEqual(max_edge_connectivity(result.best_graph), 2)
40
41 class TabuSearch(unittest.TestCase):
42     """Tabu search is the deterministic twin of the annealer: same energy and
43     neighbourhood, so it rediscovered the same small extremal values.
44     """
45
46     def test_tabu_rediscovered_directed_m2(self):
47         result = tabu_search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2,
48                                             steps=150, seed=0)
49         self.assertEqual(result.best_edge_count, 6) #  $ell_2^{dir}(4) = 6$ 
50         self.assertTrue(result.feasible_found)
51         self.assertEqual(max_edge_connectivity(result.best_graph), 1)
52
53     def test_tabu_rediscovered_undirected_m3(self):
54         result = tabu_search_for_dense_graph(SIMPLE_UNDIRECTED, n=4, m=3,
55                                             steps=150, seed=0)
56         self.assertEqual(result.best_edge_count, 4) #  $ell_3(4) = 4$ 
57         self.assertEqual(max_edge_connectivity(result.best_graph), 2)
58
59     def test_best_of_searches_accepts_tabu_method(self):
60         result = best_of_searches(SIMPLE_DIRECTED, 4, 2, restarts=2,
61                                 method="tabu", parallel=False, steps=120)
62         self.assertEqual(result.best_edge_count, 6)
63
64     def test_unknown_method_rejected(self):
65         with self.assertRaises(ValueError):
66             best_of_searches(SIMPLE_UNDIRECTED, 4, 3, method="genetic")
67
68
69 class FastPathEquivalence(unittest.TestCase):
70     """The fast (capped + monotone) search must reproduce the slow exact search
71     step for step. This is the "nothing under the rug" guarantee: the speedup
72     changes no reported value because it changes no single step of the walk.
73     """
74
75     def test_fast_path_matches_exact(self):
76         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
77                   SIMPLE_DIRECTED, MULTI_DIRECTED)
78         for variant in variants:
79             for n in (3, 4, 5):
80                 for m in (2, 3):
81                     for sep in ("edge", "vertex"):
82                         for seed in range(2):
83                             opts = dict(separation=sep, steps=300, seed=seed)
84                             ref = search_for_dense_graph(
85                                 variant, n, m, reference_mode=True, **opts)
86                             fast = search_for_dense_graph(variant, n, m, **opts)

```

```

87         tag = (f"{variant.name} n={n} m={m} sep={sep} "
88                f"seed={seed}")
89         # Bit-for-bit: same energies, same accept/reject
90         # decisions, same density at every step.
91         self.assertEqual(
92             [s.energy for s in fast.history],
93             [s.energy for s in ref.history], tag)
94         self.assertEqual(
95             [s.accepted for s in fast.history],
96             [s.accepted for s in ref.history], tag)
97         self.assertEqual(
98             [s.edge_count for s in fast.history],
99             [s.edge_count for s in ref.history], tag)
100         self.assertEqual(
101             fast.best_edge_count, ref.best_edge_count, tag)
102         self.assertTrue(
103             (fast.best_graph.mu == ref.best_graph.mu).all(),
104             tag)
105         # The returned witness is genuinely feasible by the
106         # exact checker (certified, not just flagged).
107         split = sep == "vertex"
108         self.assertLessEqual(
109             max_connectivity(fast.best_graph,
110                             vertex_split=split),
111             m - 1, tag)
112
113
114 if __name__ == "__main__":
115     unittest.main()

```

C.4.9 The prover

```

1  """The cut-counting prover: a zero-gap solve is a genuine upper-bound proof."""
2
3  import os
4  import unittest
5
6  import numpy as np
7
8  from erdos915_unified import (
9      MULTI_DIRECTED,
10     PULP_AVAILABLE,
11     Graph,
12     _canonical_form,
13     enumerate_extremal_directed_multigraphs,
14     max_edge_connectivity,
15     prove_directed_multigraph,
16 )
17
18
19 @unittest.skipUnless(PULP_AVAILABLE, "the MILP certifier needs the optional pulp")
20 class Prover(unittest.TestCase):
21     def test_small_optima_are_proved(self):
22         # M*(n) = 2(n-1), optimal with zero gap, for the reachable sizes.
23         for n in (3, 4, 5):
24             result = prove_directed_multigraph(n, time_limit=120.0)
25             self.assertTrue(result.is_proof())

```

```

26         self.assertEqual(result.status, "OPTIMAL")
27         self.assertEqual(round(result.scaled_optimum), 2 * (n - 1))
28
29     def test_one_solve_settles_every_m(self):
30         result = prove_directed_multigraph(4, time_limit=120.0)
31         self.assertTrue(result.is_proof())
32         for m in (2, 3, 4, 5):
33             #  $L_m^{\text{dir}}(4) = (m-1) * M(4) = (m-1) * 6$ .
34             self.assertEqual(result.value_for(m), (m - 1) * 2 * (4 - 1))
35
36
37 class EnumerationDedup(unittest.TestCase):
38     """The streamed canonical-form dedup keeps one representative per iso-class
39     and returns exactly the known small extremal sets.
40
41     The active path stays at n in {3, 4} (sub-second). The n=5 enumeration is
42     minutes long (the DFS, not the dedup), so its known count is checked only when
43     'RUN_SLOW_ENUM' is set; the dedup logic it exercises is identical to n=4.
44     """
45
46     # m=3 directed multigraph, target = 4(n-1) arcs: the doubled bidirected
47     # spanning trees. Known counts (claude.md): n=3 -> 1 (path), n=4 -> 2
48     # (star, path), n=5 -> 3 (star, broom, path).
49     FAST_COUNTS = {3: 1, 4: 2}
50
51     def test_known_iso_class_counts(self):
52         for n, expected in self.FAST_COUNTS.items():
53             reps = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
54             self.assertEqual(len(reps), expected, f"n={n}")
55
56     def test_representatives_are_distinct_and_feasible(self):
57         for n in self.FAST_COUNTS:
58             reps = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
59             keys = [_canonical_form(mu) for mu in reps]
60             # One representative per class: all canonical keys distinct.
61             self.assertEqual(len(set(keys)), len(keys), f"n={n}")
62             for mu in reps:
63                 g = Graph(n, MULTI_DIRECTED)
64                 g.mu = mu.copy()
65                 self.assertEqual(g.edge_count(), 4 * (n - 1))
66                 self.assertEqual(max_edge_connectivity(g), 2, f"n={n}")
67
68     def test_dedup_preserves_the_iso_classes(self):
69         # The deduped run must cover exactly the iso-classes of the full labelled
70         # run: no class invented, none dropped.
71         for n in self.FAST_COUNTS:
72             labelled = enumerate_extremal_directed_multigraphs(
73                 n, 3, 4 * (n - 1), up_to_iso=False)
74             deduped = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
75             self.assertEqual(
76                 {_canonical_form(mu) for mu in labelled},
77                 {_canonical_form(mu) for mu in deduped},
78                 f"n={n}")
79             # Every labelled matrix is itself one of the kept iso-classes.
80             kept = {_canonical_form(mu) for mu in deduped}
81             for mu in labelled:
82                 self.assertIn(_canonical_form(mu), kept, f"n={n}")
83

```

```

84     def test_canonical_form_is_permutation_invariant(self):
85         rng = np.random.default_rng(0)
86         for n in (3, 4, 5, 6):
87             mu = rng.integers(0, 3, size=(n, n))
88             np.fill_diagonal(mu, 0)
89             key = _canonical_form(mu)
90             for _ in range(5):
91                 p = rng.permutation(n)
92                 self.assertEqual(_canonical_form(mu[np.ix_(p, p)]), key, f"n={n}")
93
94     @unittest.skipUnless(os.environ.get("RUN_SLOW_ENUM"),
95                          "n=5 enumeration is minutes long; set RUN_SLOW_ENUM=1")
96     def test_n5_known_count(self):
97         reps = enumerate_extremal_directed_multigraphs(5, 3, 16)
98         self.assertEqual(len(reps), 3) # star, broom, path
99
100
101 if __name__ == "__main__":
102     unittest.main()

```

C.4.10 The random model

```

1  """The random model: the appearance threshold and the Whitney relation.
2
3  All sampling is seeded, so these outcomes are deterministic.
4  """
5
6  import unittest
7
8  from erdos915_unified import (
9      connectivity_distribution,
10     edge_vertex_distribution,
11     estimate_appearance_probability,
12     threshold_curve,
13 )
14
15
16 class MonteCarlo(unittest.TestCase):
17     def test_appearance_probability_crosses_m_over_n(self):
18         n, m = 30, 3
19         below = estimate_appearance_probability(n, 0.3 * m / n, m, trials=120,
20         ↪ seed=1)
21         above = estimate_appearance_probability(n, 1.8 * m / n, m, trials=120,
22         ↪ seed=1)
23         self.assertLess(below, 0.5)
24         self.assertGreater(above, 0.5)
25
26     def test_whitney_holds_on_every_sample(self):
27         edge = connectivity_distribution(20, 0.25, trials=40, separation="edge",
28         ↪ seed=3)
29         vertex = connectivity_distribution(20, 0.25, trials=40, separation="vertex
30         ↪ ", seed=3)
31         self.assertEqual(len(edge), 40)
32         for kappa, lam in zip(vertex, edge):
33             self.assertLessEqual(kappa, lam)
34
35     def test_edge_and_vertex_agree_at_small_n(self):

```

```

32     lam, kap = edge_vertex_distribution(5, 0.5, trials=120, seed=7)
33     self.assertEqual(sum(k < 1 for k, l in zip(kap, lam)), 0) # no gap at n =
        ↪ 5
34     self.assertTrue(all(k <= 1 for k, l in zip(kap, lam))) # Whitney
35
36     def test_edge_and_vertex_part_company_at_larger_n(self):
37         lam, kap = edge_vertex_distribution(16, 0.25, trials=200, seed=7)
38         gap = sum(k < 1 for k, l in zip(kap, lam))
39         self.assertGreater(gap, 0)
40         self.assertTrue(all(k <= 1 for k, l in zip(kap, lam)))
41
42     def test_threshold_curve_reports_the_predicted_threshold(self):
43         curve = threshold_curve(20, 2, [0.05, 0.2], trials=20, seed=0)
44         self.assertAlmostEqual(curve.predicted_threshold, 2 / 20)
45         self.assertEqual(len(curve.probabilities), 2)
46
47
48 if __name__ == "__main__":
49     unittest.main()

```

C.4.11 The solver

```

1  """The single solver: one solve() call handles every kind of question."""
2
3  import shutil
4  import unittest
5
6  import itertools
7
8  import numpy as np
9
10 from erdos915_unified import (
11     solve,
12     enumerate_extremal_directed_multigraphs as _dfs_enum,
13     enumerate_extremal_directed_multigraphs_via_generation as _gen_enum,
14     _decorate_support_worker,
15     _canonical_form,
16     PULP_AVAILABLE,
17 )
18
19
20 class Solve(unittest.TestCase):
21     def test_exhaustive_simple_directed_is_exact(self):
22         r = solve(4, 2, directed=True, simple=True, exhaustive=True, max_seconds
            ↪ =120.0)
23         self.assertTrue(r.proven)
24         self.assertEqual(r.bound, "exact")
25         self.assertEqual(r.value, 6)
26
27     def test_discovery_returns_a_lower_bound(self):
28         # verification stays on sa; tabu is the default for solving the problems
29         r = solve(4, 2, directed=True, simple=True, exhaustive=False,
30                 max_seconds=3.0, method="sa")
31         self.assertEqual(r.bound, "lower")
32         self.assertEqual(r.value, 6)
33
34     def test_exhaustive_undirected_finds_the_spanning_tree(self):

```



```

35     r = solve(5, 2, directed=False, simple=True, exhaustive=True, max_seconds
        ↪ =30.0)
36     self.assertTrue(r.proven)
37     self.assertEqual(r.value, 4) # a spanning tree on 5 vertices
38
39     @unittest.skipUnless(PULP_AVAILABLE,
40                           "solve() proves this one through the MILP certifier, "
41                           "which needs the optional pulp")
42     def test_directed_multigraph_is_proved(self):
43         r = solve(4, 3, directed=True, simple=False, exhaustive=True, max_seconds
        ↪ =120.0)
44         self.assertTrue(r.proven)
45         self.assertEqual(r.value, 12) #  $L_3^{dir}(4) = 2(n-1)(m-1) = 12$ 
46
47     def test_vertex_separation_undirected(self):
48         #  $k_2(n) = n - 1$ : no cycle is allowed, so a spanning tree is extremal.
49         r = solve(5, 2, directed=False, simple=True, separation="vertex",
50                 exhaustive=True, max_seconds=30.0)
51         self.assertTrue(r.proven)
52         self.assertEqual(r.value, 4)
53
54     def test_hypergraph_discovery(self):
55         r = solve(7, 2, hypergraph=True, r=3, exhaustive=False, max_seconds=3.0,
56                 method="sa")
57         self.assertEqual(r.bound, "lower")
58         self.assertEqual(r.value, 3)
59
60     def test_result_describe_is_readable(self):
61         r = solve(4, 2, directed=True, simple=True, exhaustive=True, max_seconds
        ↪ =120.0)
62         self.assertIn("value", r.describe())
63
64
65     @unittest.skipIf(shutil.which("geng") is None, "nauty's geng is not installed")
66     class GengGeneration(unittest.TestCase):
67         """The geng-seeded enumerator must return exactly the same isomorphism
68         classes as the DFS enumerator wherever the DFS one is a sound complete
69         search ( $n \leq 6$ ). This pins down the new generation pipeline against the
70         method whose values are already in the thesis."""
71
72         @staticmethod
73         def _classes(reps):
74             return {_canonical_form(mu) for mu in reps}
75
76     def test_matches_dfs_at_n4(self):
77         #  $n = 4$ ,  $m = 3$ : full target range and a degree cap. Fast, and the DFS
78         # enumerator is provably complete here, so equality is the gold standard.
79         for target, cap in [(12, None), (10, None), (8, None), (8, 6), (6, None)]:
80             with self.subTest(target=target, cap=cap):
81                 a = self._classes(_dfs_enum(4, 3, target, max_degree=cap))
82                 b = self._classes(_gen_enum(4, 3, target, max_degree=cap))
83                 self.assertEqual(a, b)
84
85     def test_extremal_is_the_doubled_spanning_trees(self):
86         # At  $n \leq 6$  the extremal directed multigraphs are exactly the doubled
87         # bidirected spanning trees, one per unlabelled tree: 2 on 4 vertices.
88         reps = _gen_enum(4, 3, 12) #  $L_3^{dir}(4) = 2(n-1)(m-1) = 12$ 
89         self.assertEqual(len(self._classes(reps)), 2)

```

```

90
91
92 class SupportWorker(unittest.TestCase):
93     """The per-support decorator is the unit the generation enumerator fans out
94     across processes. Over every support it must reproduce the DFS classes, so
95     the parallel refactor is covered here even without geng on PATH."""
96
97     @staticmethod
98     def _all_supports(n, min_e, max_e):
99         # Every non-isomorphic simple graph on n vertices in the edge range, the
100         # output geng would feed the enumerator, by brute force + canonical dedup.
101         pairs = [(i, j) for i in range(n) for j in range(i + 1, n)]
102         seen, out = set(), []
103         for mask in itertools.product((0, 1), repeat=len(pairs)):
104             if not (min_e <= sum(mask) <= max_e):
105                 continue
106             mu = np.zeros((n, n), dtype=int)
107             edges = []
108             for (i, j), bit in zip(pairs, mask):
109                 if bit:
110                     mu[i, j] = mu[j, i] = 1
111                     edges.append((i, j))
112             key = _canonical_form(mu)
113             if key not in seen:
114                 seen.add(key)
115                 out.append(edges)
116         return out
117
118     def test_worker_union_matches_dfs_at_n4(self):
119         n, m, target = 4, 3, 12
120         supports = self._all_supports(n, (target + 3) // 4, min(target, n * (n -
121             ↪ 1) // 2))
122         reps = []
123         for edges in supports:
124             reps.extend(_decorate_support_worker((n, m, target, None, True, edges)
125             ↪ ))
126         worker = {_canonical_form(mu) for mu in reps}
127         dfs = {_canonical_form(mu) for mu in _dfs_enum(n, m, target)}
128         self.assertEqual(worker, dfs)
129
130 if __name__ == "__main__":
131     unittest.main()

```

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